The arithmetic geometry of elliptic curves (DRAFT)

Math 714, Fall 2026

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About these notes. These notes accompany Math 714 and are a work in progress: sections are written and revised during the semester, and some are not yet finished.

Sections marked **DRAFT** are written but not yet proofread, and sections marked **TODO** still have parts to be written; beside each mark is the date the section last changed.

Last updated 9 Nov 2026.

Preface

These are the notes for Math 714, a first graduate course on the arithmetic of elliptic curves. The main reference is [Sil09], which we follow closely from Chapter 4 on.

Prerequisites

We assume the first-year graduate courses in algebra and complex analysis. More precisely, the reader should be comfortable with:

- **Algebra.** Finite and infinite Galois theory, including profinite groups and the Krull topology; modules over a principal ideal domain; basic commutative algebra (localization, integral extensions, discrete valuation rings).
- **Algebraic number theory.** Number fields and p-adic fields and their rings of integers; Dedekind domains; the decomposition of primes; and the finiteness of the class group and of the unit rank.
- **Complex analysis.** Holomorphic and meromorphic functions, Liouville's theorem, the residue theorem, and analytic continuation. Chapter 1 builds the theory of Riemann surfaces it needs.

No algebraic geometry is assumed. In the spirit of [Poo06], we use the language of scheme theory and explain how to think concretely about scheme-theoretic constructions in terms of commutative algebra. Chapters 2 and 3 collect what we use from the theory of algebraic varieties and algebraic curves.

Notation and conventions

Rings are commutative with 1. We write $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$ and $\mathbb{C}$ as usual, $\mathbb{F}_q$ for the field with q elements, $\mathbb{Z}_p$ and $\mathbb{Q}_p$ for the p-adic integers and numbers, and $\#S$ or $|S|$ for the cardinality of a set S.

Fields. We do not assume that fields are perfect, nor that residue field extensions are separable. We assume it only case by case, where it makes the theory considerably simpler, and we always say so. Each chapter states its notation and standing hypotheses in a Situation at its start. For a field k, we fix an algebraic closure $\overline{k}$, and write k^{sep} for the separable closure of k in $\overline{k}$. We write $\mathrm{Gal}_k := \mathrm{Gal}(k^{\mathrm{sep}}/k)$ for the absolute Galois group of k. The group Gal_k acts on $X(k^{\mathrm{sep}})$. Since restriction $\mathrm{Aut}(\overline{k}/k) \to \mathrm{Gal}_k$ is an isomorphism, this also determines the action on $X(\overline{k})$. Elements of Galois groups are σ, τ, γ, and they act on points through their coordinates, written $\sigma(Q)$. The letters g, h are kept for elements of an abstract group. The set of places of a global field k is $\mathcal{P}_k$.

Varieties. A k-variety is a separated scheme of finite type over k. We do not assume varieties to be integral or reduced. A curve is an integral k-variety of dimension 1, and a curve is nice if it is smooth, projective, and geometrically irreducible.

Groups. $\mathbb{G}_{\mathrm{a}}$ and $\mathbb{G}_{\mathrm{m}}$ are the additive and multiplicative groups, and $\mathbb{G}_{\mathrm{m}}^{(d)}$ is the twisted multiplicative group, the norm-one elements of a quadratic algebra L_d (never $\mathbb{G}_m[d]$, which would read as a torsion subgroup).

Local fields. For a non-archimedean place v, k_v^{unr} is the maximal unramified extension of k_v in k_v^{sep}, and $I_v \subseteq \mathrm{Gal}_{k_v}$ is the inertia group.

Coordinates. Upper case letters are homogeneous coordinates on projective space, lower case letters are affine coordinates, and the two are related by dehomogenization. On the standard patch $U_0 = \{X_0 \neq 0\}$ of $\mathbb{P}^n$ we write $x_i = X_i/X_0$. For a plane curve in $\mathbb{P}^2$ with coordinates X, Y, Z the affine chart is $Z \neq 0$, with $x = X/Z$ and $y = Y/Z$. The same rule applies to the names of polynomials: F is homogeneous and f is its dehomogenization.

Elliptic curves. For an elliptic curve E over k and an integer m, $E[m]$ is the group scheme given by the kernel of multiplication by m, and $E[m](\overline{k})$ is the m-torsion subgroup of $E(\overline{k})$. Likewise, $\mu_m(\overline{k})$ is the group of m-th roots of unity in $\overline{k}^\times$. For a nice curve C we write $\operatorname{Div} C$ and $\operatorname{Pic} C$ for the divisor group and the Picard group, and $\operatorname{Div}^0 C$, $\operatorname{Pic}^0 C$ for their degree-zero parts. Isogenies in $\operatorname{Hom}(E_1, E_2)$, $\operatorname{End}(E)$ and $\operatorname{Aut}(E)$ are defined over the ground field, and $\operatorname{Hom}(E_{1,\overline{k}}, E_{2,\overline{k}})$ denotes those defined over $\overline{k}$. For N prime to the characteristic, $\bar{\rho}_{E,N}$ is the mod N Galois representation on $E[N](k^{\mathrm{sep}})$, and $\rho_{E,\ell}$ is the ℓ-adic one on the Tate module $T_\ell(E)$.

Frobenius. In characteristic p, F is the relative Frobenius morphism and V is its dual, the Verschiebung. Here F^r is the p^r-th power Frobenius $E \to E^{(p^r)}$, and over $\mathbb{F}_q$ with $q = p^e$ the q-th power Frobenius endomorphism is F^e, also written F when the ground field is fixed.

The table below, based on [Poo06, Warning 1.9.6], compares our notation with Silverman's, for a nice curve C and elliptic curves E_1, E_2 over k.

Our notation	Notation in [Sil09]
k any field	k perfect ($k^{\mathrm{sep}} = \overline{k}$)
$\operatorname{Div} C$	$\operatorname{Div}_k(C)$
$\operatorname{Div} C_{k^{\mathrm{sep}}}$	$\operatorname{Div}(C)$
$\operatorname{Pic} C$	No notation for this.
$\operatorname{Pic} C_{k^{\mathrm{sep}}}$	$\operatorname{Pic}(C)$
$(\operatorname{Pic} C_{k^{\mathrm{sep}}})^{\mathrm{Gal}_k}$	$\operatorname{Pic}_k(C)$
$\operatorname{Hom}(E_1, E_2)$	$\operatorname{Hom}_k(E_1, E_2)$
$\operatorname{Hom}(E_{1,k^{\mathrm{sep}}}, E_{2,k^{\mathrm{sep}}})$	$\operatorname{Hom}(E_1, E_2)$
$\operatorname{End}(E)$, $\operatorname{Aut}(E)$	$\operatorname{End}_k(E)$, $\operatorname{Aut}_k(E)$
$\operatorname{End}(E_{k^{\mathrm{sep}}})$	$\operatorname{End}(E)$
$E[m](k^{\mathrm{sep}})$	$E[m]$
$\mathrm{Gal}_k = \operatorname{Gal}(k^{\mathrm{sep}}/k)$	$G_{\overline{k}/k} = \operatorname{Gal}(\overline{k}/k)$
F^r, V^r	ϕ_r, $\hat{\phi}_r$

A road map

The notes fall into four parts.

- **Foundations (Chapters 1–4).** Chapter 1 treats elliptic curves over $\mathbb{C}$ as complex tori $\mathbb{C}/\Lambda$, as a source of geometric intuition for what follows. Chapters 2 and 3 develop varieties and curves over an arbitrary field, and Chapter 4 uses them to set up the general theory: Weierstrass equations, the group law, isogenies, the Tate module and the Weil pairing.
- **Local theory (Chapters 5 and 6).** Elliptic curves over finite fields, including the Hasse bound, and over local fields: minimal models, reduction and the formal group.
- **Global theory (Chapters 7–10).** Galois cohomology, the Mordell–Weil theorem, integral points; twists, Selmer groups, and the Tate–Shafarevich group.
- **Further topics (Chapters 11–15).** Complex multiplication, the Tate curve, the L-function and the conjecture of Birch and Swinnerton-Dyer, the Sato–Tate conjecture, and an introduction to modular curves.

A thread running through the notes is the comparison of an elliptic curve with the multiplicative group $\mathbb{G}_{\mathrm{m}}$: over $\mathbb{C}$, the torus $\mathbb{C}/\Lambda$ against $\mathbb{C}^\times = \mathbb{C}/2\pi i\mathbb{Z}$; over an arbitrary field, their invariant differentials, torsion points and Tate modules; and in Chapter 12, the Tate curve, which realizes an elliptic curve over a p-adic field as a quotient of $\mathbb{G}_{\mathrm{m}}$.

Chapter 1 can be read independently of Chapters 2 and 3, and Chapter 4 relies on Chapters 2 and 3. After Chapter 4, the main line runs through Chapters 5 to 8. Chapters 9 and 10 build on Chapter 8, and each of the further topics can be read after the chapters it cites. The worksheets for the problem sessions, with their solutions, are linked from the web page of the notes. Each one says which sections it draws on.

Origin of these notes

I taught Math 714 (The Arithmetic of Elliptic Curves) at UMass in the fall of 2026. I prepared hand written notes ahead of each section, scanned them, and used Claude to transform them into this format.

DRAFT
15 Sep 2026

Acknowledgements

This text owes an enormous intellectual debt to Silverman's book [Sil09] and Poonen's notes [Poo06]. I also benefited greatly from Milne's excellent book [Mil20].

DRAFT
13 Oct 2026

AI disclosure

Throughout this project, we used Claude code for writing code, carrying out computations, formatting, and proofreading. All mathematical results (beyond the draft stage) in this manuscript, as well as the AI-generated code and computations verifying them, have been independently verified by the author, who takes full responsibility for the accuracy and content of this work.

CHAPTER 1

Elliptic curves as Riemann surfaces

Sources. This chapter follows [Mir95] (parts of chapters I, II and IV) and [DS05] (parts of Chapter I).

The main goal of this chapter is to introduce the category of **complex tori** and to develop their theory from the complex analytic point of view. Students without previous experience in algebraic geometry can use it as a source of geometric intuition when we turn to the algebraic theory later on.

SITUATION 1.0.1. There is no ground field in this chapter: everything is complex analytic. Riemann surfaces are connected (Definition 1.1.2), and Λ, Λ_1, Λ_2 denote lattices in $\mathbb{C}$ (Definition 1.2.1).

1.1. Riemann surfaces

Intuitively, a Riemann surface is a topological space that looks like $\mathbb{C}$ near each of its points. We make this intuition more precise.

DEFINITION 1.1.1. Let X be a topological space.

(1) A chart in X is a homeomorphism $\phi\colon U \to V$ between open subsets $U \subseteq X$, $V \subseteq \mathbb{C}$.
(2) For any two charts ϕ_1 and ϕ_2 we obtain two transition maps
$$\phi_2 \circ \phi_1^{-1}\colon \phi_1(U_1 \cap U_2) \to \phi_2(U_1 \cap U_2), \qquad \phi_1 \circ \phi_2^{-1}\colon \phi_2(U_1 \cap U_2) \to \phi_1(U_1 \cap U_2).$$
(3) Two charts ϕ_1 and ϕ_2 are compatible if either $U_1 \cap U_2 = \varnothing$, or the transition maps are holomorphic.
(4) A complex structure on X is a family $\{\phi_i\}_i$ of compatible charts such that $X = \bigcup_i U_i$.

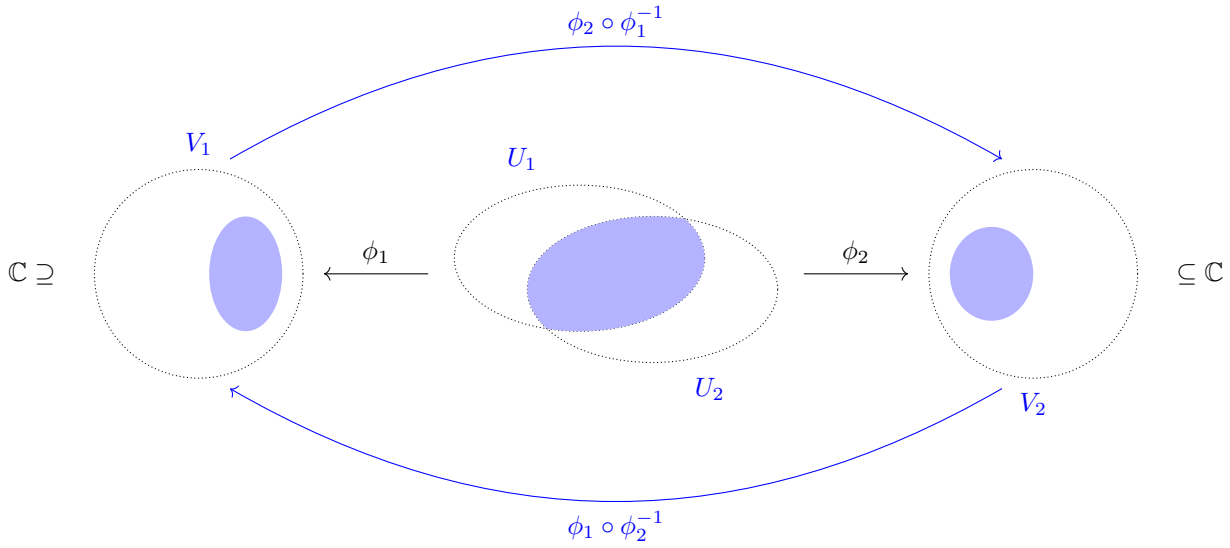


FIGURE 1.1.1. Two charts on overlapping open sets, and the transition maps between their images.

A topological space is connected if it is not the union of two disjoint non-empty open subsets, and Hausdorff if any two distinct points of the space have disjoint open neighborhoods.

DEFINITION 1.1.2. A Riemann surface is a connected Hausdorff topological space with a complex structure.

Recall that a topological space is second countable if it admits a countable basis for its topology. For Riemann surfaces this is automatic, by a theorem of Radó [For91, Theorem 23.3], but it is part of the definition of a manifold in general: the long line is connected, Hausdorff and locally homeomorphic to $\mathbb{R}$, and not second countable.

The simplest examples are open subsets of $\mathbb{C}$.

EXAMPLE 1.1.3. Every connected open $U \subseteq \mathbb{C}$ is a Riemann surface, with the complex structure consisting of the single chart $\mathrm{id}\colon U \to U$.

More interestingly, the one-point compactification of $\mathbb{C}$ is a Riemann surface.

EXAMPLE 1.1.4. Define

$$\mathbb{P}^1(\mathbb{C}) \coloneqq \big(\mathbb{C}^2 - \{(0,0)\}\big)/\mathbb{C}^\times,$$

with the quotient topology. This is the Riemann sphere, named after the sphere because it is isomorphic to the sphere S^2 with the complex structure given by stereographic projection (Example 1.1.9). A point of $\mathbb{P}^1(\mathbb{C})$ is a line through the origin:

$$(X_0 : X_1) \coloneqq \{(\lambda X_0, \lambda X_1) : \lambda \in \mathbb{C}^\times\}.$$

The subsets $U_i \coloneqq \{(X_0 : X_1) \in \mathbb{P}^1(\mathbb{C}) : X_i \neq 0\}$, for $i = 0, 1$, are open, and $\mathbb{P}^1(\mathbb{C}) = U_0 \cup U_1$. The maps

$$\phi_0\colon U_0 \to \mathbb{C}, \quad (X_0 : X_1) \mapsto X_1/X_0, \qquad \phi_1\colon U_1 \to \mathbb{C}, \quad (X_0 : X_1) \mapsto X_0/X_1,$$

are homeomorphisms. They are compatible: $\phi_0(U_0 \cap U_1) = \mathbb{C}^\times$, and on it

$$(\phi_1 \circ \phi_0^{-1})(w) = \phi_1(1 : w) = 1/w$$

is holomorphic. We identify $\mathbb{C}$ with U_0 via ϕ_0, and write $\infty \coloneqq (0 : 1)$ for the one remaining point of $\mathbb{P}^1(\mathbb{C})$.

We use charts to extend the notion of a holomorphic map between open subsets of $\mathbb{C}$ to Riemann surfaces.

DEFINITION 1.1.5. A holomorphic map of Riemann surfaces is a continuous function $f\colon X \to Y$ such that for any two charts $\phi_1\colon U_1 \to V_1$ of X and $\psi_2\colon U_2 \to V_2$ of Y with $U_1 \cap f^{-1}(U_2) \neq \varnothing$, the map

$$\psi_2 \circ f \circ \phi_1^{-1}\big|_{\phi_1(U_1 \cap f^{-1}(U_2))}$$

is holomorphic.

Figure 1.1.2 shows the maps in Definition 1.1.5.

DEFINITION 1.1.6. An isomorphism of Riemann surfaces is a bijective holomorphic map with holomorphic inverse.

EXAMPLE 1.1.7. The charts of a Riemann surface are isomorphisms onto their images.

EXAMPLE 1.1.8. Let $\mathbb{H} \coloneqq \{z \in \mathbb{C} : \mathrm{Im}(z) > 0\}$ denote the upper half plane, and $\mathbb{D} \coloneqq \{z \in \mathbb{C} : |z| < 1\}$ the open unit disk. The map

$$\phi\colon \mathbb{H} \to \mathbb{D}, \qquad z \mapsto \frac{z - i}{z + i},$$

is an isomorphism of Riemann surfaces.

EXAMPLE 1.1.9. Let $S^2 \coloneqq \{x \in \mathbb{R}^3 : x_1^2 + x_2^2 + x_3^2 = 1\}$ be the unit sphere, with north pole $\mathbf{n} \coloneqq (0, 0, 1)$ and south pole $\mathbf{s} \coloneqq (0, 0, -1)$. The stereographic projections

$$\sigma_{\mathbf{n}}\colon S^2 - \{\mathbf{n}\} \to \mathbb{C}, \quad x \mapsto \frac{x_1 + ix_2}{1 - x_3}, \qquad \sigma_{\mathbf{s}}\colon S^2 - \{\mathbf{s}\} \to \mathbb{C}, \quad x \mapsto \frac{x_1 - ix_2}{1 + x_3},$$

are homeomorphisms. Since $x_1^2 + x_2^2 = 1 - x_3^2$, we have $\sigma_{\mathbf{n}}(x)\sigma_{\mathbf{s}}(x) = 1$ for $x \neq \mathbf{n}, \mathbf{s}$. So $\sigma_{\mathbf{n}}(S^2 - \{\mathbf{n}, \mathbf{s}\}) = \mathbb{C}^\times$, and on it

$$(\sigma_{\mathbf{s}} \circ \sigma_{\mathbf{n}}^{-1})(w) = 1/w$$

is holomorphic. Thus $\{\sigma_{\mathbf{n}}, \sigma_{\mathbf{s}}\}$ is a complex structure, and S^2, a connected subspace of $\mathbb{R}^3$, is a Riemann surface.

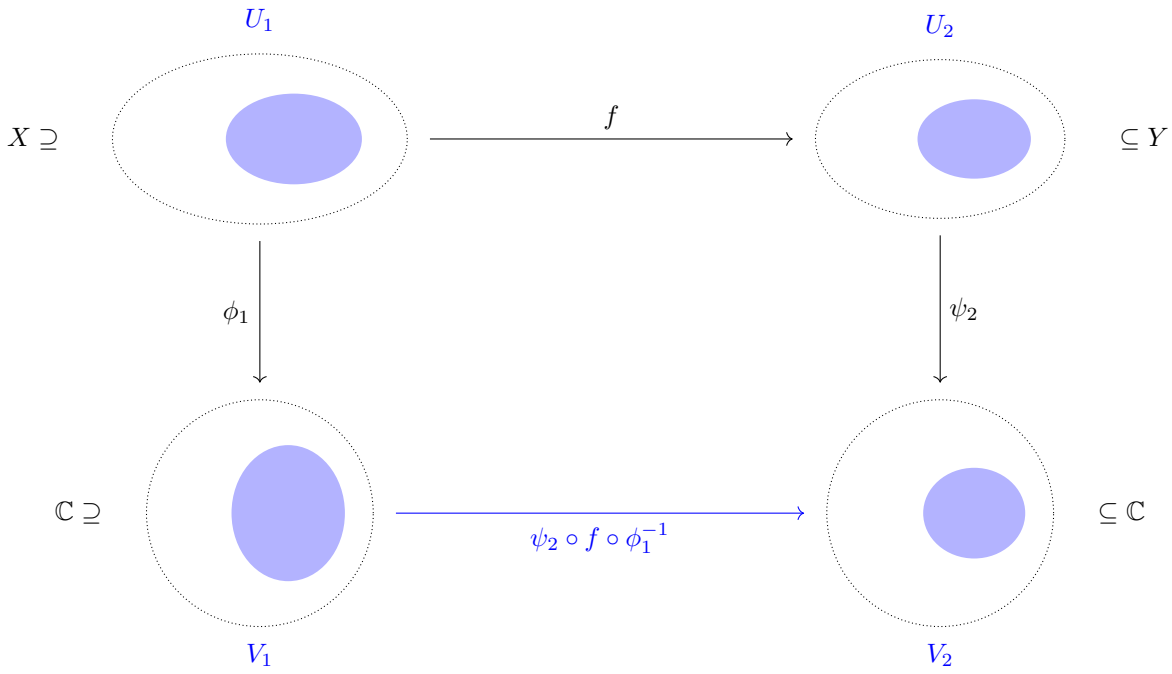


FIGURE 1.1.2. A holomorphic map read in charts: $\psi_2 \circ f \circ \phi_1^{-1}$ on $\phi_1(U_1 \cap f^{-1}(U_2))$.

Define $F\colon S^2 \to \mathbb{P}^1(\mathbb{C})$ by $F(x) \coloneqq (1 : \sigma_{\mathbf{n}}(x))$ for $x \neq \mathbf{n}$, and $F(\mathbf{n}) \coloneqq \infty$. It is a bijection, since $\sigma_{\mathbf{n}}$ is a bijection onto $\mathbb{C}$ and $U_0 = \mathbb{P}^1(\mathbb{C}) - \{\infty\}$. In the charts of Example 1.1.4,

$$\phi_0 \circ F \circ \sigma_{\mathbf{n}}^{-1} = \mathrm{id}_{\mathbb{C}}, \qquad \phi_1 \circ F \circ \sigma_{\mathbf{s}}^{-1} = \mathrm{id}_{\mathbb{C}},$$

the second because $1/\sigma_{\mathbf{n}} = \sigma_{\mathbf{s}}$ away from $\mathbf{n}$ and $\mathbf{s}$, while $\sigma_{\mathbf{s}}(\mathbf{n}) = 0 = \phi_1(\infty)$. Hence F and F^{-1} are holomorphic, and F is an isomorphism of Riemann surfaces.

Recall that a connected topological space X is simply connected if, for any base point $x \in X$, the fundamental group $\pi_1(X, x)$ is trivial. In other words, simply connected spaces are those in which any loop can be continuously deformed to a point within X.

THEOREM 1.1.10 (Uniformization theorem). *Every simply connected Riemann surface is isomorphic to exactly one of* $\mathbb{C}$, $\mathbb{P}^1(\mathbb{C})$, *or* $\mathbb{D}$ *[For91, Section 27].*

So far, we have only seen examples of simply connected Riemann surfaces. One cheap way of ruining the simply connected property is by puncturing. Indeed, $\mathbb{C}^\times$ is a Riemann surface by Example 1.1.3, and $\pi_1(\mathbb{C}^\times, 1) \cong \mathbb{Z}$. We now turn to more interesting examples of surfaces that are not simply connected: donuts.

1.2. Complex tori

DEFINITION 1.2.1. A lattice in $\mathbb{C}$ is a subgroup $\Lambda \subseteq \mathbb{C}$, free of rank 2, that spans $\mathbb{C}$ as an $\mathbb{R}$-vector space.

By definition, every lattice Λ admits a $\mathbb{Z}$-basis that is also an $\mathbb{R}$-basis of $\mathbb{C}$; that is, a pair of elements $\omega_1, \omega_2 \in \mathbb{C}$ such that $\Lambda = \mathbb{Z}\omega_1 \oplus \mathbb{Z}\omega_2$ and

$$\Lambda \otimes_{\mathbb{Z}} \mathbb{R} \cong \mathbb{R}\omega_1 \oplus \mathbb{R}\omega_2 = \mathbb{C}.$$

It follows that Λ is a discrete subset of $\mathbb{C}$.

EXAMPLE 1.2.2. $\Lambda_i \coloneqq \mathbb{Z} \oplus i\mathbb{Z}$ is a lattice, the underlying group of the ring $\mathbb{Z}[i]$ of Gaussian integers.

EXAMPLE 1.2.3. $\Lambda_\omega \coloneqq \mathbb{Z} \oplus \omega\mathbb{Z}$ for $\omega = -\frac{1}{2} + \frac{\sqrt{-3}}{2}$ a primitive cube root of 1 is a lattice, the underlying group of the ring $\mathbb{Z}[\omega]$ of Eisenstein integers.

Figure 1.2.1 shows both lattices.

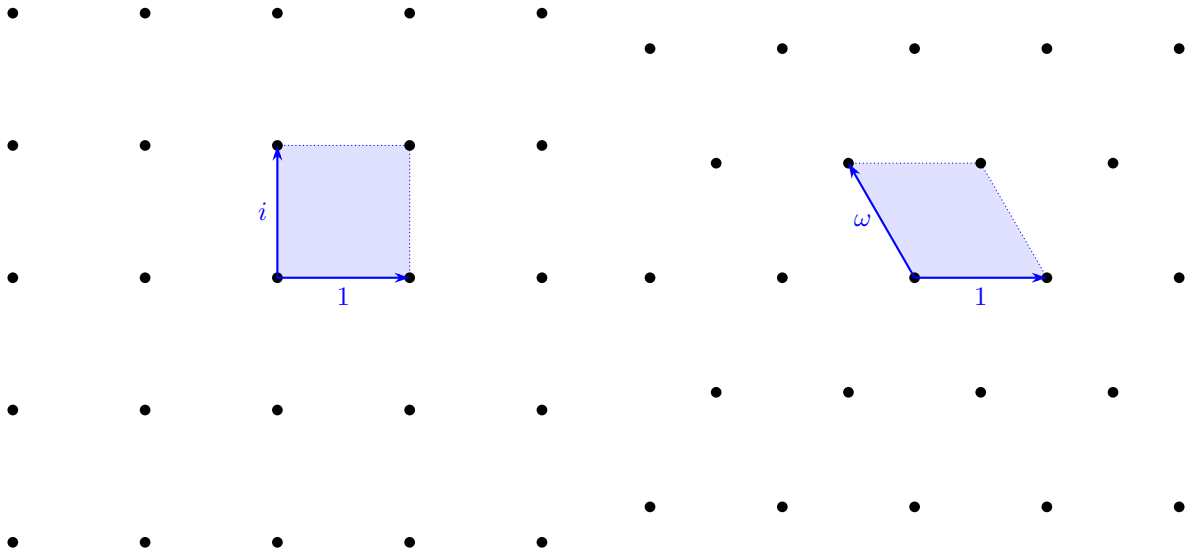


FIGURE 1.2.1. The lattices Λ_i (left) and Λ_ω (right), with the $\mathbb{Z}$-bases $\{1, i\}$ and $\{1, \omega\}$ in blue and their fundamental parallelograms shaded.

Any lattice $\Lambda \subseteq \mathbb{C}$ gives rise to a Riemann surface, namely the quotient group

$$\mathbb{C}/\Lambda \coloneqq \{z + \Lambda : z \in \mathbb{C}\}$$

with the quotient topology. We call $\mathbb{C}/\Lambda$ a complex torus.

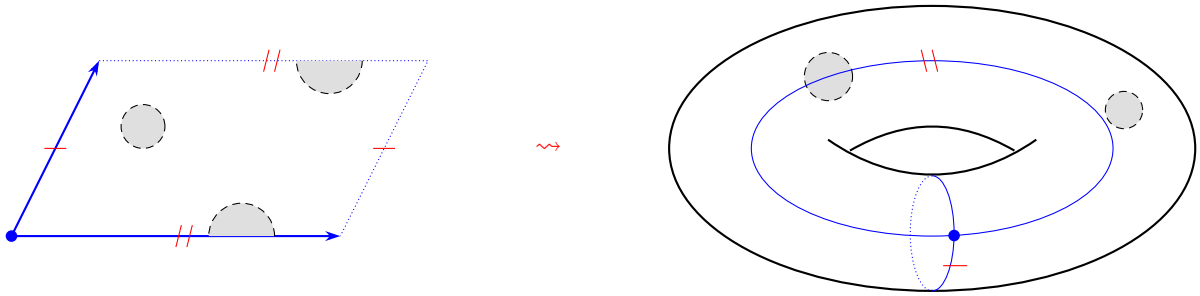


FIGURE 1.2.2. Gluing opposite sides of a fundamental parallelogram gives the torus $\mathbb{C}/\Lambda$: the two pairs of sides become the two circles marked with the same ticks, and disks in the parallelogram map to disks on the torus.

LEMMA 1.2.4. *$\mathbb{C}/\Lambda$ is a compact Riemann surface.*

PROOF. Consider the quotient map

$$\pi : \mathbb{C} \to \mathbb{C}/\Lambda.$$

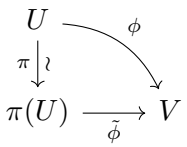
It is continuous by the definition of the quotient topology, and open: for $U \subseteq \mathbb{C}$ open, $\pi^{-1}(\pi(U)) = \bigcup_{\lambda\in\Lambda}(U + \lambda)$ is open. Since π is continuous and surjective, $\mathbb{C}/\Lambda$ is connected, as $\mathbb{C}$ is. It is Hausdorff because Λ is discrete: if $z - w \notin \Lambda$, choose $\varepsilon > 0$ with $|z - w - \lambda| > 2\varepsilon$ for every $\lambda \in \Lambda$. Then the images

under π of the ε-disks around z and w are disjoint open neighborhoods of $\pi(z)$ and $\pi(w)$. Let $\Lambda = \mathbb{Z}\omega_1 \oplus \mathbb{Z}\omega_2$, and consider the fundamental parallelogram

$$\mathcal{F}_\Lambda \coloneqq \{t_1\omega_1 + t_2\omega_2 : 0 \leqslant t_1, t_2 < 1\}.$$

Its closure $\overline{\mathcal{F}_\Lambda}$ is compact, and $\mathbb{C}/\Lambda = \pi(\overline{\mathcal{F}_\Lambda})$, so $\mathbb{C}/\Lambda$ is compact.

For the complex structure, note that no two distinct points of the open parallelogram $\mathcal{F}_\Lambda^\circ$ differ by an element of Λ, so π restricts to a homeomorphism from $\mathcal{F}_\Lambda^\circ$ onto its image, and likewise on every translate $z + \mathcal{F}_\Lambda^\circ$. Hence every chart $\phi\colon U \to V$ with $U \subseteq z + \mathcal{F}_\Lambda^\circ$ descends to a chart $\tilde{\phi}\colon \pi(U) \to V$ with $\tilde{\phi} \circ \pi = \phi$ on U.



The translates $\pi(z + \mathcal{F}_\Lambda^\circ)$ cover $\mathbb{C}/\Lambda$: translating is needed to reach the image of the boundary of the parallelogram. Two such charts are compatible because, on the overlap, $(\pi|_{U_2})^{-1} \circ \pi|_{U_1}$ sends u to $u + \lambda$ with $\lambda \in \Lambda$ depending continuously on u, hence locally constant. So the transition maps are locally $\phi_2 \circ (u \mapsto u + \lambda) \circ \phi_1^{-1}$, which are holomorphic. □

DRAFT
13 Oct 2026

1.3. Isogenies

A complex torus is both a Riemann surface and an abelian group. Holomorphic maps between complex tori are affine linear, and those respecting the group structure are linear.

PROPOSITION 1.3.1. *Suppose $\varphi\colon \mathbb{C}/\Lambda_1 \to \mathbb{C}/\Lambda_2$ is holomorphic. Then there exist $\alpha, \beta \in \mathbb{C}$ with $\alpha\Lambda_1 \subseteq \Lambda_2$ such that*

$$\varphi(z + \Lambda_1) = \alpha z + \beta + \Lambda_2$$

for every $z \in \mathbb{C}$. The map φ is an isomorphism if and only if $\alpha\Lambda_1 = \Lambda_2$.

PROOF. Let $\pi_j\colon \mathbb{C} \to \mathbb{C}/\Lambda_j$ be the quotient maps. Since $\mathbb{C}$ is simply connected and π_2 is a covering map, $\varphi \circ \pi_1$ lifts to a continuous map $\tilde{\varphi}\colon \mathbb{C} \to \mathbb{C}$ with $\pi_2 \circ \tilde{\varphi} = \varphi \circ \pi_1$. As π_2 is a local isomorphism, $\tilde{\varphi}$ is holomorphic. For any $\lambda \in \Lambda_1$ we have $\tilde{\varphi}(z + \lambda) + \Lambda_2 = \tilde{\varphi}(z) + \Lambda_2$. In other words, the function

$$f_\lambda(z) \coloneqq \tilde{\varphi}(z + \lambda) - \tilde{\varphi}(z)$$

takes values in Λ_2, which is discrete. Since f_λ is continuous and $\mathbb{C}$ is connected, f_λ is constant, so $f_\lambda' = 0$, that is, $\tilde{\varphi}'(z + \lambda) = \tilde{\varphi}'(z)$ for every $z \in \mathbb{C}$. So $\tilde{\varphi}'$ is Λ_1-periodic. It is therefore determined by its values on the compact set $\overline{\mathcal{F}_{\Lambda_1}}$ from the proof of Lemma 1.2.4, hence bounded. By Liouville's theorem, $\tilde{\varphi}' = \alpha$ is constant, so $\tilde{\varphi}(z) = \alpha z + \beta$. Moreover $\alpha\lambda = f_\lambda(z) \in \Lambda_2$ for every $\lambda \in \Lambda_1$, that is, $\alpha\Lambda_1 \subseteq \Lambda_2$.

If $\alpha = 0$, then φ is constant, so not injective. If $\alpha \neq 0$ and $\alpha\Lambda_1 \subsetneq \Lambda_2$, pick $w \in \Lambda_2 - \alpha\Lambda_1$. Then $w/\alpha \notin \Lambda_1$, and $\varphi(w/\alpha + \Lambda_1) = w + \beta + \Lambda_2 = \beta + \Lambda_2 = \varphi(0 + \Lambda_1)$, so φ is not injective. If $\alpha\Lambda_1 = \Lambda_2$, then $\psi(z + \Lambda_2) \coloneqq \frac{1}{\alpha}(z - \beta) + \Lambda_1$ is a well defined holomorphic inverse of φ. □

COROLLARY 1.3.2. *A map $\varphi\colon \mathbb{C}/\Lambda_1 \to \mathbb{C}/\Lambda_2$ is a holomorphic group homomorphism if and only if $\varphi(z + \Lambda_1) = \alpha z + \Lambda_2$ for some $\alpha \in \mathbb{C}$ with $\alpha\Lambda_1 \subseteq \Lambda_2$.*

PROOF. In Proposition 1.3.1, φ is a homomorphism if and only if $\varphi(0 + \Lambda_1) = 0 + \Lambda_2$, that is, $\beta \in \Lambda_2$. □

DEFINITION 1.3.3. An isogeny is a non-zero holomorphic homomorphism between complex tori. An isomorphism of complex tori is an invertible isogeny.

EXAMPLE 1.3.4. Let Λ be any lattice, and let $m \geqslant 1$. Then $m\Lambda \subseteq \Lambda$, and the multiplication by m map

$$[m]\colon \mathbb{C}/\Lambda \to \mathbb{C}/\Lambda, \qquad z + \Lambda \mapsto mz + \Lambda$$

is an isogeny. Its kernel is the m-torsion subgroup of $\mathbb{C}/\Lambda$,

$$(\mathbb{C}/\Lambda)[m] \coloneqq \{z+\Lambda : mz \in \Lambda\} = \tfrac{1}{m}\Lambda/\Lambda \cong (\mathbb{Z}/m\mathbb{Z})^2.$$

Figure 1.3.1 shows the 4-torsion points in the fundamental parallelogram. This is the analogue, one dimension up, of the m-torsion subgroup of the circle

$$\mathbb{R}/\mathbb{Z} \cong \{z \in \mathbb{C} : |z| = 1\} \cong \mathbb{C}^\times/\mathbb{R}_{>0},$$

which is $\frac{1}{m}\mathbb{Z}/\mathbb{Z} \cong \mathbb{Z}/m\mathbb{Z}$, identified via $t \mapsto e^{2\pi i t}$ with the group of complex m-th roots of unity

$$\mu_m(\mathbb{C}) \coloneqq \{z \in \mathbb{C} : z^m = 1\}.$$

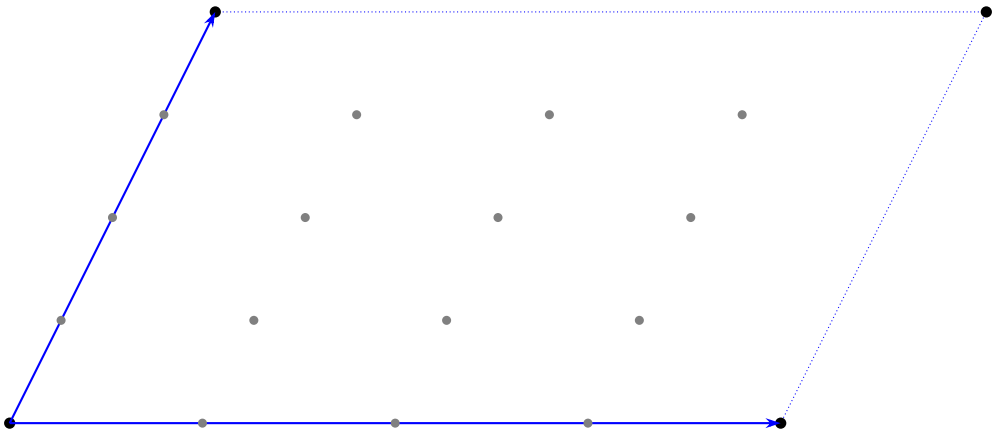


FIGURE 1.3.1. The 4-torsion points of $\mathbb{C}/\Lambda$ (gray) in the fundamental parallelogram, whose vertices are lattice points (black).

EXAMPLE 1.3.5. Let $\Lambda = \mathbb{Z}\omega_1 \oplus \mathbb{Z}\omega_2$ be a lattice, and let $N \geqslant 1$. The lattice $\Lambda' \coloneqq \mathbb{Z}\frac{\omega_1}{N} \oplus \mathbb{Z}\omega_2$ contains Λ, and $\Lambda'/\Lambda \cong \mathbb{Z}/N\mathbb{Z}$ is generated by $\frac{\omega_1}{N} + \Lambda$. By Corollary 1.3.2 with $\alpha = 1$, the map

$$\varphi\colon \mathbb{C}/\Lambda \to \mathbb{C}/\Lambda', \qquad z + \Lambda \mapsto z + \Lambda'$$

is an isogeny. Its kernel Λ'/Λ is a cyclic subgroup of order N of $(\mathbb{C}/\Lambda)[N]$, so φ is a cyclic isogeny. Figure 1.3.2 shows the case $N = 4$. Conversely, every cyclic subgroup $C \subseteq \mathbb{C}/\Lambda$ of order N is the kernel of such an isogeny: take $\Lambda' \coloneqq \pi^{-1}(C)$, where $\pi\colon \mathbb{C} \to \mathbb{C}/\Lambda$ is the quotient map.
Since $N\Lambda' = \mathbb{Z}\omega_1 \oplus \mathbb{Z}N\omega_2 \subseteq \Lambda$, multiplication by N also defines an isogeny $\hat{\varphi}\colon \mathbb{C}/\Lambda' \to \mathbb{C}/\Lambda$, $z+\Lambda' \mapsto Nz+\Lambda$. Both composites $\hat{\varphi} \circ \varphi$ and $\varphi \circ \hat{\varphi}$ are multiplication by N (Example 1.3.4).

In Example 1.3.5 the isogeny φ has a partner $\hat{\varphi}$ going the other way, and both composites are multiplication by the order of the kernel. Every isogeny has such a partner. For a lattice Λ, write $\operatorname{area}(\Lambda)$ for the area of the fundamental parallelogram $\mathcal{F}_\Lambda$ from the proof of Lemma 1.2.4. It does not depend on the basis, since a change of basis has determinant ± 1.

LEMMA 1.3.6. *Let $\varphi\colon \mathbb{C}/\Lambda_1 \to \mathbb{C}/\Lambda_2$, $z + \Lambda_1 \mapsto \alpha z + \Lambda_2$, be an isogeny. Then $\ker\varphi = \alpha^{-1}\Lambda_2/\Lambda_1$ is finite, and*

$$\#\ker\varphi = [\Lambda_2 : \alpha\Lambda_1] = |\alpha|^2 \frac{\operatorname{area}(\Lambda_1)}{\operatorname{area}(\Lambda_2)}.$$

PROOF. We have $z + \Lambda_1 \in \ker\varphi$ if and only if $\alpha z \in \Lambda_2$, that is, $z \in \alpha^{-1}\Lambda_2$. Since $\varphi \neq 0$, $\alpha \neq 0$, and multiplication by α is a group isomorphism $\alpha^{-1}\Lambda_2/\Lambda_1 \to \Lambda_2/\alpha\Lambda_1$. Now $\alpha\Lambda_1 \subseteq \Lambda_2$ are both lattices. For lattices $\Lambda' \subseteq \Lambda$, write a basis of Λ' in terms of a basis of Λ by a matrix $M \in \mathrm{M}_2(\mathbb{Z})$ with $\det M \neq 0$. By the Smith normal form, $[\Lambda : \Lambda'] = |\det M|$, and the $\mathbb{R}$-linear map of $\mathbb{C}$ taking the one basis to the other scales areas by $|\det M|$. So $[\Lambda : \Lambda'] = \operatorname{area}(\Lambda')/\operatorname{area}(\Lambda)$. Finally, $z \mapsto \alpha z$ is $\mathbb{R}$-linear with determinant $|\alpha|^2$, so $\operatorname{area}(\alpha\Lambda_1) = |\alpha|^2 \operatorname{area}(\Lambda_1)$. □

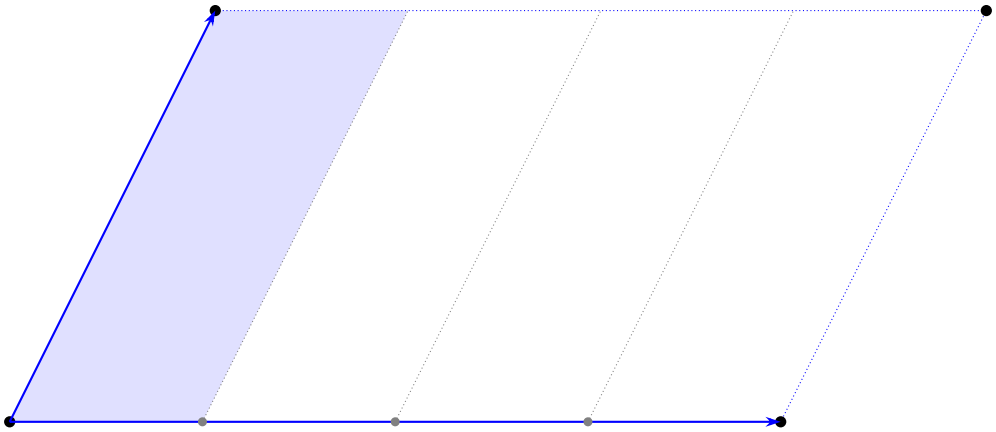


FIGURE 1.3.2. The kernel of the cyclic isogeny $\mathbb{C}/\Lambda \to \mathbb{C}/\Lambda'$ for $N = 4$ (gray) in the fundamental parallelogram of Λ, whose vertices are lattice points (black). A fundamental parallelogram of Λ' is shaded, and its translates by the kernel are dotted.

DEFINITION 1.3.7. The degree of an isogeny φ of complex tori is $\deg \varphi := \#\ker \varphi$.

For instance, $\deg[m] = m^2$ by Example 1.3.4, the cyclic isogeny of Example 1.3.5 has degree N, and by Proposition 1.3.1 the isogenies of degree 1 are exactly the isomorphisms. If $\lambda\colon \mathbb{C}/\Lambda_2 \to \mathbb{C}/\Lambda_3$ is another isogeny, given by $z \mapsto \beta z$, then $\lambda \circ \varphi$ is given by $z \mapsto \beta\alpha z$, and the area formula of Lemma 1.3.6 gives $\deg(\lambda\circ\varphi) = \deg\lambda\cdot\deg\varphi$. The scalar α is determined by φ: if $\alpha z - \alpha' z \in \Lambda_2$ for every $z \in \mathbb{C}$, then $\alpha - \alpha' = 0$, because a continuous map from $\mathbb{C}$ to the discrete set Λ_2 is constant.

THEOREM 1.3.8. *Let $\varphi\colon \mathbb{C}/\Lambda_1 \to \mathbb{C}/\Lambda_2$, $z + \Lambda_1 \mapsto \alpha z + \Lambda_2$, be an isogeny of degree d.*

(1) There is a unique isogeny $\hat\varphi\colon \mathbb{C}/\Lambda_2 \to \mathbb{C}/\Lambda_1$, the dual isogeny, with $\hat\varphi \circ \varphi = [d]$. It is $\hat\varphi(z + \Lambda_2) = \frac{d}{\alpha} z + \Lambda_1$.
(2) $\varphi \circ \hat\varphi = [d]$ on $\mathbb{C}/\Lambda_2$.
(3) $\widehat{\lambda \circ \varphi} = \hat\varphi \circ \hat\lambda$ for an isogeny $\lambda\colon \mathbb{C}/\Lambda_2 \to \mathbb{C}/\Lambda_3$.
(4) $\widehat{\varphi + \psi} = \hat\varphi + \hat\psi$ for an isogeny $\psi\colon \mathbb{C}/\Lambda_1 \to \mathbb{C}/\Lambda_2$ with $\varphi + \psi \neq 0$.
(5) $\widehat{[m]} = [m]$ for all $m \geqslant 1$.
(6) $\deg\hat\varphi = \deg\varphi$ and $\hat{\hat\varphi} = \varphi$.

PROOF. By Lemma 1.3.6, $\Lambda_2/\alpha\Lambda_1$ is a group of order d, so it is killed by d: $d\Lambda_2 \subseteq \alpha\Lambda_1$, that is, $\frac{d}{\alpha}\Lambda_2 \subseteq \Lambda_1$. By Corollary 1.3.2, $\hat\varphi$ is an isogeny, and $\hat\varphi \circ \varphi$ is $z \mapsto dz$, which is $[d]$. Composition of isogenies multiplies their scalars, and the scalar determines the isogeny, so an isogeny $z \mapsto \gamma z$ with this property has $\gamma\alpha = d$. This proves part 1, and part 2 follows from $\alpha \cdot \frac{d}{\alpha} = d$.

The rest is arithmetic with scalars. Set $c := \operatorname{area}(\Lambda_1)/\operatorname{area}(\Lambda_2)$. By Lemma 1.3.6, $d = c|\alpha|^2$, so

$$\frac{d}{\alpha} = c\,\bar\alpha.$$

For part 3, let λ be given by $z \mapsto \beta z$, of degree d'. Then $\lambda \circ \varphi$ has scalar $\beta\alpha$ and degree dd', so its dual has scalar $\frac{dd'}{\beta\alpha} = \frac{d}{\alpha} \cdot \frac{d'}{\beta}$, which is the scalar of $\hat\varphi \circ \hat\lambda$. For part 4, let ψ be given by $z \mapsto \beta z$. Then $\varphi + \psi$ has scalar $\alpha + \beta$, and since c depends only on Λ_1 and Λ_2, its dual has scalar $c\,\overline{(\alpha+\beta)} = c\,\bar\alpha + c\,\bar\beta$. For part 5, $[m]$ has scalar m and degree m^2, so its dual has scalar $m^2/m = m$. For part 6, Lemma 1.3.6 gives $\deg\hat\varphi = |d/\alpha|^2/c = d^2/d = d$, and so the dual of $\hat\varphi$ has scalar $d/(d/\alpha) = \alpha$. □

In Example 1.3.5, $\alpha = 1$ and $d = N$, so the isogeny $\hat\varphi$ there is the dual of φ. The kernel of the dual is

$$\ker\hat\varphi = \tfrac{\alpha}{d}\Lambda_1/\Lambda_2 = \varphi\big((\mathbb{C}/\Lambda_1)[d]\big),$$

since $(\mathbb{C}/\Lambda_1)[d] = \frac{1}{d}\Lambda_1/\Lambda_1$. The formula $d/\alpha = c\,\bar{\alpha}$ says that, up to the positive constant c, taking the dual is complex conjugation of the scalar. In particular, if we allow the zero map and set $\deg 0 := 0$, then $\varphi \mapsto \deg\varphi = c|\alpha|^2$ is a positive definite quadratic form on the group of holomorphic homomorphisms $\mathbb{C}/\Lambda_1 \to \mathbb{C}/\Lambda_2$. For an endomorphism of $\mathbb{C}/\Lambda$ we have $c = 1$, so the dual is $\bar{\alpha}$ and the degree is $|\alpha|^2$. These statements return for elliptic curves over an arbitrary field in Theorem 4.3.10 and Corollary 4.3.11, where there is no scalar to conjugate and the proofs are harder.

EXAMPLE 1.3.9. Let $\Lambda_\omega = \mathbb{Z} \oplus \omega\mathbb{Z}$ be the lattice of Example 1.2.3. Since $\mathbb{Z}[\omega]$ is a ring, $\alpha\Lambda_\omega \subseteq \Lambda_\omega$ for every $\alpha \in \mathbb{Z}[\omega]$, and $z \mapsto \alpha z$ is an endomorphism of degree $|\alpha|^2$ with dual $z \mapsto \bar{\alpha} z$.

(1) For $\alpha = \omega$, the degree is 1, so $z \mapsto \omega z$ is an automorphism, and its dual $z \mapsto \bar{\omega} z = \omega^2 z$ is its inverse.
(2) For $\alpha = 1 - \omega$, the degree is $|1 - \omega|^2 = 3$, and the dual is $z \mapsto (1 - \omega^2)z$. Indeed $(1 - \omega)(1 - \omega^2) = 3$. The kernel is cyclic of order 3, generated by $\frac{1}{1-\omega} + \Lambda_\omega = \frac{2+\omega}{3} + \Lambda_\omega$.

Compare part 3 of Proposition 4.8.11, where the same identity appears for the curves C_d. Similarly, on $\mathbb{C}/\Lambda_i$ the endomorphism $z \mapsto (1 + i)z$ has degree 2 and dual $z \mapsto (1 - i)z$.

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1.4. Meromorphic functions

DEFINITION 1.4.1. Let X be a Riemann surface. A meromorphic function on X is a holomorphic map $X \to \mathbb{P}^1(\mathbb{C})$ that is not identically ∞. Denote the set of meromorphic functions on X by $\mathbb{C}(X)$.

Equivalently, a meromorphic function is a function f with values in $\mathbb{C}$, defined and holomorphic away from a discrete set of points of X, at each of which f has a pole. There we set $f = \infty$ [Mir95, Proposition II.3.13].

EXAMPLE 1.4.2. Let $F_0, F_1 \in \mathbb{C}[X_0, X_1]$ be homogeneous polynomials of equal degree with no common zero on $\mathbb{P}^1(\mathbb{C})$, and $F_0 \neq 0$. Then the map

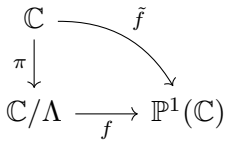
$$F\colon \mathbb{P}^1(\mathbb{C}) \to \mathbb{P}^1(\mathbb{C}), \qquad (X_0 : X_1) \mapsto (F_0 : F_1)$$

is a meromorphic function. Moreover, every meromorphic function on $\mathbb{P}^1(\mathbb{C})$ is of this form.

EXAMPLE 1.4.3. Let $\Lambda \subseteq \mathbb{C}$ be a lattice. Then there is a bijective correspondence between

- meromorphic functions on $\mathbb{C}/\Lambda$, and
- Λ-periodic meromorphic functions on $\mathbb{C}$,

given by $f \mapsto \tilde{f} := f \circ \pi$.



Given $f, g \in \mathbb{C}(X)$, let $S \subseteq X$ be the set of poles of f or g. It is discrete [Mir95, Theorem II.1.33], and on $X - S$ the pointwise sum $f + g$ and product fg are holomorphic. At each point of S they have either a removable singularity or a pole [Mir95, Example II.1.17]. Filling in the removable singularities and setting the value ∞ at the poles defines $f + g, fg \in \mathbb{C}(X)$. The constant maps $0, 1 \in \mathbb{C}(X)$ serve as additive and multiplicative identities, respectively.

LEMMA 1.4.4. *Let X be a Riemann surface. With the operations defined above, the set $\mathbb{C}(X)$ of meromorphic functions on X is a field.*

PROOF. The ring axioms hold pointwise away from a discrete set, hence everywhere by continuity. If $f \neq 0$, then since X is connected the zeros of f form a discrete set [Mir95, Theorem II.1.33], so $1/f$ is a meromorphic function [Mir95, Example II.1.17] and $f \cdot (1/f) = 1$. □

1.5. Invariant differentials

On an open subset of $\mathbb{C}$ we can write down differentials $f(z)\,dz$. A Riemann surface has no global coordinate z, so we write a differential in every chart and ask that the expressions agree under the chain rule [Mir95, Section IV.1].

DEFINITION 1.5.1. Let X be a Riemann surface. A meromorphic differential ω on X assigns to every chart $\phi\colon U \to V$ of X a meromorphic function f_ϕ on V, written $\omega = f_\phi(z)\,dz$, such that for any two charts ϕ_1 and ϕ_2, with transition map $T := \phi_2 \circ \phi_1^{-1}$,

$$f_{\phi_1}(z) = f_{\phi_2}\big(T(z)\big)\,T'(z) \qquad \text{for } z \in \phi_1(U_1 \cap U_2).$$

If every f_ϕ is holomorphic, then ω is a holomorphic differential.

By the chain rule, it is enough to give functions f_ϕ satisfying the transformation rule for the charts of a single complex structure on X. They extend uniquely to all charts. Since T' does not vanish, whether f_ϕ has a zero or a pole at $\phi(p)$ depends only on p, so a non-zero meromorphic differential has a discrete set of zeros and poles. If $\eta \neq 0$ is a meromorphic differential, then the quotients f_ϕ/h_ϕ, where $\eta = h_\phi(z)\,dz$, glue to a meromorphic function, because the factors T' cancel. So every meromorphic differential is $g\eta$ for a unique $g \in \mathbb{C}(X)$.

Differentials pull back along holomorphic maps. Let $F\colon X \to Y$ be a non-constant holomorphic map and ω a meromorphic differential on Y. For charts $\phi\colon U \to V$ of X and $\psi\colon U' \to V'$ of Y with $F(U) \subseteq U'$, write $F_{\psi\phi} := \psi \circ F \circ \phi^{-1}$ and $\omega = f_\psi(w)\,dw$, and set

$$F^*\omega := f_\psi\big(F_{\psi\phi}(z)\big)\,F'_{\psi\phi}(z)\,dz.$$

Such charts ϕ cover X, and the chain rule shows that $F^*\omega$ is a meromorphic differential on X. If ω is holomorphic, the same formula makes sense for constant F, and gives $F^*\omega = 0$. Pullback satisfies $(G \circ F)^* = F^* \circ G^*$ and $F^*(g\omega) = (g \circ F)\,F^*\omega$ for $g \in \mathbb{C}(Y)$. For instance, if F is a map between open subsets of $\mathbb{C}$, then $F^*(dw) = F'(z)\,dz$, which is the familiar substitution $w = F(z)$.

The Riemann surfaces $\mathbb{C}^\times$ and $\mathbb{C}/\Lambda$ are also abelian groups, and for every element a the translation τ_a, namely $z \mapsto az$ on $\mathbb{C}^\times$ and $z + \Lambda \mapsto z + a + \Lambda$ on $\mathbb{C}/\Lambda$, is an isomorphism of Riemann surfaces.

DEFINITION 1.5.2. Let G be $\mathbb{C}^\times$ or a complex torus. A meromorphic differential ω on G is invariant if $\tau_a^*\omega = \omega$ for every $a \in G$.

LEMMA 1.5.3. *Let G be $\mathbb{C}^\times$ or a complex torus, and let η be a non-zero invariant differential on G. Then η has no zeros or poles, and every invariant differential on G is $c\eta$ for a unique $c \in \mathbb{C}$.*

PROOF. Since $\tau_a^*\eta = \eta$, a point p of G is a zero of η if and only if $\tau_a(p)$ is, and likewise for poles. The translations act transitively on G, so if η had a zero or a pole it would have one everywhere, contradicting discreteness. Let ω be invariant, and write $\omega = g\eta$ with $g \in \mathbb{C}(G)$. Then

$$g\eta = \omega = \tau_a^*\omega = (g \circ \tau_a)\,\tau_a^*\eta = (g \circ \tau_a)\,\eta,$$

so $g \circ \tau_a = g$ for every $a \in G$. Hence g is constant, and its value $c \neq \infty$, since g is not identically ∞. □

EXAMPLE 1.5.4. On $\mathbb{C}^\times$, with its single chart id, consider the holomorphic differential

$$\omega := \frac{dz}{z}.$$

For $a \in \mathbb{C}^\times$, the translation $\tau_a(z) = az$ gives

$$\tau_a^*\omega = \frac{a\,dz}{az} = \frac{dz}{z} = \omega,$$

so ω is invariant, and by Lemma 1.5.3 every invariant differential on $\mathbb{C}^\times$ is a multiple of dz/z. By contrast, dz is also a holomorphic differential on $\mathbb{C}^\times$, but $\tau_a^*dz = a\,dz$, so it is not invariant. The holomorphic differentials $f(z)\,dz$, with f holomorphic on $\mathbb{C}^\times$, form an infinite-dimensional space. Only a line of them is invariant.

(1) **Endomorphisms.** For $m \in \mathbb{Z}$, the map $[m]\colon z \mapsto z^m$ is a holomorphic group homomorphism $\mathbb{C}^\times \to \mathbb{C}^\times$, with kernel $\mu_m(\mathbb{C})$ when $m \neq 0$ (Example 1.3.4). It pulls ω back to

$$[m]^*\omega = \frac{mz^{m-1}\,dz}{z^m} = m\,\frac{dz}{z} = m\omega.$$

(2) **Uniformization.** The exponential $\exp\colon \mathbb{C} \to \mathbb{C}^\times$ is a surjective holomorphic homomorphism with kernel $2\pi i\mathbb{Z}$, so it induces an isomorphism $\mathbb{C}/2\pi i\mathbb{Z} \cong \mathbb{C}^\times$ of Riemann surfaces and of groups. In the coordinate w on $\mathbb{C}$,

$$\exp^*\omega = \frac{e^w\,dw}{e^w} = dw,$$

so the invariant differential on $\mathbb{C}^\times$ corresponds to the translation-invariant differential dw on $\mathbb{C}$.

(3) **Periods.** If γ is a loop in $\mathbb{C}^\times$, then $\int_\gamma dz/z$ is $2\pi i$ times the winding number of γ around 0. So the set of $\int_\gamma \omega$, as γ runs over loops, is $2\pi i\mathbb{Z}$, the kernel of exp, and it is generated by the integral over the unit circle, the image of the segment from 0 to $2\pi i$. The inverse of the isomorphism in part 2 is $z \mapsto \int_1^z \omega \bmod 2\pi i\mathbb{Z}$, the logarithm, which is well defined modulo $2\pi i\mathbb{Z}$.

(4) **At 0 and ∞.** In the chart ϕ_1 of Example 1.1.4, with coordinate $u = 1/z$, the transformation rule gives $\omega = -du/u$. So ω is a meromorphic differential on $\mathbb{P}^1(\mathbb{C})$ with simple poles at 0 and ∞. Unlike dz on $\mathbb{C}/\Lambda$ below, it does not extend to a holomorphic differential on this compactification.

Algebraically, $\mathbb{C}^\times$ is the group of complex points of the multiplicative group $\mathbb{G}_\mathrm{m}$ (Example 2.1.6), and ω is dx/x. See Examples 3.4.7 and 3.4.8 for the algebraic version over an arbitrary field.

Example 1.5.5. Let $\Lambda \subseteq \mathbb{C}$ be a lattice, and $\pi\colon \mathbb{C} \to \mathbb{C}/\Lambda$ the quotient map. Among the charts constructed in the proof of Lemma 1.2.4, take those coming from $\phi = \mathrm{id}$, that is, the inverses of the restrictions of π to small open subsets of $\mathbb{C}$. Their transition maps are locally $u \mapsto u + \lambda$ with $\lambda \in \Lambda$, whose derivative is 1. So setting $f_\phi \coloneqq 1$ on each of them satisfies the transformation rule, and defines a holomorphic differential

$$dz \quad \text{on } \mathbb{C}/\Lambda, \qquad \text{with } \pi^*dz = dz.$$

In these charts the translation τ_a reads $u \mapsto u + a$, up to adding an element of Λ, so $\tau_a^*dz = dz$. Hence dz is invariant, it has no zeros or poles, and every invariant differential on $\mathbb{C}/\Lambda$ is a multiple of dz, by Lemma 1.5.3.

(1) **All holomorphic differentials are invariant.** Let ω be a holomorphic differential on $\mathbb{C}/\Lambda$, and write $\omega = g\,dz$ with $g \in \mathbb{C}(\mathbb{C}/\Lambda)$. Since dz has no zeros, g has no poles. By Example 1.4.3, $g \circ \pi$ is a Λ-periodic entire function. It is bounded on the compact set $\overline{\mathcal{F}_\Lambda}$, hence bounded, hence constant by Liouville's theorem. So the holomorphic differentials on $\mathbb{C}/\Lambda$ are exactly the multiples of dz, unlike on $\mathbb{C}^\times$. More generally, the meromorphic differentials on $\mathbb{C}/\Lambda$ are the $g\,dz$ with $g \circ \pi$ a Λ-periodic meromorphic function on $\mathbb{C}$.

(2) **Isogenies.** By Proposition 1.3.1, a holomorphic map $\varphi\colon \mathbb{C}/\Lambda_1 \to \mathbb{C}/\Lambda_2$ has the form $z+\Lambda_1 \mapsto \alpha z+\beta+\Lambda_2$. In the charts above it reads $u \mapsto \alpha u + \beta$, up to adding an element of Λ_2, so

$$\varphi^*dz = \alpha\,dz.$$

In particular $[m]^*dz = m\,dz$ for the multiplication by m map of Example 1.3.4, exactly as $[m]^*\omega = m\omega$ on $\mathbb{C}^\times$. An isogeny is determined by its effect on the invariant differential: by Corollary 1.3.2, $\varphi \mapsto \alpha$ identifies the holomorphic homomorphisms $\mathbb{C}/\Lambda_1 \to \mathbb{C}/\Lambda_2$ with $\{\alpha \in \mathbb{C} : \alpha\Lambda_1 \subseteq \Lambda_2\}$, and it turns composition into multiplication.

(3) **Periods.** A path γ in $\mathbb{C}/\Lambda$ lifts along π to a path $\tilde\gamma$ in $\mathbb{C}$, unique up to translation by Λ, and we set $\int_\gamma dz \coloneqq \int_{\tilde\gamma} dz$, the difference of the endpoints of $\tilde\gamma$. If γ is a loop, then $\tilde\gamma$ runs from some z to $z+\lambda$ with $\lambda \in \Lambda$, and $\int_\gamma dz = \lambda$. Every λ occurs, as the image of the segment from 0 to λ. So the periods of dz form the lattice Λ itself, the kernel of π, and $p \mapsto \int_0^p dz \bmod \Lambda$ is the identity of $\mathbb{C}/\Lambda$.

For $c \in \mathbb{C}^\times$, the periods of $c\,dz$ form the lattice $c\Lambda$, and the isomorphism $\mathbb{C}/\Lambda \to \mathbb{C}/c\Lambda$, $z+\Lambda \mapsto cz+c\Lambda$, of Proposition 1.3.1 pulls dz back to $c\,dz$. So a complex torus determines its lattice only up to scaling, but a complex torus together with a non-zero holomorphic differential determines the lattice itself. Under the

isomorphism of Theorem 1.7.1, dz becomes $dx/(2y)$ on $E_{A,B}(\mathbb{C})$, as we will see in Remark 4.1.8. This is the invariant differential of an elliptic curve in Chapter 4.

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1.6. Projective plane curves as Riemann surfaces

We follow [Mir95, Section I.3]. The projective plane is

$$\mathbb{P}^2(\mathbb{C}) \coloneqq \big(\mathbb{C}^3 - \{(0,0,0)\}\big)/\mathbb{C}^\times,$$

with the quotient topology, and we write its points as $(X : Y : Z)$. It is compact, being the image of the unit sphere in $\mathbb{C}^3$, and it is covered by the open sets U_X, U_Y and U_Z where X, Y or Z does not vanish, each homeomorphic to $\mathbb{C}^2$ by dividing by that coordinate. If $F \in \mathbb{C}[X,Y,Z]$ is homogeneous of degree d, then $F(\lambda z) = \lambda^d F(z)$, so the zero locus

$$C_F \coloneqq \{(X : Y : Z) \in \mathbb{P}^2(\mathbb{C}) : F(X,Y,Z) = 0\}$$

is a well defined closed subset of $\mathbb{P}^2(\mathbb{C})$.

Definition 1.6.1. A homogeneous polynomial $F \in \mathbb{C}[X,Y,Z]$ is nonsingular if F, $\partial F/\partial X$, $\partial F/\partial Y$ and $\partial F/\partial Z$ have no common zero in $\mathbb{P}^2(\mathbb{C})$. In that case C_F is a smooth projective plane curve.

Proposition 1.6.2. *Let F be a nonsingular homogeneous polynomial. Then C_F is a compact Riemann surface. Near every point of C_F, some ratio of two homogeneous coordinates is a local coordinate.*

Proof. See [Mir95, Proposition I.3.6]. The idea: $C_F \cap U_Z$ is the affine curve $f(x,y) \coloneqq F(x,y,1) = 0$, where $x = X/Z$ and $y = Y/Z$. At a point $(x : y : 1)$ of C_F we have $\partial f/\partial x = \partial F/\partial X$ and $\partial f/\partial y = \partial F/\partial Y$, and Euler's formula

$$X\frac{\partial F}{\partial X} + Y\frac{\partial F}{\partial Y} + Z\frac{\partial F}{\partial Z} = dF$$

gives $\partial F/\partial Z = dF - x\,\partial F/\partial X - y\,\partial F/\partial Y$. Hence the singular points of F in U_Z are exactly the common zeros of f, $\partial f/\partial x$ and $\partial f/\partial y$, and likewise in U_X and U_Y. Where $\partial f/\partial y \neq 0$, the implicit function theorem writes C_F locally as a graph $y = g(x)$, so x is a local coordinate. Similarly y is a local coordinate where $\partial f/\partial x \neq 0$. The transition maps between these charts are holomorphic, being built from such functions g and ratios of coordinates. As a subspace of $\mathbb{P}^2(\mathbb{C})$, C_F is Hausdorff, and it is compact, being closed.

That C_F is connected is less elementary. Here is the idea. First, F is irreducible: if $F = GH$ with G and H nonconstant, then the curves C_G and C_H meet at some point p of $\mathbb{P}^2(\mathbb{C})$ by Bézout's theorem, and every partial derivative of $F = GH$ vanishes at p by the product rule, so F is singular at p. After permuting the coordinates we may assume $Z \nmid F$. Then f is irreducible too, and C_F meets the line $Z = 0$ in finitely many points. Second, the zero locus of an irreducible f is connected [Mir95, Theorem I.2.3]. Away from finitely many points, the projection $(x,y) \mapsto x$ makes it a finite covering of $\mathbb{C}$ minus a finite set. A connected component would pick out a subset of the roots y of $f(x,\cdot) = 0$ that is preserved as x moves. The elementary symmetric functions of those roots are then rational functions of x, and they give a proper factor of f. Finally, C_F is the closure of the connected set $C_F \cap U_Z$, because the finitely many remaining points are not isolated in C_F. Hence C_F is connected. □

Example 1.6.3. Let $A, B \in \mathbb{C}$ and

$$F \coloneqq Y^2Z - X^3 - AXZ^2 - BZ^3,$$

$$E_{A,B}(\mathbb{C}) \coloneqq C_F = \{(X : Y : Z) \in \mathbb{P}^2(\mathbb{C}) : Y^2Z = X^3 + AXZ^2 + BZ^3\}.$$

Then F is nonsingular if and only if $4A^3 + 27B^2 \neq 0$. Indeed:

- On $Z = 0$, the equation forces $X = 0$, so the only point of C_F there is $(0 : 1 : 0)$, where $\partial F/\partial Z = Y^2 - 2AXZ - 3BZ^2 = 1 \neq 0$.
- On U_Z, with $x = X/Z$, $y = Y/Z$ and $p(x) \coloneqq x^3 + Ax + B$, we have $f = y^2 - p(x)$. By Euler's formula, a singular point is a common zero of f, $\partial f/\partial x = -p'(x)$ and $\partial f/\partial y = 2y$, that is, $y = 0$ and x a repeated root of p. Such an x exists if and only if the discriminant of p, which is $-(4A^3 + 27B^2)$, vanishes.

So if $4A^3 + 27B^2 \neq 0$, then $E_{A,B}(\mathbb{C})$ is a compact Riemann surface by Proposition 1.6.2.

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1.7. The uniformization theorem

Given any lattice $\Lambda \subseteq \mathbb{C}$, the associated Weierstrass $\wp$-function is the meromorphic function

$$\wp_\Lambda \colon \mathbb{C} \to \mathbb{P}^1(\mathbb{C}), \qquad z \mapsto \frac{1}{z^2} + \sum_{\substack{\omega \in \Lambda \\ \omega \neq 0}} \left[\frac{1}{(z-\omega)^2} - \frac{1}{\omega^2} \right].$$

Its derivative is

$$\wp'_\Lambda \colon \mathbb{C} \to \mathbb{P}^1(\mathbb{C}), \qquad z \mapsto -2 \sum_{\omega \in \Lambda} \frac{1}{(z-\omega)^3}.$$

For even $k > 2$, we have the Eisenstein series

$$G_k(\Lambda) \coloneqq \sum_{\substack{\omega \in \Lambda \\ \omega \neq 0}} \frac{1}{\omega^k}.$$

We have the following beautiful result.

THEOREM 1.7.1. *Given $A, B \in \mathbb{C}$ with $4A^3 + 27B^2 \neq 0$, there exists a lattice $\Lambda \subseteq \mathbb{C}$ such that the map*

$$\phi \colon \mathbb{C}/\Lambda \to E_{A,B}(\mathbb{C}), \qquad z + \Lambda \mapsto \begin{cases} (\wp_\Lambda(z) : \frac{1}{2}\wp'_\Lambda(z) : 1) & \text{if } z \notin \Lambda, \\ (0:1:0) & \text{if } z \in \Lambda, \end{cases}$$

is an isomorphism of Riemann surfaces. Moreover,

$$A = -15G_4(\Lambda), \qquad B = -35G_6(\Lambda).$$

QUESTION 1.7.2 (Diamond–Shurman). This is [DS05, Exercise 1.4.2]. Let $\Lambda \subseteq \mathbb{C}$ be a lattice.

(1) Show that $\wp_\Lambda$ is even and that $\wp'_\Lambda$ is Λ-periodic.
(2) Let $\{\omega_1, \omega_2\}$ be a $\mathbb{Z}$-basis for Λ. For $j = 1, 2$, show that $\wp_\Lambda(z+\omega_j) - \wp_\Lambda(z)$ is some constant c_j. Substitute $z = -\omega_j/2$ to conclude that $c_j = 0$, and deduce that $\wp_\Lambda$ is Λ-periodic.

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1.8. Isomorphism classes of complex tori

Proposition 1.3.1 lets us classify complex tori up to isomorphism. Start with an arbitrary lattice $\Lambda \subseteq \mathbb{C}$, and choose a $\mathbb{Z}$-basis $\{\omega_1, \omega_2\}$. Since $\omega_1/\omega_2 \notin \mathbb{R}$, after possibly swapping ω_1 and ω_2 we have $\tau \coloneqq \omega_1/\omega_2 \in \mathbb{H}$. Define $\Lambda_\tau \coloneqq \mathbb{Z} \oplus \tau\mathbb{Z}$. Since $\frac{1}{\omega_2}\Lambda = \Lambda_\tau$, Proposition 1.3.1 implies that the map

$$\varphi_\tau \colon \mathbb{C}/\Lambda \to \mathbb{C}/\Lambda_\tau, \qquad z + \Lambda \mapsto \frac{z}{\omega_2} + \Lambda_\tau$$

is an isomorphism of complex tori. So every complex torus is isomorphic to one of the form $\mathbb{C}/\Lambda_\tau$ with $\tau \in \mathbb{H}$. The number τ depends on the choice of basis, and the group $\mathrm{SL}_2(\mathbb{Z})$ accounts for this ambiguity. Recall that it acts on $\mathbb{H}$ by

$$\gamma \cdot \tau \coloneqq \frac{a\tau + b}{c\tau + d}, \qquad \gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}).$$

PROPOSITION 1.8.1. *Let $\tau, \tau' \in \mathbb{H}$. Then $\mathbb{C}/\Lambda_\tau \cong \mathbb{C}/\Lambda_{\tau'}$ if and only if $\tau' = \gamma \cdot \tau$ for some $\gamma \in \mathrm{SL}_2(\mathbb{Z})$. Consequently, $\tau \mapsto \mathbb{C}/\Lambda_\tau$ induces a bijection between $\mathrm{SL}_2(\mathbb{Z})\backslash\mathbb{H}$ and the set of isomorphism classes of complex tori.*

PROOF. By Proposition 1.3.1, $\mathbb{C}/\Lambda_\tau \cong \mathbb{C}/\Lambda_{\tau'}$ if and only if $\alpha\Lambda_\tau = \Lambda_{\tau'}$ for some $\alpha \in \mathbb{C}^\times$.

Suppose $\tau' = \gamma \cdot \tau$, and set $\alpha \coloneqq 1/(c\tau + d)$, which is non-zero since $\tau \notin \mathbb{R}$. Since γ is invertible over $\mathbb{Z}$, $\{a\tau + b, c\tau + d\}$ is a $\mathbb{Z}$-basis of Λ_τ, so $\alpha\Lambda_\tau = \mathbb{Z}\tau' \oplus \mathbb{Z} = \Lambda_{\tau'}$.

Conversely, suppose $\alpha\Lambda_\tau = \Lambda_{\tau'}$. Then $\{\alpha\tau, \alpha\}$ is a $\mathbb{Z}$-basis of $\Lambda_{\tau'} = \mathbb{Z}\tau' \oplus \mathbb{Z}$, so there is $\delta = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{GL}_2(\mathbb{Z})$ with $\alpha\tau = a\tau' + b$ and $\alpha = c\tau' + d$. Dividing, $\tau = (a\tau' + b)/(c\tau' + d)$, and the identity

$$\operatorname{Im}\left(\frac{a\tau' + b}{c\tau' + d}\right) = \frac{\det(\delta)\,\operatorname{Im}(\tau')}{|c\tau' + d|^2}$$

together with $\tau, \tau' \in \mathbb{H}$ forces $\det(\delta) = 1$. Hence $\tau = \delta \cdot \tau'$, that is, $\tau' = \delta^{-1} \cdot \tau$ with $\delta^{-1} \in \mathrm{SL}_2(\mathbb{Z})$. The last statement follows, since every complex torus is isomorphic to some $\mathbb{C}/\Lambda_\tau$. □

The region in Figure 1.8.1, including its solid boundary and excluding its dotted boundary, contains exactly one point of every $\mathrm{SL}_2(\mathbb{Z})$-orbit in $\mathbb{H}$. By Proposition 1.8.1, it is in bijection with the set of isomorphism classes of complex tori.

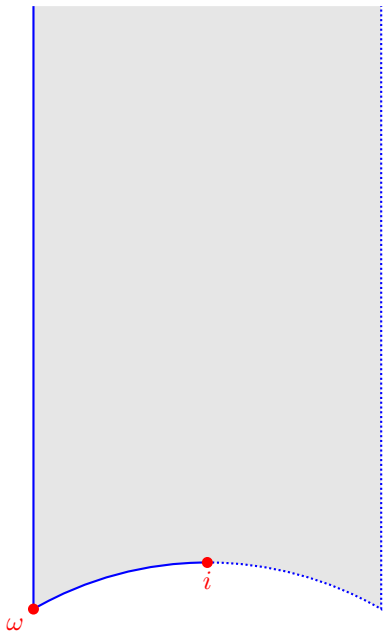


FIGURE 1.8.1. A fundamental domain for the action of $\mathrm{SL}_2(\mathbb{Z})$ on $\mathbb{H}$.

CHAPTER 2

Algebraic varieties

Sources. This chapter follows [Poo06, Chapter 1] and [Sil09, Chapter I]. Unlike these sources, we do not assume that the ground field is perfect.

The goal of this chapter is to describe the categories of affine and projective varieties over a field k through their *functor of points*. The goal is to give a quick and easy way to think about scheme theoretic concepts concretely in the context of varieties.

SITUATION 2.0.1. The field k is arbitrary, and as in the preface $k^{\text{sep}} \subseteq \overline{k}$ are a separable and an algebraic closure, and $\text{Gal}_k := \text{Gal}(k^{\text{sep}}/k)$. We do not assume k perfect: the difference between a property of X and the same property of $X_{\overline{k}}$ is one of the subjects of the chapter.

2.1. Varieties and their points

Recall that a k-algebra A is of finite type if it admits a surjective homomorphism from a polynomial ring $k[x_1, \dots, x_n] \to A$. In other words, every such k-algebra is isomorphic to

$$k[x_1, \dots, x_n]/I$$

for some ideal I. Since polynomial rings are Noetherian, the ideal I is generated by a finite number of polynomials $f_1, \dots, f_m$.

DEFINITION 2.1.1. An affine k-variety is a scheme over k isomorphic to $\operatorname{Spec} A$ for some k-algebra A of finite type.

Recall that a topological space X is Hausdorff if and only if the diagonal subset $\Delta(X) = \{(x_1, x_2) \in X \times X : x_1 = x_2\}$ is closed. Affine k-varieties are usually not Hausdorff (the opens in the Zariski topology are too big), but they always satisfy that the diagonal morphism $\Delta\colon X \to X \times_k X$, which is induced by the multiplication homomorphism $A \otimes_k A \to A$, is a closed embedding. Schemes with this property are called separated.

DEFINITION 2.1.2. A k-variety is a separated scheme of finite type over k. A morphism of k-varieties is a morphism of k-schemes.

Whatever the precise definition of these words is, the bottom line is that a k-variety is a topological space X that can be glued from open affine subvarieties in such a way that the end result is separated. Likewise, a morphism of k-schemes $f\colon X \to Y$ between k-varieties is one that can be glued from morphisms of the form $\operatorname{Spec}(\phi)\colon \operatorname{Spec} B \to \operatorname{Spec} A$, where $\phi\colon A \to B$ is a k-algebra homomorphism.

If you don't know what Spec means yet (go read this chapter in Vakil [Vak23] as soon as possible!), it will be enough for the time being to understand k-varieties X through their sets of R-points, denoted $X(R)$, as we range over all k-algebras R. For an affine k-variety $X = \operatorname{Spec} A$ there is a bijection, natural in R,

$$X(R) \cong \text{Hom}_k(A, R) \cong \{\mathbf{r} \in R^n : f_1(\mathbf{r}) = \cdots = f_m(\mathbf{r}) = 0\}.$$

In general,

DEFINITION 2.1.3. For a k-algebra R, the set of R-points of a k-variety X is

$$X(R) := \text{Hom}_k(\operatorname{Spec} R, X).$$

The elements of $X(k)$ are the rational points of X.

Observe that this assignment is functorial: given a k-algebra homomorphism $\psi\colon R \to S$, we obtain a function $X(R) \to X(S)$. Indeed, if $\mathbf{r} \in X(R)$, since the defining equations $f_1, \dots, f_m$ have coefficients in k, we see that $f(\psi(\mathbf{r})) = \psi(f(\mathbf{r})) = \psi(0) = 0$ for each $f \in I$, so $\psi(\mathbf{r}) \in X(S)$.

EXAMPLE 2.1.4. Given $n \geqslant 1$, affine n-space is $\mathbb{A}^n_k \coloneqq \operatorname{Spec} k[x_1, \dots, x_n]$. A k-algebra homomorphism $k[x_1, \dots, x_n] \to R$ is determined by the images of the variables x_i, so $\mathbb{A}^n(R) = R^n$ for every k-algebra R. An ideal $I \subseteq k[x_1, \dots, x_n]$ defines the closed subvariety $V(I) \coloneqq \operatorname{Spec} k[x_1, \dots, x_n]/I$ of $\mathbb{A}^n_k$, with

$$V(I)(R) = \{\mathbf{r} \in R^n : f(\mathbf{r}) = 0 \text{ for all } f \in I\}.$$

EXAMPLE 2.1.5. Let $X = \operatorname{Spec} k[x]$, let $Y = \operatorname{Spec} k[y]$, and let $f\colon X \to Y$ be the map of k-varieties induced by the ring homomorphism $\varphi\colon k[y] \to k[x]$, $y \mapsto x^2$. The induced function on R-points, denoted $f(R)\colon X(R) \to Y(R)$, sends the R-point $r \in R = X(R)$ to $r^2 \in R = Y(R)$. Indeed, the R-point r is the k-algebra homomorphism $k[x] \to R$, $x \mapsto r$, and composing it with φ gives the commutative diagram of k-algebras

$$k[y] \xrightarrow{\varphi} k[x] \xrightarrow{x \mapsto r} R, \qquad y \mapsto r^2 \text{ (composite)}$$

whose composite is the R-point r^2 of Y.

EXAMPLE 2.1.6. The multiplicative group is $\mathbb{G}_{\mathrm{m}} \coloneqq \operatorname{Spec} k[x, x^{-1}]$. Since

$$k[x, x^{-1}] \cong k[x, y]/(xy - 1),$$

$\mathbb{G}_{\mathrm{m}}$ can be embedded in the affine plane as $\mathbb{G}_{\mathrm{m}} \cong V(xy-1) \subseteq \mathbb{A}^2_k$. It is also the open subvariety $\{x \neq 0\}$ of $\mathbb{A}^1_k$. A k-algebra homomorphism $k[x, x^{-1}] \to R$ is determined by the image of x, which can be any unit of R, so

$$\mathbb{G}_{\mathrm{m}}(R) = R^\times$$

for every k-algebra R. The group structure on $R^\times$ is natural in R. This is the sense in which $\mathbb{G}_{\mathrm{m}}$ is a group.

The multiplicative group is our first example of a group variety [Poo06, Section 1.11].

DEFINITION 2.1.7. A group variety over k is a k-variety G with morphisms $m\colon G \times G \to G$, $i\colon G \to G$ and $e\colon \operatorname{Spec} k \to G$ satisfying the group axioms, written as commutative diagrams of morphisms. A homomorphism of group varieties is a morphism that respects m, i and e.

Equivalently, by Yoneda's lemma, a group variety is a k-variety G with a group structure on $G(R)$ for every k-algebra R, natural in R. This is how we saw that $\mathbb{G}_{\mathrm{m}}$ is a group variety in Example 2.1.6.

The group varieties of dimension 1 that are not elliptic curves are affine, and there are few of them [Poo08, Section 6.2]. To list them we need a quadratic algebra. Fix $d \in k$, with $d \neq 0$ if $\operatorname{char} k \neq 2$, and let

$$L_d \coloneqq \begin{cases} k[t]/(t^2 - d) & \text{if } \operatorname{char} k \neq 2, \\ k[t]/(t^2 + t + d) & \text{if } \operatorname{char} k = 2. \end{cases} \tag{2.1.1}$$

In both cases the polynomial is separable, so L_d is either a separable quadratic extension field of k or isomorphic to $k \times k$. Its non-trivial automorphism sends t to $-t$, respectively $t+1$, and the norm of $x + yt$ is

$$\operatorname{Nm}(x + yt) = \begin{cases} x^2 - dy^2 & \text{if } \operatorname{char} k \neq 2, \\ x^2 + xy + dy^2 & \text{if } \operatorname{char} k = 2. \end{cases} \tag{2.1.2}$$

EXAMPLE 2.1.8. The following are group varieties of dimension 1 over k: the additive group $\mathbb{G}_{\mathrm{a}}$, the multiplicative group $\mathbb{G}_{\mathrm{m}}$ of Example 2.1.6, and the twisted multiplicative group $\mathbb{G}_{\mathrm{m}}^{(d)}$.

G	variety	$m\big((x_1, y_1), (x_2, y_2)\big)$	$G(k)$
$\mathbb{G}_{\mathrm{a}}$	$\mathbb{A}^1$	$x_1 + x_2$	k, under $+$
$\mathbb{G}_{\mathrm{m}}$	$xy = 1$	(x_1x_2, y_1y_2)	$k^\times$
$\mathbb{G}_{\mathrm{m}}^{(d)}$	$\operatorname{Nm}(x + yt) = 1$	$(x_1x_2 + dy_1y_2,\ x_1y_2 + x_2y_1)$	$\ker(\operatorname{Nm}\colon L_d^\times \to k^\times)$

The group law of $\mathbb{G}_{\mathrm{m}}^{(d)}$ is multiplication in L_d, with (x, y) standing for $x + yt$. In characteristic 2, where $t^2 = t + d$, its second coordinate is $x_1y_2 + x_2y_1 + y_1y_2$. The identity is $(1, 0)$ and the inverse is conjugation. More generally $\mathbb{G}_{\mathrm{a}}(R) = R$, $\mathbb{G}_{\mathrm{m}}(R) = R^\times$ and

$$\mathbb{G}_{\mathrm{m}}^{(d)}(R) = \ker\bigl(\mathrm{Nm}\colon (L_d \otimes_k R)^\times \to R^\times\bigr)$$

for every k-algebra R. An element of norm 1 is a unit, its inverse being its conjugate. Over $\mathbb{R}$, $\mathbb{G}_{\mathrm{m}}^{(-1)}$ is the circle group $x^2 + y^2 = 1$, and over $\mathbb{Q}$, $\mathbb{G}_{\mathrm{m}}^{(2)}$ is the "Pell equation torus" $x^2 - 2y^2 = 1$ of [Poo06, Example 1.11.3].

PROPOSITION 2.1.9. *Let d and L_d be as in (2.1.1).*

(1) If $L_d \cong k \times k$, then $\mathbb{G}_m^{(d)} \cong \mathbb{G}_m$. In particular $\mathbb{G}_m^{(1)} \cong \mathbb{G}_m$ if $\operatorname{char} k \neq 2$, and $\mathbb{G}_m^{(0)} \cong \mathbb{G}_m$ if $\operatorname{char} k = 2$.

(2) $(\mathbb{G}_m^{(d)})_{L_d} \cong (\mathbb{G}_m)_{L_d}$. The isomorphism class of $\mathbb{G}_m^{(d)}$ depends only on the class of d in $k^\times/k^{\times 2}$, respectively in $k/\{c^2 + c : c \in k\}$.

(3) If L_d is a field, then $\mathbb{G}_m^{(d)} \not\cong \mathbb{G}_m$ over k.

(4) If $k = \mathbb{F}_q$ and L_d is a field, then $\mathbb{G}_m^{(d)}(\mathbb{F}_q)$ is cyclic of order $q + 1$.

PROOF. For part 1, $L_d \cong k \times k$ means that $t^2 - d$, respectively $t^2 + t + d$, has a root $\delta \in k$. Let δ' be the other one, $-\delta$, respectively $\delta + 1$, so that $\delta' \neq \delta$. Then $\mathrm{Nm}(x + yt) = (x + \delta y)(x + \delta' y)$, and $(x, y) \mapsto x + \delta y$, the restriction of the ring homomorphism $L_d \to k$, $t \mapsto \delta$, is a homomorphism $\mathbb{G}_{\mathrm{m}}^{(d)} \to \mathbb{G}_{\mathrm{m}}$. Its inverse is

$$u \mapsto \left(\frac{\delta' u - \delta u^{-1}}{\delta' - \delta},\ \frac{u^{-1} - u}{\delta' - \delta}\right),$$

since $x + \delta' y = (x + \delta y)^{-1}$ on $\mathbb{G}_{\mathrm{m}}^{(d)}$.

For part 2, the image of t is a root in L_d, and $L_d \otimes_k L_d \cong L_d \times L_d$, so part 1 applies over L_d. Replacing d by $c^2 d$ with $c \in k^\times$, respectively by $d + c^2 + c$ with $c \in k$, gives an isomorphic group variety, through $(x, y) \mapsto (x, y/c)$, respectively $(x, y) \mapsto (x + cy, y)$: both are the isomorphisms $L_{c^2 d} \to L_d$, $t \mapsto ct$, respectively $L_{d+c^2+c} \to L_d$, $t \mapsto t + c$, written in coordinates.

For part 3, the projective closure $\overline{G} \coloneqq V_+(\mathrm{Nm}(X + YT) - Z^2) \subseteq \mathbb{P}_k^2$ of $\mathbb{G}_{\mathrm{m}}^{(d)}$, where $\mathrm{Nm}(X + YT)$ is the quadratic form (2.1.2) in X and Y, is a smooth conic: its partial derivatives vanish together only at $(0, 0, 0)$. The complement $\overline{G} - \mathbb{G}_{\mathrm{m}}^{(d)} = V_+(\mathrm{Nm}(X + YT), Z)$ is a single closed point with residue field L_d, because $\mathrm{Nm}(X + YT)$ is irreducible over k. The complement of $\mathbb{G}_{\mathrm{m}}$ in its projective closure $\mathbb{P}_k^1$ is two rational points, 0 and ∞. An isomorphism $\mathbb{G}_{\mathrm{m}}^{(d)} \to \mathbb{G}_{\mathrm{m}}$ and its inverse extend to morphisms between the projective closures (Proposition 3.2.1), inverse to each other, which would match the complements.

For part 4, $L_d = \mathbb{F}_{q^2}$, whose non-trivial automorphism is $\alpha \mapsto \alpha^q$, so the norm is $\alpha \mapsto \alpha^{q+1}$. Its kernel in the cyclic group $\mathbb{F}_{q^2}^\times$ of order $q^2 - 1$ is cyclic of order $q + 1$. □

DEFINITION 2.1.10. A group variety T over k is a torus if $T_{\overline{k}} \cong (\mathbb{G}_{\mathrm{m}}^n)_{\overline{k}}$ for some $n \geqslant 0$, and a split torus if $T \cong \mathbb{G}_{\mathrm{m}}^n$ over k [Poo06, Definition 1.11.4].

By Proposition 2.1.9, $\mathbb{G}_{\mathrm{m}}^{(d)}$ is a torus of dimension 1, split exactly when $L_d \cong k \times k$. For $d = 2$ over $\mathbb{Q}$ this is [Poo06, Exercise 1.7]. Over a perfect field of characteristic not 2, every connected group variety of dimension 1 is an elliptic curve, $\mathbb{G}_{\mathrm{a}}$, $\mathbb{G}_{\mathrm{m}}$ or some $\mathbb{G}_{\mathrm{m}}^{(d)}$ [Poo08, Section 6.2], [Mil20, Section II.3]. In Chapter 4 the last three appear as the nonsingular parts of singular Weierstrass cubics.

In the complex analytic world, $\mathbb{P}^n(\mathbb{C})$ is the set of lines through the origin in $\mathbb{C}^{n+1}$. For $n = 1, 2$ we saw that it is covered by $n + 1$ copies of $\mathbb{C}^n$: see Example 1.1.4 and Section 1.6. In the algebraic world, we want to define $\mathbb{P}_k^n$ in a similar way, as the quotient of $\mathbb{A}_k^{n+1} - \{0\}$ by the scaling action of $\mathbb{G}_{\mathrm{m}}$ (Example 2.1.6). This is what the Proj construction achieves. Still, it is quite tricky to describe the set $\mathbb{P}^n(R)$.

EXAMPLE 2.1.11. Projective n-space is $\mathbb{P}_k^n \coloneqq \operatorname{Proj} k[X_0, \ldots, X_n]$. A homogeneous ideal $I \subseteq k[X_0, \ldots, X_n]$ defines the closed subvariety $V_+(I) \coloneqq \operatorname{Proj} k[X_0, \ldots, X_n]/I$ of $\mathbb{P}_k^n$.

Let R be a k-algebra. To describe a point of $\mathbb{P}^n(R)$, consider the tuples $(P, s_0, \dots, s_n)$, where P is an invertible R-module – a finitely generated projective R-module of constant rank 1 – and $s_0, \dots, s_n$ are elements generating P. Call $(P, s_0, \dots, s_n)$ and $(P', s'_0, \dots, s'_n)$ equivalent if some isomorphism of R-modules $\varphi\colon P \to P'$ satisfies $\varphi(s_i) = s'_i$ for every i. A point of $\mathbb{P}^n(R)$ is then an equivalence class $[P, s_0, \dots, s_n]$ [Har77, Theorem II.7.1].

To evaluate a homogeneous polynomial at such a point, note that the tensor power $P^{\otimes d} \coloneqq P \otimes_R \cdots \otimes_R P$ is again invertible. Write $\mathbf{X} \coloneqq (X_0, \dots, X_n)$ and $\mathbf{s} \coloneqq (s_0, \dots, s_n)$, and for a multi-index $\mathbf{a} = (a_0, \dots, a_n)$ of non-negative integers write $|\mathbf{a}| \coloneqq a_0 + \cdots + a_n$. A monomial $\mathbf{X}^{\mathbf{a}} = X_0^{a_0} \cdots X_n^{a_n}$ with $|\mathbf{a}| = d$ has a value

$$\mathbf{s}^{\mathbf{a}} \coloneqq \underbrace{s_0 \otimes \cdots \otimes s_0}_{a_0} \otimes \cdots \otimes \underbrace{s_n \otimes \cdots \otimes s_n}_{a_n} \in P^{\otimes d}.$$

So a homogeneous $F = \sum_{|\mathbf{a}|=d} c_{\mathbf{a}} \mathbf{X}^{\mathbf{a}} \in k[X_0, \dots, X_n]$ of degree d has a value

$$F(\mathbf{s}) \coloneqq \sum_{|\mathbf{a}|=d} c_{\mathbf{a}} \mathbf{s}^{\mathbf{a}} \in P^{\otimes d},$$

and $\varphi^{\otimes d}$ carries $F(\mathbf{s})$ to $F(\mathbf{s}')$, so whether this value vanishes depends only on the R-point. The R-point lies in $V_+(I)(R)$ if and only if $F(\mathbf{s}) = 0$ for every homogeneous $F \in I$.

If every invertible R-module is free, e.g. if R is a field, a local ring or a principal ideal domain, then choosing a basis of P gives

$$\mathbb{P}^n(R) = \{(a_0, \dots, a_n) \in R^{n+1} : a_0 R + \cdots + a_n R = R\}/R^\times$$

and

$$V_+(I)(R) = \{(a_0 : \cdots : a_n) \in \mathbb{P}^n(R) : F(a) = 0 \text{ for all homogeneous } F \in I\}.$$

When R is a field, the condition $a_0 R + \cdots + a_n R = R$ says that $a \neq 0$, so

$$\mathbb{P}^n(R) = \big(R^{n+1} - \{0\}\big)/R^\times.$$

For general R there is no such description: an R-point whose P is not free has no homogeneous coordinates. The open subschemes $U_i \coloneqq \{X_i \neq 0\}$ of $\mathbb{P}^n_k$ are isomorphic to $\mathbb{A}^n_k$, and cover it. The affine coordinates on U_0 are the ratios $x_i = X_i/X_0$. For a closed subvariety $X \subseteq \mathbb{P}^n_k$, the $X \cap U_i$ are the standard affine patches of X. The equations of $X \cap U_0$ are obtained by setting $X_0 = 1$ (dehomogenization).

Conversely, identify $\mathbb{A}^n_k$ with the patch $U_0 \subseteq \mathbb{P}^n_k$. The projective closure of a closed subvariety $X = V(I) \subseteq \mathbb{A}^n_k$ is the Zariski closure $\overline{X}$ of X in $\mathbb{P}^n_k$. It is cut out by the homogenizations of the elements of I. The homogenization of $f \in k[x_1, \dots, x_n]$ is the homogeneous polynomial $F \in k[X_0, \dots, X_n]$ of the same degree given by

$$F(X_0, \dots, X_n) \coloneqq X_0^{\deg f}\, f(X_1/X_0, \dots, X_n/X_0),$$

and $\overline{X} = V_+(J)$, where J is the homogeneous ideal generated by the homogenizations of all elements of I. For a single non-zero f this says that the projective closure of $V(f)$ is $V_+(F)$. For several equations one must homogenize every element of the ideal, not only a set of generators.

Example 2.1.12. The twisted cubic is the affine curve $C \coloneqq V(I) \subseteq \mathbb{A}^3_k$, where

$$I \coloneqq (x_2 - x_1^2,\ x_3 - x_1^3),$$

the image of $t \mapsto (t, t^2, t^3)$. Homogenizing the two generators of I gives the quadric surface $V_+(X_0X_2 - X_1^2)$ and the cubic surface $V_+(X_0^2X_3 - X_1^3)$ in $\mathbb{P}^3_k$. On the hyperplane $X_0 = 0$ both equations reduce to $X_1 = 0$, so their intersection contains the whole line $\{X_0 = X_1 = 0\}$. But $\overline{C}$ meets $X_0 = 0$ only in $(0 : 0 : 0 : 1)$, the limit of the points $(1 : t : t^2 : t^3)$ of C for large t. The projective closure is therefore strictly smaller.

The identity

$$x_1x_3 - x_2^2 = x_1(x_3 - x_1^3) + (x_1^2 - x_2)(x_1^2 + x_2)$$

shows that $x_1x_3 - x_2^2$ lies in I. Its homogenization $X_1X_3 - X_2^2$ is again of degree 2, and the quadric surface $V_+(X_1X_3 - X_2^2)$ it defines cuts the spurious line down to the rational point $(0 : 0 : 0 : 1)$: on $\{X_0 = X_1 = 0\}$ the form $X_1X_3 - X_2^2$ restricts to $-X_2^2$, which forces $X_2 = 0$. So this quadric is needed to cut out $\overline{C}$, although it is not in the ideal generated by the homogenizations of the two generators of I.

Definition 2.1.13. Let X be a k-variety.

(1) X is irreducible if its underlying topological space is non-empty and is not the union of two proper closed subsets.
(2) X is reduced if for every open $U \subseteq X$, the ring $\mathcal{O}_X(U)$ has no non-zero nilpotent elements.
(3) X is integral if for every non-empty open $U \subseteq X$, the ring $\mathcal{O}_X(U)$ is an integral domain.

A k-variety is integral if and only if it is reduced and irreducible [Har77, Proposition II.3.1]. For an affine variety $X = V(I) \subseteq \mathbb{A}^n_k$: X is reduced if and only if I is radical, irreducible if and only if $\sqrt{I}$ is prime, and integral if and only if I is prime.

Warning 2.1.14. Varieties need not be reduced or irreducible (Definition 2.1.13). The variety $V(x^2) \subseteq \mathbb{A}^1_k$ is irreducible but not reduced, and $V(xy) \subseteq \mathbb{A}^2_k$ is reduced but not irreducible. For every field L, both $V(x^2)(L)$ and $V(x)(L)$ consist of the single L-point 0, but $V(x^2) \neq V(x)$: their points with values in $R = k[\varepsilon]/(\varepsilon^2)$ differ.

The rational points do not determine a variety either.

Example 2.1.15. Consider the three closed subvarieties

$$C_1 \coloneqq V_+(X^2 + Y^2 + Z^2), \qquad C_2 \coloneqq V_+(X^2 + Y^2 - 3Z^2), \qquad C_3 \coloneqq V_+(1)$$

of $\mathbb{P}^2_{\mathbb{Q}}$. None of them has a rational point.
For C_1 there is nothing to prove over $\mathbb{Q}$: already $C_1(\mathbb{R}) = \varnothing$, since a sum of three squares of real numbers vanishes only when all three do. The variety C_3 is empty, so it has no points with values in any $\mathbb{Q}$-algebra at all.
For C_2 we argue by descent. A point of $C_2(\mathbb{Q})$ can be scaled to a solution $(X_0, Y_0, Z_0) \in \mathbb{Z}^3$ with $\gcd(X_0, Y_0, Z_0) = 1$. Reduce the equation $X_0^2 + Y_0^2 = 3Z_0^2$ modulo 3. Since -1 is not a square modulo 3, the congruence $X_0^2 + Y_0^2 \equiv 0 \pmod 3$ forces $3 \mid X_0$ and $3 \mid Y_0$. Then $9 \mid 3Z_0^2$, so $3 \mid Z_0$, contradicting $\gcd(X_0, Y_0, Z_0) = 1$.
The three varieties are nevertheless pairwise distinct: their homogeneous ideals differ, and this is already visible on points with values in larger fields. Indeed $C_3(\mathbb{C}) = \varnothing$, while $C_1(\mathbb{C}) \neq \varnothing$; and $C_1(\mathbb{R}) = \varnothing$, while $(\sqrt{3} : 0 : 1) \in C_2(\mathbb{R})$.

Example 2.1.16. The cubic $E \coloneqq V_+(Y^2Z - X^3 - 17Z^3) \subseteq \mathbb{P}^2_{\mathbb{Q}}$ is the projective closure of the affine curve $y^2 = x^3 + 17$, where $x = X/Z$ and $y = Y/Z$. Setting $Z = 0$ leaves $X^3 = 0$, so E has a single point at infinity, $(0 : 1 : 0)$, and

$$E(\mathbb{Q}) = \{(x, y) \in \mathbb{Q}^2 : y^2 = x^3 + 17\} \cup \{(0 : 1 : 0)\},$$

which contains for instance $(-2, 3)$ and $(5234, 378661)$ [Sil09, Examples I.1.3.3 and I.2.8].

The Galois group Gal_k acts on $X(k^{\mathrm{sep}})$ through its action on k^{sep}. On k^{sep}-points of $\mathbb{A}^n$ or $\mathbb{P}^n$ it acts on coordinates.

Proposition 2.1.17. *Let X be a k-variety. Then $X(k) = X(k^{\mathrm{sep}})^{\mathrm{Gal}_k}$.*

Proof. Every k^{sep}-point of X lies in $U(k^{\mathrm{sep}})$ for some affine open subvariety $U \subseteq X$, and $U(k^{\mathrm{sep}})$ is stable under Gal_k. So it suffices to treat closed subvarieties of $\mathbb{A}^n_k$, where the claim follows from Galois theory: $(k^{\mathrm{sep}})^{\mathrm{Gal}_k} = k$.

For a closed subvariety of $\mathbb{P}^n_k$ one can also argue in homogeneous coordinates: let $P = (a_0 : \cdots : a_n) \in X(k^{\mathrm{sep}})$ be fixed by Gal_k. Then $\sigma(a_i) = \lambda_\sigma a_i$ for some $\lambda_\sigma \in (k^{\mathrm{sep}})^\times$, and $\sigma \mapsto \lambda_\sigma$ is a 1-cocycle. By Hilbert's Theorem 90 there is $\alpha \in (k^{\mathrm{sep}})^\times$ with $\lambda_\sigma = \alpha/\sigma(\alpha)$, and then all αa_i lie in k. □

Warning 2.1.18. [Sil09] works with $\overline{k}$ instead of k^{sep}. The fixed field of $\mathrm{Aut}(\overline{k}/k)$ is the perfect closure k^{perf}, so $X(\overline{k})^{\mathrm{Aut}(\overline{k}/k)} = X(k^{\mathrm{perf}})$, which can be larger than $X(k)$ when k is imperfect. For example, if $k = \mathbb{F}_p(t)$ and $X = V(x^p - t) \subseteq \mathbb{A}^1_k$, then $X(k) = \varnothing$, while the unique point $t^{1/p}$ of $X(\overline{k})$ is fixed by $\mathrm{Aut}(\overline{k}/k)$.

Remark 2.1.19. In [Sil09, Section I.1], whose ground field K is our k, a variety V/K is a subset of $\mathbb{A}^n(\overline{k})$ or $\mathbb{P}^n(\overline{k})$, cut out by an ideal $I(V/K)$ with coefficients in K. It corresponds to the k-variety $X \coloneqq V(I(V/K))$ or $V_+(I(V/K))$: then $V = X(\overline{k})$ and $V(K) = X(k)$. The standing assumptions of [Sil09] (that $I(V/K)$ generates the ideal of V, and that it is prime over $\overline{k}$) say that X is geometrically integral (Definition 2.2.2).

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2.2. Base extension and geometric properties

Definition 2.2.1. Let X be a k-variety and $L \supseteq k$ a field extension. The base extension of X to L is the L-variety $X_L \coloneqq X \times_{\operatorname{Spec} k} \operatorname{Spec} L$.

Concretely, X_L is defined by the same equations as X, with the coefficients viewed in L: if $X = V(f_1, \dots, f_m) \subseteq \mathbb{A}^n_k$, then $X_L = V(f_1, \dots, f_m) \subseteq \mathbb{A}^n_L$, and likewise for projective varieties. We have $X(L) = X_L(L)$ [Poo06, Section 1.3].

Definition 2.2.2. A k-variety X is geometrically reduced, geometrically irreducible or geometrically integral if $X_{\overline{k}}$ is reduced, irreducible or integral (Definition 2.1.13), respectively.

Definition 2.2.3. Let X be an integral k-variety. Its function field $k(X)$ is the local ring at the generic point of X: for $X = \operatorname{Spec} A$ it is $\operatorname{Frac} A$, and in general it is the function field of any non-empty affine open subvariety. Its elements are the rational functions on X [Poo06, Definitions 1.4.4 and 1.4.8].

Proposition 2.2.4. *Let X be an integral k-variety.*
(1) X is geometrically irreducible if and only if k is separably closed in $k(X)$.
(2) X is geometrically reduced if and only if the extension $k(X)/k$ is separable, i.e. separably generated.
(3) If k is perfect, then X is geometrically reduced.

Proof. Part 1 is [Poo06, Remarks 1.4.6 and 1.4.7]. Part 3 follows from part 2, since every finitely generated extension of a perfect field is separably generated. □

Example 2.2.5. The curve $X \coloneqq V(x^2 - 2y^2) \subseteq \mathbb{A}^2_{\mathbb{Q}}$ is integral, since $x^2 - 2y^2$ is irreducible in $\mathbb{Q}[x, y]$. Over $\mathbb{Q}(\sqrt{2})$ it becomes the union of the lines $x = \pm\sqrt{2}\, y$, so X is not geometrically irreducible. Accordingly $x/y \in \mathbb{Q}(X)$ satisfies $(x/y)^2 = 2$ [Poo06, Example 1.4.3].

Example 2.2.6. Let $k \coloneqq \mathbb{F}_p(a)$ and $f \coloneqq x^p + ay^p$. Then f is irreducible in $k[x, y]$, so $X \coloneqq V(f) \subseteq \mathbb{A}^2_k$ is integral. Over $\overline{k}$, $f = (x + \alpha y)^p$ with $\alpha^p = a$, so X is not geometrically reduced [Mil20, Example I.1.2]. Accordingly $(x/y)^p = -a$ in $k(X)$, so $k(X)$ contains the inseparable extension $k(a^{1/p})$ of k.

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2.3. Morphisms and rational maps

Definition 2.3.1. Let X be an integral k-variety and Y a k-variety. A rational map $\varphi\colon X \dashrightarrow Y$ is an equivalence class of morphisms $U \to Y$ with $U \subseteq X$ a non-empty open subvariety, two being equivalent if they agree on the intersection of their domains. It is dominant if its image is dense. Integral k-varieties X and Y are birational if there are dominant rational maps $X \dashrightarrow Y$ and $Y \dashrightarrow X$ inverse to each other [Poo06, Definitions 1.5.2, 1.5.3 and 1.5.5].

Explicitly, a rational map $\varphi\colon X \dashrightarrow \mathbb{P}^n_k$ is a tuple $(f_0 : \cdots : f_n)$ of elements of $k(X)$, not all zero, modulo multiplication by $k(X)^\times$. It is regular at a point P of X if there is $g \in k(X)^\times$ such that every gf_i lies in $\mathcal{O}_{X,P}$ and some $(gf_i)(P) \in k(P)$ is non-zero. Then $\varphi(P) \coloneqq ((gf_0)(P) : \cdots : (gf_n)(P))$. A rational map to a closed subvariety $Y \subseteq \mathbb{P}^n_k$ is one whose image lies in Y, and it is a morphism if it is regular at every point [Poo06, Definitions 1.5.8 and 1.5.9], [Sil09, Section I.3]. For $X \subseteq \mathbb{P}^m_k$, clearing denominators writes φ as a tuple of homogeneous polynomials of the same degree, not all in the ideal of X [Sil09, Remark I.3.2].

Remark 2.3.2. A morphism of k-varieties is automatically defined over k. A map "defined over $\overline{k}$" in the sense of [Sil09, Section I.3] is a morphism $X_{\overline{k}} \to Y_{\overline{k}}$ [Poo06, Warning 1.5.7]. Varieties X and Y are isomorphic over L if $X_L \cong Y_L$.

THEOREM 2.3.3. *The functor $X \mapsto k(X)$ is an equivalence between the category of integral k-varieties with dominant rational maps and the opposite of the category of finitely generated field extensions of k with k-algebra homomorphisms. In particular, integral k-varieties X and Y are birational if and only if $k(X) \cong k(Y)$ over k.*

PROOF. See [Poo06, equation (1.5.4) and Definition 1.5.6]. □

EXAMPLE 2.3.4. Suppose $\operatorname{char} k \neq 2$, and let $C \coloneqq V_+(X^2+Y^2-Z^2) \subseteq \mathbb{P}^2_k$. The rational map $\varphi \coloneqq (X+Z : Y)\colon C \dashrightarrow \mathbb{P}^1_k$ is regular away from the rational point $(1:0:-1)$. Since $(X+Z)(X-Z) = -Y^2$ on C, also $\varphi = (-Y : X-Z)$, which is regular at $(1:0:-1)$ with value $(0:1)$. So φ is a morphism. The morphism

$$\psi\colon \mathbb{P}^1_k \to C, \qquad (S:T) \mapsto (S^2-T^2 : 2ST : S^2+T^2)$$

is inverse to φ, so $C \cong \mathbb{P}^1_k$ [Sil09, Example I.3.5].

EXAMPLE 2.3.5. Suppose $\operatorname{char} k \neq 2$, and let $C \coloneqq V_+(Y^2Z - X^3 - X^2Z) \subseteq \mathbb{P}^2_k$. The morphism

$$\psi\colon \mathbb{P}^1_k \to C, \qquad (S:T) \mapsto \big((S^2-T^2)T : (S^2-T^2)S : T^3\big)$$

and the rational map $\varphi \coloneqq (Y : X)\colon C \dashrightarrow \mathbb{P}^1_k$ are inverse to each other where defined, so C is birational to $\mathbb{P}^1_k$. But φ is not regular at $(0:0:1)$, and ψ sends both $(1:1)$ and $(1:-1)$ to $(0:0:1)$, so $C \ncong \mathbb{P}^1_k$. The rational point $(0:0:1)$ is singular (Example 2.4.5) [Sil09, Example I.3.7].

EXAMPLE 2.3.6. Let $C \coloneqq V_+(X^2+Y^2-Z^2)$ and $C_2 \coloneqq V_+(X^2+Y^2-3Z^2)$ in $\mathbb{P}^2_{\mathbb{Q}}$. By Examples 2.3.4 and 2.1.15, $C(\mathbb{Q})$ is infinite and $C_2(\mathbb{Q})$ is empty, so $C \ncong C_2$. But $(X:Y:Z) \mapsto (X:Y:\sqrt{3}\,Z)$ is an isomorphism $(C_2)_{\mathbb{Q}(\sqrt{3})} \cong C_{\mathbb{Q}(\sqrt{3})}$ [Sil09, Example I.3.8].

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2.4. Dimension, regularity and smoothness

The dimension of a k-variety is its dimension as a topological space. If X is integral, then $\dim X = \operatorname{trdeg}_k k(X)$, and $\dim X_L = \dim X$ for every field extension $L \supseteq k$ [Poo06, Section 1.6].

DEFINITION 2.4.1. Let X be a k-variety.

(1) X is regular at a point P if the local ring $\mathcal{O}_{X,P}$ is regular, i.e. $\dim_{k(P)} \mathfrak{m}_P/\mathfrak{m}_P^2 = \dim \mathcal{O}_{X,P}$. It is regular if it is regular at every point.
(2) Suppose $X = V(f_1,\dots,f_m) \subseteq \mathbb{A}^n_k$ has pure dimension d. Then X is smooth at $P \in X(\overline{k})$ if the Jacobian matrix $(\partial f_i/\partial x_j(P))$ has rank $n-d$. A projective variety is smooth at P if some (equivalently, every) affine patch containing P is. X is smooth if it is smooth at every $P \in X(\overline{k})$ [Poo06, Definition 1.7.1].
(3) X is nice if it is smooth, projective and geometrically integral [Poo06, Definition 1.7.8].

Smoothness is tested at $\overline{k}$-points, so it is a geometric property: X is smooth if and only if X_L is, for any field $L \supseteq k$ [Poo06, Remark 1.7.6]. Regularity is not.

PROPOSITION 2.4.2. *Let X be a k-variety.*

(1) X is smooth if and only if $X_{\overline{k}}$ is regular. In particular, a smooth variety is regular.
(2) If k is perfect, then X is smooth if and only if it is regular.
(3) Let X be integral, and $P \in X(\overline{k})$. Then X is smooth at P if and only if $\dim_{\overline{k}} \mathfrak{m}_P/\mathfrak{m}_P^2 = \dim X$, where $\mathfrak{m}_P$ is the maximal ideal of $\mathcal{O}_{X_{\overline{k}},P}$.

PROOF. Part 3 is [Sil09, Proposition I.1.7], and it gives part 1 at closed points. Over an imperfect field, regular does not imply smooth: see Example 2.4.6. □

REMARK 2.4.3. The "smooth curves" of [Sil09, Chapter II] are the nice k-varieties of dimension 1.

A singular point of the hypersurface $V(f) \subseteq \mathbb{A}^n_k$ is a common zero of f and its partial derivatives [Poo06, Example 1.7.4].

EXAMPLE 2.4.4. Suppose $\operatorname{char} k \neq 2$. On $y^2 = x^3 + x$, a singular point satisfies $2y = 3x^2 + 1 = 0$ and $x(x^2 + 1) = 0$, which is impossible. So this curve is smooth [Sil09, Example I.1.6].

EXAMPLE 2.4.5. Suppose $\operatorname{char} k \neq 2$. On $y^2 = x^3 + x^2$, the conditions $2y = 3x^2 + 2x = 0$ and $x^2(x + 1) = 0$ leave only $(0, 0)$, which is a singular point [Sil09, Example I.1.6].

EXAMPLE 2.4.6. Let $k \coloneqq \mathbb{F}_3(t)$, let $F \coloneqq Y^2Z - X^3 + tZ^3$, and let $E \coloneqq V_+(F) \subseteq \mathbb{P}^2_k$, the curve $y^2 = x^3 - t$. Let $s \in \overline{k}$ with $s^3 = t$. Since $\operatorname{char} k = 3$,

$$\frac{\partial F}{\partial X} = 0, \qquad \frac{\partial F}{\partial Y} = 2YZ, \qquad \frac{\partial F}{\partial Z} = Y^2,$$

and F vanishes at $(s : 0 : 1)$ too, so this is a singular point and E is not smooth. Over $\overline{k}$ the equation is $y^2 = (x - s)^3$, a cuspidal cubic, which is integral. So E is geometrically integral. Yet E is regular. Every other $\overline{k}$-point is smooth, and the closed point $P \coloneqq V(y, x^3 - t)$, with residue field $k(s)$ of degree 3, has maximal ideal $(y, x^3 - t) = (y)$ in $\mathcal{O}_{E,P}$, because $x^3 - t = y^2$ there. So $\mathcal{O}_{E,P}$ is a discrete valuation ring.

CHAPTER 3

Algebraic curves

Sources. This chapter follows [Poo06, Chapter 2] and [Sil09, Chapter II], with material on plane curves from [Mil20, Sections I.1 and I.6]. Unlike these sources, we do not assume that the ground field is perfect.

DEFINITION 3.0.1. A curve over k is an integral k-variety of dimension 1. A nice curve is a curve that is nice (Definition 2.4.1).

SITUATION 3.0.2. We keep Situation 2.0.1: k is an arbitrary field, and $\mathrm{Gal}_k := \mathrm{Gal}(k^{\mathrm{sep}}/k)$. Unless stated otherwise, C, C_1 and C_2 denote nice curves over k.

Remarks point out where a curve that is only regular behaves differently. The basic example is Example 2.4.6.

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3.1. Closed points and local rings

DEFINITION 3.1.1. A closed point of a k-variety X is a point whose residue field $k(P)$ is a finite extension of k. Its degree is $\deg P := [k(P) : k]$ [Poo06, Definition 2.2.1].

The rational points of X are the closed points of degree 1.

PROPOSITION 3.1.2. *Let X be a k-variety. Sending a $\overline{k}$-point to its image is a surjection from $X(\overline{k})$ onto the set of closed points of X, whose fibers are the $\mathrm{Aut}(\overline{k}/k)$-orbits. The fiber over P has $[k(P) : k]_s$ elements, and it meets $X(k^{\mathrm{sep}})$ if and only if $k(P)/k$ is separable.*

PROOF. A $\overline{k}$-point over P is the same as a k-embedding $k(P) \hookrightarrow \overline{k}$. There are $[k(P) : k]_s$ of these, permuted transitively by $\mathrm{Aut}(\overline{k}/k)$, and they land in k^{sep} exactly when $k(P)/k$ is separable. □

EXAMPLE 3.1.3. On $E\colon y^2 = x^3 + 17$ over $\mathbb{Q}$, the closed point $P := V(x, y^2 - 17)$ has residue field $\mathbb{Q}(\sqrt{17})$ and degree 2. It corresponds to the orbit $\{(0, \sqrt{17}), (0, -\sqrt{17})\}$ in $E(\overline{\mathbb{Q}})$.

EXAMPLE 3.1.4. Let $k := \mathbb{F}_p(t)$. The closed point $P := V(x^p - t)$ of $\mathbb{A}^1_k$ has degree p and residue field $k(t^{1/p})$. It has a single $\overline{k}$-point above it, and no k^{sep}-point.

PROPOSITION 3.1.5. *Let X be a curve over k, and P a closed point. The local ring $\mathcal{O}_{X,P}$ is a discrete valuation ring if and only if X is regular at P. In particular this holds at every closed point of a nice curve.*

PROOF. A noetherian local domain of dimension 1 is a discrete valuation ring if and only if its maximal ideal is principal, i.e. it is regular [Sil09, Proposition II.1.1 and Exercise 2.1]. □

DEFINITION 3.1.6. Let P be a closed point of C. The normalized valuation of $\mathcal{O}_{C,P}$ extends to $\mathrm{ord}_P\colon k(C)^\times \to \mathbb{Z}$. A uniformizer at P is a $t \in k(C)$ with $\mathrm{ord}_P(t) = 1$. If $\mathrm{ord}_P(f) > 0$, then f has a zero at P; if $\mathrm{ord}_P(f) < 0$, a pole. If $\mathrm{ord}_P(f) \geqslant 0$, then f is regular at P, with value $f(P) \in k(P)$ [Sil09, Section II.1].

PROPOSITION 3.1.7. *Let $f \in k(C)^\times$. Then f has only finitely many zeros and poles. If f has no poles, then $f \in k$.*

PROOF. See [Sil09, Proposition II.1.2]. For the last statement over k, note that k is algebraically closed in $k(C)$, since C is geometrically integral (Proposition 2.2.4). □

EXAMPLE 3.1.8. Let $f(x, y) = 0$ be an affine patch of C, and $P = (a, b) \in C(k)$. If $\partial f/\partial x(P) \neq 0$, then $y - b$ is a uniformizer at P. If $\partial f/\partial y(P) \neq 0$, then $x - a$ is [Poo06, Example 1.8.5]. For instance, on $y^2 = x^3 + x$ with $\operatorname{char} k \neq 2$, the function y is a uniformizer at $P = (0, 0)$. Since $x = y^2/(x^2 + 1)$, we get $\operatorname{ord}_P(x) = 2$, and $\operatorname{ord}_P(2y^2 - x) = \operatorname{ord}_P\big(x(2x^2 + 1)\big) = 2$ [Sil09, Example II.1.3].

PROPOSITION 3.1.9. *Let $P \in C(k)$, and let t be a uniformizer at P. Then $k(C)$ is a finite separable extension of $k(t)$.*

PROOF. [Sil09, Proposition II.1.4] proves this for perfect k, by taking p-th roots of coefficients. In general: since $k(P) = k$, the map $\mathfrak{m}_P/\mathfrak{m}_P^2 \to \Omega_{C/k,P} \otimes k(P)$ is an isomorphism [Har77, Proposition II.8.7]. So dt generates the stalk $\Omega_{C/k,P}$, and in particular $dt \neq 0$ in $\Omega_{k(C)/k}$. Conclude by Proposition 3.4.2. □

The hypothesis $P \in C(k)$ cannot be dropped: see Warning 3.4.5.

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3.2. Morphisms of curves

PROPOSITION 3.2.1. *Let X be a curve over k, let P be a closed point at which X is regular, and let $Y \subseteq \mathbb{P}^n_k$ be a projective variety. Then every rational map $\varphi\colon X \dashrightarrow Y$ is regular at P. In particular, if X is regular, then φ is a morphism.*

PROOF. Write $\varphi = (f_0 : \cdots : f_n)$, let t be a uniformizer at P and $m := \min_i \operatorname{ord}_P(f_i)$. Then every $t^{-m} f_i$ is regular at P, and some $(t^{-m} f_i)(P) \in k(P)$ is non-zero [Sil09, Proposition II.2.1]. The argument only uses that $\mathcal{O}_{X,P}$ is a discrete valuation ring, so it applies to closed points of any degree. □

If $f\colon X \to Y$ is a holomorphic map between Riemann surfaces and X is compact, then f is either constant or surjective. The algebraic analog is the following.

PROPOSITION 3.2.2. *A morphism from a smooth projective curve to a curve is either constant or surjective.*

PROOF. See [Har77, Proposition II.6.8]. □

THEOREM 3.2.3. *The functor $X \mapsto k(X)$ is an equivalence between the category of regular projective curves over k with non-constant morphisms, and the opposite of the category of finitely generated extensions K/k of transcendence degree 1 with k-embeddings. Under this equivalence, X is geometrically integral if and only if k is algebraically closed in $k(X)$ and $k(X)/k$ is separable.*

PROOF. See [Poo06, Section 2.1.2 and Remark 2.1.3] and [Mil20, Theorem I.6.3] for the equivalence. The last statement follows from Proposition 2.2.4. If $k(X)/k$ is separable, an element of $k(X)$ that is algebraic over k is separable over k. □

REMARK 3.2.4. [Sil09, Remark II.2.5] states the equivalence for smooth curves and function fields in which k is algebraically closed. Over an imperfect field, the regular projective model of such a function field need not be smooth. The function field $\mathbb{F}_3(t)(x, y)$, with $y^2 = x^3 - t$, satisfies both conditions of Theorem 3.2.3, but its regular projective model is the curve of Example 2.4.6, which is not smooth. So no nice curve has this function field.

Definition 3.2.5. Let $\varphi\colon C_1 \to C_2$ be a non-constant morphism, and $\varphi^*\colon k(C_2) \hookrightarrow k(C_1)$ the induced embedding. The degree of φ is $\deg\varphi := [k(C_1) : \varphi^*k(C_2)]$. A constant morphism has degree 0. The morphism φ is separable, inseparable or purely inseparable if the extension $k(C_1)/\varphi^*k(C_2)$ is, and $\deg_s\varphi$ and $\deg_i\varphi$ are its separable and inseparable degrees [Sil09, Section II.2], [Poo06, Definition 2.1.4]. The norm gives $\varphi_* := (\varphi^*)^{-1} \circ \mathrm{Nm}_{k(C_1)/\varphi^*k(C_2)}\colon k(C_1) \to k(C_2)$.

Corollary 3.2.6. *A morphism of degree 1 between regular projective curves is an isomorphism.*

Proof. This follows from Theorem 3.2.3. Compare [Sil09, Corollary II.2.4.1]. □

Definition 3.2.7. Let $\varphi\colon C_1 \to C_2$ be non-constant, P a closed point of C_1, and $Q := \varphi(P)$ with uniformizer t_Q. The ramification index of φ at P is $e_\varphi(P) := \mathrm{ord}_P(\varphi^*t_Q) \geqslant 1$, and the residue degree is $f_\varphi(P) := [k(P) : k(Q)]$. The morphism φ is unramified at P if $e_\varphi(P) = 1$ [Sil09, Section II.2].

Proposition 3.2.8. *Let $\varphi\colon C_1 \to C_2$ be non-constant.*
(1) For every closed point Q of C_2, $\sum_{P\mapsto Q} e_\varphi(P)f_\varphi(P) = \deg\varphi$.
(2) For all but finitely many $Q \in C_2(\overline{k})$, the fiber of $\varphi_{\overline{k}}$ over Q has $\deg_s\varphi$ elements.
(3) If $\psi\colon C_2 \to C_3$ is non-constant, then $e_{\psi\circ\varphi}(P) = e_\varphi(P)\, e_\psi(\varphi(P))$.

Proof. Over $\overline{k}$ all residue degrees are 1, and this is [Sil09, Proposition II.2.6]. Over k, part 1 is the formula $\sum e_i f_i = n$ for the finite extension of Dedekind domains $\mathcal{O}_{C_2,Q} \subseteq \mathcal{O}_{C_1}$ above Q [Poo06, Remark 2.5.3]. □

Example 3.2.9. The morphism $\varphi\colon \mathbb{P}^1_k \to \mathbb{P}^1_k$ given by $\varphi = (X^3(X-Y)^2 : Y^5)$ has degree 5. The fiber over $(0:1)$ consists of $(0:1)$, with $e_\varphi = 3$, and $(1:1)$, with $e_\varphi = 2$. And indeed $3+2=5$ [Sil09, Example II.2.9].

Example 3.2.10. Suppose $\mathrm{char}\, k \neq 2$. Let $E\colon y^2 = p(x)$ with p a separable cubic, a smooth plane cubic by Example 3.7.9 below, and let $O = (0:1:0)$. The morphism $x\colon E \to \mathbb{P}^1_k$ has degree 2, since $k(E) = k(x)(y)$ with $y^2 \in k(x)$, and it is separable. Above ∞ lies only O, with $e_x(O) = 2$. Above the closed point of $\mathbb{A}^1_k$ given by an irreducible factor g of p lies only the closed point $V(g,y)$, with ramification index 2. All other closed points are unramified. In characteristic 2 the extension $k(x,y)/k(x)$ defined by $y^2 = p(x)$ is purely inseparable [Mil20, Example I.4.25], but in that case this curve is singular, again by Example 3.7.9. See Example 3.6.4 instead.

Definition 3.2.11. Suppose $\mathrm{char}\, k = p > 0$, and let $q = p^r$. Let $C \subseteq \mathbb{P}^n_k$ be defined by homogeneous polynomials $G_1, \dots, G_m$, and let $C^{(q)}$ be defined by the polynomials $G_i^{(q)}$ obtained by raising every coefficient to the q-th power. Equivalently, $C^{(q)}$ is the base change of C along $k \to k$, $a \mapsto a^q$. The relative Frobenius morphism is

$$\mathsf{F}\colon C \to C^{(q)}, \qquad (X_0 : \cdots : X_n) \mapsto (X_0^q : \cdots : X_n^q).$$

It lands in $C^{(q)}$ because $G^{(q)}(X_0^q, \dots, X_n^q) = G(X_0, \dots, X_n)^q$ [Sil09, Section II.2].

Proposition 3.2.12. *Let $\mathsf{F}\colon C \to C^{(q)}$ be the relative Frobenius morphism.*
(1) $\mathsf{F}^*k(C^{(q)}) = k \cdot k(C)^q$, the compositum of k and $k(C)^q$ in $k(C)$.
(2) F is purely inseparable of degree q.

Proof. For perfect k, part 1 reads $\mathsf{F}^*k(C^{(q)}) = k(C)^q$, and this is [Sil09, Proposition II.2.11]. In general, suppose X_0 does not vanish on C, and let $x_i = X_i/X_0$ be the affine coordinates on the patch U_0. Then $k(C)$ and $k(C^{(q)})$ are generated over k by the x_i, and $\mathsf{F}^*(x_i) = x_i^q$. Since F^* is k-linear, its image is $k(x_i^q)$. This equals $k \cdot k(C)^q$: for $g \in k(C)$, written as a rational function in the x_i with coefficients in k, we have $g^q = g^{(q)}(x_i^q)$.

For part 2, pure inseparability follows from part 1. By induction on r it suffices to treat $q = p$. Since C is geometrically reduced, $k(C)/k$ is separably generated (Proposition 2.2.4): choose $t \in k(C)$ with $k(C)/k(t)$

finite separable. A separable algebraic extension L/K satisfies $L = KL^p$, so $k(C) = (k \cdot k(C)^p)(t)$, and $t^p \in k \cdot k(C)^p$. So the degree is at most p. Finally $t \notin k \cdot k(C)^p$, since d vanishes on $k \cdot k(C)^p$ but $dt \neq 0$ (Proposition 3.4.2). □

Warning 3.2.13. If k is imperfect, then $k(C)^q \subsetneq k \cdot k(C)^q$, because an element $a \in k - k^q$ is not a q-th power in $k(C)$. Indeed, k is algebraically closed in $k(C)$. So [Sil09, Proposition II.2.11(a)] must be read with $k \cdot k(C)^q$ in place of $k(C)^q$.

Corollary 3.2.14. *Suppose* $\operatorname{char} k = p > 0$. *Let* $\psi\colon C_1 \to C_2$ *be non-constant, and* $q \coloneqq \deg_i \psi$. *Then* $\psi = \lambda \circ \mathsf{F}$, *with* $\mathsf{F}\colon C_1 \to C_1^{(q)}$ *the relative Frobenius morphism and* λ *separable.*

Proof. Let K be the separable closure of $\psi^* k(C_2)$ in $k(C_1)$. Then $k(C_1)/K$ is purely inseparable of degree q, so $k(C_1)^q \subseteq K$. Since also $k \subseteq K$, we get $k \cdot k(C_1)^q \subseteq K$. Both have index q in $k(C_1)$, by Proposition 3.2.12, so they are equal. Theorem 3.2.3 then gives λ. Compare [Sil09, Corollary II.2.12]. □

Example 3.2.15. If $E\colon Y^2Z = X^3 + AXZ^2 + BZ^3$, then $E^{(q)}\colon Y^2Z = X^3 + A^qXZ^2 + B^qZ^3$ [Sil09, Example II.2.10]. If $A, B \in \mathbb{F}_q$, then $E^{(q)} = E$, and the relative Frobenius morphism is an endomorphism of E of degree q.

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3.3. Divisors

Definition 3.3.1. The group $\operatorname{Div} C$ of divisors on C is the free abelian group on the closed points of C. The degree of $D = \sum n_P P$ is $\deg D \coloneqq \sum n_P \deg P$, and $\operatorname{Div}^0 C$ is the kernel of the degree. D is effective, $D \geqslant 0$, if all $n_P \geqslant 0$. For $f \in k(C)^\times$, the principal divisor of f is $\operatorname{div}(f) \coloneqq \sum_P \operatorname{ord}_P(f) P$. Divisors are linearly equivalent if their difference is principal. The Picard group is $\operatorname{Pic} C \coloneqq \operatorname{Div} C / \operatorname{div}(k(C)^\times)$, and $\operatorname{Pic}^0 C$ is the image of $\operatorname{Div}^0 C$ [Poo06, Sections 1.9 and 2.2].

Proposition 3.3.2. *Let* $f \in k(C)^\times$. *Then* $\operatorname{div}(f) = 0$ *if and only if* $f \in k^\times$, *and* $\deg \operatorname{div}(f) = 0$. *So there is an exact sequence*

$$1 \to k^\times \to k(C)^\times \xrightarrow{\operatorname{div}} \operatorname{Div}^0 C \to \operatorname{Pic}^0 C \to 0.$$

Proof. See [Sil09, Proposition II.3.1 and Remark II.3.4], and [Poo06, Section 2.2.2]. The first statement is Proposition 3.1.7. □

Definition 3.3.3. Let $f \in \overline{k}(C)^\times$ and let $D = \sum n_P P \in \operatorname{Div} C_{\overline{k}}$ be a divisor whose support is disjoint from that of $\operatorname{div}(f)$. The value of f at D is

$$f(D) \coloneqq \prod_P f(P)^{n_P} \in \overline{k}^\times,$$

the product over the points $P \in C(\overline{k})$ in the support of D [Poo06, Section 1.9.2].

Each factor is a well defined element of $\overline{k}^\times$: the function f has neither a zero nor a pole at P, since P is not in the support of $\operatorname{div}(f)$. Two properties are immediate from the definition. First, $D \mapsto f(D)$ is a homomorphism on the group of divisors supported away from $\operatorname{div}(f)$, and $f \mapsto f(D)$ is a homomorphism on the functions whose divisors are supported away from D. Second, evaluation commutes with the Galois action: for $\sigma \in \operatorname{Aut}(\overline{k}/k)$,

$$\sigma\big(f(D)\big) = f^\sigma(\sigma D),$$

because $\sigma(f(P)) = f^\sigma(\sigma(P))$ on each factor. In particular, if f is defined over k, then $\sigma(f(D)) = f(\sigma D)$.

Theorem 3.3.4 (Weil reciprocity). *Let* $f, g \in \overline{k}(C)^\times$ *be functions on a nice curve* C *such that* $\operatorname{div}(f)$ *and* $\operatorname{div}(g)$ *have disjoint supports. Then*

$$f(\operatorname{div} g) = g(\operatorname{div} f).$$

Proof. See [Poo06, Theorem 1.9.5] and [Sil09, Exercise 2.11]. Poonen states Weil reciprocity over an algebraically closed field. Since divisors on $C_{\overline{k}}$ are supported on the points of $C(\overline{k})$, that is exactly the statement above. □

The content of the theorem is already visible on $\mathbb{P}^1$. If f and g are polynomials in $k[x]$ with no common root, then the affine zeros contribute $\prod_{g(\beta)=0} f(\beta)$ to the left side and $\prod_{f(\alpha)=0} g(\alpha)$ to the right, and both products are the resultant $\operatorname{Res}(f,g)$ up to the sign $(-1)^{\deg f \deg g}$ and the leading coefficients. The poles at ∞ supply exactly those factors. So reciprocity is a geometric form of the symmetry of the resultant. We use Theorem 3.3.4 to construct the Weil pairing in Section 4.6.

Definition 3.3.5. Let $\varphi\colon C_1 \to C_2$ be non-constant. On closed points, set

$$\varphi^* Q := \sum_{P \mapsto Q} e_\varphi(P)\, P, \qquad \varphi_* P := f_\varphi(P)\, \varphi(P),$$

and extend linearly to $\varphi^*\colon \operatorname{Div} C_2 \to \operatorname{Div} C_1$ and $\varphi_*\colon \operatorname{Div} C_1 \to \operatorname{Div} C_2$ [Poo06, Section 2.2.3].

Proposition 3.3.6. *Let $\varphi\colon C_1 \to C_2$ be non-constant.*
(1) $\deg \varphi^* D = (\deg \varphi)(\deg D)$ *and* $\deg \varphi_* D = \deg D$.
(2) $\varphi^* \operatorname{div}(f) = \operatorname{div}(\varphi^* f)$ *and* $\varphi_* \operatorname{div}(f) = \operatorname{div}(\varphi_* f)$.
(3) $\varphi_* \circ \varphi^*$ *is multiplication by* $\deg \varphi$ *on* $\operatorname{Div} C_2$.
(4) $(\psi \circ \varphi)^* = \varphi^* \circ \psi^*$ *and* $(\psi \circ \varphi)_* = \psi_* \circ \varphi_*$.
So φ^* and φ_* induce homomorphisms between $\operatorname{Pic} C_1$ *and* $\operatorname{Pic} C_2$, *and between* $\operatorname{Pic}^0 C_1$ *and* $\operatorname{Pic}^0 C_2$.

Proof. Part 1 follows from Proposition 3.2.8. For the rest see [Sil09, Proposition II.3.6] and [Poo06, Section 2.2.3]. □

[Sil09, Section II.3] works with divisors on $C_{\overline{k}}$. His $\operatorname{Div}_k(C)$ consists of the $\operatorname{Aut}(\overline{k}/k)$-invariant divisors on $C_{\overline{k}}$, and $\operatorname{Pic}_k(C) := (\operatorname{Pic} C_{\overline{k}})^{\operatorname{Aut}(\overline{k}/k)}$. The dictionary with our groups is as follows. Compare [Poo06, Warning 1.9.6].

Proposition 3.3.7.
(1) Pulling back along $C_{k^{\mathrm{sep}}} \to C$ identifies $\operatorname{Div} C$ *with* $(\operatorname{Div} C_{k^{\mathrm{sep}}})^{\mathrm{Gal}_k}$, *compatibly with degrees and principal divisors. A closed point P goes to the sum, with multiplicity* 1, *of the closed points of $C_{k^{\mathrm{sep}}}$ above P.*
(2) Pulling back along $C_{\overline{k}} \to C$ sends a closed point P to $[k(P):k]_i$ times the sum of the points of $C(\overline{k})$ above P. This is injective, compatible with degrees and principal divisors, and lands in $\operatorname{Div}_k(C)$. *It is onto* $\operatorname{Div}_k(C)$ *if k is perfect, but not in general.*
(3) The homomorphism $\operatorname{Pic} C \to (\operatorname{Pic} C_{k^{\mathrm{sep}}})^{\mathrm{Gal}_k}$ *is injective, but need not be surjective.*

Proof. For part 1: $k(P) \otimes_k k^{\mathrm{sep}}$ is reduced, since k^{sep}/k is separable, so every closed point of $C_{k^{\mathrm{sep}}}$ above P has multiplicity 1. They form one Gal_k-orbit. For part 2: $k(P) \otimes_k \overline{k}$ has length $[k(P):k]_i$ at each of its $[k(P):k]_s$ $\overline{k}$-points (Proposition 3.1.2). Part 3 follows from Hilbert's Theorem 90 [Poo06, Section 1.9.3 and Exercise 1.6]. □

Warning 3.3.8. In general the injection $\operatorname{Pic} C \to (\operatorname{Pic} C_{k^{\mathrm{sep}}})^{\mathrm{Gal}_k}$ of Proposition 3.3.7 is not surjective: its cokernel embeds in the Brauer group $\operatorname{Br} k$ [Poo17, Corollary 6.7.8]. If C has a rational point, then the obstruction vanishes and $\operatorname{Pic} C \to (\operatorname{Pic} C_{k^{\mathrm{sep}}})^{\mathrm{Gal}_k}$ is an isomorphism, so the two readings of $\operatorname{Pic}^0 C$ agree. In particular Silverman's $\operatorname{Pic}_k(C)$ need not be the quotient of $\operatorname{Div}_k(C)$ by principal divisors, although it is when C has genus 1 and a rational point [Sil09, Exercise 2.13]. Whenever the distinction matters we say which rational, closed or $\overline{k}$-points are meant.

EXAMPLE 3.3.9. The degree is an isomorphism $\operatorname{Pic}\mathbb{P}^1_k \cong \mathbb{Z}$. Indeed, the closed points of $\mathbb{P}^1_k$ other than ∞ correspond to monic irreducible $g_P \in k[x]$, with $\deg P = \deg g_P$. A divisor $D = \sum n_P P + n_\infty \infty$ of degree 0 is $\operatorname{div}\left(\prod g_P^{n_P}\right)$, since $\operatorname{ord}_\infty(g_P) = -\deg g_P$ [Sil09, Example II.3.2].

EXAMPLE 3.3.10. Suppose $\operatorname{char} k \neq 2$. Let $E\colon y^2 = (x-e_1)(x-e_2)(x-e_3)$ with distinct $e_i \in k$, let $P_i \coloneqq (e_i, 0)$ and $O \coloneqq (0:1:0)$. Then

$$\operatorname{div}(x - e_i) = 2P_i - 2O, \qquad \operatorname{div}(y) = P_1 + P_2 + P_3 - 3O$$

[Sil09, Example II.3.3]. If instead $y^2 = p(x)$ with p irreducible of degree 3, then $\operatorname{div}(y) = P - 3O$, where $P \coloneqq V(y, p(x))$ is a closed point of degree 3.

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3.4. Differentials

DEFINITION 3.4.1. Let X be a curve over k. The space $\Omega_{k(X)/k}$ of meromorphic differentials on X is the $k(X)$-vector space generated by symbols dx, $x \in k(X)$, subject to $d(x+y) = dx + dy$, $d(xy) = x\,dy + y\,dx$ and $da = 0$ for $a \in k$. A non-constant morphism $\varphi\colon X_1 \to X_2$ induces $\varphi^*\colon \Omega_{k(X_2)/k} \to \Omega_{k(X_1)/k}$ [Sil09, Section II.4], [Poo06, Section 2.3.1].

PROPOSITION 3.4.2. *Let X be a curve over k with $k(X)/k$ separable, e.g. a geometrically reduced curve.*

(1) $\Omega_{k(X)/k}$ has dimension 1 over $k(X)$.
(2) For $x \in k(X)$, the differential dx is a basis if and only if it is non-zero, if and only if $k(X)/k(x)$ is finite separable.
(3) A non-constant morphism $\varphi\colon X_1 \to X_2$ of such curves is separable if and only if $\varphi^* \neq 0$ on differentials.

PROOF. See [Sil09, Proposition II.4.2] and [Poo06, Proposition 2.3.1]. In characteristic p, the last condition of [Poo06, Proposition 2.3.1], $t \notin k(X)^p$, must be read as $t \notin k \cdot k(X)^p$ when k is imperfect. □

Since C is smooth of dimension 1, the sheaf $\Omega_{C/k}$ is invertible. So for every closed point P the stalk $\Omega_{C/k,P} \subseteq \Omega_{k(C)/k}$ is free of rank 1 over $\mathcal{O}_{C,P}$.

DEFINITION 3.4.3. Let $\omega \in \Omega_{k(C)/k}$ be non-zero, P a closed point, and η a generator of $\Omega_{C/k,P}$. Write $\omega = g\eta$, and set $\operatorname{ord}_P(\omega) \coloneqq \operatorname{ord}_P(g)$. The divisor of ω is $\operatorname{div}(\omega) \coloneqq \sum_P \operatorname{ord}_P(\omega) P$. The differential ω is regular if $\operatorname{div}(\omega) \geqslant 0$. A canonical divisor is a divisor $K_C \coloneqq \operatorname{div}(\omega)$. Its class in $\operatorname{Pic} C$ does not depend on ω, since any other non-zero differential is $f\omega$ with $f \in k(C)^\times$ [Sil09, Section II.4], [Poo06, Definitions 2.3.2 to 2.3.4].

PROPOSITION 3.4.4. *Let P be a closed point of C with $k(P)/k$ separable, e.g. $P \in C(k)$, and let t be a uniformizer at P. Then dt generates $\Omega_{C/k,P}$, so $\operatorname{ord}_P(\omega) = \operatorname{ord}_P(\omega/dt)$, as in [Sil09, Proposition II.4.3].*

PROOF. For $P \in C(k)$ see the proof of Proposition 3.1.9. □

WARNING 3.4.5. If $k(P)/k$ is inseparable, a uniformizer need not work. Let $k \coloneqq \mathbb{F}_p(a)$ and $P \coloneqq V(x^p - a)$ on $\mathbb{P}^1_k$. Then $t \coloneqq x^p - a$ is a uniformizer at P, but $dt = px^{p-1}\,dx - da = 0$. The stalk $\Omega_{\mathbb{P}^1/k,P}$ is generated by dx, so $\operatorname{ord}_P(dx) = 0$. This is why Definition 3.4.3 uses a generator of the stalk, rather than dt as in [Poo06, Definition 2.3.2].

EXAMPLE 3.4.6. Let t be the coordinate on $\mathbb{P}^1_k$. On $\mathbb{A}^1_k = \operatorname{Spec} k[t]$, the module of differentials is free on dt, so $\operatorname{ord}_P(dt) = 0$ for every closed point $P \neq \infty$. At ∞, the function $s \coloneqq 1/t$ is a uniformizer at a rational point, and $dt = -s^{-2}\,ds$, so $\operatorname{div}(dt) = -2\infty$. In particular $\mathbb{P}^1_k$ has no non-zero regular differentials [Sil09, Example II.4.5].

The next two examples start the comparison of elliptic curves with the multiplicative group $\mathbb{G}_m$ of Example 2.1.6, which runs through Chapter 4.

EXAMPLE 3.4.7. Let t be the coordinate on $\mathbb{G}_{\mathrm{m}} = \mathbb{A}^1_k - \{0\} \subseteq \mathbb{P}^1_k$. Then $k(\mathbb{G}_{\mathrm{m}}) = k(\mathbb{P}^1_k) = k(t)$, so a differential on $\mathbb{G}_{\mathrm{m}}$ is the same thing as a differential on $\mathbb{P}^1_k$. Let $\omega \coloneqq dt/t$.

(1) **Invariance.** For $a \in k^\times$, the translation $\tau_a \colon \mathbb{G}_{\mathrm{m}} \to \mathbb{G}_{\mathrm{m}}$, $t \mapsto at$, gives $\tau_a^*\omega = d(at)/(at) = \omega$. Conversely, suppose that $g \in \overline{k}(t)$ and $\tau_a^*(g\,dt) = g\,dt$ for every $a \in \overline{k}^\times$. Then $a\,g(at) = g(t)$, so $g(a) = g(1)/a$ for all but finitely many $a \in \overline{k}^\times$, and hence $g\,dt = g(1)\,\omega$. So the invariant differentials on $(\mathbb{G}_{\mathrm{m}})_{\overline{k}}$ form the line spanned by ω.

(2) **Divisor.** Since t is a unit of $k[t, t^{-1}]$, ω generates $\Omega_{\mathbb{G}_{\mathrm{m}}/k}$, so ω has no zeros or poles on $\mathbb{G}_{\mathrm{m}}$. At $t = 0$ it has a simple pole. At ∞, with the uniformizer $s \coloneqq 1/t$,

$$\omega = \frac{dt}{t} = -\frac{ds}{s^2} \cdot s = -\frac{ds}{s},$$

again a simple pole. So $\operatorname{div}(\omega) = -(0) - (\infty)$, of degree -2, as Example 3.4.6 requires. The coefficients of dt/t and ds/s, the residues, are 1 and -1, and they sum to 0.

This is the algebraic form of Example 1.5.4. The invariant differential of $\mathbb{G}_{\mathrm{m}}$ has poles at the two points of $\mathbb{P}^1_k - \mathbb{G}_{\mathrm{m}}$. By contrast, the invariant differential of an elliptic curve has no poles at all (Example 3.4.9), because an elliptic curve is already projective.

EXAMPLE 3.4.8. Let m be a non-zero integer, and let $[m] \colon \mathbb{P}^1_k \to \mathbb{P}^1_k$ be the morphism extending $t \mapsto t^m$ (Proposition 3.2.1). It maps $\mathbb{G}_{\mathrm{m}}$ to $\mathbb{G}_{\mathrm{m}}$, and on L-points it is the homomorphism $L^\times \to L^\times$, $x \mapsto x^m$. The polynomial $T^{|m|} - t^{|m|}$ is Eisenstein over $k[t^{|m|}]$, so $\deg[m] = [k(t) : k(t^m)] = |m|$.

(1) **Pullback.** $[m]^*\omega = d(t^m)/t^m = m\,t^{m-1}\,dt/t^m = m\omega$. By Proposition 3.4.2, $[m]$ is separable if and only if $\operatorname{char} k \nmid m$.

(2) **Additivity.** By the Leibniz rule, $d(uv)/(uv) = du/u + dv/v$. Since $t^{m+n} = t^m \cdot t^n$, this gives $[m+n]^*\omega = [m]^*\omega + [n]^*\omega$ without computing either side. The same statement for elliptic curves is Theorem 4.4.2, whose proof is much harder.

(3) **Kernel.** The fiber of $[m]$ over 1 is $\mu_m \coloneqq \operatorname{Spec} k[t]/(t^m - 1)$ for $m > 0$. If $\operatorname{char} k \nmid m$, then $t^m - 1$ is separable, so $\mu_m(\overline{k})$ has $m = \deg[m]$ elements. If $\operatorname{char} k = p$ and $m = p^r$, then $t^{p^r} - 1 = (t-1)^{p^r}$, so $\mu_{p^r}(\overline{k}) = \{1\}$, with $e_{[p^r]}(1) = p^r$: all of the degree is inseparable degree. Indeed, $[p]$ is the relative Frobenius morphism of $\mathbb{P}^1_k = (\mathbb{P}^1_k)^{(p)}$ (Definition 3.2.11).

(4) **Ramification.** Above 0 and ∞ lie only 0 and ∞, each with ramification index $|m|$. If $\operatorname{char} k \nmid m$, then $[m]$ is unramified elsewhere, by part 3 and translation, and the Hurwitz formula (Theorem 3.6.2) reads $-2 = |m|(-2) + 2(|m| - 1)$.

For an elliptic curve, too, $[m]^*\omega = m\omega$ (Corollary 4.4.3), but $\deg[m] = m^2$ (Theorem 4.3.10), and $[m]$ is unramified everywhere.

EXAMPLE 3.4.9. Suppose $\operatorname{char} k \neq 2$. Let $E \colon y^2 = p(x)$ with p a separable cubic, and $O = (0 : 1 : 0)$. On the affine part, $2y\,dy = p'(x)\,dx$, and y and $p'(x)$ have no common zero. So $dx/y = 2\,dy/p'(x)$ generates $\Omega_{E/k}$ there. At O, $\operatorname{ord}_O(x) = -2$ and $\operatorname{ord}_O(y) = -3$, so $\operatorname{ord}_O(dx) = -3$ and $\operatorname{div}(dx/y) = 0$ [Sil09, Example II.4.6].

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3.5. The Riemann–Roch theorem

DEFINITION 3.5.1. For $D \in \operatorname{Div} C$, let

$$\mathcal{L}(D) \coloneqq \{f \in k(C)^\times : \operatorname{div}(f) + D \geqslant 0\} \cup \{0\},$$

a finite-dimensional k-vector space, and $\ell(D) \coloneqq \dim_k \mathcal{L}(D)$. If $\deg D < 0$, then $\mathcal{L}(D) = 0$. If $D \sim D'$, then $\ell(D) = \ell(D')$ [Poo06, Definition 2.4.1], [Sil09, Proposition II.5.2].

PROPOSITION 3.5.2. *Let $D \in \operatorname{Div} C$, let $L \supseteq k$ be a field extension, and D_L the pullback of D to C_L. Then the natural map $\mathcal{L}(D) \otimes_k L \to \mathcal{L}(D_L)$ is an isomorphism, and $\deg D_L = \deg D$. In particular $\ell(D_L) = \ell(D)$.*

PROOF. $\mathcal{L}(D)$ is the space of global sections of the invertible sheaf $\mathcal{O}_C(D)$, and $\mathcal{O}_C(D)$ pulls back to $\mathcal{O}_{C_L}(D_L)$. Taking global sections commutes with the flat base change $\operatorname{Spec} L \to \operatorname{Spec} k$ [Har77, Proposition

III.9.3]. For $L = \overline{k}$ and perfect k, [Sil09, Proposition II.5.8] proves this with Galois descent instead, and [Poo06, Section 2.4.1] states it. □

DEFINITION 3.5.3. The genus of C is $g \coloneqq \ell(K_C)$, the dimension of the k-vector space of regular differentials on C [Poo06, Definition 2.4.5].

THEOREM 3.5.4 (Riemann–Roch). *Let C be a nice curve of genus g, and $D \in \operatorname{Div} C$. Then*

$$\ell(D) - \ell(K_C - D) = \deg D - g + 1.$$

PROOF. See [Sil09, Theorem II.5.4] and [Poo06, Theorem 2.4.7]. To reduce to $k = \overline{k}$, note that $\Omega_{C_{\overline{k}}/\overline{k}}$ is the pullback of $\Omega_{C/k}$, so K_C pulls back to a canonical divisor. Then apply Proposition 3.5.2. □

COROLLARY 3.5.5. *Let C be a nice curve of genus g.*

(1) $\deg K_C = 2g - 2$.
(2) If $\deg D > 2g - 2$, *then* $\ell(D) = \deg D - g + 1$.
(3) For every field extension $L \supseteq k$, the nice curve C_L has genus g.

PROOF. Parts 1 and 2 are [Sil09, Corollary II.5.5] and [Poo06, Corollaries 2.4.8 and 2.4.9]. Part 3 follows from Proposition 3.5.2, since K_C pulls back to a canonical divisor. □

EXAMPLE 3.5.6. By Example 3.4.6, $\mathbb{P}^1_k$ has genus 0, and $\ell(D) = \deg D + 1$ whenever $\deg D \geqslant -1$ [Sil09, Example II.5.6].

EXAMPLE 3.5.7. Let $E\colon y^2 = p(x)$ be as in Example 3.4.9. We may take $K_E = 0$, so E has genus $\ell(0) = 1$, and $\ell(D) = \deg D$ whenever $\deg D \geqslant 1$. In particular $\ell(P) = 1$ for $P \in E(k)$, so no function has a single simple pole. For $O = (0 : 1 : 0)$, the functions $1, x$ form a basis of $\mathcal{L}(2O)$, and $1, x, y$ a basis of $\mathcal{L}(3O)$. The seven functions $1, x, y, x^2, xy, x^3, y^2$ lie in the 6-dimensional space $\mathcal{L}(6O)$, and the linear relation between them is the equation of E [Sil09, Example II.5.7].

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3.6. The Hurwitz formula

DEFINITION 3.6.1. Let $\varphi\colon C_1 \to C_2$ be a non-constant separable morphism, and ω a non-zero differential on C_2, so that $\varphi^*\omega \neq 0$ (Proposition 3.4.2). The ramification divisor of φ is

$$R_\varphi \coloneqq \operatorname{div}(\varphi^*\omega) - \varphi^* \operatorname{div}(\omega) \in \operatorname{Div} C_1.$$

It does not depend on ω [Poo06, Definition 2.5.1].

THEOREM 3.6.2 (Hurwitz). *Let $\varphi\colon C_1 \to C_2$ be a non-constant separable morphism of nice curves of genera g_1 and g_2. Then*

$$2g_1 - 2 = (\deg \varphi)(2g_2 - 2) + \deg R_\varphi.$$

For every $P \in C_1(\overline{k})$, the base change $\varphi_{\overline{k}}$ satisfies $\operatorname{ord}_P(R_{\varphi_{\overline{k}}}) \geqslant e(P) - 1$, where $e(P) \coloneqq e_{\varphi_{\overline{k}}}(P)$, with equality if and only if $\operatorname{char} k \nmid e(P)$. So if no ramification index is divisible by $\operatorname{char} k$, then

$$2g_1 - 2 = (\deg \varphi)(2g_2 - 2) + \sum_{P \in C_1(\overline{k})} \big(e(P) - 1\big).$$

PROOF. Take degrees in $\operatorname{div}(\varphi^*\omega) = \varphi^* \operatorname{div}(\omega) + R_\varphi$, and use Proposition 3.3.6 and Corollary 3.5.5 [Poo06, Theorem 2.5.7]. Base change to $\overline{k}$ preserves separability, degrees and genera, and over $\overline{k}$ the rest is [Poo06, Proposition 2.5.4 and Corollary 2.5.8] and [Sil09, Theorem II.5.9]. □

EXAMPLE 3.6.3. Let $x\colon E \to \mathbb{P}^1_k$ be the morphism of Example 3.2.10, with $\operatorname{char} k \neq 2$. Over $\overline{k}$ it is ramified exactly at the three roots $(e_i, 0)$ of p and at O, each with index 2. Hurwitz gives $2g - 2 = 2 \cdot (-2) + 4 = 0$, so $g = 1$, in agreement with Example 3.5.7.

EXAMPLE 3.6.4. Suppose $\operatorname{char} k = 2$, and let $C\colon Y^2Z + YZ^2 = X^3$, the curve $y^2 + y = x^3$. The partial derivatives are X^2, Z^2 and Y^2, with no common zero, so C is a smooth cubic, of genus 1 by the degree–genus formula (Corollary 3.7.7 below). The morphism $x\colon C \to \mathbb{P}^1_k$ has degree 2, and it is separable, since $y^2 + y - x^3$ has derivative 1 in y. Over each $a \in \overline{k}$ the equation $y^2 + y = a^3$ has two distinct roots. So the only ramification is at $O = (0:1:0)$, with index 2, which is wild. One computes $R_x = 4O$ [Poo06, Example 2.5.6], and Hurwitz gives $2g - 2 = 2 \cdot (-2) + 4 = 0$. The formula for the tame case would give the impossible value $2 \cdot (-2) + 1$.

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3.7. Plane curves

We follow [Mil20, Section I.1], and use coordinates $(X : Y : Z)$ on $\mathbb{P}^2_k$, with $x = X/Z$ and $y = Y/Z$ on the affine patch $Z \neq 0$.

DEFINITION 3.7.1. Let $F \in k[X, Y, Z]$ be a non-constant homogeneous polynomial of degree d with no repeated factor in $\overline{k}[X, Y, Z]$. The projective plane curve defined by F is $C_F \coloneqq V_+(F) \subseteq \mathbb{P}^2_k$, and d is its degree. Affine plane curves $V(f) \subseteq \mathbb{A}^2_k$ are defined in the same way.

The condition on F says that C_F is geometrically reduced [Mil20, Aside I.1.7]. The affine patches of C_F are the affine plane curves $F(x, y, 1) = 0$, $F(x, 1, z) = 0$ and $F(1, y, z) = 0$.

PROPOSITION 3.7.2. *Let $C = V_+(F) \subseteq \mathbb{P}^2_k$ be a projective plane curve of degree d. Its singular points are the common zeros in $\mathbb{P}^2(\overline{k})$ of F, $\partial F/\partial X$, $\partial F/\partial Y$ and $\partial F/\partial Z$. If $\operatorname{char} k \nmid d$, the condition $F = 0$ may be omitted, by the Euler relation*

$$dF = X\frac{\partial F}{\partial X} + Y\frac{\partial F}{\partial Y} + Z\frac{\partial F}{\partial Z}.$$

PROOF. See [Poo06, Example 1.7.5] and [Mil20, Exercise I.1.17]. □

When $\operatorname{char} k \mid d$ the condition $F = 0$ cannot be dropped: for $F = Y^2Z - X^3 + tZ^3$ over $\mathbb{F}_3(t)$ (Example 2.4.6) the three partial derivatives vanish at $(0:0:1)$, which is not a $\overline{k}$-point of the curve.

DEFINITION 3.7.3. Let $C = V_+(F)$ and $D = V_+(G)$ be projective plane curves with no common component. Then $C \cap D \coloneqq V_+(F, G)$ is a finite k-scheme. For a closed point P of $C \cap D$, choose an affine patch containing P, with dehomogenizations f and g of F and G. The intersection number of C and D at P is

$$I_P(C, D) \coloneqq \dim_k \mathcal{O}_{\mathbb{P}^2_k, P}/(f, g).$$

For $P \in C(k) \cap D(k)$ this is the intersection number of [Mil20, Proposition I.1.8], and it does not change under base extension [Mil20, Remark I.1.9].

THEOREM 3.7.4 (Bézout). *Let C and D be projective plane curves over k of degrees m and n, with no common component. Then*

$$\sum_{P \in C \cap D} I_P(C, D) = mn.$$

PROOF. Base change to $\overline{k}$, and apply [Mil20, Theorem I.1.18]. □

COROLLARY 3.7.5. *A smooth projective plane curve is geometrically integral, hence nice.*

PROOF. It is geometrically reduced by definition. Over $\overline{k}$, two irreducible components would meet by Theorem 3.7.4, and a $\overline{k}$-point on two components is singular [Mil20, Remarks I.1.4 and I.1.21]. □

PROPOSITION 3.7.6. *Let $C = V_+(F) \subseteq \mathbb{P}^2_k$ be a smooth plane curve of degree d, not the line $Z = 0$, and let $f(x, y) \coloneqq F(x, y, 1)$. Let H be the effective divisor cut out by the line $Z = 0$. Then H has degree d,*

and

$$\operatorname{div}\left(\frac{dx}{\partial f/\partial y}\right) = (d-3)H.$$

Here $dx/(\partial f/\partial y)$ stands for $-dy/(\partial f/\partial x)$ if $\partial f/\partial y$ vanishes identically on C.

PROOF. On the affine part, $\frac{\partial f}{\partial x}\,dx + \frac{\partial f}{\partial y}\,dy = 0$, and the partial derivatives have no common zero (Proposition 3.7.2). So $\omega \coloneqq dx/(\partial f/\partial y) = -dy/(\partial f/\partial x)$ generates $\Omega_{C/k}$ there, and $\operatorname{ord}_P(\omega) = 0$ at the closed points P of the affine part.

At a closed point P on the line $Z = 0$, say with $Y \neq 0$, use the coordinates $u \coloneqq X/Y$ and $v \coloneqq Z/Y$, so that $x = u/v$ and $y = 1/v$, and let $g(u, v) \coloneqq F(u, 1, v)$. As before, $\eta \coloneqq du/(\partial g/\partial v) = -dv/(\partial g/\partial u)$ generates $\Omega_{C/k}$ near P. The Euler relation gives $u\,\partial g/\partial u + v\,\partial g/\partial v = -\frac{\partial F}{\partial Y}(u, 1, v)$ on C, and $\partial f/\partial y = v^{1-d}\,\frac{\partial F}{\partial Y}(u, 1, v)$ because $\partial F/\partial Y$ is homogeneous of degree $d - 1$. Substituting into $dx = (v\,du - u\,dv)/v^2$ gives

$$\omega = -v^{d-3}\,\eta,$$

so $\operatorname{ord}_P(\omega) = (d-3)\operatorname{ord}_P(v)$, where v is a local equation of the line $Z = 0$. The case $X \neq 0$ is symmetric. Hence $\operatorname{div}(\omega) = (d-3)H$. The degree of H is d by Theorem 3.7.4: on a smooth curve, $I_P(C, L) = \operatorname{ord}_P(\ell)\deg P$ for a local equation ℓ of the line L. □

For $d = 3$ and $f = y^2 - p(x)$ this recovers Example 3.4.9. For $d = 1$, with $f = y$, it recovers Example 3.4.6.

COROLLARY 3.7.7. *A smooth plane curve of degree d has genus $(d-1)(d-2)/2$.*

PROOF. By Proposition 3.7.6, $\deg K_C = d(d-3)$, and $d(d-3) = 2g - 2$ by Corollary 3.5.5. See also [Sil09, Exercise 2.7] and [Poo06, Section 2.9.2]. □

REMARK 3.7.8. Smoothness cannot be weakened to regularity here. The curve $E\colon y^2 = x^3 - t$ over $k = \mathbb{F}_3(t)$ of Example 2.4.6 is regular and geometrically integral, and as a plane cubic it has arithmetic genus 1. Over $k' \coloneqq k(s)$, where $s^3 = t$, it becomes the cuspidal cubic $y^2 = (x - s)^3$, whose normalization $u \mapsto (s + u^2, u^3)$ is $\mathbb{P}^1_{k'}$. So the genus drops under base extension, and Corollary 3.5.5 fails for E. Also $\Omega_{E/k}$ is not invertible at the non-smooth point, so Definition 3.4.3 does not apply.

EXAMPLE 3.7.9. Let $h \in k[x]$ have degree $n \geqslant 3$ and leading coefficient c, let $H(X, Z) \coloneqq Z^n h(X/Z)$ be its homogenization, and let $C \coloneqq V_+(F) \subseteq \mathbb{P}^2_k$, where

$$F \coloneqq Y^2 Z^{n-2} - H(X, Z),$$

the projective closure of the affine curve $y^2 = h(x)$. Both F and C have degree n.

(1) **C is a plane curve.** Since $F \equiv -cX^n \pmod{Z}$, the polynomial Z does not divide F, so a repeated factor of F is one of $y^2 - h(x)$. A factorization $y^2 - h(x) = (y - g)(y + g)$ over $\overline{k}$ has a repeated factor only if $2g = 0$, so F has one exactly when $\operatorname{char} k = 2$ and $h \in k[x^2]$. Exclude that case, so that Definition 3.7.1 applies.

(2) **Singular points of the affine part.** Here $f = y^2 - h(x)$, so $\partial f/\partial y = 2y$ and $\partial f/\partial x = -h'(x)$, and the singular points of $y^2 = h(x)$ are the pairs $(\alpha, \beta) \in \overline{k}^2$ with

$$\beta^2 = h(\alpha), \qquad 2\beta = 0, \qquad h'(\alpha) = 0.$$

If $\operatorname{char} k \neq 2$, then $\beta = 0$, and the conditions $h(\alpha) = h'(\alpha) = 0$ say that α is a repeated root of h. So the affine curve is singular exactly at the $\overline{k}$-points $(e, 0)$ with e a repeated root of h in $\overline{k}$, and it is smooth if and only if h is separable. For $h = x^3 + Ax + B$ this says $4A^3 + 27B^2 \neq 0$ [Mil20, Example I.1.5]. If $\operatorname{char} k = 2$, the condition $2\beta = 0$ is vacuous. When n is odd, h' has degree $n - 1 \geqslant 2$, hence a root $\alpha \in \overline{k}$, and any β with $\beta^2 = h(\alpha)$ gives a singular point. For $h = x^3 + Ax + B$ it is the $\overline{k}$-point with $\alpha^2 = A$ and $\beta^2 = \alpha^3 + A\alpha + B$.

(3) **The point at infinity.** Since $n \geqslant 3$, we have $F(X, Y, 0) = -cX^n$, so C meets the line $L \coloneqq V_+(Z)$ only at $O \coloneqq (0 : 1 : 0)$, and $I_O(C, L) = n$ by Theorem 3.7.4. Now

$$\frac{\partial F}{\partial X} = -\frac{\partial H}{\partial X}, \qquad \frac{\partial F}{\partial Y} = 2YZ^{n-2}, \qquad \frac{\partial F}{\partial Z} = (n-2)Y^2Z^{n-3} - \frac{\partial H}{\partial Z}.$$

The partial derivatives of H are homogeneous of degree $n-1 \geqslant 2$ in (X, Z), so they vanish at O, as does $\partial F/\partial Y$. For $n = 3$ the remaining derivative is $\partial F/\partial Z = Y^2 - \partial H/\partial Z$, with value 1 at O. For $n \geqslant 4$ it vanishes at O as well. So O is a smooth point of C if $n = 3$, and a singular point if $n \geqslant 4$, in every characteristic.

(4) **Conclusion.** If $\operatorname{char} k \neq 2$, then C is smooth if and only if $n = 3$ and h is separable. It is then nice by Corollary 3.7.5 and of genus 1 by Corollary 3.7.7. If $\operatorname{char} k = 2$, then C is singular: by part 2 when n is odd, and by part 3 when n is even, since then $n \geqslant 4$. One needs the general Weierstrass equation instead [Sil09, Section III.1].

So a curve $y^2 = h(x)$ with $\deg h \geqslant 4$ has no smooth plane model, and must be studied through some other embedding. The case $n = 3$ is the one we pursue, in Section 4.1.

CHAPTER 4

Elliptic curves as algebraic curves

Sources. This chapter follows [Sil09, Chapter III], with some material from [Mil20, Sections II.1, II.2, II.5 and II.6], and [Voi21, Chapters 2, 3, 5 and 10] for quaternion algebras and their orders. As in the previous two chapters, we do not assume that the ground field is perfect.

In Chapter 1 we met complex tori $\mathbb{C}/\Lambda$: compact Riemann surfaces that are also abelian groups. The uniformization theorem (Theorem 1.7.1) identifies each of them with a smooth plane cubic $E_{A,B}(\mathbb{C})$, and under this identification the isogenies of Section 1.3 and the torsion subgroups $(\mathbb{C}/\Lambda)[m] \cong (\mathbb{Z}/m\mathbb{Z})^2$ of Example 1.3.4 become statements about cubic curves. The goal of this chapter is to recover that whole picture over an arbitrary field k, where there is no lattice to divide by. The tools are the ones developed in Chapters 2 and 3: the Riemann–Roch theorem replaces the Weierstrass $\wp$-function, the Picard group replaces the quotient $\mathbb{C}/\Lambda$, and the invariant differential replaces dz. Alongside, we compare E with the multiplicative group $\mathbb{G}_{\mathrm{m}}$, whose invariant differential, torsion points and Tate module are elementary (Examples 3.4.7 and 3.4.8). The comparison is summarized at the end of Section 4.5.

SITUATION 4.0.1. We keep Situation 2.0.1: k is an arbitrary field, not assumed perfect, $k^{\mathrm{sep}} \subseteq \overline{k}$ are a separable and an algebraic closure, and $\mathrm{Gal}_k := \mathrm{Gal}(k^{\mathrm{sep}}/k)$. For a k-variety X and a field $L \supseteq k$, recall that X_L is the base extension.

4.1. Weierstrass equations

DEFINITION 4.1.1. An elliptic curve over k is a pair (E, O), where E is a nice curve over k of genus 1 and $O \in E(k)$. We usually write E for the pair. A morphism of elliptic curves $(E, O) \to (E', O')$ is a morphism of curves $E \to E'$ sending O to O'.

The rational point is part of the data. A nice curve of genus 1 need not have one: by a theorem of Selmer [Sel51], the smooth plane cubic $3X^3 + 4Y^3 + 5Z^3 = 0$ over $\mathbb{Q}$, which has genus 1 by Corollary 3.7.7, has no rational points, although it has points over $\mathbb{R}$ and over every $\mathbb{Q}_p$. Such curves will reappear in Section 4.8 and in Chapter 10.

The model for every elliptic curve is the smooth plane cubic $y^2 = h(x)$ of Example 3.7.9. In characteristic 2 that equation is always singular, by the same example, so we need a more general one.

DEFINITION 4.1.2. A Weierstrass equation over k is a projective plane cubic

$$Y^2Z + a_1XYZ + a_3YZ^2 = X^3 + a_2X^2Z + a_4XZ^2 + a_6Z^3 \tag{4.1.1}$$

with $a_1, a_2, a_3, a_4, a_6 \in k$, which on the affine patch $Z \neq 0$ reads $y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$. Intersecting with the line at infinity $Z = 0$ leaves $X^3 = 0$, so the only point at infinity is the flex $O := (0 : 1 : 0)$. We define the b-invariants and c-invariants of the Weierstrass equation by

$$\begin{aligned} b_2 &:= a_1^2 + 4a_2, & b_4 &:= 2a_4 + a_1a_3, \\ b_6 &:= a_3^2 + 4a_6, & b_8 &:= a_1^2a_6 + 4a_2a_6 - a_1a_3a_4 + a_2a_3^2 - a_4^2, \\ c_4 &:= b_2^2 - 24b_4, & c_6 &:= -b_2^3 + 36b_2b_4 - 216b_6, \end{aligned}$$

the discriminant $\Delta := -b_2^2b_8 - 8b_4^3 - 27b_6^2 + 9b_2b_4b_6$, and, when $\Delta \neq 0$, the j-invariant $j := c_4^3/\Delta$. The invariant differential is

$$\omega := \frac{dx}{2y + a_1x + a_3} = \frac{dy}{3x^2 + 2a_2x + a_4 - a_1y}.$$

These quantities satisfy $4b_8 = b_2b_6 - b_4^2$ and $1728\Delta = c_4^3 - c_6^2$.

A flex of a plane cubic is a smooth $\overline{k}$-point P whose tangent line meets the cubic at P with multiplicity 3. The tangent line at O is $Z = 0$. The two expressions for ω agree because $f := y^2 + a_1xy + a_3y - x^3 - a_2x^2 - a_4x - a_6$ satisfies $\frac{\partial f}{\partial x}\,dx + \frac{\partial f}{\partial y}\,dy = 0$, and $\partial f/\partial y = 2y + a_1x + a_3$.

EXAMPLE 4.1.3 (Recalling the formulas in Magma). Over the field $\mathbb{Q}(a_1, a_2, a_3, a_4, a_6)$ of rational functions in the coefficients, (4.1.1) is an elliptic curve, and Magma prints the quantities of Definition 4.1.2 for it as polynomials in the a_i [BCP97].

```
K<a1,a2,a3,a4,a6> := FunctionField(Rationals(), 5);
E := EllipticCurve([a1, a2, a3, a4, a6]);
aInvariants(E);            // [a1, a2, a3, a4, a6]
bInvariants(E);            // [b2, b4, b6, b8]
cInvariants(E);            // [c4, c6]
```

Printing `Discriminant(E)` and `jInvariant(E)` gives expressions too long to read, but the identities of Definition 4.1.2 fit on one line each. A `true` here is an identity of rational functions with integer coefficients, so it holds for every Weierstrass equation over every field.

```
b2, b4, b6, b8 := Explode(bInvariants(E));
c4, c6 := Explode(cInvariants(E));
4*b8 eq b2*b6 - b4^2;
1728*Discriminant(E) eq c4^3 - c6^2;
jInvariant(E) eq c4^3/Discriminant(E);
```

LEMMA 4.1.4 (Short Weierstrass form). *If* $\operatorname{char} k \neq 2$, *completing the square in* y, *and then, if also* $\operatorname{char} k \neq 3$, *completing the cube in* x, *brings the equation to the short form*

$$y^2 = x^3 + Ax + B, \tag{4.1.2}$$

whose discriminant, j-invariant and invariant differential are

$$\Delta = -16(4A^3 + 27B^2), \qquad j = 1728\,\frac{4A^3}{4A^3 + 27B^2}, \qquad \omega = \frac{dx}{2y}.$$

PROOF. Both steps are instances of the change of variables $x = u^2x' + r$, $y = u^3y' + su^2x' + t$ with $u \in k^\times$ and $r, s, t \in k$, which reappears in Theorem 4.1.6. Magma performs one with `Transformation`, in the inverse convention: our (u, r, s, t) is Magma's tuple $[-r/u^2,\ -s/u,\ (rs - t)/u^3,\ 1/u]$. We translate once and for all, and use `subst` again in the proofs below. The blocks of this section run in one session, in order.

```
// Weierstrass change of variables, as above:
//   x = u^2*x' + r,  y = u^3*y' + s*u^2*x' + t.
// Magma's Transformation inverts it, so translate the tuple.
subst := func< E,u,r,s,t | Transformation(E,
  [BaseRing(E) | -r/u^2, -s/u, (r*s - t)/u^3, 1/u]) >;
```

If $\operatorname{char} k \neq 2$, take $u = 1$, $r = 0$, $s = -a_1/2$ and $t = -a_3/2$. This makes the new a_1 and a_3 vanish, and the equation becomes $y^2 = x^3 + \frac{b_2}{4}x^2 + \frac{b_4}{2}x + \frac{b_6}{4}$.

```
K<a1,a2,a3,a4,a6> := FunctionField(Rationals(), 5);
E := EllipticCurve([a1,a2,a3,a4,a6]);
b2, b4, b6, b8 := Explode(bInvariants(E));
c4, c6 := Explode(cInvariants(E));
E1 := subst(E, 1, 0, -a1/2, -a3/2);
```

```
aInvariants(E1) eq [K | 0, b2/4, 0, b4/2, b6/4];   // true
```

If also $\operatorname{char} k \neq 3$, take $u = 1$ and $r = -b_2/12$. This makes the new a_2 vanish and leaves (4.1.2), with $A = -c_4/48$ and $B = -c_6/864$.

```
E2 := subst(E1, 1, -b2/12, 0, 0);
A := -c4/48;  B := -c6/864;
aInvariants(E2) eq [K | 0, 0, 0, A, B];          // true
Discriminant(E2) eq -16*(4*A^3 + 27*B^2);        // true
jInvariant(E2) eq 1728*4*A^3/(4*A^3 + 27*B^2);   // true
Discriminant(E2) eq Discriminant(E);             // true
```

The last line records that neither substitution changes Δ, as both have $u = 1$. The same holds for j and for ω. The displayed formulas are those of Definition 4.1.2 read off the short form. □

Weierstrass equations with non-zero discriminant, together with the point at infinity, define elliptic curves over any field k.

PROPOSITION 4.1.5. *Let E be the plane cubic (4.1.1).*
(1) E is smooth if and only if $\Delta \neq 0$.
(2) If $\Delta \neq 0$, then (E, O) is an elliptic curve over k, and the invariant differential satisfies $\operatorname{div}(\omega) = 0$.

PROOF. Write F for the cubic form in (4.1.1). At O we have $\partial F/\partial Z = Y^2 = 1$, so O is always a smooth point. That a singular point of the affine part exists if and only if $\Delta = 0$ is [Sil09, Proposition III.1.4(a)], a computation over $\overline{k}$. When $\operatorname{char} k \neq 2, 3$ it is Example 3.7.9, by (4.1.2).

If E is smooth, it is nice by Corollary 3.7.5, of genus 1 by Corollary 3.7.7, and $O \in E(k)$. Finally, let $\omega = dx/(\partial f/\partial y)$, so $\operatorname{div}(\omega) = (3 - 3)O = 0$ by Proposition 3.7.6. □

The converse is the first application of the Riemann–Roch theorem: every elliptic curve is a Weierstrass cubic. Example 3.5.7 already contains the idea, run backwards. The following is [Sil09, Proposition III.3.1].

THEOREM 4.1.6. *Let (E, O) be an elliptic curve over k.*
(1) There are $x, y \in k(E)$ such that the morphism $E \to \mathbb{P}^2_k$ given by $(x : y : 1)$ is an isomorphism onto a smooth Weierstrass cubic (4.1.1), sending O to $(0 : 1 : 0)$. Moreover $k(E) = k(x, y)$ and $[k(E) : k(x)] = 2$.
(2) Any two such pairs, (x, y) and (x', y'), are related by

$$x = u^2 x' + r, \qquad y = u^3 y' + s u^2 x' + t, \tag{4.1.3}$$

for some $u \in k^\times$ and $r, s, t \in k$.

The functions x and y are Weierstrass coordinates on E.

PROOF. Since E has genus 1, Corollary 3.5.5 gives $\ell(nO) = n$ for every $n \geqslant 1$. So $\mathcal{L}(O) = k$, and we may choose $x \in \mathcal{L}(2O)$ and $y \in \mathcal{L}(3O)$ with $\{1, x\}$ and $\{1, x, y\}$ bases. Then $\operatorname{ord}_O(x) = -2$ and $\operatorname{ord}_O(y) = -3$. The seven functions $1, x, y, x^2, xy, y^2, x^3$ lie in the 6-dimensional space $\mathcal{L}(6O)$, so they satisfy a linear relation over k. The only functions in the list with a pole of order 6 are y^2 and x^3, so both appear with non-zero coefficients, and rescaling x and y by elements of $k^\times$ gives a Weierstrass equation. Let C be the cubic it defines. The rational map $\varphi := (x : y : 1) \colon E \dashrightarrow C$ is a morphism by Proposition 3.2.1, and $\varphi(O) = (x/y : 1 : 1/y)(O) = (0 : 1 : 0)$.

The morphism $x \colon E \to \mathbb{P}^1_k$ pulls ∞ back to $2O$, so it has degree 2 by Proposition 3.2.8. Likewise y has degree 3. Hence $[k(E) : k(x, y)]$ divides 2 and 3, so $k(E) = k(x, y)$ and φ is birational onto C. If C were singular, then $C_{\overline{k}}$, and so $E_{\overline{k}}$, would be birational to $\mathbb{P}^1_{\overline{k}}$ [Sil09, Proposition III.1.6]. Compare Example 2.3.5. By Corollary 3.2.6 this would make the nice curve $E_{\overline{k}}$ isomorphic to $\mathbb{P}^1_{\overline{k}}$, which has genus 0, whereas $E_{\overline{k}}$ has

genus 1 by Corollary 3.5.5. Since the genus is a geometric invariant, C is smooth, and φ is an isomorphism by Corollary 3.2.6.

For part 2, $\{1, x\}$ and $\{1, x'\}$ are both bases of $\mathcal{L}(2O)$, and $\{1, x, y\}$ and $\{1, x', y'\}$ of $\mathcal{L}(3O)$. So $x = u_1x' + r$ and $y = u_2y' + s_2x' + t$ with $u_1, u_2 \in k^\times$. Comparing the coefficients of x'^3 and y'^2 in the two equations gives $u_1^3 = u_2^2$, and $u := u_2/u_1$ does the job. □

PROPOSITION 4.1.7. *Under the change of variables (4.1.3), the quantities of the primed equation satisfy*

$$\begin{aligned}
ua_1' &= a_1 + 2s,\\
u^2a_2' &= a_2 - sa_1 + 3r - s^2,\\
u^3a_3' &= a_3 + ra_1 + 2t,\\
u^4a_4' &= a_4 - sa_3 + 2ra_2 - (t + rs)a_1 + 3r^2 - 2st,\\
u^6a_6' &= a_6 + ra_4 + r^2a_2 + r^3 - ta_3 - t^2 - rta_1,\\
u^2b_2' &= b_2 + 12r,\\
u^4b_4' &= b_4 + rb_2 + 6r^2,\\
u^6b_6' &= b_6 + 2rb_4 + r^2b_2 + 4r^3,\\
u^8b_8' &= b_8 + 3rb_6 + 3r^2b_4 + r^3b_2 + 3r^4,
\end{aligned} \tag{4.1.4}$$

and

$$u^4c_4' = c_4, \qquad u^6c_6' = c_6, \qquad u^{12}\Delta' = \Delta, \qquad j' = j, \qquad u^{-1}\omega' = \omega.$$

In particular the j-invariant and the class of Δ in $k^\times/k^{\times 12}$ are invariants of the elliptic curve E. In the short form (4.1.2), the only changes of variables preserving the form are $x = u^2x'$, $y = u^3y'$, and they give $A' = u^{-4}A$ and $B' = u^{-6}B$.

PROOF. This is a direct computation, which we can do in Magma: over the function field in the five coefficients and the four parameters of (4.1.3), a `true` is an identity of rational functions, hence holds for every Weierstrass equation over every field.

```
L<a1,a2,a3,a4,a6,u,r,s,t> := FunctionField(Rationals(), 9);
E := EllipticCurve([a1,a2,a3,a4,a6]);
Ep := subst(E, u, r, s, t);
a1p, a2p, a3p, a4p, a6p := Explode(aInvariants(Ep));
u*a1p eq a1 + 2*s;                                                  // true
u^2*a2p eq a2 - s*a1 + 3*r - s^2;                                   // true
u^3*a3p eq a3 + r*a1 + 2*t;                                         // true
u^4*a4p eq a4 - s*a3 + 2*r*a2 - (t + r*s)*a1 + 3*r^2 - 2*s*t;       // true
u^6*a6p eq a6 + r*a4 + r^2*a2 + r^3 - t*a3 - t^2 - r*t*a1;          // true
b2, b4, b6, b8 := Explode(bInvariants(E));
b2p, b4p, b6p, b8p := Explode(bInvariants(Ep));
u^2*b2p eq b2 + 12*r;                                               // true
u^4*b4p eq b4 + r*b2 + 6*r^2;                                       // true
u^6*b6p eq b6 + 2*r*b4 + r^2*b2 + 4*r^3;                            // true
u^8*b8p eq b8 + 3*r*b6 + 3*r^2*b4 + r^3*b2 + 3*r^4;                 // true
c4, c6 := Explode(cInvariants(E));
c4p, c6p := Explode(cInvariants(Ep));
u^4*c4p eq c4;                              // true
u^6*c6p eq c6;                              // true
u^12*Discriminant(Ep) eq Discriminant(E);   // true
jInvariant(Ep) eq jInvariant(E);            // true
```

For ω, differentiating $x = u^2x' + r$ gives $dx = u^2\,dx'$, while $2y + a_1x + a_3 = u^3(2y' + a_1'x' + a_3')$, so $\omega = u^{-1}\omega'$.

A change of variables preserving the short form (4.1.2) must have $a_1' = a_3' = 0$, forcing $s = t = 0$, and then $a_2' = 0$ forces $r = 0$. What is left rescales A and B.

```
M<A,B,u> := FunctionField(Rationals(), 3);
E := EllipticCurve([M | 0, 0, 0, A, B]);
aInvariants(subst(E, u, 0, 0, 0));   // [0, 0, 0, A/u^4, B/u^6]
```

That is $A' = u^{-4}A$ and $B' = u^{-6}B$. □

REMARK 4.1.8. Chapter 1 uses the short form (4.1.2) (Example 1.6.3), and the discriminant of Lemma 4.1.4 explains the condition $4A^3+27B^2 \neq 0$ there. Under the isomorphism of Theorem 1.7.1 we have $x = \wp_\Lambda(z)$ and $y = \frac{1}{2}\wp'_\Lambda(z)$, so $\omega = \wp'_\Lambda(z)\,dz/\wp'_\Lambda(z) = dz$: the invariant differential is the translation-invariant differential dz on $\mathbb{C}/\Lambda$. With $A = -15G_4(\Lambda)$ and $B = -35G_6(\Lambda)$, the j-invariant becomes a function of the lattice, and by Proposition 4.1.7 a function on $\mathrm{SL}_2(\mathbb{Z})\backslash\mathbb{H}$ (Proposition 1.8.1).

The j-invariant classifies elliptic curves up to isomorphism over $\overline{k}$, but not over k. The failure is already visible in the short form. Suppose $\operatorname{char} k \neq 2$, let $E\colon y^2 = x^3 + Ax + B$ and let $d \in k^\times$. The quadratic twist of E by d is

$$E^d\colon y^2 = x^3 + d^2Ax + d^3B,$$

which is $dy^2 = x^3 + Ax + B$ in the coordinates dx and d^2y. So $(x, y) \mapsto (dx, d^{3/2}y)$ is an isomorphism $E \to E^d$ over $k(\sqrt{d})$, and $j(E^d) = j(E)$. Suppose now that $j(E) \neq 0, 1728$, that is $AB \neq 0$. By Proposition 4.1.7 an isomorphism $E \to E^d$ over k would be a change of variables $x = u^2x'$, $y = u^3y'$ with $u^4 = d^{-2}$ and $u^6 = d^{-3}$, forcing $u^2 = d^{-1}$. So E and E^d are isomorphic over k exactly when $d \in k^{\times 2}$, and the quadratic twists of E are indexed by $d \in k^\times/k^{\times 2}$. They all share one j-invariant. For $j = 0$ and $j = 1728$ the curve has extra automorphisms and there are more twists than these. See Theorem 4.7.11 and Chapter 10.

EXAMPLE 4.1.9. Take $E\colon y^2 = x^3 + x + 1$ over $\mathbb{Q}$ and $d = -1$, so that $E^{-1}\colon y^2 = x^3 + x - 1$. Here $d^6 = 1$, so the two curves have the same discriminant -496 as well as the same j-invariant $6912/31$. They are nonetheless not isomorphic over $\mathbb{Q}$, because $-1 \notin \mathbb{Q}^{\times 2}$. They become isomorphic over $\mathbb{Q}(i)$. So Δ and j together do not determine an elliptic curve up to isomorphism over k.

```
E := EllipticCurve([Rationals() | 0, 0, 0, 1, 1]);
Ed := QuadraticTwist(E, -1);
aInvariants(Ed);                    // [0, 0, 0, 1, -1]
jInvariant(Ed) eq jInvariant(E);    // true
Discriminant(Ed) eq Discriminant(E); // true
IsIsomorphic(E, Ed);                // false
K<i> := QuadraticField(-1);
IsIsomorphic(ChangeRing(E, K), ChangeRing(Ed, K));   // true
```

The curves $y^2 = x^3 - x$ and $y^2 = x^3 + x$, which also share a j-invariant, are **not** quadratic twists of one another: each is its own twist by -1. They are quartic twists, the extra twists that $j = 1728$ admits.

Twists in general are the subject of Chapter 7, and those of elliptic curves of Chapter 10. The cubic twists of Section 4.8 are the other case we meet in this chapter.

PROPOSITION 4.1.10.

(1) Two elliptic curves over k are isomorphic over $\overline{k}$ if and only if they have the same j-invariant.

(2) Every $j_0 \in k$ is the j-invariant of an elliptic curve over k, namely the one listed here, where $c := 1/(j_0 - 1728)$.

j_0	*Weierstrass equation*	Δ	*needs*
$\neq 0, 1728$	$y^2 + xy = x^3 - 36cx - c$	$j_0^2c^3$	
0	$y^2 + y = x^3$	-27	$\operatorname{char} k \neq 3$
1728	$y^2 = x^3 + x$	-64	$\operatorname{char} k \neq 2$

In characteristic 2 and 3 we have $0 = 1728$, and exactly one of the last two rows applies, so the single special value is covered.

PROOF. In part 1, an isomorphism over $\overline{k}$ is a change of variables (4.1.3) over $\overline{k}$, so it preserves j by Proposition 4.1.7. For the converse, suppose $\operatorname{char} k \neq 2, 3$ and put both curves in the short form (4.1.2). The changes of variables preserving that form are $x = u^2x'$, $y = u^3y'$, giving $A' = u^{-4}A$ and $B' = u^{-6}B$, so we must produce $u \in \overline{k}^\times$ with $u^4 = A/A'$ and $u^6 = B/B'$. Now $j = j'$ means $A^3B'^2 = A'^3B^2$. If $A = 0$, then $A' = 0$, and $u^6 = B/B'$ will do. If $B = 0$, then $B' = 0$, and $u^4 = A/A'$ will do. Otherwise take $u^4 = A/A'$. Then $(u^6)^2 = (A/A')^3 = (B/B')^2$, and replacing u by $\sqrt{-1}\,u$ if necessary gives $u^6 = B/B'$. The characteristics 2 and 3 need the normal forms of Proposition 4.1.12, and are finished after it.

For part 2, the first row has $b_2 = 1$, $b_4 = -72c$, $b_6 = -4c$ and $b_8 = -c - 1296c^2$, whence $c_4 = 1 + 1728c$ and $\Delta = j_0^2c^3$. Since $j_0 = 1728 + 1/c$ we have $c_4 = j_0c$, so $j = c_4^3/\Delta = j_0^3c^3/(j_0^2c^3) = j_0$. The second row has $b_6 = 1$ as its only non-zero b, so $\Delta = -27$ and $c_4 = 0$, giving $j = 0$. The third has $b_4 = 2$ and $b_8 = -1$, so $\Delta = -64$ and $c_4 = -48$, giving $j = 1728$. Each row asks only that its own Δ be non-zero, which is the condition listed. □

REMARK 4.1.11. If $\Delta = 0$, the cubic (4.1.1) has exactly one singular point S. It is a node (two distinct tangent directions) if $c_4 \neq 0$, and a cusp (one tangent direction) if $c_4 = 0$ [Sil09, Proposition III.1.4(a)]. In both cases the curve is birational to $\mathbb{P}^1$, as for the nodal cubic of Example 2.3.5 [Sil09, Proposition III.1.6]. These curves matter to us because an elliptic curve over $\mathbb{Q}$ acquires such a singularity when its equation is reduced modulo a prime dividing Δ (Chapter 6). See Proposition 4.2.11.

In characteristic 2 and 3 the short form (4.1.2) is not available, and the general equation (4.1.1) carries too many coefficients to compute with. The substitutions (4.1.3) still cut it down to a short list of normal forms: two in each of these characteristics, according as $j = 0$ or not.

PROPOSITION 4.1.12. *Let E be an elliptic curve over k. Then the Weierstrass coordinates of Theorem 4.1.6 can be chosen so that the model (4.1.1) is the short form (4.1.2) when $\operatorname{char} k \neq 2, 3$, and otherwise is the one listed here.*

$\operatorname{char} k$	*case*	*Weierstrass equation*	Δ	j
3	$j(E) \neq 0$	$y^2 = x^3 + a_2x^2 + a_6$	$-a_2^3a_6$	$-a_2^3/a_6$
3	$j(E) = 0$	$y^2 = x^3 + a_4x + a_6$	$-a_4^3$	0
2	$j(E) \neq 0$	$y^2 + xy = x^3 + a_2x^2 + a_6$	a_6	$1/a_6$
2	$j(E) = 0$	$y^2 + a_3y = x^3 + a_4x + a_6$	a_3^4	0

PROOF. The case $\operatorname{char} k \neq 2, 3$ is the completion of the square and of the cube that produced (4.1.2). In the two remaining characteristics we start from a model (4.1.1) and apply a substitution (4.1.3), reading off the new coefficients from the transformation table [Sil09, Table 3.1]. Nothing deep goes into the choice of u, r, s, t: one only asks that certain coefficients become 0 or 1. Each step below can be watched in Magma over the function field in the coefficients, in the characteristic at hand, with `subst` as defined above. The blocks run in one session, in order, and each redefines the a_i it needs.

Suppose $\operatorname{char} k = 3$. Completing the square in y is (4.1.3) with $u = 1$, $r = 0$, $s = -a_1/2$ and $t = -a_3/2$, and it gives $y^2 = x^3 + a_2x^2 + a_4x + a_6$.

```
K<a1,a2,a3,a4,a6> := FunctionField(GF(3), 5);
E := EllipticCurve([a1,a2,a3,a4,a6]);
aInvariants(subst(E, 1, 0, -a1/2, -a3/2));   // a1 = a3 = 0
```

For such an equation a direct computation in characteristic 3 gives $c_4 = a_2^2$ and $\Delta = a_2^2a_4^2 - a_2^3a_6 - a_4^3$. So $j = a_2^6/\Delta$, and $j = 0$ if and only if $a_2 = 0$.

```
L<a2,a4,a6> := FunctionField(GF(3), 3);
D := EllipticCurve([L | 0, a2, 0, a4, a6]);
cInvariants(D)[1] eq a2^2;                    // true
Discriminant(D) eq a2^2*a4^2 - a2^3*a6 - a4^3;  // true
jInvariant(D) eq a2^6/Discriminant(D);        // true
```

If $j = 0$, the equation is already the one listed, with $\Delta = -a_4^3$. If $j \neq 0$, then $a_2 \neq 0$, and $x = x' + a_4/a_2$ turns the coefficient of x' into $2a_2(a_4/a_2) + a_4 = 3a_4 = 0$, leaving the equation listed for $j \neq 0$. Its discriminant is $-a_2^3 a_6$, so $j = a_2^6/(-a_2^3 a_6) = -a_2^3/a_6$.

```
D2 := subst(D, 1, a4/a2, 0, 0);
a6new := aInvariants(D2)[5];
aInvariants(D2);                    // [0, a2, 0, 0, a6new]
Discriminant(D2) eq -a2^3*a6new;    // true
jInvariant(D2) eq -a2^3/a6new;      // true
```

Suppose $\operatorname{char} k = 2$. Now $c_4 = a_1^4$, so $j = a_1^{12}/\Delta$, and $j = 0$ if and only if $a_1 = 0$. If $j \neq 0$, the substitution

$$x = a_1^2 x' + \frac{a_3}{a_1}, \qquad y = a_1^3 y' + \frac{a_1^2 a_4 + a_3^2}{a_1^3}$$

makes the new a_1 equal to 1 and kills the new a_3 and a_4, leaving the equation listed for $j \neq 0$. There $b_2 = 1$, $b_4 = b_6 = 0$ and $b_8 = a_6$, so $\Delta = a_6$ and $j = 1/a_6$.

```
K<a1,a2,a3,a4,a6> := FunctionField(GF(2), 5);
E := EllipticCurve([a1,a2,a3,a4,a6]);
cInvariants(E)[1] eq a1^4;          // true
F := subst(E, a1, a3/a1, 0, (a1^2*a4 + a3^2)/a1^3);
a6new := aInvariants(F)[5];
aInvariants(F);                     // [1, a2new, 0, 0, a6new]
Discriminant(F) eq a6new;           // true
jInvariant(F) eq 1/a6new;           // true
```

If $j = 0$, then $a_1 = 0$, and $x = x' + a_2$, $y = y'$ kills the new a_2, since $4a_2 = 0$, leaving the last equation of the table.

```
M<a2,a3,a4,a6> := FunctionField(GF(2), 4);
G := EllipticCurve([M | 0, a2, a3, a4, a6]);
G2 := subst(G, 1, a2, 0, 0);
aInvariants(G2);            // [0, 0, a3, a2^2 + a4, a2*a4 + a6]
Discriminant(G2) eq a3^4;   // true
jInvariant(G2) eq 0;        // true
```

There $b_2 = b_4 = 0$ and $b_6 = a_3^2$, so $\Delta = -27a_3^4 = a_3^4$, while $c_4 = 0$, so $j = 0$. This is [Sil09, Proposition A.1.1]. □

Since $\Delta \neq 0$, the two characteristic 2 rows force $a_6 \neq 0$ and $a_3 \neq 0$ respectively. In characteristic 2 there is no smooth Weierstrass equation with $a_1 = a_3 = 0$.

The normal forms finish the converse in Proposition 4.1.10, which we left open in characteristic 2 and 3. Let E and E' be elliptic curves over $\overline{k}$ with $j(E) = j(E')$, both put in the form of the same row of the table. We must find a change of variables (4.1.3) over $\overline{k}$ carrying one to the other. In each case the substitutions preserving the row's shape are constrained first, by asking that the coefficients the row sets to 0 or 1 stay that way.

- **Characteristic 3, $j \neq 0$.** In $y^2 = x^3 + a_2x^2 + a_6$, the condition $a_1' = 0$ forces $s = 0$, then $a_3' = 0$ forces $t = 0$, and then $a_4' = 0$ reads $2ra_2 = 0$, so $r = 0$ because $a_2 \neq 0$. What is left is $x = u^2x'$, $y = u^3y'$, giving $a_2' = a_2/u^2$ and $a_6' = a_6/u^6$. Now $j = -a_2^3/a_6$, so $j = j'$ says $a_2^3 a_6' = a_2'^3 a_6$, and $u^2 := a_2/a_2'$ does the job.
- **Characteristic 3, $j = 0$.** In $y^2 = x^3 + a_4x + a_6$ we again get $s = t = 0$, while $a_2' = 0$ is automatic since $3r = 0$. So $x = u^2x' + r$, $y = u^3y'$, with $a_4' = a_4/u^4$ and $a_6' = (a_6 + ra_4 + r^3)/u^6$. Here $a_4 \neq 0$, so choose u with $u^4 = a_4/a_4'$, and then r as a root of $r^3 + a_4r + a_6 - u^6a_6' = 0$.

- **Characteristic 2, $j \neq 0$.** In $y^2 + xy = x^3 + a_2x^2 + a_6$, the condition $a_1' = 1$ forces $u = 1$, then $a_3' = 0$ forces $r = 0$ and $a_4' = 0$ forces $t = 0$. So $x = x'$, $y = y' + sx'$, with $a_6' = a_6$ and $a_2' = a_2 + s + s^2$. Since $j = 1/a_6$, the hypothesis $j = j'$ already gives $a_6' = a_6$, and it remains to take s a root of $s^2+s+a_2+a_2' = 0$.
- **Characteristic 2, $j = 0$.** In $y^2 + a_3y = x^3 + a_4x + a_6$ the condition $a_1' = 0$ is automatic and $a_2' = 0$ forces $r = s^2$. Then $a_3' = a_3/u^3$, $a_4' = (a_4 + sa_3 + s^4)/u^4$ and $a_6' = (a_6 + s^2a_4 + s^6 + ta_3 + t^2)/u^6$. Here $a_3 \neq 0$, so choose u with $u^3 = a_3/a_3'$, then s a root of $s^4 + a_3s + a_4 - u^4a_4' = 0$, and then t a root of $t^2 + a_3t + a_6 + s^2a_4 + s^6 - u^6a_6' = 0$.

Each equation has a root because $\overline{k}$ is algebraically closed, so Proposition 4.1.10 holds in every characteristic. Reading the same four computations with $E = E'$ counts the automorphisms, which is Theorem 4.7.11.

DRAFT
13 Oct 2026

4.2. The group law

On $\mathbb{C}/\Lambda$ the group law comes for free from addition in $\mathbb{C}$. On an elliptic curve over k it comes from the group $\operatorname{Pic}^0 E$ of degree-0 divisors modulo principal divisors (Proposition 3.3.2): for a curve of genus 1, the map $P \mapsto [P - O]$ is a bijection $E(k) \to \operatorname{Pic}^0 E$.

LEMMA 4.2.1. *Let E be a nice curve of genus 1, and $P, Q \in E(k)$. If $P \sim Q$, then $P = Q$.*

PROOF. Say $\operatorname{div}(f) = P - Q$. Then $f \in \mathcal{L}(Q)$. Since $\deg Q = 1 > 2g - 2 = 0$, Corollary 3.5.5 gives $\ell(Q) = 1$, so $\mathcal{L}(Q) = k$ and f is constant. Hence $P - Q = \operatorname{div}(f) = 0$. □

This is what distinguishes genus 1 from genus 0: on $\mathbb{P}^1_k$ any two rational points are linearly equivalent (Example 3.3.9).

PROPOSITION 4.2.2. *Let (E, O) be an elliptic curve over k.*
(1) For every $D \in \operatorname{Div}^0 E$ there is a unique $P \in E(k)$ with $D \sim P - O$. Write $\sigma(D) \coloneqq P$.
(2) The map $\kappa\colon E(k) \to \operatorname{Pic}^0 E$, $P \mapsto [P - O]$, is a bijection, with inverse induced by σ.

PROOF. Uniqueness in part 1 is Lemma 4.2.1. For existence, $\deg(D + O) = 1$, so $\ell(D + O) = 1$ by Corollary 3.5.5. Let f span $\mathcal{L}(D + O)$. Then $\operatorname{div}(f) + D + O$ is an effective divisor of degree 1, so it is a single closed point P of degree 1, that is, a rational point, and $D \sim P - O$. Part 2 follows [Sil09, Proposition III.3.4]. □

Note that the argument works over k itself, with no Galois descent: an effective divisor of degree 1 is automatically a rational point. Since O is a rational point, $\operatorname{Pic}^0 E$ is also $(\operatorname{Pic}^0 E_{k^{\mathrm{sep}}})^{\mathrm{Gal}_k}$ by Warning 3.3.8, so both readings of Pic^0 give $E(k)$. This fails for a nice curve of genus 1 with no rational point, such as $3X^3 + 4Y^3 + 5Z^3 = 0$ (Remark 4.8.14). We use κ to transport the group structure of $\operatorname{Pic}^0 E$ to $E(k)$. So $P + Q$ is the unique rational point with

$$(P + Q) - O \sim (P - O) + (Q - O),$$

and O is the identity. For every field $L \supseteq k$, the same construction on E_L makes $E(L)$ a group, and $E(k) \subseteq E(L)$ is a subgroup. If L/k is Galois, then $\mathrm{Gal}(L/k)$ acts on $E(L)$ by group automorphisms, with fixed points $E(k)$ (Proposition 2.1.17).

COROLLARY 4.2.3. *Let $D = \sum n_P P$ be a divisor on E supported on points of $E(k)$. Then D is principal if and only if $\sum n_P = 0$ and $\sum [n_P]P = O$ in $E(k)$. More generally, there is an exact sequence*

$$1 \to k^\times \to k(E)^\times \xrightarrow{\operatorname{div}} \operatorname{Div}^0 E \xrightarrow{\sigma} E(k) \to 0.$$

PROOF. Combine Proposition 3.3.2 with Proposition 4.2.2 [Sil09, Corollary III.3.5]. □

For a smooth plane cubic whose base point is a flex, such as a Weierstrass cubic, this group law is the classical chord-and-tangent construction, illustrated in Figure 4.2.1. By Definition 4.1.2, the point at infinity of a Weierstrass cubic is a flex, with tangent line $Z = 0$.

PROPOSITION 4.2.4. *Let $E \subseteq \mathbb{P}^2_k$ be a smooth plane cubic, let $O \in E(k)$ be a flex, and give $E(\overline{k})$ the group law with identity O.*

(1) If a line meets E in $P, Q, R \in E(\overline{k})$, counted with intersection multiplicity, then $P + Q + R = O$.

(2) If E is a Weierstrass cubic (4.1.1) and $O = (0 : 1 : 0)$, then $-P = (x_0, -y_0 - a_1 x_0 - a_3)$ for $P = (x_0, y_0)$.

So $P + Q$ is obtained by drawing the line through P and Q (the tangent line if $P = Q$), taking its third intersection point R with E, and reflecting: $P + Q$ is the third intersection point of the line through R and O.

PROOF. We may assume that $k = \overline{k}$. Let L_O be a linear form defining the tangent line at O, which meets E only at O, with multiplicity 3. For a Weierstrass cubic, $L_O = Z$ (Definition 4.1.2). Let L be a linear form defining the line. On a smooth curve the intersection number at a closed point P is $\operatorname{ord}_P(L/L_O) \deg P$ away from $L_O = 0$ (see the proof of Proposition 3.7.6), so by Theorem 3.7.4

$$\operatorname{div}(L/L_O) = P + Q + R - 3O.$$

So $(P - O) + (Q - O) + (R - O) \sim 0$, which is part 1. For part 2, the vertical line $X - x_0 Z$ passes through O, and its other two intersection points with E are the roots in y of $f(x_0, y) = 0$, whose sum is $-a_1 x_0 - a_3$. Part 1 with $R = O$ gives part 2. □

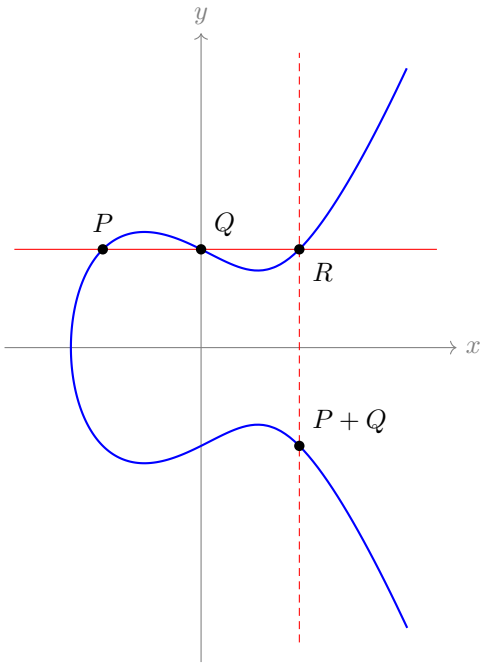


FIGURE 4.2.1. The chord-and-tangent law on $y^2 = x^3 - x + 1$ over $\mathbb{R}$: the line through $P = (-1, 1)$ and $Q = (0, 1)$ meets the curve again at $R = (1, 1)$, and $P + Q = -R = (1, -1)$.

To compute the third intersection point we expand the cubic form along a line. For vectors $P, Q \in k^3$ we write $\nabla F(P) \cdot Q \coloneqq \sum_i \frac{\partial F}{\partial X_i}(P)\, Q_i$.

LEMMA 4.2.5 (Polar expansion). *Let $F \in k[X, Y, Z]$ be a cubic form, and $P, Q \in k^3$. Then, for $s, t \in k$,*

$$F(sP + tQ) = s^3 F(P) + s^2 t\, \nabla F(P) \cdot Q + s t^2\, \nabla F(Q) \cdot P + t^3 F(Q). \tag{4.2.1}$$

PROOF. Since F is homogeneous of degree 3, $F(sP+tQ)$ is a form of degree 3 in s and t, say $\sum_{m=0}^{3} c_m s^{3-m}t^m$, with $c_0 = F(P)$ and $c_3 = F(Q)$. Setting $s=1$, c_1 is the coefficient of t in the polynomial $F(P+tQ) \in k[t]$, which is its derivative at $t=0$. By the chain rule for polynomials, this is $\nabla F(P)\cdot Q$. Exchanging the roles of P and Q gives $c_2 = \nabla F(Q)\cdot P$. □

The tangent line to E at a smooth point P is $\{\nabla F(P)\cdot V = 0\}$, so the coefficient $\nabla F(P)\cdot Q$ vanishes exactly when Q lies on the tangent line at P. This turns the chord-and-tangent construction into formulas in homogeneous coordinates.

PROPOSITION 4.2.6. *Let $E = V_+(F) \subseteq \mathbb{P}^2_k$ be a smooth plane cubic with a flex $O \in E(k)$, and give $E(\overline{k})$ the group law with identity O, as in Proposition 4.2.4. We write $\overline{k}$-points as vectors in $\overline{k}^3$, and the formulas below are homogeneous.*

(1) ***Chord.*** *Let $P_1 \neq P_2$ be points of $E(\overline{k})$, and set $A := \nabla F(P_1)\cdot P_2$ and $B := \nabla F(P_2)\cdot P_1$. Then A and B do not both vanish, the line through P_1 and P_2 meets E in the third intersection point $R := BP_1 - AP_2$, and $P_1+P_2 = -R$.*

(2) ***Tangent.*** *Let $P \in E(\overline{k})$, and let $Q \in \overline{k}^3$ be a vector on the tangent line at P, that is $\nabla F(P)\cdot Q = 0$, which is not a multiple of P. Then $F(Q)$ and $\nabla F(Q)\cdot P$ do not both vanish, the tangent line meets E again in $R := F(Q)P - (\nabla F(Q)\cdot P)\,Q$, and $[2]P = -R$.*

(3) ***Weierstrass cubics.*** *If E is the Weierstrass cubic (4.1.1) with $O = (0:1:0)$, then $-(X:Y:Z) = (X:-Y-a_1X-a_3Z:Z)$, and for $P \neq O$ the vector $Q := \left(\frac{\partial F}{\partial Y}(P), -\frac{\partial F}{\partial X}(P), 0\right)$ satisfies the hypotheses of part 2.*

PROOF. For part 1, $F(P_1) = F(P_2) = 0$, so (4.2.1) reads $F(sP_1+tP_2) = st(As+Bt)$. If $A = B = 0$, this vanishes identically, so the line would be a component of E, which is impossible since E is geometrically integral (Corollary 3.7.5). Hence the line meets E at $(s:t) = (1:0)$, $(0:1)$ and $(B:-A)$, that is, at P_1, P_2 and R. If $A = 0$, then P_2 lies on the tangent line at P_1, and $R = P_1$ is the tangency, counted twice. Now $P_1+P_2+R = O$ by Proposition 4.2.4.

For part 2, $F(P) = 0$ and $\nabla F(P)\cdot Q = 0$, so $F(sP+tQ) = t^2\big(s\,\nabla F(Q)\cdot P + tF(Q)\big)$. The double root at $t=0$ is the tangency at P, and the remaining root $(s:t) = \big(F(Q) : -\nabla F(Q)\cdot P\big)$ is R. As before the two coefficients cannot both vanish. Then $P+P+R = O$.

For part 3, negation is part 2 of Proposition 4.2.4, homogenized. The vector Q satisfies $\nabla F(P)\cdot Q = \frac{\partial F}{\partial X}\frac{\partial F}{\partial Y} - \frac{\partial F}{\partial Y}\frac{\partial F}{\partial X} = 0$. It is non-zero: if both partial derivatives vanished at P, the Euler relation $X\frac{\partial F}{\partial X} + Y\frac{\partial F}{\partial Y} + Z\frac{\partial F}{\partial Z} = 3F$ evaluated at P would give $Z_0\frac{\partial F}{\partial Z}(P) = 0$, so, as $Z_0 \neq 0$ for $P \neq O$, the $\overline{k}$-point P would be singular. Finally, Q has third coordinate 0 and P does not, so Q is not a multiple of P. □

For $P = O$ there is nothing to compute, since $[2]O = O$. Reading these formulas on the affine patch $Z = 1$, with the line written as $y = \lambda x + \nu$, gives the classical algorithm.

ALGORITHM 4.2.7 (Group law). Let E be a smooth Weierstrass cubic (4.1.1), and $P_i = (x_i, y_i) \in E(k)$ for $i = 1, 2$.

(1) If $x_1 = x_2$ and $y_1 + y_2 + a_1x_2 + a_3 = 0$, then $P_1 + P_2 = O$.

(2) Otherwise set

$$\lambda := \frac{y_2 - y_1}{x_2 - x_1}, \qquad \nu := \frac{y_1x_2 - y_2x_1}{x_2 - x_1} \qquad \text{if } x_1 \neq x_2,$$

$$\lambda := \frac{3x_1^2 + 2a_2x_1 + a_4 - a_1y_1}{2y_1 + a_1x_1 + a_3}, \qquad \nu := \frac{-x_1^3 + a_4x_1 + 2a_6 - a_3y_1}{2y_1 + a_1x_1 + a_3} \qquad \text{if } x_1 = x_2,$$

so that $y = \lambda x + \nu$ is the line through P_1 and P_2, or the tangent line if $P_1 = P_2$. Then $P_1 + P_2 = (x_3, y_3)$ with

$$x_3 = \lambda^2 + a_1\lambda - a_2 - x_1 - x_2, \qquad y_3 = -(\lambda + a_1)x_3 - \nu - a_3.$$

In particular, for the short form (4.1.2) and $P = (x, y)$ with $y \neq 0$, the duplication formula is

$$x([2]P) = \frac{x^4 - 2Ax^2 - 8Bx + A^2}{4(x^3 + Ax + B)}.$$

PROOF. Substitute $y = \lambda x + \nu$ into the equation of E. This is (4.2.1) on the patch $Z = 1$, with the line parametrized by x instead of $(s : t)$. The resulting cubic in x has roots x_1, x_2 and x_3, and the coefficient of x^2 gives $x_1 + x_2 + x_3 = \lambda^2 + a_1\lambda - a_2$. Then negate, by part 3 of Proposition 4.2.6 [Sil09, Algorithm III.2.3]. □

EXAMPLE 4.2.8. On $E\colon y^2 = x^3 + 17$ over $\mathbb{Q}$, let $P_1 \coloneqq (-2, 3)$ and $P_2 \coloneqq (-1, 4)$. The line through them has $\lambda = 1$, $\nu = 5$, so $x_3 = 1 + 2 + 1 = 4$, $y_3 = -4 - 5 = -9$, and $P_1 + P_2 = (4, -9)$. Likewise $[2]P_1 = (8, -23)$ and $[2]P_2 = (137/64, -2651/512)$. Denominators appear quickly. The Mordell–Weil theorem (Chapter 8) says that nonetheless $E(\mathbb{Q})$ is finitely generated.

EXAMPLE 4.2.9. A point $P \neq O$ of $E(\bar{k})$ satisfies $[2]P = O$ if and only if $P = -P$, i.e. if and only if $2y + a_1x + a_3 = 0$ at P. Suppose $\operatorname{char} k \neq 2$, and let $E\colon y^2 = (x - e_1)(x - e_2)(x - e_3)$ with distinct $e_i \in k$, and $P_i \coloneqq (e_i, 0)$, as in Example 3.3.10. The divisors computed there,

$$\operatorname{div}(x - e_i) = 2P_i - 2O, \qquad \operatorname{div}(y) = P_1 + P_2 + P_3 - 3O,$$

say, by Corollary 4.2.3, that $[2]P_i = O$ and $P_1 + P_2 + P_3 = O$. So

$$\{O, P_1, P_2, P_3\} \cong (\mathbb{Z}/2\mathbb{Z})^2,$$

exactly as for $(\mathbb{C}/\Lambda)[2]$ in Example 1.3.4.

The group law is given by morphisms.

THEOREM 4.2.10. *Let E be an elliptic curve over k. The formulas of Algorithm 4.2.7 define morphisms of k-varieties*

$$+\colon E \times E \to E, \qquad -\colon E \to E.$$

In particular, for $Q \in E(k)$, translation $\tau_Q\colon E \to E$, $P \mapsto P + Q$, is an isomorphism of curves.

PROOF. Negation is a rational map $E \dashrightarrow E$, hence a morphism by Proposition 3.2.1. For addition, see [Sil09, Theorem III.3.6]: one shows that τ_Q is a morphism for each Q, and then that the rational map $E \times E \dashrightarrow E$ is regular everywhere by translating. □

So an elliptic curve is a group variety (Definition 2.1.7), an abelian variety of dimension 1. We end the section with the singular Weierstrass cubics of Remark 4.1.11: removing the singular point leaves one of the group varieties $\mathbb{G}_a$, $\mathbb{G}_m$, $\mathbb{G}_m^{(d)}$ of Example 2.1.8. We follow [Poo08, Section 6.3] and [Mil20, Section II.3].

Let E be a Weierstrass cubic (4.1.1) over k with $\Delta = 0$, let S be its singular point, and let $E^{\mathrm{ns}} \coloneqq E - \{S\}$. The point O is nonsingular (Proposition 4.1.5), so S is affine. Suppose that $S \in E(k)$. The change of variables (4.1.3) with $u = 1$, $r = x(S)$, $s = 0$ and $t = y(S)$ moves S to $(0, 0)$ and does not change c_4. Then $a_6 = 0$ because S lies on E, and $a_4 = a_3 = 0$ because the partial derivatives vanish at S, so

$$E\colon y^2 + a_1xy = x^3 + a_2x^2, \tag{4.2.2}$$
$$b_2 = a_1^2 + 4a_2, \qquad b_4 = 0, \qquad c_4 = b_2^2.$$

The tangent cone at S is $y^2 + a_1xy - a_2x^2 = 0$, and the slopes of the tangent lines are the roots of $T^2 + a_1T - a_2$. Let

$$L \coloneqq k[T]/(T^2 + a_1T - a_2)$$

be the algebra of tangent slopes, and let α and $\alpha' = -a_1 - \alpha$ be the images of the roots. Its discriminant is b_2, so L is separable exactly when $c_4 \neq 0$: then S is a node, with two tangent directions, and otherwise a cusp (Remark 4.1.11). If $c_4 \neq 0$, then $L \cong L_d$ (2.1.1), with

$$d \coloneqq b_2 = a_1^2 + 4a_2 \quad \text{if } \operatorname{char} k \neq 2, \qquad d \coloneqq a_2/a_1^2 \quad \text{if } \operatorname{char} k = 2, \tag{4.2.3}$$

through $t \mapsto 2\alpha + a_1$, respectively $t \mapsto \alpha/a_1$. If $c_4 = 0$, then $T^2 + a_1 T - a_2 = (T - \beta)^2$ over $\overline{k}$, and the tangent line at S is $y = \beta x$.

PROPOSITION 4.2.11. *Let E be a Weierstrass cubic over k with $\Delta = 0$, whose singular point S is a rational point, put in the form (4.2.2). For every field $K \supseteq k$, the chord-and-tangent construction of Proposition 4.2.4 makes $E^{\mathrm{ns}}(K)$ an abelian group with identity O, and E^{ns} is a group variety.*

(1) ***Node.*** *If $c_4 \neq 0$, then*

$$P \mapsto \frac{y - \alpha x}{y - \alpha' x}$$

is an isomorphism of group varieties $E^{\mathrm{ns}} \to \mathbb{G}_m^{(d)}$, with d as in (4.2.3). In particular $E^{\mathrm{ns}} \cong \mathbb{G}_m$ if the slopes of the tangent lines at S lie in k.

(2) ***Cusp.*** *If $c_4 = 0$ and the tangent line $y = \beta x$ at S is defined over k, then*

$$P \mapsto \frac{x}{y - \beta x}$$

is an isomorphism of group varieties $E^{\mathrm{ns}} \to \mathbb{G}_a$.

Both hypotheses on S hold if k is perfect, and the one on the tangent line holds if $\operatorname{char} k \neq 2$, where $\beta = -a_1/2$. Over $\overline{k}$, these are the isomorphisms $E^{\mathrm{ns}}(\overline{k}) \cong \overline{k}^{\times}$ and $E^{\mathrm{ns}}(\overline{k}) \cong \overline{k}$ of [Sil09, Proposition III.2.5].

PROOF. **The composition law.** Let F be the cubic form of E. A line through S meets E at S with multiplicity at least 2, since all partial derivatives of F vanish at S. So a line through two points of $E^{\mathrm{ns}}(K)$, or tangent to E at one, does not pass through S, and its third intersection point with E, counted with multiplicity as a root of the restriction of F to the line, lies in $E^{\mathrm{ns}}(K)$. As in Proposition 4.2.4, define $-P$ as the third intersection point of the line through P and O, and $P + Q \coloneqq -R$, where R is the third intersection point of the line through P and Q. Suppose that $\varphi\colon E^{\mathrm{ns}}(K) \to G(K)$ is a bijection onto the K-points of a group variety G, written additively, with $\varphi(O) = 0$, and such that whenever $P_1, P_2, P_3 \in E^{\mathrm{ns}}(K)$ are the three intersection points of a line with E,

$$\varphi(P_1) + \varphi(P_2) + \varphi(P_3) = 0. \tag{4.2.4}$$

Then $\varphi(-P) = -\varphi(P)$ and $\varphi(P + Q) = -\varphi(R) = \varphi(P) + \varphi(Q)$. So the composition law is transported from $G(K)$: it is associative, $E^{\mathrm{ns}}(K)$ is an abelian group, and φ is an isomorphism. If moreover φ is an isomorphism of varieties $E^{\mathrm{ns}} \to G$, then the group law of E^{ns} is given by morphisms, and φ is an isomorphism of group varieties.

The cusp. Substituting $y + \beta x$ for y, a change of variables over k, turns (4.2.2) into $y^2 = x^3$, that is, $Y^2 Z = X^3$, and the map into $P \mapsto x/y = X/Y$. On E^{ns} we have $Y \neq 0$, since $Y = 0$ forces $X = 0$, which is S. So $t \mapsto (t : 1 : t^3)$ is an isomorphism $\mathbb{G}_{\mathrm{a}} \to E^{\mathrm{ns}}$, inverse to X/Y, with $t = 0$ going to O. A line $aX + bY + cZ = 0$ not through $S = (0 : 0 : 1)$ has $c \neq 0$, and it meets E at the $\overline{k}$-points with parameters the roots of $ct^3 + at + b$. Since this cubic has no t^2 term, the three roots add up to 0, which is (4.2.4) [Poo08, Section 6.3].

The node, split. Suppose first that $\alpha, \alpha' \in k$. The coordinates $U \coloneqq Y - \alpha X$, $W \coloneqq Y - \alpha' X$ and Z are projective coordinates on $\mathbb{P}^2$, since $\alpha \neq \alpha'$. In them $S = (0 : 0 : 1)$, $O = (1 : 1 : 0)$, $X = (U - W)/(\alpha' - \alpha)$, and E is

$$c\,UWZ = (U - W)^3, \qquad c \coloneqq (\alpha' - \alpha)^3,$$

because $Y^2 + a_1 XY - a_2 X^2 = UW$. On E^{ns}, U and W are non-zero, since either one vanishing forces the other to, which is S. So $\varphi \coloneqq U/W$ is a morphism $E^{\mathrm{ns}} \to \mathbb{G}_{\mathrm{m}}$, and

$$r \mapsto \left(c\,r^2 : c\,r : (r - 1)^3\right)$$

is its inverse, with $r = 1$ going to O. A line not through S is $Z = \lambda U + \mu W$, and it meets E at the $\overline{k}$-points with $r = U/W$ a root of $(r - 1)^3 - c\,r(\lambda r + \mu)$. This cubic is monic with constant term -1, so the product of the three roots is 1, which is (4.2.4), written multiplicatively [Mil20, Section II.3]. Then $\mathbb{G}_{\mathrm{m}} \cong \mathbb{G}_{\mathrm{m}}^{(d)}$ by

part 1 of Proposition 2.1.9, through $x + yt \mapsto x + \delta y$ with $\delta = 2\alpha + a_1$, respectively α/a_1, and the composite is the map of part 1.

The node, in general. Since $(y - \alpha x)(y - \alpha' x) = x^3$ on E,

$$\frac{y - \alpha x}{y - \alpha' x} = \frac{(y - \alpha x)^2}{x^3} = \frac{y^2 + a_2 x^2}{x^3} - \frac{2y + a_1 x}{x^2}\,\alpha,$$

using $\alpha^2 = a_2 - a_1\alpha$. Its coordinates in the basis $1, \alpha$ of L are rational functions defined over k, and its norm is 1. So it is a morphism $\Phi\colon E^{\mathrm{ns}} \to \mathbb{G}_{\mathrm{m}}^{(d)}$ defined over k. Over L, where α, α' are rational, Φ is the isomorphism of the split case. Its inverse is invariant under $\mathrm{Gal}(L/k)$, hence defined over k, and Φ is an isomorphism. For a field $K \supseteq k$, the identity (4.2.4) for Φ holds in $\mathbb{G}_{\mathrm{m}}^{(d)}(K) \subseteq \mathbb{G}_{\mathrm{m}}^{(d)}(K \otimes_k L)$, where it holds by the split case. □

DRAFT
13 Oct 2026

4.3. Isogenies

By Proposition 1.3.1, every holomorphic map of complex tori is a homomorphism followed by a translation, and holomorphic homomorphisms are the maps $z \mapsto \alpha z$ with $\alpha\Lambda_1 \subseteq \Lambda_2$ (Corollary 1.3.2). We now prove the algebraic analogue.

Definition 4.3.1. Let E_1 and E_2 be elliptic curves over k. An isogeny is a morphism $\varphi\colon E_1 \to E_2$ with $\varphi(O_1) = O_2$. By Proposition 3.2.2, either $\varphi = 0$ is constant, or φ is surjective and has a degree (Definition 3.2.5). We set $\deg 0 := 0$. The curves are isogenous if there is a non-zero isogeny between them. We write $\mathrm{Hom}(E_1, E_2)$ for the set of isogenies, $\mathrm{End}(E) := \mathrm{Hom}(E, E)$ and $\mathrm{Aut}(E)$ for the invertible elements of $\mathrm{End}(E)$. For a field $L \supseteq k$, the isogenies after base extension to L are the elements of $\mathrm{Hom}(E_{1,L}, E_{2,L})$, $\mathrm{End}(E_L)$ and $\mathrm{Aut}(E_L)$.

Unlike Definition 1.3.3, we allow the zero map. By Remark 2.3.2, elements of $\mathrm{Hom}(E_1, E_2)$ are morphisms over k, and $\mathrm{Hom}(E_1, E_2) \subseteq \mathrm{Hom}(E_{1,\overline{k}}, E_{2,\overline{k}})$. Silverman writes $\mathrm{Hom}(E_1, E_2)$ for the latter. By Theorem 4.2.10, $\mathrm{Hom}(E_1, E_2)$ is a group under $(\varphi + \psi)(P) := \varphi(P) + \psi(P)$, and $\mathrm{End}(E)$ is a ring under composition.

Example 4.3.2. For $m \in \mathbb{Z}$, the multiplication by m map $[m]\colon E \to E$ is an isogeny, by Theorem 4.2.10. Its kernel is the group scheme $E[m]$, and $E[m](\overline{k})$ is the m-torsion subgroup of $E(\overline{k})$.

Example 4.3.3. Suppose $\mathrm{char}\, k = p > 0$ and $q = p^r$. The relative Frobenius morphism $\mathsf{F}\colon E \to E^{(q)}$ of Definition 3.2.11 sends $(0 : 1 : 0)$ to $(0 : 1 : 0)$, and $E^{(q)}$ is the Weierstrass cubic with coefficients a_i^q, which has discriminant $\Delta^q \neq 0$. So F is an isogeny, purely inseparable of degree q by Proposition 3.2.12. If E is defined over $\mathbb{F}_q$, then $\mathsf{F} \in \mathrm{End}(E)$ (Example 3.2.15).

Example 4.3.4. If $\mathrm{char}\, k \neq 2$ and $i \in k$ is a square root of -1, then $[i](x, y) := (-x, iy)$ is an automorphism of $y^2 = x^3 + x$ with $[i]^2 = [-1]$.

Example 4.3.5. For $Q \in E(k)$, $Q \neq O$, the translation τ_Q is a morphism but not an isogeny. Every morphism $\psi\colon E_1 \to E_2$ is $\tau_{\psi(O_1)} \circ \varphi$ with $\varphi := \tau_{-\psi(O_1)} \circ \psi$ an isogeny, as in Proposition 1.3.1.

Theorem 4.3.6. *An isogeny $\varphi\colon E_1 \to E_2$ is a group homomorphism $E_1(L) \to E_2(L)$, for every field $L \supseteq k$.*

Proof. After base extension we may assume $L = k$, and that $\varphi \neq 0$. By Proposition 3.3.6, $\varphi_*\colon \mathrm{Pic}^0 E_1 \to \mathrm{Pic}^0 E_2$ is a homomorphism, and $\varphi_*(P - O_1) = \varphi(P) - O_2$ for $P \in E_1(k)$, because rational points have residue degree 1. So $\varphi = \kappa_2^{-1} \circ \varphi_* \circ \kappa_1$ on $E_1(k)$, a composition of homomorphisms (Proposition 4.2.2) [Sil09, Theorem III.4.8]. □

Proposition 4.3.7.
(1) If $m \neq 0$, then $[m] \neq 0$.
(2) $\mathrm{Hom}(E_1, E_2)$ is a torsion-free $\mathbb{Z}$-module.

(3) $\operatorname{End}(E)$ *is a ring of characteristic* 0 *without zero divisors.*

PROOF. Part 1 is [Sil09, Proposition III.4.2(a)]. When $\operatorname{char} k \nmid m$ it also follows from Corollary 4.4.3 below. Given part 1, $[m] \circ \varphi = 0$ forces $\varphi = 0$, since deg is multiplicative and non-constant morphisms compose to non-constant morphisms. This gives parts 2 and 3. □

We write $\ker \varphi$ for the kernel of φ on $E_1(\overline{k})$. The kernel of a non-zero isogeny is finite, being a fiber. The next theorem is the analogue of Lemma 1.3.6. It is the Galois theory of function fields of elliptic curves.

THEOREM 4.3.8. *Let* $\varphi\colon E_1 \to E_2$ *be a non-zero isogeny.*

(1) For every $Q \in E_2(\overline{k})$*, the fiber* $\varphi^{-1}(Q) \subseteq E_1(\overline{k})$ *has* $\deg_s \varphi$ *elements, and* $e_\varphi(P) = \deg_i \varphi$ *for every* $P \in E_1(\overline{k})$*.*

(2) Suppose φ *is separable. Then* φ *is unramified,* $\#\ker\varphi = \deg\varphi$*, and* $\overline{k}(E_1)/\varphi^*\overline{k}(E_2)$ *is a Galois extension, with Galois group* $\ker\varphi$ *acting by* $T \mapsto \tau_T^*$*.*

PROOF. By Proposition 3.2.8, all but finitely many fibers have $\deg_s \varphi$ elements. But for $Q, Q' \in E_2(\overline{k})$ with $\varphi(P) = Q$ and $\varphi(P') = Q'$, translation by $P' - P$ is a bijection $\varphi^{-1}(Q) \to \varphi^{-1}(Q')$, by Theorem 4.3.6, and it preserves ramification indices. So all fibers have the same size and the same ramification indices, and part 1 follows from $\sum_{P\mapsto Q} e_\varphi(P) = \deg\varphi$. In part 2, $e_\varphi = 1$, so $\#\ker\varphi = \deg\varphi$. The translations τ_T, $T \in \ker\varphi$, give $\deg\varphi$ distinct automorphisms of $\overline{k}(E_1)$ over $\varphi^*\overline{k}(E_2)$, so the extension is Galois [Sil09, Theorem III.4.10]. □

If φ is separable, the scheme-theoretic fiber $\varphi^{-1}(O_2)$ is étale over k, so $\ker\varphi \subseteq E_1(k^{\mathrm{sep}})$, and it is a Gal_k-stable subgroup. So an isogeny determines a finite Gal_k-module. Conversely, every finite subgroup is a kernel. This replaces the lattice $\Lambda_2 \supseteq \Lambda_1$ of Chapter 1.

PROPOSITION 4.3.9. *Let* $\Phi \subseteq E(\overline{k})$ *be a finite subgroup. There are an elliptic curve* E' *over* $\overline{k}$*, unique up to isomorphism, and a separable isogeny* $\varphi\colon E_{\overline{k}} \to E'$ *with* $\ker\varphi = \Phi$*. We write* $E/\Phi := E'$*. If* $\Phi \subseteq E(k^{\mathrm{sep}})$ *is* Gal_k*-stable, then* E' *and* φ *can be defined over* k*. Moreover, if* $\varphi\colon E_1 \to E_2$ *is separable and* $\psi\colon E_1 \to E_3$ *is an isogeny with* $\ker\varphi \subseteq \ker\psi$*, then* $\psi = \lambda\circ\varphi$ *for a unique isogeny* $\lambda\colon E_2 \to E_3$*.*

PROOF. Take E' to be the nice curve with function field $\overline{k}(E)^\Phi$, the fixed field of the translations τ_T^*, $T \in \Phi$ (Theorem 3.2.3). See [Sil09, Proposition III.4.12, Remarks III.4.13.1 and III.4.13.2, and Corollary III.4.11]. □

Over $\mathbb{C}$, the structure $(\mathbb{C}/\Lambda)[N] \cong (\mathbb{Z}/N\mathbb{Z})^2$ was immediate from $(\mathbb{C}/\Lambda)[N] = \frac{1}{N}\Lambda/\Lambda$. Over k we know from Theorem 4.3.8 that $\#E[N](\overline{k}) = \deg_s[N]$, so what we need is $\deg[N] = N^2$. The tool is the dual isogeny, the algebraic analogue of Theorem 1.3.8.

THEOREM 4.3.10. *Let* $\varphi\colon E_1 \to E_2$ *be an isogeny of degree* d*.*

(1) There is a unique isogeny $\hat\varphi\colon E_2 \to E_1$*, the dual isogeny, with* $\hat\varphi\circ\varphi = [d]$*. If* $\varphi \neq 0$*, then on rational points* $\hat\varphi = \sigma\circ\varphi^*\circ\kappa_2$*, that is,* $Q \mapsto \sigma\big(\varphi^*(Q - O_2)\big)$*, with* σ *as in Proposition 4.2.2. We set* $\hat 0 := 0$*.*

(2) $\varphi\circ\hat\varphi = [d]$ *on* E_2*.*

(3) $\widehat{\lambda\circ\varphi} = \hat\varphi\circ\hat\lambda$ *for an isogeny* $\lambda\colon E_2 \to E_3$*.*

(4) $\widehat{\varphi+\psi} = \hat\varphi + \hat\psi$ *for* $\psi \in \operatorname{Hom}(E_1, E_2)$*.*

(5) $\widehat{[m]} = [m]$ *and* $\deg[m] = m^2$ *for all* $m \in \mathbb{Z}$*.*

(6) $\deg\hat\varphi = \deg\varphi$ *and* $\hat{\hat\varphi} = \varphi$*.*

PROOF. See [Sil09, Theorems III.6.1 and III.6.2]. We indicate two steps.

For a separable φ and $P \in E_1(\overline{k})$, Theorem 4.3.8 gives $\varphi^*(\varphi(P)) = \sum_{T \in \ker \varphi}(P+T)$ and $\varphi^*(O_2) = \sum_{T \in \ker \varphi} T$. So $\sigma\big(\varphi^*(\varphi(P) - O_2)\big) = \sum_T (P+T) - \sum_T T = [d]P$, which is $\hat{\varphi} \circ \varphi = [d]$. The inseparable case reduces to this one and to Frobenius, using Corollary 3.2.14.

The formula in part 1 shows that $\hat{\varphi}$ is defined over k. Apply it to the generic point of E_2, that is, to the point $\eta \in E_2(k(E_2))$ given by the identity. It gives a point of $E_1(k(E_2))$, which is a rational map $E_2 \dashrightarrow E_1$ over k, hence a morphism by Proposition 3.2.1. Part 5 follows from part 4 by induction on m, and then $[\deg[m]] = \widehat{[m]} \circ [m] = [m^2]$, so $\deg[m] = m^2$ by Proposition 4.3.7. □

COROLLARY 4.3.11. *The degree* $\deg\colon \operatorname{Hom}(E_1, E_2) \to \mathbb{Z}$ *is a positive definite quadratic form:* $(\varphi, \psi) \mapsto \deg(\varphi + \psi) - \deg\varphi - \deg\psi$ *is bilinear, and* $\deg\varphi > 0$ *for* $\varphi \neq 0$.

PROOF. Viewing integers in $\operatorname{End}(E_1)$ via $m \mapsto [m]$, the pairing is $\hat{\varphi}\psi + \hat{\psi}\varphi$, which is bilinear by part 4 of Theorem 4.3.10 [Sil09, Corollary III.6.3]. □

In characteristic p, the dual of the Frobenius morphism has a name of its own.

DEFINITION 4.3.12. Suppose $\operatorname{char} k = p > 0$, and let E be an elliptic curve over k. For $r \geqslant 1$, write

$$\mathsf{F}^r\colon E \to E^{(p^r)}$$

for the p^r-th power relative Frobenius morphism (Definition 3.2.11 with $q = p^r$), and

$$\mathsf{V}^r := \widehat{\mathsf{F}^r}\colon E^{(p^r)} \to E$$

for its dual isogeny, the Verschiebung.

The exponent is not an abuse: F^r is the composite $E \to E^{(p)} \to \cdots \to E^{(p^r)}$ of p-th power Frobenius morphisms, one for each of the curves $E^{(p^i)}$. By Example 4.3.3 it is purely inseparable of degree p^r, so Theorem 4.3.10 gives

$$\mathsf{V}^r \circ \mathsf{F}^r = [p^r] \text{ on } E, \qquad \mathsf{F}^r \circ \mathsf{V}^r = [p^r] \text{ on } E^{(p^r)}, \qquad \deg \mathsf{V}^r = p^r,$$

by parts 1, 2 and 6. If E is defined over $\mathbb{F}_{p^r}$, then $E^{(p^r)} = E$ and both F^r and V^r lie in $\operatorname{End}(E)$. The name is the German word for "shift".

EXAMPLE 4.3.13. Suppose $\operatorname{char} k \neq 2$. Let $a, b \in k$ with $b(a^2 - 4b) \neq 0$, and

$$E_1\colon y_1^2 = x_1^3 + ax_1^2 + bx_1, \qquad E_2\colon y_2^2 = x_2^3 - 2ax_2^2 + (a^2 - 4b)x_2,$$

with x_i, y_i the affine coordinates of E_i. Both cubics have distinct roots, so both curves are smooth, and $T := (0,0)$ is a point of order 2 on E_1. Let

$$\varphi\colon E_1 \to E_2, \qquad \varphi(x_1, y_1) := \Big(\frac{y_1^2}{x_1^2},\ \frac{y_1\,(x_1^2 - b)}{x_1^2}\Big)$$

[Sil09, Example III.4.5].

(1) **Kernel and degree.** The function $\varphi^* x_2 = y_1^2/x_1^2 = x_1 + a + b/x_1$ has poles only at T and O_1, so $\ker\varphi = \{O_1, T\}$. Since $y_1 = x_1^2\,\varphi^* y_2/(x_1^2 - b)$ and $x_1^2 + (a - \varphi^* x_2)x_1 + b = 0$, the field $k(E_1)$ has degree at most 2 over $\varphi^* k(E_2)$. By Theorem 4.3.8, $\deg\varphi \geqslant \#\ker\varphi$, so $\deg\varphi = 2$, and φ is separable.

(2) **The dual.** The same construction applied to E_2, whose coefficients are $-2a$ and $a^2 - 4b$, gives an isogeny $\psi\colon E_2 \to E_3$ with $E_3\colon y_3^2 = x_3^3 + 4ax_3^2 + 16bx_3$, and $\iota(x_3, y_3) := (x_3/4, y_3/8)$ is an isomorphism $E_3 \to E_1$. A direct computation with Algorithm 4.2.7 shows that $\iota \circ \psi \circ \varphi = [2]$, so $\hat{\varphi} = \iota \circ \psi$. Its kernel is $\{O_2, (0,0)\}$, which is $\varphi(E_1[2](\overline{k}))$: the other two points $(e, 0)$ of $E_1[2](\overline{k})$ have $e^2 + ae + b = 0$, and $\varphi(e, 0) = (0, 0)$.

(3) **Example.** For $y_1^2 = x_1^3 - x_1$, that is $a = 0$ and $b = -1$, we get $E_2\colon y_2^2 = x_2^3 + 4x_2$ and $\varphi(x_1, y_1) = \big((x_1^2 - 1)/x_1,\ y_1(x_1^2 + 1)/x_1^2\big)$.

The invariant differential detects the sign in part 2 without computing [2] (Example 4.4.7).

Remark 4.3.14. Compare the multiplicative group. A morphism $(\mathbb{G}_{\mathrm{m}})_{\overline{k}} \to (\mathbb{G}_{\mathrm{m}})_{\overline{k}}$ is $t \mapsto ct^m$, since the units of $\overline{k}[t, t^{-1}]$ are the ct^m, and it is a homomorphism only if $c = 1$. So the endomorphisms of $\mathbb{G}_{\mathrm{m}}$ are the maps $[m]$, and $\deg[m] = |m|$ (Example 3.4.8). There is no dual in the sense of Theorem 4.3.10: for $m \geqslant 1$, the only $[n]$ with $[n] \circ [m] = [\deg[m]]$ is [1], and the rule $[m] \mapsto [1]$ is not additive. The identity $\deg[m] = m^2$ of part 5 is what makes the dual of an elliptic curve possible. Over $\mathbb{C}$, it reflects the fact that $\mathbb{C}/\Lambda$ is the quotient of $\mathbb{C}$ by a lattice of rank 2, whereas $\mathbb{C}^\times \cong \mathbb{C}/2\pi i\mathbb{Z}$ is the quotient by a subgroup of rank 1 (Example 1.5.4).

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4.4. The invariant differential

On $\mathbb{C}/\Lambda$, the differential dz is invariant under translation, and an isogeny $z \mapsto \alpha z$ pulls it back to $\alpha\, dz$, so pullback of differentials is additive (Example 1.5.5). By Remark 4.1.8 the invariant differential ω is dz, and all of this survives over k. The consequences in positive characteristic are striking, because pullback detects separability (Proposition 3.4.2).

Throughout, ω is the invariant differential of a Weierstrass model of E. By Proposition 4.1.5, $\operatorname{div}(\omega) = 0$. Any non-zero differential with divisor 0 is $a\omega$ with $a \in k^\times$, by Proposition 3.1.7, so up to a scalar ω does not depend on the model.

Proposition 4.4.1. *For every $Q \in E(k)$, $\tau_Q^*\omega = \omega$.*

Proof. Since τ_Q is an isomorphism, $\operatorname{div}(\tau_Q^*\omega) = \tau_Q^* \operatorname{div}(\omega) = 0$, so $\tau_Q^*\omega = a_Q\omega$ with $a_Q \in k^\times$. After base extension to $\overline{k}$, the map $E_{\overline{k}} \to \mathbb{P}^1_{\overline{k}}$, $Q \mapsto a_Q$, is a morphism that omits 0 and ∞, so it is constant by Proposition 3.2.2, equal to $a_O = 1$ [Sil09, Proposition III.5.1]. □

Theorem 4.4.2. *Let $\varphi, \psi\colon E' \to E$ be isogenies. Then $(\varphi + \psi)^*\omega = \varphi^*\omega + \psi^*\omega$.*

Proof. See [Sil09, Theorem III.5.2]. The idea: on $E \times E$, the differential $\mathrm{sum}^*\omega$ is a combination of the pullbacks of ω along the two projections, and translation invariance forces both coefficients to be 1. □

The two plus signs have different meanings: the first is addition in $\operatorname{Hom}(E', E)$, the second is addition of differentials.

Corollary 4.4.3. *For every $m \in \mathbb{Z}$, $[m]^*\omega = m\omega$. So if $\operatorname{char} k \nmid m$, then $[m]$ is non-zero and separable.*

Proof. Induction on m, starting from $[0]^*\omega = 0$ and $[1]^*\omega = \omega$, using Theorem 4.4.2. If $m \neq 0$ in k, then $[m]^*\omega \neq 0$, so $[m] \neq 0$, and $[m]$ is separable by Proposition 3.4.2 [Sil09, Corollaries III.5.3 and III.5.4]. □

Corollary 4.4.4. *Let E be an elliptic curve over $\mathbb{F}_q$, of characteristic p, and $\mathsf{F} \in \operatorname{End}(E)$ its q-th power Frobenius endomorphism (Example 3.2.15). For $m, n \in \mathbb{Z}$, the isogeny $m + n\mathsf{F}$ is separable if and only if $p \nmid m$. In particular $1 - \mathsf{F}$ is separable.*

Proof. F is inseparable, so $\mathsf{F}^*\omega = 0$ by Proposition 3.4.2. By Theorem 4.4.2 and Corollary 4.4.3, $(m + n\mathsf{F})^*\omega = m\omega$, which vanishes if and only if $p \mid m$ [Sil09, Corollary III.5.5]. □

Since $E(\mathbb{F}_q)$ is the set of fixed points of F on $E(\overline{k})$, it is the kernel of $1 - \mathsf{F}$. By Theorem 4.3.8 and Corollary 4.4.4,

$$\#E(\mathbb{F}_q) = \deg(1 - \mathsf{F}).$$

Every isogeny acts on invariant differentials by a scalar.

Proposition 4.4.5. *Let ω_1, ω_2 and ω_3 be invariant differentials of Weierstrass models of elliptic curves E_1, E_2 and E_3.*

(1) For every isogeny $\varphi\colon E_1 \to E_2$ there is a unique $c_\varphi \in k$ with $\varphi^*\omega_2 = c_\varphi\,\omega_1$.
(2) $c_{\lambda\circ\varphi} = c_\lambda c_\varphi$ for $\lambda\colon E_2 \to E_3$, and $c_{\varphi+\psi} = c_\varphi + c_\psi$ for $\psi \in \operatorname{Hom}(E_1, E_2)$.
(3) Replacing ω_1 by $u\omega_1$, with $u \in k^\times$, replaces c_φ by c_φ/u.
(4) φ is separable if and only if $c_\varphi \neq 0$.
(5) $c_{\hat\varphi}\,c_\varphi = \deg\varphi$, where $c_{\hat\varphi}$ is computed with ω_2 and ω_1.

PROOF. For part 1 we may assume $\varphi \neq 0$. Over $\overline{k}$, for $P \in E_1(\overline{k})$, we have $\varphi\circ\tau_P = \tau_{\varphi(P)}\circ\varphi$ by Theorem 4.3.6, so by Proposition 4.4.1

$$\tau_P^*\varphi^*\omega_2 = \varphi^*\tau_{\varphi(P)}^*\omega_2 = \varphi^*\omega_2.$$

Write $\varphi^*\omega_2 = g\omega_1$ with $g \in k(E_1)$. Since ω_1 is invariant too, $g\circ\tau_P = g$ for every P. If $g \neq 0$ had a pole, it would have a pole at every point of $E_1(\overline{k})$. So g is constant, and $g \in k$ by Proposition 3.1.7. Part 2 follows from $(\lambda\circ\varphi)^* = \varphi^*\circ\lambda^*$ and Theorem 4.4.2, part 3 is immediate, and part 4 is Proposition 3.4.2. For part 5, by parts 2 and 3 and Corollary 4.4.3, $c_{\hat\varphi}c_\varphi = c_{\hat\varphi\circ\varphi} = c_{[\deg\varphi]} = \deg\varphi$ [Sil09, Section III.5]. □

COROLLARY 4.4.6. *For $\varphi \in \operatorname{End}(E)$, compute c_φ with $\omega_1 = \omega_2 = \omega$, so that $\varphi^*\omega = c_\varphi\,\omega$. Then $\varphi \mapsto c_\varphi$ is a ring homomorphism $\operatorname{End}(E) \to k$ whose kernel consists of 0 and the inseparable endomorphisms. If $\operatorname{char} k = 0$, then $\operatorname{End}(E_{\overline{k}})$ embeds in $\overline{k}$, and so it is commutative.*

PROOF. It is a ring homomorphism by part 2 of Proposition 4.4.5, with the stated kernel by part 4. In characteristic 0 every non-zero endomorphism is separable. Apply this to $E_{\overline{k}}$ [Sil09, Corollary III.5.6]. □

Over $\mathbb{C}$, the homomorphism $\varphi \mapsto c_\varphi$ sends $z \mapsto \alpha z$ to α: it identifies $\operatorname{End}(\mathbb{C}/\Lambda)$ with $\{\alpha \in \mathbb{C} : \alpha\Lambda \subseteq \Lambda\}$, as in Corollary 1.3.2.

EXAMPLE 4.4.7. Take the isogenies of Example 4.3.13, with $\omega_i := dx_i/(2y_i)$ on E_i. Since $\varphi^*y_2 = y_1(x_1^2-b)/x_1^2$ and $\varphi^*x_2 = x_1 + a + b/x_1$, we have $\varphi^*dx_2 = (x_1^2 - b)\,dx_1/x_1^2$, so

$$\varphi^*\omega_2 = \frac{(x_1^2 - b)\,dx_1/x_1^2}{2y_1\,(x_1^2 - b)/x_1^2} = \omega_1, \qquad c_\varphi = 1.$$

The same computation gives $c_\psi = 1$, and $\iota^*\omega_1 = (dx_3/4)/(2y_3/8) = 2\,\omega_3$, so $c_\iota = 2$. Hence $c_{\hat\varphi} = 2$ and $c_{\hat\varphi}c_\varphi = 2 = \deg\varphi$, as part 5 of Proposition 4.4.5 requires. If we only knew that $\iota\circ\psi\circ\varphi = \pm[2]$, this would decide the sign, since $c_{[-2]} = -2 \neq 2$.

COROLLARY 4.4.8. *Suppose $\operatorname{char} k = p > 0$.*
(1) If $p \mid \deg\varphi$, then φ or $\hat\varphi$ is inseparable.
(2) $[p]$ is inseparable.

PROOF. Part 1 is part 5 of Proposition 4.4.5 with $\deg\varphi = 0$ in k. For part 2, $c_{[p]} = p = 0$. □

For $\mathbb{G}_{\mathrm{m}}$ the same two facts are elementary (Examples 3.4.7 and 3.4.8). In both cases $c_{[m]} = m$ is the derivative of $[m]$ at the identity: of $t \mapsto t^m$ at $t = 1$, and of $z \mapsto mz$ on $\mathbb{C}/\Lambda$. The difference is the degree. In characteristic p, $[p]$ on $\mathbb{G}_{\mathrm{m}}$ is inseparable of degree p, so it has no room for a kernel. On E it is inseparable of degree p^2, and $E[p](\overline{k})$ can have p elements.

WARNING 4.4.9. If k is imperfect, the points of $E[p](\overline{k})$ need not lie in $E(k^{\mathrm{sep}})$. Let $k := \mathbb{F}_5(u)$ and $E\colon y^2 = x^3 + 3x + u$, with $j = 2/(u^2+4)$. Its 5-division polynomial is $(x^5 - c_1)(x^5 - c_2)$, with $c_1 = 4u^2 + 2u + 1$ and $c_2 = u^2 + 2u + 4$. So the four non-zero points of $E[5](\overline{k})$ have x-coordinates $c_1^{1/5}$ and $c_2^{1/5}$, which are purely inseparable over k and not in k. Hence $E[5](\overline{k}) \cong \mathbb{Z}/5\mathbb{Z}$, but $E[5](\overline{k}) \cap E(k^{\mathrm{sep}}) = \{O\}$. This does not contradict the discussion after Theorem 4.3.8, since [5] is inseparable.

4.5. The Tate module

Corollary 4.5.1. *Let E be an elliptic curve over k, and $N \geqslant 1$.*
(1) If $\operatorname{char} k \nmid N$, then $E[N](\overline{k}) \subseteq E(k^{\mathrm{sep}})$, and $E[N](\overline{k}) \cong (\mathbb{Z}/N\mathbb{Z})^2$.
(2) If $\operatorname{char} k = p > 0$, then either $E[p^e](\overline{k}) = 0$ for all $e \geqslant 1$, or $E[p^e](\overline{k}) \cong \mathbb{Z}/p^e\mathbb{Z}$ for all $e \geqslant 1$.
Here $E[N](\overline{k})$ is the kernel of $[N]$ on $E(\overline{k})$.

Proof.

(1) By Corollary 4.4.3, $[N]$ is separable, so $\#E[N](\overline{k}) = \deg[N] = N^2$ by Theorems 4.3.8 and 4.3.10. The same holds for every divisor of N. A finite abelian group of order N^2 with d^2 elements killed by d, for every $d \mid N$, is $(\mathbb{Z}/N\mathbb{Z})^2$. The points of $E[N](\overline{k})$ lie in $E(k^{\mathrm{sep}})$ by the discussion after Theorem 4.3.8.
(2) Write $[p] = \mathsf{V} \circ \mathsf{F}$, with F the p-th power relative Frobenius and V its Verschiebung (Definition 4.3.12). Here F is purely inseparable of degree p. Then $\#E[p^e](\overline{k}) = \deg_s[p^e] = (\deg_s \mathsf{V})^e$, and $\deg_s \mathsf{V} \in \{1, p\}$ [Sil09, Corollary III.6.4].

□

Compare with Example 1.3.4. In case 2, E is ordinary if $E[p](\overline{k}) \cong \mathbb{Z}/p\mathbb{Z}$, and supersingular if $E[p](\overline{k}) = 0$ (Theorem 5.4.1). By part 1, Gal_k acts on $E[N](\overline{k}) \cong (\mathbb{Z}/N\mathbb{Z})^2$ whenever $\operatorname{char} k \nmid N$, giving the mod N Galois representation

$$\bar{\rho}_{E,N} \colon \mathrm{Gal}_k \to \mathrm{Aut}(E[N](\overline{k})) \cong \mathrm{GL}_2(\mathbb{Z}/N\mathbb{Z}),$$

the second isomorphism depending on a choice of basis. For instance, for $y^2 = (x-e_1)(x-e_2)(x-e_3)$ as in Example 4.2.9, $E[2](\overline{k})$ consists of rational points and $\bar{\rho}_{E,2}$ is trivial. If $y^2 = p(x)$ with p irreducible, then Gal_k permutes the three roots of p, and $\bar{\rho}_{E,2}$ has image containing an element of order 3 in $\mathrm{GL}_2(\mathbb{F}_2) \cong S_3$.

For $\mathbb{G}_{\mathrm{m}}$ the analogue of Corollary 4.5.1 is elementary (Example 3.4.8). If $\operatorname{char} k \nmid N$, then $\mu_N(\overline{k}) \subseteq k^{\mathrm{sep}}$ is a subgroup of order N of $(k^{\mathrm{sep}})^\times$, hence cyclic: $\mu_N(\overline{k}) \cong \mathbb{Z}/N\mathbb{Z}$, and an isomorphism is the choice of a primitive N-th root of unity. If $\operatorname{char} k = p$, then $\mu_{p^e}(\overline{k}) = \{1\}$. Torsion of rank 1 for $\mathbb{G}_{\mathrm{m}}$, of rank 2 for E, matches $\deg[N] = N$ versus N^2.

Example 4.5.2. Let $k := \mathbb{F}_7$. Since $\mathbb{F}_{7^m}^\times$ is cyclic of order $7^m - 1$, we have $\mu_{3^n}(\overline{k}) \subseteq \mathbb{F}_{7^m}$ if and only if $3^n \mid 7^m - 1$. By the lifting the exponent lemma, $v_3(7^m - 1) = v_3(7-1) + v_3(m) = 1 + v_3(m)$. So $\mu_3(\overline{k}) \subseteq \mathbb{F}_7$, and the field generated by $\mu_{3^n}(\overline{k})$ is $\mathbb{F}_{7^{3^{n-1}}}$. Equivalently, the Frobenius $\zeta \mapsto \zeta^7$ generates the action of $\mathrm{Gal}_{\mathbb{F}_7}$ on $\mu_{3^n}(\overline{k}) \cong \mathbb{Z}/3^n\mathbb{Z}$, and 7 has order 3^{n-1} in $(\mathbb{Z}/3^n\mathbb{Z})^\times$.

The groups $E[\ell^n](k^{\mathrm{sep}})$ for varying n fit together.

Situation 4.5.3. In this section and the next, ℓ is a prime different from $\operatorname{char} k$.

Definition 4.5.4. The (ℓ-adic) Tate module of E is the inverse limit

$$T_\ell(E) := \varprojlim_n E[\ell^n](k^{\mathrm{sep}}),$$

taken with respect to the maps $[\ell] \colon E[\ell^{n+1}](k^{\mathrm{sep}}) \to E[\ell^n](k^{\mathrm{sep}})$. The ℓ-adic representation of E is the continuous homomorphism $\rho_{E,\ell} \colon \mathrm{Gal}_k \to \mathrm{Aut}(T_\ell(E))$ obtained from the action of Gal_k on each $E[\ell^n](k^{\mathrm{sep}})$.

Likewise, $T_\ell(\mathbb{G}_{\mathrm{m}}) := \varprojlim_n \mu_{\ell^n}(k^{\mathrm{sep}})$, with the maps $\zeta \mapsto \zeta^\ell$, is also written $\mathbb{Z}_\ell(1)$. Choosing primitive ℓ^n-th roots of unity ζ_{ℓ^n} with $\zeta_{\ell^{n+1}}^\ell = \zeta_{\ell^n}$, which is possible because $\zeta \mapsto \zeta^\ell$ maps $\mu_{\ell^{n+1}}(k^{\mathrm{sep}})$ onto $\mu_{\ell^n}(k^{\mathrm{sep}})$, gives an isomorphism $\mathbb{Z}_\ell \to T_\ell(\mathbb{G}_{\mathrm{m}})$, $(a_n)_n \mapsto (\zeta_{\ell^n}^{a_n})_n$. Over $\mathbb{C}$ one can take $\zeta_{\ell^n} = e^{2\pi i/\ell^n}$. The group Gal_k acts on $T_\ell(\mathbb{G}_{\mathrm{m}})$ through the ℓ-adic cyclotomic character $\chi_\ell \colon \mathrm{Gal}_k \to \mathbb{Z}_\ell^\times$, defined by $\sigma(\zeta) = \zeta^{\chi_\ell(\sigma)}$. In characteristic p, $T_p(\mathbb{G}_{\mathrm{m}}) = 0$.

By Corollary 4.5.1, $T_\ell(E) \cong \mathbb{Z}_\ell^2$ as a $\mathbb{Z}_\ell$-module, so $\rho_{E,\ell}$ takes values in $\mathrm{GL}_2(\mathbb{Z}_\ell)$ after choosing a basis. And $T_p(E)$ is 0 or $\mathbb{Z}_p$ if $p = \operatorname{char} k > 0$ [Sil09, Proposition III.7.1]. Concretely, $[\ell] \colon E[\ell^{n+1}](k^{\mathrm{sep}}) \to E[\ell^n](k^{\mathrm{sep}})$ is surjective, because $[\ell]$ is surjective on $E(\overline{k})$. Choose a basis (P_1, Q_1) of $E[\ell](k^{\mathrm{sep}})$, and inductively P_{n+1}, Q_{n+1} with $[\ell]P_{n+1} = P_n$ and $[\ell]Q_{n+1} = Q_n$. If $[a]P_n + [b]Q_n = O$, then applying $[\ell^{n-1}]$ and induction show that $\ell^n \mid a$ and $\ell^n \mid b$, so (P_n, Q_n) is a basis of $E[\ell^n](k^{\mathrm{sep}}) \cong (\mathbb{Z}/\ell^n\mathbb{Z})^2$,

and $(a_n, b_n)_n \mapsto ([a_n]P_n + [b_n]Q_n)_n$ is an isomorphism $\mathbb{Z}_\ell^2 \to T_\ell(E)$. Over $\mathbb{C}$ this is familiar: since $E[\ell^n](\mathbb{C}) = \ell^{-n}\Lambda/\Lambda \cong \Lambda/\ell^n\Lambda$ compatibly in n,

$$T_\ell(\mathbb{C}/\Lambda) \cong \Lambda \otimes_{\mathbb{Z}} \mathbb{Z}_\ell, \qquad T_\ell(\mathbb{C}^\times) \cong 2\pi i\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}_\ell,$$

and Λ and $2\pi i\mathbb{Z}$ are the first homology groups of $\mathbb{C}/\Lambda$ and $\mathbb{C}^\times$ (Examples 1.5.4 and 1.5.5). So the Tate module is an algebraic substitute for H_1, and unlike H_1 it carries an action of Gal_k.

EXAMPLE 4.5.5. Let $E\colon y^2 = x^3 - x$ over $\mathbb{Q}$, and $\ell = 2$. Then $E[2](\overline{\mathbb{Q}}) = \{O, (0,0), (1,0), (-1,0)\}$. By Algorithm 4.2.7, $[2]^*x := x \circ [2] = (x^2+1)^2/\big(4(x^3-x)\big)$, and

$$[2]^*x - e = \frac{g_e(x)^2}{4(x^3-x)}, \qquad g_0 = x^2+1, \quad g_1 = x^2 - 2x - 1, \quad g_{-1} = x^2 + 2x - 1.$$

So the points of $E(\overline{\mathbb{Q}})$ halving $(0,0)$ have $x = \pm i$, those halving $(1,0)$ have $x = 1 \pm \sqrt{2}$, and those halving $(-1,0)$ have $x = -1 \pm \sqrt{2}$. For instance $P_2 := (i, 1-i)$ and $Q_2 := (1+\sqrt{2}, 2+\sqrt{2})$ lie on E, because $i^3 - i = (1-i)^2$ and $x^3 - x = 2x^2 = (2+\sqrt{2})^2$ for $x = 1+\sqrt{2}$. They satisfy $[2]P_2 = (0,0)$ and $[2]Q_2 = (1,0)$, so, with $P_1 := (0,0)$ and $Q_1 := (1,0)$, (P_2, Q_2) is a basis of $E[4](\overline{\mathbb{Q}})$ as above. Hence $E[4](\overline{\mathbb{Q}}) \subseteq E(\mathbb{Q}(i, \sqrt{2}))$, and $\mathbb{Q}(i, \sqrt{2}) = \mathbb{Q}(\zeta_8)$ is the field generated by $\mu_8(\overline{\mathbb{Q}})$, which it must contain by Corollary 4.6.5. Complex conjugation σ fixes Q_2 and sends P_2 to $(-i, 1+i) = [3]P_2 + [2]Q_2$, so in this basis

$$\bar\rho_{E,4}(\sigma) = \begin{pmatrix} 3 & 0 \\ 2 & 1 \end{pmatrix}, \qquad \det \bar\rho_{E,4}(\sigma) = 3 \equiv \chi_2(\sigma) \pmod 4,$$

since $\sigma(i) = i^3$. Continuing to halve gives compatible sequences $(P_1, P_2, P_3, \dots)$ and $(Q_1, Q_2, Q_3, \dots)$, a $\mathbb{Z}_2$-basis of $T_2(E)$, just as $(-1, i, \zeta_8, \dots)$ is a $\mathbb{Z}_2$-basis of $T_2(\mathbb{G}_{\mathrm{m}})$.

An isogeny $\varphi\colon E_1 \to E_2$ restricts to maps $E_1[\ell^n](k^{\mathrm{sep}}) \to E_2[\ell^n](k^{\mathrm{sep}})$ compatible with $[\ell]$, and so induces a $\mathbb{Z}_\ell$-linear map $\varphi_\ell\colon T_\ell(E_1) \to T_\ell(E_2)$. Since φ is a morphism of k-varieties, φ_ℓ commutes with Gal_k. The construction respects composition and sums, and $[m]_\ell = m$, so Theorem 4.3.10 gives $\hat\varphi_\ell \circ \varphi_\ell = \deg\varphi$. Since multiplication by $\deg\varphi \neq 0$ is injective on the torsion-free module $T_\ell(E_1)$, φ_ℓ is injective if $\varphi \neq 0$, and an isomorphism if $\ell \nmid \deg\varphi$. For $\mathbb{G}_{\mathrm{m}}$, $[m]_\ell$ is multiplication by m on $\mathbb{Z}_\ell(1)$. Note that $c_{\hat\varphi}c_\varphi = \deg\varphi$ and $\hat\varphi_\ell \circ \varphi_\ell = \deg\varphi$ both come from $\hat\varphi \circ \varphi = [\deg\varphi]$: pullback of differentials reverses the order of composition and $\varphi \mapsto \varphi_\ell$ does not, but scalars commute. In characteristic p, the scalar $c_{[p]} = 0$ detects that $[p]$ is inseparable, while T_p is too small to see it.

THEOREM 4.5.6. *The map* $\mathrm{Hom}(E_1, E_2) \otimes \mathbb{Z}_\ell \to \mathrm{Hom}_{\mathbb{Z}_\ell}(T_\ell(E_1), T_\ell(E_2))$, $\varphi \mapsto \varphi_\ell$, *is injective. Consequently* $\mathrm{Hom}(E_1, E_2)$ *is a free* $\mathbb{Z}$*-module of rank at most* 4.

PROOF. See [Sil09, Theorem III.7.4 and Corollary III.7.5]. The key input is that $\ker[\ell^n] \subseteq \ker\varphi$ forces φ to factor through $[\ell^n]$ (Proposition 4.3.9). The rank bound follows because the target is free of rank 4 over $\mathbb{Z}_\ell$, and $\mathrm{Hom}(E_1, E_2)$ is torsion-free (Proposition 4.3.7). □

For number fields and finite fields the image of $\mathrm{Hom}(E_1, E_2) \otimes \mathbb{Z}_\ell$ is exactly the Gal_k-equivariant homomorphisms. This is the Tate conjecture, a theorem of Tate over finite fields and of Faltings over number fields [Sil09, Section III.7].

We summarize the comparison between the multiplicative group and an elliptic curve. Here $\ell \neq \mathrm{char}\, k$, $p = \mathrm{char}\, k > 0$ in the row about $[p^e]$, and kernels are taken on $\overline{k}$-points.

	$\mathbb{G}_{\mathrm{m}}$	E
points over $\overline{k}$	$\overline{k}^\times$	$E(\overline{k})$
over $\mathbb{C}$	$\mathbb{C}/2\pi i\mathbb{Z}$	$\mathbb{C}/\Lambda$
invariant differential	dt/t, poles at 0 and ∞	$dx/(2y)$, no zeros or poles
$[m]^*\omega$	$m\omega$	$m\omega$
$\deg[m]$	$\lvert m\rvert$	m^2
$\ker[\ell^n]$	$\mathbb{Z}/\ell^n\mathbb{Z}$	$(\mathbb{Z}/\ell^n\mathbb{Z})^2$
$\ker[p^e]$	0	0 or $\mathbb{Z}/p^e\mathbb{Z}$
T_ℓ	$\mathbb{Z}_\ell(1) \cong \mathbb{Z}_\ell$	$\mathbb{Z}_\ell^2$
T_ℓ over $\mathbb{C}$	$2\pi i\mathbb{Z} \otimes \mathbb{Z}_\ell$	$\Lambda \otimes \mathbb{Z}_\ell$

4.6. The Weil pairing

In Example 1.3.4 we compared $(\mathbb{C}/\Lambda)[N]$ with the group $\mu_N(\mathbb{C})$ of roots of unity. There is a deeper link: if $\Lambda = \mathbb{Z}\omega_1 \oplus \mathbb{Z}\omega_2$ with $\omega_1/\omega_2 \in \mathbb{H}$, the pairing

$$\left(\frac{a\omega_1 + b\omega_2}{N}, \frac{c\omega_1 + d\omega_2}{N}\right) \mapsto e^{2\pi i(ad-bc)/N}$$

is a well defined, bilinear, alternating and non-degenerate map $(\mathbb{C}/\Lambda)[N] \times (\mathbb{C}/\Lambda)[N] \to \mu_N(\mathbb{C})$. Over an arbitrary field there is no $e^{2\pi it}$, but the pairing can be built from functions. Its values lie in $\mu_N(k^{\mathrm{sep}})$, where $\mu_N = \mathbb{G}_{\mathrm{m}}[N]$, so the Weil pairing relates the torsion of E to the torsion of the multiplicative group.

Fix $N \geqslant 2$ with $\operatorname{char} k \nmid N$, so that $E[N](\overline{k}) = E[N](k^{\mathrm{sep}}) \cong (\mathbb{Z}/N\mathbb{Z})^2$ by Corollary 4.5.1. Let $S, T \in E[N](k^{\mathrm{sep}})$. Choose divisors

$$D_S \sim (S) - (O), \qquad D_T \sim (T) - (O)$$

in $\operatorname{Div} E_{k^{\mathrm{sep}}}$ with disjoint supports. This is possible because D_T may be replaced by $(T+R)-(R)$ for any $R \in E(k^{\mathrm{sep}})$ outside the support of D_S, and $E(k^{\mathrm{sep}})$ is infinite. The divisor ND_S is linearly equivalent to $N(S) - N(O)$, which is principal by Corollary 4.2.3, since $[N]S = O$. So there are $f_S, f_T \in k^{\mathrm{sep}}(E)^\times$ with

$$\operatorname{div}(f_S) = ND_S, \qquad \operatorname{div}(f_T) = ND_T,$$

and the supports of $\operatorname{div}(f_S)$ and D_T are disjoint, as are those of $\operatorname{div}(f_T)$ and D_S. Both values $f_S(D_T)$ and $f_T(D_S)$ are therefore defined in the sense of Definition 3.3.3.

Definition 4.6.1. The Weil pairing is

$$e_N \colon E[N](k^{\mathrm{sep}}) \times E[N](k^{\mathrm{sep}}) \to \mu_N(k^{\mathrm{sep}}), \qquad e_N(S,T) \coloneqq \frac{f_S(D_T)}{f_T(D_S)},$$

for any choice of D_S, D_T, f_S and f_T as above.

Lemma 4.6.2. *The quantity $e_N(S,T)$ lies in $\mu_N(k^{\mathrm{sep}})$ and does not depend on the choices of D_S, D_T, f_S and f_T.*

Proof. Everything follows from Weil reciprocity (Theorem 3.3.4), applied over $\overline{k}$ to functions that happen to be defined over k^{sep}.

The function f_S is determined by D_S up to a constant $c \in k^{\mathrm{sep}\times}$ by Proposition 3.3.2, and replacing f_S by cf_S multiplies $f_S(D_T)$ by $c^{\deg D_T} = c^0 = 1$. So $e_N(S,T)$ depends only on D_S and D_T.

Now replace D_S by $D_S + \operatorname{div}(h)$, which is the general divisor linearly equivalent to D_S. Moving D_T if necessary, we may assume the supports stay disjoint. Then f_S is replaced by $f_S h^N$, and $e_N(S,T)$ is multiplied by

$$\frac{h(D_T)^N}{f_T(\operatorname{div} h)} = \frac{h(ND_T)}{f_T(\operatorname{div} h)} = \frac{h(\operatorname{div} f_T)}{f_T(\operatorname{div} h)} = 1,$$

the last equality by reciprocity. The same argument applies to D_T.

Finally, $\deg D_S = \deg D_T = 0$ gives

$$e_N(S,T)^N = \frac{f_S(D_T)^N}{f_T(D_S)^N} = \frac{f_S(ND_T)}{f_T(ND_S)} = \frac{f_S(\operatorname{div} f_T)}{f_T(\operatorname{div} f_S)} = 1,$$

again by reciprocity. So $e_N(S,T) \in \mu_N(\overline{k})$, and it lies in $\mu_N(k^{\mathrm{sep}})$ because $\operatorname{char} k \nmid N$. □

The values are N-th roots of unity for the same reason that the pairing is well defined, and neither fact needs $e^{2\pi i t}$.

PROPOSITION 4.6.3. *The Weil pairing is*

(1) bilinear;
(2) alternating: $e_N(T,T) = 1$, so $e_N(S,T) = e_N(T,S)^{-1}$;
(3) non-degenerate: if $e_N(S,T) = 1$ for all $S \in E[N](k^{\mathrm{sep}})$, then $T = O$;
(4) Galois equivariant: $\sigma(e_N(S,T)) = e_N(\sigma(S),\sigma(T))$ for $\sigma \in \mathrm{Gal}_k$;
(5) compatible: $e_{NN'}(S,T) = e_N([N']S,T)$ for $S \in E[NN'](k^{\mathrm{sep}})$ and $T \in E[N](k^{\mathrm{sep}})$;
(6) adjoint for isogenies: $e_N(\varphi S,T) = e_N(S,\hat{\varphi}T)$ for $\varphi\colon E_1 \to E_2$, $S \in E_1[N](k^{\mathrm{sep}})$ and $T \in E_2[N](k^{\mathrm{sep}})$.

PROOF. Parts 1 and 4 come straight from the definition. For additivity in the first variable, take D_{S_1}, D_{S_2} and $D_{S_1+S_2}$ supported away from D_T. Their difference is linearly equivalent to $(S_1)+(S_2)-(S_1+S_2)-(O)$, a divisor of degree 0 that sums to O in $E(k^{\mathrm{sep}})$, hence is $\operatorname{div}(h)$ for some h by Corollary 4.2.3. So $f_{S_1}f_{S_2}/f_{S_1+S_2}$ and h^N have the same divisor, and reciprocity kills the resulting factor exactly as in Lemma 4.6.2. Part 4 is the Galois equivariance of evaluation noted after Definition 3.3.3: applying σ carries a choice of D_S, D_T, f_S, f_T for the pair (S,T) to one for $(\sigma(S),\sigma(T))$. Swapping S and T inverts the defining ratio, which gives the second half of part 2. The first half, $e_N(T,T) = 1$, is stronger in characteristic 2 and is proved from the construction of Remark 4.6.4. For that, and for parts 3, 5 and 6, see [Sil09, Propositions III.8.1 and III.8.2]. □

REMARK 4.6.4. Silverman [Sil09] defines the pairing differently, by pulling back along $[N]$. Given $T \in E[N](k^{\mathrm{sep}})$, choose f with $\operatorname{div}(f) = N(T) - N(O)$, and choose $T' \in E[N^2](k^{\mathrm{sep}})$ with $[N]T' = T$. The divisor $\sum_{R \in E[N](k^{\mathrm{sep}})} \big((T'+R)-(R)\big)$ has degree 0 and sums to $[N^2]T' = O$, so it is $\operatorname{div}(g)$ for some g, and then $f \circ [N]$ and g^N have the same divisor. Rescaling f gives $f \circ [N] = g^N$. For $S \in E[N](k^{\mathrm{sep}})$ and $X \in E(k^{\mathrm{sep}})$ we get $g(X+S)^N = f([N]X) = g(X)^N$, so $g(X+S)/g(X)$ is an N-th root of unity, and as a function of X it is a morphism $E \to \mathbb{P}^1$ that is not surjective, hence constant. That constant is $e_N(S,T)$ [Sil09, Exercise 3.16]. This construction makes the analogy with $\mathbb{G}_{\mathrm{m}}$ visible — the pairing is built from the N-th power map — and it is the form in which part 2 of Proposition 4.6.3 is proved. The definition above is the one to compute with: evaluating f_S at D_T is what Miller's algorithm does.

COROLLARY 4.6.5. *The Weil pairing is surjective onto $\mu_N(k^{\mathrm{sep}})$. In particular, if $E[N](k^{\mathrm{sep}}) \subseteq E(k)$, then $\mu_N(k^{\mathrm{sep}}) \subseteq k$.*

PROOF. The image is a subgroup $\mu_d(k^{\mathrm{sep}})$. Then $e_N([d]S,T) = e_N(S,T)^d = 1$ for all S and T, so $[d]S = O$ for all $S \in E[N](k^{\mathrm{sep}})$ by non-degeneracy, and $d = N$ by Corollary 4.5.1. If $E[N](k^{\mathrm{sep}}) \subseteq E(k)$, then Gal_k fixes every value $e_N(S,T)$ by Galois equivariance [Sil09, Corollary III.8.1.1]. □

EXAMPLE 4.6.6. If E is an elliptic curve over $\mathbb{Q}$ and $N \geqslant 3$, then $E[N](\overline{\mathbb{Q}}) \not\subseteq E(\mathbb{Q})$, since $\mu_N(\overline{\mathbb{Q}}) \not\subseteq \mathbb{Q}$. So $E(\mathbb{Q})_{\mathrm{tors}}$ never contains a copy of $(\mathbb{Z}/N\mathbb{Z})^2$ with $N \geqslant 3$, while $(\mathbb{Z}/2\mathbb{Z})^2$ does occur (Example 4.2.9).

By compatibility, the pairings e_{ℓ^n} fit into a pairing $e\colon T_\ell(E) \times T_\ell(E) \to \mathbb{Z}_\ell(1) = T_\ell(\mathbb{G}_{\mathrm{m}})$, which is bilinear, alternating, non-degenerate and Galois equivariant [Sil09, Proposition III.8.3]. A bilinear alternating form on a free module of rank 2 transforms under an endomorphism by its determinant. So $e(\varphi_\ell x, \varphi_\ell y) = e(x,y)^{\det \varphi_\ell}$, while by adjointness $e(\varphi_\ell x, \varphi_\ell y) = e(x, \hat{\varphi}_\ell \varphi_\ell y) = e(x,y)^{\deg \varphi}$. Similarly, Galois equivariance says that $\det \rho_{E,\ell}$ is the ℓ-adic cyclotomic character.

PROPOSITION 4.6.7. *Let $\varphi \in \mathrm{End}(E)$. Then*

$$\det \varphi_\ell = \deg \varphi, \qquad \operatorname{tr} \varphi_\ell = 1 + \deg \varphi - \deg(1-\varphi).$$

In particular, the characteristic polynomial of φ_ℓ has integer coefficients independent of ℓ.

PROOF. The determinant was computed above. For the trace, apply $\det(1-M) = 1 - \operatorname{tr} M + \det M$, valid for 2×2 matrices, to $M = \varphi_\ell$ [Sil09, Proposition III.8.6]. □

For the Frobenius endomorphism of an elliptic curve over $\mathbb{F}_q$, this gives the characteristic polynomial of F_ℓ (Theorem 5.3.8), and positivity of the degree (Corollary 4.3.11) gives Hasse's bound (Theorem 5.2.1).

4.7. Endomorphisms and automorphisms

For complex tori, $\operatorname{End}(\mathbb{C}/\Lambda) = \{\alpha \in \mathbb{C} : \alpha\Lambda \subseteq \Lambda\}$ (Corollary 1.3.2). Such an α satisfies a monic quadratic equation over $\mathbb{Z}$, by the Cayley–Hamilton theorem applied to multiplication by α on $\Lambda \cong \mathbb{Z}^2$. If $\alpha \in \mathbb{R}$, then multiplication by α on $\Lambda \otimes_{\mathbb{Z}} \mathbb{R} = \mathbb{C}$ is the scalar α, so its characteristic polynomial $(T-\alpha)^2$ has integer coefficients, and $\alpha \in \mathbb{Z}$. So if $\alpha \notin \mathbb{Z}$, then $\mathbb{Q}(\alpha)$ is an imaginary quadratic field. Hence $\operatorname{End}(\mathbb{C}/\Lambda)$ is either $\mathbb{Z}$ or an order in an imaginary quadratic field. The Gaussian and Eisenstein lattices of Examples 1.2.2 and 1.2.3 have the extra endomorphisms i and ω. In positive characteristic a third possibility appears, a non-commutative one, and to name it we first collect what we need about quaternion algebras.

4.7.1. Quaternion algebras. We follow [Voi21]. Throughout this subsection, F is a field with $\operatorname{char} F \neq 2$.

DEFINITION 4.7.1. A quaternion algebra over F is an F-algebra B with a basis $1, i, j, ij$ such that

$$i^2 = a, \qquad j^2 = b, \qquad ji = -ij$$

for some $a, b \in F^\times$. These relations determine the multiplication, and for every $a, b \in F^\times$ they define a quaternion algebra, denoted $\left(\frac{a,b}{F}\right)$ or $(a, b \mid F)$ [Voi21, Definition 2.2.1].

EXAMPLE 4.7.2.

(1) Hamilton's quaternions are $\mathbf{H} := (-1, -1 \mid \mathbb{R})$ [Voi21, Example 2.2.3].
(2) The matrix algebra $M_2(F)$ is $(1, 1 \mid F)$, via $i \mapsto \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$ and $j \mapsto \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ [Voi21, Example 2.2.4].
(3) For a field extension $K \supseteq F$, $(a, b \mid F) \otimes_F K = (a, b \mid K)$, with the same basis.

Every $\alpha = t + xi + yj + zij \in B$ has a conjugate $\bar\alpha := t - xi - yj - zij$. The map $\alpha \mapsto \bar\alpha$ is F-linear, fixes 1, and satisfies $\bar{\bar\alpha} = \alpha$ and $\overline{\alpha\beta} = \bar\beta\bar\alpha$. It is the standard involution of B. The reduced trace and the reduced norm are

$$\operatorname{trd}(\alpha) := \alpha + \bar\alpha = 2t, \qquad \operatorname{nrd}(\alpha) := \alpha\bar\alpha = t^2 - ax^2 - by^2 + abz^2,$$

both in F, and every $\alpha \in B$ satisfies

$$\alpha^2 - \operatorname{trd}(\alpha)\,\alpha + \operatorname{nrd}(\alpha) = 0$$

[Voi21, Sections 3.2 and 3.3]. The reduced norm is multiplicative, and α is a unit if and only if $\operatorname{nrd}(\alpha) \neq 0$, with inverse $\bar\alpha/\operatorname{nrd}(\alpha)$. On $M_2(F)$, the reduced trace and norm are the trace and the determinant. More generally, a standard involution on an F-algebra B is an F-linear map $\alpha \mapsto \bar\alpha$ with $\bar 1 = 1$, $\bar{\bar\alpha} = \alpha$, $\overline{\alpha\beta} = \bar\beta\bar\alpha$ and $\alpha\bar\alpha \in F$ for every α [Voi21, Definitions 3.2.1 and 3.2.4]. The field F, and a quadratic extension of F with its conjugation, have one.

THEOREM 4.7.3. *Let B be a division algebra over F with a standard involution. Then B is F, a quadratic field extension of F, or a division quaternion algebra over F.*

PROOF. See [Voi21, Theorem 3.5.1 and Corollary 3.5.4]. The involution makes every $\alpha \in B$ a root of $T^2 - \operatorname{trd}(\alpha)T + \operatorname{nrd}(\alpha) \in F[T]$, where $\operatorname{trd}(\alpha) := \alpha + \bar\alpha$ and $\operatorname{nrd}(\alpha) := \alpha\bar\alpha$. The theorem says that a division algebra all of whose elements are quadratic over F has dimension at most 4, with the three shapes listed. □

DEFINITION 4.7.4. A quaternion algebra B over F is split if $B \cong M_2(F)$, and a field $K \supseteq F$ is a splitting field for B if $B \otimes_F K$ is split [Voi21, Definition 5.4.5]. A quaternion algebra B over $\mathbb{Q}$ is definite if $B \otimes_{\mathbb{Q}} \mathbb{R} \cong \mathbf{H}$, that is, if $\mathbb{R}$ is not a splitting field [Voi21, Section 14.1].

Proposition 4.7.5. *Let $B = (a, b \mid F)$. The following are equivalent: B is split; B is not a division ring; the quadratic form* nrd *on B has a non-trivial zero; b is a norm from $F(\sqrt{a})$. In particular, every quaternion algebra over an algebraically closed field is split, $(a, b \mid \mathbb{R})$ is split if and only if $a > 0$ or $b > 0$, and $(a, b \mid \mathbb{Q})$ is definite if and only if $a < 0$ and $b < 0$.*

Proof. See [Voi21, Main Theorem 5.4.4 and Section 14.1]. □

The endomorphism ring of an elliptic curve is not a $\mathbb{Q}$-algebra but a ring of integers inside one.

Definition 4.7.6. Let B be a finite-dimensional $\mathbb{Q}$-algebra. A lattice in B is a finitely generated $\mathbb{Z}$-submodule $M \subseteq B$ that spans B over $\mathbb{Q}$. It is then free of rank $\dim_{\mathbb{Q}} B$ [Voi21, Definition 9.3.1]. An order of B is a lattice that is also a subring of B, and a maximal order is an order not properly contained in another order [Voi21, Definitions 10.2.1 and 10.4.1].

Every element α of an order $\mathcal{O}$ is integral over $\mathbb{Z}$: the subring $\mathbb{Z}[\alpha] \subseteq \mathcal{O}$ is a finitely generated $\mathbb{Z}$-module, so α is a root of a monic polynomial with integer coefficients. In a quaternion algebra this says that $\mathrm{trd}(\alpha)$ and $\mathrm{nrd}(\alpha)$ are integers for every $\alpha \in \mathcal{O}$ [Voi21, Section 10.3].

Example 4.7.7.

(1) In an imaginary quadratic field K, the orders are the rings $\mathbb{Z} + f\mathcal{O}_K$ with $f \geqslant 1$, and the ring of integers $\mathcal{O}_K$ is the unique maximal order. For instance $\mathbb{Z}[\sqrt{-3}]$ is an order of $\mathbb{Q}(\sqrt{-3})$, properly contained in the maximal order $\mathbb{Z}[\omega]$ of Eisenstein integers (Example 1.2.3).
(2) In $(-1, -1 \mid \mathbb{Q})$, the Lipschitz order $\mathbb{Z} + \mathbb{Z}i + \mathbb{Z}j + \mathbb{Z}ij$ is not maximal: it is properly contained in the Hurwitz order, generated over it by $\frac{1+i+j+ij}{2}$, which is maximal [Voi21, Section 11.1]. Unlike a number field, a quaternion algebra has in general several maximal orders, none containing the others.

4.7.2. The endomorphism ring.

Theorem 4.7.8. *Let E be an elliptic curve over k. The ring* $\mathrm{End}(E)$ *is isomorphic to one of:*
(1) $\mathbb{Z}$;
(2) an order in an imaginary quadratic field;
(3) an order in a definite quaternion algebra over $\mathbb{Q}$.
If $\mathrm{char}\, k = 0$*, case 3 does not occur.*

Proof. The ring $\mathrm{End}(E)$ has no zero divisors (Proposition 4.3.7), it is free of rank at most 4 over $\mathbb{Z}$ (Theorem 4.5.6), and it has the involution $\varphi \mapsto \hat{\varphi}$, with $\hat{\varphi}\varphi = [\deg \varphi]$ a positive integer for $\varphi \neq 0$ (Theorem 4.3.10). These force the three cases, as follows. The degree is a positive definite quadratic form on $\mathrm{End}(E)$ (Corollary 4.3.11), so its associated pairing

$$\langle \varphi, \psi \rangle := \deg(\varphi + \psi) - \deg \varphi - \deg \psi = \hat{\varphi}\psi + \hat{\psi}\varphi$$

is $\mathbb{Z}$-bilinear. Taking $\psi = 1$ shows that the trace $t(\varphi) := \varphi + \hat{\varphi}$ is the integer $1 + \deg \varphi - \deg(1 - \varphi)$, and multiplying $\varphi + \hat{\varphi} = [t(\varphi)]$ by φ gives the monic integral quadratic relation

$$\varphi^2 - [t(\varphi)]\varphi + [\deg \varphi] = 0.$$

Its discriminant satisfies $(\varphi - \hat{\varphi})^2 = t(\varphi)^2 - 4 \deg \varphi = -\deg(\varphi - \hat{\varphi})$, so $t(\varphi)^2 - 4 \deg \varphi \leqslant 0$, with equality only when $\varphi = \hat{\varphi}$. In that case $([2]\varphi - [t(\varphi)])^2 = t(\varphi)^2 - 4 \deg \varphi = 0$, so $[2]\varphi = [t(\varphi)]$ because $\mathrm{End}(E)$ has no zero divisors. Taking degrees gives $t(\varphi)^2 = 4 \deg \varphi$, so $t(\varphi) = 2s$ is even, and $\varphi = [s] \in \mathbb{Z}$ because $\mathrm{End}(E)$ is torsion-free. So every element of $B := \mathrm{End}(E) \otimes_{\mathbb{Z}} \mathbb{Q}$ outside $\mathbb{Q}$ generates an imaginary quadratic field.

Now B is a division algebra: it has no zero divisors and is finite-dimensional over $\mathbb{Q}$, so multiplication by a non-zero element is injective, hence bijective. The map $\varphi \mapsto \hat{\varphi}$ extends $\mathbb{Q}$-linearly to B, and by Theorem 4.3.10 it is a standard involution, with $\mathrm{trd}(\varphi) = t(\varphi)$ and $\mathrm{nrd}(\varphi) = \deg \varphi$. Theorem 4.7.3 leaves three possibilities for B: the field $\mathbb{Q}$; a quadratic field, which is imaginary because $t(\varphi)^2 - 4 \deg \varphi < 0$ for $\varphi \notin \mathbb{Q}$; or a quaternion algebra $(a, b \mid \mathbb{Q})$, in which $a = i^2 = -\mathrm{nrd}(i) = -\deg i < 0$ because $\mathrm{trd}(i) = 0$, and likewise $b < 0$, so that B is definite by Proposition 4.7.5. Finally, $\mathrm{End}(E)$ is a subring of B, and a free $\mathbb{Z}$-module of finite rank (Theorem 4.5.6) spanning B. So it is an order (Definition 4.7.6). Compare [Sil09, Theorem

III.9.3 and Corollary III.9.4]. In characteristic 0, $\mathrm{End}(E)$ is commutative by Corollary 4.4.6, which rules out case 3. □

The endomorphism ring of $E_{\overline{k}}$ will be the subject of Chapter 11 in characteristic 0, where E has complex multiplication if $\mathrm{End}(E_{\overline{k}}) \neq \mathbb{Z}$, and of Section 5.4 in characteristic p, where case 3 happens exactly for supersingular curves.

Example 4.7.9. Suppose $\operatorname{char} k \neq 2, 3$. On $y^2 = x^3 + x$, the automorphism $[i]$ of Example 4.3.4 satisfies $[i]^2 = [-1]$, so $\mathbb{Z}[i] \subseteq \mathrm{End}(E_{\overline{k}})$.

Example 4.7.10. Suppose $\operatorname{char} k \neq 2, 3$. On $y^2 = x^3 + 1$, let $\zeta \in \overline{k}$ be a primitive cube root of 1. Then $[\zeta](x, y) \coloneqq (\zeta x, y)$ satisfies $[\zeta]^2 + [\zeta] + 1 = 0$, so $\mathbb{Z}[\zeta] \subseteq \mathrm{End}(E_{\overline{k}})$.

These are the algebraic counterparts of Examples 1.2.2 and 1.2.3. Indeed, since $i\Lambda_i = \Lambda_i$, we have $G_6(\Lambda_i) = G_6(i\Lambda_i) = i^{-6} G_6(\Lambda_i)$, so $G_6(\Lambda_i) = 0$, and in Theorem 1.7.1 $B = 0$. Similarly $G_4(\Lambda_\omega) = 0$, so $A = 0$.

Automorphisms are the units of $\mathrm{End}(E_{\overline{k}})$, and so there are very few of them.

Theorem 4.7.11. *Let E be an elliptic curve over k. Then $\mathrm{Aut}(E_{\overline{k}})$ is finite, of order*

$$\#\operatorname{Aut}(E_{\overline{k}}) = \begin{cases} 2 & \text{if } j(E) \neq 0, 1728, \\ 4 & \text{if } j(E) = 1728 \text{ and } \operatorname{char} k \neq 2, 3, \\ 6 & \text{if } j(E) = 0 \text{ and } \operatorname{char} k \neq 2, 3, \\ 12 & \text{if } j(E) = 0 = 1728 \text{ and } \operatorname{char} k = 3, \\ 24 & \text{if } j(E) = 0 = 1728 \text{ and } \operatorname{char} k = 2. \end{cases}$$

If $\operatorname{char} k \neq 2, 3$, let $n \coloneqq \#\operatorname{Aut}(E_{\overline{k}})$. Then $\operatorname{Aut}(E_{\overline{k}}) = \operatorname{Aut}(E_{k^{\mathrm{sep}}})$, and $[\zeta](x, y) \coloneqq (\zeta^2 x, \zeta^3 y)$ defines an isomorphism of Gal_k-modules $\mu_n(k^{\mathrm{sep}}) \to \operatorname{Aut}(E_{k^{\mathrm{sep}}})$ for a short Weierstrass model.

Proof. We treat $\operatorname{char} k \neq 2, 3$. For the rest see [Sil09, Theorem III.10.1]. By Theorem 4.1.6 and Proposition 4.1.7, an automorphism of $y^2 = x^3 + Ax + B$ is $(x, y) \mapsto (u^2 x, u^3 y)$ with $u \in \overline{k}^\times$, $u^4 A = A$ and $u^6 B = B$. If $AB \neq 0$, this forces $u = \pm 1$. If $B = 0$, then $j = 1728$ and $u^4 = 1$. If $A = 0$, then $j = 0$ and $u^6 = 1$. In each case the solutions form $\mu_n(\overline{k})$, which lies in k^{sep} since $\operatorname{char} k \nmid n$, and $\sigma(u)$ corresponds to the conjugate automorphism [Sil09, Corollary III.10.2]. □

The last two cases are the non-abelian ones, and they are no accident: in characteristic 2 and 3 the curve with $j = 0$ is supersingular, so $\mathrm{End}(E_{\overline{k}})$ is a quaternion order by case 3 of Theorem 4.7.8, and its unit group need not be cyclic. The description of $\mathrm{Aut}(E_{\overline{k}})$ as $\mu_n(k^{\mathrm{sep}})$ holds exactly in the cases where $\mathrm{End}(E_{\overline{k}})$ is commutative.

Remark 4.7.12. Case 3 of Theorem 4.7.8 occurs for $E_{\overline{k}}$ exactly when E is supersingular (Theorem 5.4.1), and then $\mathrm{End}(E_{\overline{k}})$ is an order in a quaternion algebra. By a theorem of Deuring it is a maximal order in the definite quaternion algebra over $\mathbb{Q}$ that is split at every prime other than $\operatorname{char} k$ [Voi21, Theorem 42.1.9]. The worksheet on the Weil pairing, endomorphisms and automorphisms works this out for $y^2 = x^3 - x$ over $\mathbb{F}_9$: its endomorphism ring is the maximal order $\mathbb{Z} + \mathbb{Z}i + \mathbb{Z}\frac{1+j}{2} + \mathbb{Z}i\frac{1+j}{2}$ of $(-1, -3 \mid \mathbb{Q})$ [Voi21, Exercise 11.11], whose unit group is the group of order 12 in Theorem 4.7.11.

Remark 4.7.13. Over $\mathbb{C}$, the values $j = 1728$ and $j = 0$ come from the lattices Λ_i and Λ_ω (Examples 4.7.9 and 4.7.10), that is, from the points i and ω of the fundamental domain in Figure 1.8.1. These are exactly the points of $\mathbb{H}$ whose stabilizers in $\mathrm{SL}_2(\mathbb{Z})$ are larger than $\{\pm 1\}$: the stabilizer of i has order 4 and that of ω has order 6, matching Theorem 4.7.11. Indeed, by the proof of Proposition 1.8.1, an automorphism of $\mathbb{C}/\Lambda_\tau$ is multiplication by some α with $\alpha\Lambda_\tau = \Lambda_\tau$, and these α correspond to the stabilizer of τ.

4.8. Selmer elliptic curves

We end the chapter by running its results on a single family of plane cubics, where everything can be computed by hand. These curves are not given by Weierstrass equations: the base point is a flex, but the line $Z = 0$ is not its tangent, and finding a Weierstrass model is part of the work.

SITUATION 4.8.1. In this section $\operatorname{char} k \neq 3$, $d \in k^\times$, and $\zeta \in k^{\mathrm{sep}}$ is a primitive cube root of 1.

DEFINITION 4.8.2. The Selmer elliptic curve with parameter d is the plane cubic

$$C_d \colon X^3 + Y^3 = dZ^3, \qquad O \coloneqq (1 : -1 : 0).$$

We write $F \coloneqq X^3 + Y^3 - dZ^3$, and $Q_1 \coloneqq (1 : -\zeta : 0)$, $Q_2 \coloneqq (1 : -\zeta^2 : 0)$.

PROPOSITION 4.8.3.
(1) (C_d, O) is an elliptic curve over k.
(2) The tangent line to C_d at O is $X + Y = 0$, and it meets C_d only at O, with multiplicity 3: the base point O is a flex. The line $Z = 0$ meets C_d in the three distinct k^{sep}-points O, Q_1 and Q_2.

PROOF. The partial derivatives $3X^2$, $3Y^2$ and $-3dZ^2$ have no common zero in $\mathbb{P}^2(\overline{k})$, because 3 and d are units. So C_d is smooth, hence nice by Corollary 3.7.5 and of genus 1 by Corollary 3.7.7, and $O \in C_d(k)$. This is part 1.

For part 2, $\nabla F(O) = (3, 3, 0)$. Substituting $Y = -X$ into F leaves $-dZ^3$, which vanishes to order 3 at O. On $Z = 0$, F leaves $X^3 + Y^3$, whose roots $X/Y = -1, -\zeta, -\zeta^2$ are distinct since $\operatorname{char} k \neq 3$. □

So O plays the role of the point at infinity of a Weierstrass cubic, with the flex tangent $X + Y = 0$ in place of the line $Z = 0$, and Proposition 4.2.4 applies: if a line meets C_d in $P, Q, R \in C_d(\overline{k})$, counted with multiplicity, then $P + Q + R = O$, and by its proof $\operatorname{div}\big(L/(X + Y)\big) = P + Q + R - 3O$ for a linear form L defining the line.

PROPOSITION 4.8.4. *Let $w \coloneqq Z/(X + Y)$ and $v \coloneqq X/(X + Y)$ in $k(C_d)$.*
(1) $\operatorname{div}(w) = Q_1 + Q_2 - 2O$, and v has a pole of order 3 at O and no other pole. So $\{1, w\}$ is a basis of $\mathcal{L}(2O)$, and $\{1, w, v\}$ is a basis of $\mathcal{L}(3O)$.
(2) $3v^2 - 3v + 1 = dw^3$.
(3) The functions $x \coloneqq 3dw$ and $y \coloneqq 9dv$ are Weierstrass coordinates: $\varphi \coloneqq (x : y : 1)$ is an isomorphism from C_d onto

$$E_d \colon y^2 - 9dy = x^3 - 27d^2, \tag{4.8.1}$$

with inverse $(x : y : 1) \mapsto (y : 9d - y : 3x)$. This equation has $a_3 = -9d$, $a_6 = -27d^2$ and $a_1 = a_2 = a_4 = 0$, and $c_4 = 0$, $\Delta = -3^9 d^4$, $j = 0$.
(4) Negation on C_d is $(X_0 : Y_0 : Z_0) \mapsto (Y_0 : X_0 : Z_0)$.
(5) Let $\xi \coloneqq X/Z$ and $\eta \coloneqq Y/Z$. The invariant differential of (4.8.1) is $\omega = d\xi/(3\eta^2)$.
(6) If $\operatorname{char} k \neq 2$, then $12dw$ and $36d(X - Y)/(X + Y)$ are Weierstrass coordinates for $y^2 = x^3 - 432d^2$, which has $\Delta = -2^{12}3^9 d^4$.

PROOF. For every linear form h, the discussion after Proposition 4.8.3 gives $\operatorname{div}\big(h/(X+Y)\big) = (h \cap C_d) - 3O$. For $h = Z$ this is $\operatorname{div}(w)$, by part 2 of that proposition. The line $X = 0$ does not pass through O, so v has a pole of order exactly 3 at O. Since $\ell(nO) = n$ by Corollary 3.5.5, and $1, w, v$ have pole orders $0, 2, 3$, this proves part 1. Part 2 is the equation of C_d divided by $(X+Y)^3$, since $X^3+Y^3 = (X+Y)^3 - 3XY(X+Y)$ and $Y/(X + Y) = 1 - v$.

For part 3, part 2 gives $y^2 - 9dy = 81d^2(v^2 - v) = 27d^2(dw^3 - 1) = x^3 - 27d^2$, and the proof of Theorem 4.1.6 shows that φ is an isomorphism onto (4.8.1). For the inverse, $X : Y : Z = v : 1 - v : w$. The invariants follow from Definition 4.1.2: $b_2 = b_4 = b_8 = 0$ and $b_6 = a_3^2 + 4a_6 = -27d^2$, so $c_4 = 0$ and $\Delta = -27b_6^2 = -3^9 d^4$.

For part 4, φ is a homomorphism by Theorem 4.3.6, and by Proposition 4.2.4 negation on E_d is $(x, y) \mapsto (x, 9d - y)$. The inverse of φ turns this into $(y : 9d - y : 3x) \mapsto (9d - y : y : 3x)$.

For part 5, $\omega = dx/(2y-9d)$. On the affine patch $Z \neq 0$ we have $x = 3d/(\xi+\eta)$, $2y-9d = 9d(\xi-\eta)/(\xi+\eta)$ and $\xi^2\,d\xi + \eta^2\,d\eta = 0$, so

$$dx = -\frac{3d\,(d\xi + d\eta)}{(\xi+\eta)^2} = \frac{3d\,(\xi-\eta)}{(\xi+\eta)\,\eta^2}\,d\xi, \qquad \omega = \frac{d\xi}{3\eta^2}.$$

For part 6, completing the square in (4.8.1) gives $(2y-9d)^2 = 4x^3 - 27d^2$, and multiplying by 16 gives $(8y-36d)^2 = (4x)^3 - 432d^2$, with $4x = 12dw$ and $8y - 36d = 36d(X-Y)/(X+Y)$. This is the change of variables (4.1.3) with $u = 1/2$, so the discriminant is $2^{12}\Delta$ by Proposition 4.1.7. □

The differential in part 5 is $d\xi/(\partial f/\partial\eta)$ for $f \coloneqq \xi^3 + \eta^3 - d$, the differential that Proposition 3.7.6 attaches to the plane cubic C_d, with divisor $(3-3)H = 0$. The equation (4.8.1) is valid in characteristic 2, where $\Delta = -3^9d^4 = d^4$ is still non-zero, but it has $a_3 \neq 0$, and there is no way to complete the square. Part 6 needs $\operatorname{char} k \neq 2$ twice: to define $X - Y$ as a function with a pole of order 3 at O, and to complete the square. This is why the general Weierstrass equation keeps the coefficients a_1 and a_3. Finally, the divisor in part 1 is written in $\operatorname{Div} C_{d,\overline{k}}$. If $\zeta \notin k$, then Q_1 and Q_2 are conjugate, and in $\operatorname{Div} C_d$ they form a single closed point of degree 2.

The torsion points of C_d are cut out by lines.

Proposition 4.8.5.

(1) The non-zero points of $C_d[2](\overline{k})$ are the points $(c : c : 1)$ with $2c^3 = d$.
(2) A point $P \in C_d(\overline{k})$ satisfies $[3]P = O$ if and only if P is a flex.
(3) Let $\delta \in \overline{k}$ with $\delta^3 = d$. Then $C_d[3](\overline{k}) = \big(C_d \cap \{XYZ = 0\}\big)(\overline{k})$ consists of the nine $\overline{k}$-points

$$O,\ Q_1,\ Q_2, \qquad (0 : \zeta^i\delta : 1), \qquad (\zeta^i\delta : 0 : 1), \qquad i = 0, 1, 2.$$

Proof. For part 1, by part 4 of Proposition 4.8.4, $P = -P$ if and only if $(X_0 : Y_0 : Z_0) = (Y_0 : X_0 : Z_0)$. Either $X_0 = Y_0$, or $X_0 = -Y_0$ and $Z_0 = 0$, which is $P = O$. If $X_0 = Y_0$ and $Z_0 = 0$, then $2X_0^3 = 0$, so $\operatorname{char} k = 2$ and again $P = (1 : 1 : 0) = O$.

For part 2, if the tangent line L at P meets C_d in $3P$, then $P + P + P = O$ by Proposition 4.2.4. Conversely, if $[3]P = O$, then $3P - 3O = \operatorname{div}(f)$ for some $f \in \overline{k}(C_d)$ (Corollary 4.2.3). By part 1 of Proposition 4.8.4 and Proposition 3.5.2, f is a $\overline{k}$-linear combination of $1, w, v$, that is, $f = h/(X+Y)$ for a linear form h. So h cuts out $3P$, and h is the tangent line at P, meeting C_d with multiplicity 3.

For part 3, let $P_0 = (X_0 : Y_0 : Z_0) \in C_d(\overline{k})$ with $X_0 = 0$. The tangent line $Y_0^2Y = dZ_0^2Z$ at P_0 is the line $Z_0Y = Y_0Z$ through P_0 and $(1 : 0 : 0)$, and on it $F = X^3$, which vanishes to order 3 at P_0. So P_0 is a flex. The same computation with the roles of the coordinates exchanged applies when $Y_0 = 0$ or $Z_0 = 0$. This gives nine distinct flexes, and $\#C_d[3](\overline{k}) = 9$ by Corollary 4.5.1, so part 2 shows that they are all of $C_d[3](\overline{k})$. □

In characteristic 2, part 1 says that $C_d[2](\overline{k}) = 0$. So by Corollary 4.5.1 every Selmer curve is supersingular in characteristic 2, which fits with $j = 0$.

Example 4.8.6. Over $\mathbb{Q}$, C_d has a rational point of order 2 if and only if $d = 2c^3$ with $c \in \mathbb{Q}^\times$, and then it is $(c : c : 1)$. For instance $(1 : 1 : 1) \in C_2(\mathbb{Q})$. Otherwise the three points of order 2 are permuted transitively by $\mathrm{Gal}_{\mathbb{Q}}$, and $\mathbb{Q}(C_d[2](\overline{\mathbb{Q}}))$ is the splitting field $\mathbb{Q}(\zeta, \sqrt[3]{d/2})$ of $2T^3 - d$, of degree 6. So $\bar\rho_{C_d,2}\colon \mathrm{Gal}_{\mathbb{Q}} \to \mathrm{GL}_2(\mathbb{F}_2) \cong S_3$ is surjective.

Example 4.8.7. Over $\mathbb{Q}$, the $\overline{\mathbb{Q}}$-points Q_1 and Q_2 are never rational, and $(0 : \zeta^i\delta : 1)$ is rational only if $i = 0$ and $d = \delta^3$ is a cube. So $C_d(\mathbb{Q})[3]$ is $\{O, (0 : \delta : 1), (\delta : 0 : 1)\} \cong \mathbb{Z}/3\mathbb{Z}$ if $d = \delta^3$, and trivial otherwise. In either case $\mathbb{Q}(C_d[3](\overline{\mathbb{Q}})) = \mathbb{Q}(\zeta, \sqrt[3]{d})$ contains $\mu_3(\overline{\mathbb{Q}})$, as Corollary 4.6.5 requires, and $C_d[3](\overline{\mathbb{Q}})$ is never contained in $C_d(\mathbb{Q})$ (Example 4.6.6).

Example 4.8.8. C_1 is the Fermat cubic $X^3 + Y^3 = Z^3$. Fermat's Last Theorem for exponent 3 says exactly that $C_1(\mathbb{Q}) = \{O, (1 : 0 : 1), (0 : 1 : 1)\}$: every rational point is a flex.

For C_d, the polar expansion of Lemma 4.2.5 takes a symmetric form. For $P = (X_P, Y_P, Z_P)$, Q and R in $\overline{k}^3$, let

$$\langle P, Q, R\rangle \coloneqq X_P X_Q X_R + Y_P Y_Q Y_R - d Z_P Z_Q Z_R.$$

This form is symmetric and trilinear, with $\langle P, P, P\rangle = F(P)$ and $\nabla F(P)\cdot Q = 3\langle P, P, Q\rangle$, so (4.2.1) reads

$$F(sP + tQ) = s^3 F(P) + 3s^2 t\,\langle P, P, Q\rangle + 3st^2\,\langle P, Q, Q\rangle + t^3 F(Q). \tag{4.8.2}$$

Proposition 4.8.9. *Let $P_1 = (X_1 : Y_1 : Z_1)$ and $P_2 = (X_2 : Y_2 : Z_2)$ be points of $C_d(\overline{k})$.*
(1) If $P_1 \neq P_2$, let $A \coloneqq \langle P_1, P_1, P_2\rangle$ and $B \coloneqq \langle P_1, P_2, P_2\rangle$. Then

$$P_1 + P_2 = (BY_1 - AY_2 : BX_1 - AX_2 : BZ_1 - AZ_2). \tag{4.8.3}$$

(2) For every $P = (X_0 : Y_0 : Z_0) \in C_d(\overline{k})$,

$$[2]P = \big(-Y_0(2X_0^3 + Y_0^3) : X_0(X_0^3 + 2Y_0^3) : Z_0(X_0^3 - Y_0^3)\big). \tag{4.8.4}$$

Proof. Proposition 4.2.6 applies to C_d with its flex O. For part 1, the coefficients there are $\nabla F(P_1)\cdot P_2 = 3A$ and $\nabla F(P_2)\cdot P_1 = 3B$, and since $3 \in k^\times$ the third intersection point of the line through P_1 and P_2 is $BP_1 - AP_2$. Its negative is (4.8.3) by part 4 of Proposition 4.8.4.

For part 2, if $Z_0 = 0$, then $P \in \{O, Q_1, Q_2\} \subseteq C_d[3](\overline{k})$, so $[2]P = -P$, and (4.8.4) gives $(Y_0 : X_0 : 0)$ because $X_0^3 = -Y_0^3$. If $Z_0 \neq 0$, let $Q \coloneqq (Y_0^2, -X_0^2, 0)$, which is not a multiple of P. Then $\nabla F(P)\cdot Q = 3\langle P, P, Q\rangle = 0$, so Q lies on the tangent line at P, and part 2 of Proposition 4.2.6 says that the tangent line meets C_d again at $R \coloneqq F(Q)P - 3\langle P, Q, Q\rangle Q$. Here $F(Q) = Y_0^6 - X_0^6$ and $\langle P, Q, Q\rangle = X_0 Y_0 (X_0^3 + Y_0^3) = dX_0 Y_0 Z_0^3$. Substituting $dZ_0^3 = X_0^3 + Y_0^3$, which is non-zero, gives

$$R = -(X_0^3 + Y_0^3)\big(X_0(X_0^3 + 2Y_0^3),\ -Y_0(2X_0^3 + Y_0^3),\ Z_0(X_0^3 - Y_0^3)\big),$$

and $[2]P = -R$ is (4.8.4). □

On the line $X = Y$, (4.8.4) gives $(-3X_0^4 : 3X_0^4 : 0) = O$, recovering part 1 of Proposition 4.8.5. Note that the derivation collapses in characteristic 3, where the middle terms of (4.8.2) vanish, and $F = (X + Y - \delta Z)^3$ over $\overline{k}$ is a triple line.

Example 4.8.10. On C_7, let $P \coloneqq (2 : -1 : 1)$, since $2^3 + (-1)^3 = 7$. By (4.8.4), $[2]P = (15 : 12 : 9) = (5 : 4 : 3)$, and indeed $5^3 + 4^3 = 189 = 7 \cdot 3^3$. With $P_1 = P$ and $P_2 = [2]P$, (4.8.3) has $A = 3$ and $B = -29$, so

$$[3]P = (-17 : 73 : 38), \qquad 73^3 - 17^3 = 7 \cdot 38^3,$$

and doubling $[2]P$ gives $[4]P = (-1256 : 1265 : 183)$. By Examples 4.8.6 and 4.8.7, $C_7(\mathbb{Q})$ has no points of order 2 or 3. The growth of the coordinates suggests that P has infinite order. This will follow from the Nagell–Lutz theorem in Chapter 8. Under φ, P maps to $(21, 126) \in E_7(\mathbb{Q})$.

The cube roots of unity act on C_d by scaling one coordinate, and this is where the endomorphisms beyond $\mathbb{Z}$ come from.

Proposition 4.8.11. *Let $\alpha(X : Y : Z) \coloneqq (X : Y : \zeta Z)$, a morphism $C_{d,k(\zeta)} \to C_{d,k(\zeta)}$.*
(1) α is an automorphism of $C_{d,k(\zeta)}$ with $\alpha^2 + \alpha + 1 = 0$, so $\mathbb{Z}[\alpha] \cong \mathbb{Z}[\zeta]$ is a subring of $\mathrm{End}(C_{d,k(\zeta)})$. In the coordinates of (4.8.1), $\alpha(x, y) = (\zeta x, y)$, and $\alpha^*\omega = \zeta\omega$.
(2) If $\mathrm{char}\, k \neq 2$, then $\mathrm{Aut}(C_{d,\overline{k}}) = \{\pm 1, \pm\alpha, \pm\alpha^2\}$. If moreover $\zeta \notin k$, then $\mathrm{Aut}(C_d) = \{\pm 1\}$.
(3) The endomorphism $1 - \alpha$ is separable of degree 3, with dual $1 - \alpha^2$ and kernel $\{O, Q_1, Q_2\}$.

Proof. For part 1, $F(X, Y, \zeta Z) = F$ and $\alpha(O) = O$, so α is an isogeny with inverse α^2. Since $\alpha \neq 1$ and $(\alpha - 1)(\alpha^2 + \alpha + 1) = \alpha^3 - 1 = 0$, Proposition 4.3.7 gives $\alpha^2 + \alpha + 1 = 0$. The kernel of $\mathbb{Z}[T]/(T^2 + T + 1) \to \mathrm{End}(C_d)$ is an ideal of $\mathbb{Z}[\zeta]$, with finite quotient if it is non-zero, which would contradict Proposition 4.3.7. Next, $\alpha^* w = \zeta w$ and $\alpha^* v = v$, so $\alpha^* x = \zeta x$ and $\alpha^* y = y$, and $\alpha^*\omega = \zeta\, dx/(2y - 9d) = \zeta\omega$.

For part 2, $j = 0$, so $\#\operatorname{Aut}(C_{d,\overline{k}}) = 6$ by Theorem 4.7.11. The six automorphisms listed pull ω back to $\pm\zeta^i\omega$, six distinct multiples, so they are distinct. An element φ of $\operatorname{Aut}(C_d)$ has $c_\varphi \in k$ by Proposition 4.4.5, which rules out $\pm\alpha$ and $\pm\alpha^2$ if $\zeta \notin k$.

For part 3, α has degree 1, so $\hat{\alpha} = \alpha^{-1} = \alpha^2$ by Theorem 4.3.10, and $\widehat{1-\alpha} = 1-\alpha^2$. Then $(1-\alpha^2)(1-\alpha) = 2-\alpha-\alpha^2 = 3$, so $\deg(1-\alpha) = 3$. It is separable by Proposition 4.4.5, since $c_{1-\alpha} = 1-\zeta \neq 0$. So $\#\ker(1-\alpha) = 3$ by Theorem 4.3.8. The kernel is the set of fixed points of α, and $(X : Y : \zeta Z) = (X : Y : Z)$ forces $Z = 0$, since $X = Y = 0$ is not on C_d. □

In characteristic 0, $\operatorname{End}(C_{d,\overline{k}})$ is an order in an imaginary quadratic field by Theorem 4.7.8, and it contains the maximal order $\mathbb{Z}[\zeta]$, so $\operatorname{End}(C_{d,\overline{k}}) = \mathbb{Z}[\zeta]$. The automorphism α is the $[\zeta]$ of Example 4.7.10 in another model, and $1-\alpha$ is the algebraic counterpart of multiplication by $1-\zeta$ on the Eisenstein lattice. Note also $(1-\alpha)^2 = -3\alpha$, so $\ker(1-\alpha) \subseteq C_d[3](\overline{k})$, in accordance with Proposition 4.8.5.

EXAMPLE 4.8.12. The kernel $\{O, Q_1, Q_2\}$ of $1-\alpha$ is cut out by the line $Z = 0$, so it is Gal_k-stable, even though $1-\alpha$ is only an endomorphism of $C_{d,k(\zeta)}$. By Proposition 4.3.9, there is a 3-isogeny over k with this kernel. Suppose $\operatorname{char} k \neq 2$, and use the model $y^2 = x^3 + D$ with $D = -432d^2$ of Proposition 4.8.4, in which Q_1 and Q_2 become the $\overline{k}$-points $(0, \pm\sqrt{D})$. The isogeny is

$$\psi(x,y) \coloneqq \left(\frac{x^3+4D}{x^2},\ \frac{y\,(x^3-8D)}{x^3}\right), \qquad \psi\colon \big(y^2 = x^3 + D\big) \to \big(y^2 = x^3 - 27D\big),$$

which lands on the target because $(u+D)(u-8D)^2 = (u+4D)^3 - 27Du^2$ with $u = x^3$. Since $-27D = 3^6 \cdot 16d^2$, the target is isomorphic to $y^2 = x^3 + 16d^2$, by Proposition 4.1.7 with $u = 3$. Over $k(\zeta)$ this curve is isomorphic to C_d, because $-432d^2/(16d^2) = -27 = (\sqrt{-3})^6$ and $\sqrt{-3} = 2\zeta + 1$. Composed with such an isomorphism, ψ has the same kernel as $1-\alpha$, so by Proposition 4.3.9 it agrees with $1-\alpha$ up to an automorphism of C_d. If $\zeta \notin k$, the target is not isomorphic over k to any $C_{d'}$: that would need $-27(d'/d)^2 \in k^{\times 6}$, and so $-3 \in k^{\times 2}$.

Finally, the Selmer curves are twists of one another.

PROPOSITION 4.8.13. *Let $d, d' \in k^\times$.*

(1) If $d' = c^3 d$ with $c \in k^\times$, then $(X : Y : Z) \mapsto (X : Y : Z/c)$ is an isomorphism of elliptic curves $C_d \to C_{d'}$ over k.

(2) If $\operatorname{char} k \neq 2$ and $C_d \cong C_{d'}$ over k, then $d'/d \in k^{\times 3}$.

So the Selmer curves all become isomorphic over $\overline{k}$, and if $\operatorname{char} k \neq 2$ their isomorphism classes over k are indexed by $k^\times/(k^\times)^3$.

PROOF. Part 1 is immediate, since $d'(Z/c)^3 = dZ^3$ and O is fixed. For part 2, use the models $y^2 = x^3 - 432d^2$ and $y^2 = x^3 - 432d'^2$ of Proposition 4.8.4. By Proposition 4.1.7, an isomorphism between them has $d^2 = u^6 d'^2$ for some $u \in k^\times$, so $d/d' = \pm u^3$, and $-1 = (-1)^3$. They become isomorphic over $\overline{k}$ by Proposition 4.1.10, since $j = 0$. □

The C_d are the cubic twists of the Fermat cubic C_1. Twists in general are the subject of Chapter 7, and these return in Chapter 10.

REMARK 4.8.14. Selmer studied the diagonal cubics $aX^3 + bY^3 + cZ^3 = 0$ [Sel51]. Among them is the curve $3X^3 + 4Y^3 + 5Z^3 = 0$ of Section 4.1, which is a nice curve of genus 1 with no rational point, so not an elliptic curve over $\mathbb{Q}$. It has a Jacobian, an elliptic curve over $\mathbb{Q}$, and in general the Jacobian of $aX^3 + bY^3 + cZ^3 = 0$ is $y^2 = x^3 - 432(abc)^2$, that is, the Selmer curve C_{abc}. So Selmer's counterexample to the Hasse principle is a torsor under C_{60}. We return to this in Chapter 10, where the C_d are the first examples for descent.

CHAPTER 5

Elliptic curves over finite fields

Sources. This chapter follows [Sil09, Chapter V] and [Poo06, Chapter 3], with [Mil22] for the Galois theory of finite fields, [AH17] for the Cartier operator, and [Voi21] for the mass formula.

SITUATION 5.0.1. Let p be a prime, let $q \coloneqq p^e$ for some $e \geqslant 1$, and let $k \coloneqq \mathbb{F}_q$ be the finite field with q elements. Since k is perfect, $k^{\mathrm{sep}} = \overline{k} = \overline{\mathbb{F}}_q$.

DRAFT
13 Oct 2026

5.1. The Frobenius automorphism

We are in Situation 5.0.1. The algebraic closure of k is

$$\overline{\mathbb{F}}_q = \varinjlim_r \mathbb{F}_{q^r}.$$

Here $\mathbb{F}_{q^r}$ is the unique subfield of $\overline{\mathbb{F}}_q$ with q^r elements, and $\mathbb{F}_{q^r} \subseteq \mathbb{F}_{q^s}$ if and only if $r \mid s$. The direct limit is the union $\bigcup_{r \geqslant 1} \mathbb{F}_{q^r}$.

The Galois group $\mathrm{Gal}_{\mathbb{F}_q} = \mathrm{Gal}(\overline{\mathbb{F}}_q/\mathbb{F}_q)$ is topologically generated by the Frobenius automorphism

$$\sigma \colon \overline{\mathbb{F}}_q \to \overline{\mathbb{F}}_q, \qquad x \mapsto x^q,$$

that is, the powers of σ are dense in $\mathrm{Gal}_{\mathbb{F}_q} = \varprojlim_r \mathrm{Gal}(\mathbb{F}_{q^r}/\mathbb{F}_q)$.

In particular, for every $r > 0$,

$$\mathrm{Gal}(\mathbb{F}_{q^r}/\mathbb{F}_q) = \langle \sigma|_{\mathbb{F}_{q^r}} \rangle \cong \mathbb{Z}/r\mathbb{Z},$$

the isomorphism given by $(n \bmod r) \mapsto (\sigma|_{\mathbb{F}_{q^r}})^n$. Assembling these maps for all $r \geqslant 1$ gives

$$\widehat{\mathbb{Z}} \coloneqq \varprojlim_r \mathbb{Z}/r\mathbb{Z} \xrightarrow{\sim} \mathrm{Gal}_{\mathbb{F}_q}, \qquad (n_r \bmod r)_{r \geqslant 1} \mapsto \left((\sigma|_{\mathbb{F}_{q^r}})^{n_r}\right)_{r \geqslant 1}. \tag{5.1.1}$$

For all of this, see [Mil22].

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5.2. Hasse's bound

Let E be an elliptic curve over $\mathbb{F}_q$. We know from Theorem 4.1.6 that E is given by a Weierstrass equation (4.1.1), that is, by $f(x, y) = 0$ with

$$f \coloneqq y^2 + a_1xy + a_3y - (x^3 + a_2x^2 + a_4x + a_6) \in \mathbb{F}_q[x, y],$$

and that the set $E(\mathbb{F}_q)$ of rational points contains at least one rational point: $O \coloneqq (0 : 1 : 0)$.

On the other hand,

$$E(\mathbb{F}_q) = \{O\} \cup \{(x, y) \in \mathbb{F}_q^2 : f(x, y) = 0\}.$$

If $x \in \mathbb{F}_q$ is fixed, the polynomial $f(x, y) \in \mathbb{F}_q[y]$ has at most two roots in $\mathbb{F}_q$, so we obtain the trivial upper bound

$$\#E(\mathbb{F}_q) \leqslant 2q + 1.$$

Let $\mathsf{F} \in \mathrm{End}(E)$ be the q-th power Frobenius endomorphism (Definition 3.2.11 and Example 4.3.3). On $\overline{\mathbb{F}}_q$-points it acts as the Frobenius automorphism σ on coordinates:

$$\mathsf{F} \colon E(\overline{\mathbb{F}}_q) \to E(\overline{\mathbb{F}}_q), \qquad O \mapsto O, \qquad (x, y) \mapsto (x^q, y^q).$$

E. Artin conjectured, in his thesis, and H. Hasse proved the optimal upper bound.

THEOREM 5.2.1 (Hasse). *Let E be an elliptic curve over $\mathbb{F}_q$. Then for every $r \geqslant 1$,*

$$|\#E(\mathbb{F}_{q^r}) - (q^r + 1)| \leqslant 2q^{r/2}.$$

PROOF. (Sketch. See [Sil09, Theorem V.1.1].) There are three key observations that go into the proof of Hasse's bound.

(1) $E(\mathbb{F}_{q^r}) = \ker(1 - \mathsf{F}^r)(\overline{\mathbb{F}}_q)$. Indeed, since $\mathbb{F}_{q^r} = \overline{\mathbb{F}}_q^{\langle \sigma^r \rangle}$, we know that $P \in E(\mathbb{F}_{q^r}) \iff \mathsf{F}^r(P) = P$.
(2) The map $\deg\colon \operatorname{End}(E) \to \mathbb{Z}$ is a positive definite quadratic form (Corollary 4.3.11).
(3) For any abelian group A and any positive definite quadratic form $d\colon A \to \mathbb{Z}$, the Cauchy–Schwarz inequality gives

$$|d(a_1 - a_2) - d(a_1) - d(a_2)| \leqslant 2\sqrt{d(a_1)\, d(a_2)}.$$

To combine them: $1 - \mathsf{F}^r$ is separable (Corollary 4.4.4, applied over $\mathbb{F}_{q^r}$), so observation 1 and Theorem 4.3.8 give $\#E(\mathbb{F}_{q^r}) = \deg(1 - \mathsf{F}^r)$, and $\deg \mathsf{F}^r = q^r$ (Proposition 3.2.12). Now apply observation 3 with $d = \deg$, $a_1 = 1$ and $a_2 = \mathsf{F}^r$. □

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5.3. The Weil conjectures

In 1949, Weil made a series of remarkable conjectures about the numbers $\#X(\mathbb{F}_{q^n})$, $n \geqslant 1$, for a variety X over $\mathbb{F}_q$ [Wei49]. Their proof was one of the main motivations for the scheme-theoretic foundations of algebraic geometry and for étale cohomology (Remark 5.3.6). In this section we state the conjectures for curves, check them on $\mathbb{P}^1$, and prove them for elliptic curves. We keep Situation 5.0.1, and C denotes a nice curve over $\mathbb{F}_q$ (Definition 2.4.1).

5.3.1. The zeta function. Each set $C(\mathbb{F}_{q^n})$ is finite, and we encode all of their sizes in a single generating function.

DEFINITION 5.3.1. The zeta function of $C/\mathbb{F}_q$ is the power series

$$Z(C/\mathbb{F}_q; T) \coloneqq \exp\left(\sum_{n=1}^{\infty} \#C(\mathbb{F}_{q^n}) \frac{T^n}{n}\right) \in \mathbb{Q}[[T]]. \tag{5.3.1}$$

Here $\exp F(T) \coloneqq \sum_{m \geqslant 0} F(T)^m/m!$ for a power series $F(T)$ with no constant term, so $Z(C/\mathbb{F}_q; 0) = 1$. The zeta function determines the numbers $\#C(\mathbb{F}_{q^n})$: $\#C(\mathbb{F}_{q^n})$ is the coefficient of T^{n-1} in $\frac{d}{dT} \log Z(C/\mathbb{F}_q; T)$. The exponential in (5.3.1) looks odd at first: the more natural series is $\sum_n \#C(\mathbb{F}_{q^n}) T^n$. The choice makes $Z(C/\mathbb{F}_q; T)$ an Euler product over the closed points of C.

PROPOSITION 5.3.2. *For $d \geqslant 1$, let N_d be the number of closed points of C of degree d.*
(1) Each N_d is finite, and $\#C(\mathbb{F}_{q^n}) = \sum_{d|n} dN_d$ for every $n \geqslant 1$.
(2) In $\mathbb{Z}[[T]]$,

$$Z(C/\mathbb{F}_q; T) = \prod_P \left(1 - T^{\deg P}\right)^{-1} = \prod_{d \geqslant 1} \left(1 - T^d\right)^{-N_d}, \tag{5.3.2}$$

where the first product ranges over the closed points P of C. In particular $Z(C/\mathbb{F}_q; T) \in 1 + T\mathbb{Z}[[T]]$.
(3) For every $d \geqslant 1$, $N_d = \frac{1}{d} \sum_{e|d} \mu(d/e)\, \#C(\mathbb{F}_{q^e})$, where μ is the Möbius function.

PROOF. By Proposition 3.1.2, the points of $C(\overline{\mathbb{F}}_q)$ above a closed point P form a single Galois orbit with $\deg P$ elements, because $\mathbb{F}_q$ is perfect. Such an $\overline{\mathbb{F}}_q$-point lies in $C(\mathbb{F}_{q^n})$ exactly when the residue field $\mathbb{F}_q(P) \cong \mathbb{F}_{q^{\deg P}}$ embeds in $\mathbb{F}_{q^n}$, that is, when $\deg P \mid n$. So $N_d \leqslant \#C(\mathbb{F}_{q^d}) < \infty$, and $C(\mathbb{F}_{q^n})$ is the disjoint union of the orbits of size $d \mid n$, which proves part 1.

For part 2, only the finitely many factors with $\deg P < m$ are $\not\equiv 1 \pmod{T^m}$, so the product converges in $\mathbb{Z}[[T]]$. Taking its logarithm and expanding $-\log(1-x) = \sum_{k\geqslant 1} x^k/k$,

$$\sum_{d\geqslant 1} N_d \sum_{k\geqslant 1} \frac{T^{dk}}{k} = \sum_{n\geqslant 1} \left(\sum_{d\mid n} dN_d \right) \frac{T^n}{n} = \sum_{n\geqslant 1} \#C(\mathbb{F}_{q^n}) \frac{T^n}{n},$$

where we set $n = dk$. Part 3 is Möbius inversion of part 1, applied to $n \mapsto \#C(\mathbb{F}_{q^n})$ and $d \mapsto dN_d$. □

So the numbers of rational points over the extensions of $\mathbb{F}_q$ and the numbers of closed points of each degree determine each other. Substituting $T = q^{-s}$, and writing $NP := \#\mathbb{F}_q(P) = q^{\deg P}$ for the norm of a closed point, gives

$$\zeta_C(s) := Z(C/\mathbb{F}_q; q^{-s}) = \prod_P \left(1 - NP^{-s}\right)^{-1},$$

which has the shape of the Euler product $\zeta(s) = \prod_p (1-p^{-s})^{-1}$ of the Riemann zeta function, with closed points in place of primes.

EXAMPLE 5.3.3. Let $C := \mathbb{P}^1$ over $\mathbb{F}_q$. Then $\#\mathbb{P}^1(\mathbb{F}_{q^n}) = q^n + 1$, so

$$\log Z(\mathbb{P}^1/\mathbb{F}_q; T) = \sum_{n\geqslant 1} \frac{T^n}{n} + \sum_{n\geqslant 1} \frac{(qT)^n}{n}, \qquad Z(\mathbb{P}^1/\mathbb{F}_q; T) = \frac{1}{(1-T)(1-qT)}$$

[Sil09, Example V.2.1]. The Euler product gives the same answer in another way. The closed points of $\mathbb{P}^1$ are the rational point ∞, of degree 1, and the closed points of $\mathbb{A}^1$, which are the monic irreducible polynomials $f \in \mathbb{F}_q[t]$, with $\deg P = \deg f$. Since $\mathbb{F}_q[t]$ is a unique factorization domain, expanding each factor of (5.3.2) as a geometric series gives

$$(1-T)^{-1} \prod_f \left(1 - T^{\deg f}\right)^{-1} = \frac{1}{1-T} \sum_{g \text{ monic}} T^{\deg g} = \frac{1}{1-T} \sum_{d\geqslant 0} q^d T^d,$$

because there are q^d monic polynomials of degree d. Part 3 of Proposition 5.3.2 counts them: since $\sum_{e\mid d} \mu(d/e) = 0$ for $d > 1$, the $+1$ in $q^e + 1$ drops out, and the number of monic irreducible polynomials of degree d over $\mathbb{F}_q$ is $\frac{1}{d}\sum_{e\mid d} \mu(d/e) q^e$ for every $d \geqslant 1$. Over $\mathbb{F}_2$, in degree 2, this is $\frac{1}{2}(4-2) = 1$: the polynomial $t^2 + t + 1$, whose roots form the Galois orbit $\{\omega, \omega^2\}$ of a primitive cube root of unity in $\mathbb{F}_4$.

The zeta function of $\mathbb{P}^1$ is a rational function, and this is no accident. The next lemma turns a formula for $\#C(\mathbb{F}_{q^n})$ as a trace of n-th powers into a formula for $Z(C/\mathbb{F}_q; T)$.

LEMMA 5.3.4. *Let A be a square matrix over a field K of characteristic 0. Then*

$$\exp\left(\sum_{n\geqslant 1} \operatorname{tr}(A^n) \frac{T^n}{n} \right) = \frac{1}{\det(1 - AT)} \qquad \text{in } K[[T]]. \tag{5.3.3}$$

PROOF. (See [Sil09, Exercise 5.3].) Over an algebraic closure $\bar{K}$, triangularize A, with its eigenvalues $\lambda_1, \dots, \lambda_m$ on the diagonal. Then $\operatorname{tr}(A^n) = \sum_i \lambda_i^n$ and $\det(1 - AT) = \prod_i (1 - \lambda_i T)$. Since $\exp(F+G) = \exp F \exp G$, both sides of (5.3.3) are the products over i of the corresponding sides for the 1×1 matrix (λ_i), and for these

$$\sum_{n\geqslant 1} \lambda^n \frac{T^n}{n} = -\log(1 - \lambda T), \qquad \exp\bigl(-\log(1-\lambda T)\bigr) = (1-\lambda T)^{-1}$$

in $\bar{K}[[T]]$, which needs characteristic 0. Both sides of (5.3.3) have coefficients in K. □

For example, if there are $\alpha_1, \dots, \alpha_r \in \mathbb{C}$ and signs $\varepsilon_i \in \{\pm 1\}$ with $\#C(\mathbb{F}_{q^n}) = \sum_i \varepsilon_i \alpha_i^n$ for all $n \geqslant 1$, then Lemma 5.3.4, applied to the 1×1 matrices (α_i), gives $Z(C/\mathbb{F}_q; T) = \prod_{i=1}^r (1 - \alpha_i T)^{-\varepsilon_i}$.

5.3.2. The statement.

THEOREM 5.3.5 (Weil conjectures for curves). *Let C be a nice curve of genus g over $\mathbb{F}_q$ (Definition 3.5.3).*

(1) (Rationality) There is a polynomial $P \in 1 + T\mathbb{Z}[T]$ of degree $2g$ such that

$$Z(C/\mathbb{F}_q; T) = \frac{P(T)}{(1-T)(1-qT)}.$$

(2) (Functional equation)

$$Z\left(C/\mathbb{F}_q; \frac{1}{qT}\right) = q^{1-g}T^{2-2g}Z(C/\mathbb{F}_q; T).$$

(3) (Riemann hypothesis) Over $\mathbb{C}$,

$$P(T) = \prod_{i=1}^{2g}(1-\alpha_i T), \qquad |\alpha_i| = \sqrt{q} \quad \text{for every } i.$$

PROOF. Rationality and the functional equation are due to F. K. Schmidt, from the Riemann–Roch theorem (Theorem 3.5.4). The Riemann hypothesis is due to Weil [Wei49]. See Remark 5.3.6. We prove the theorem for elliptic curves in Theorem 5.3.9. □

Taking the logarithm of the formula for $Z(C/\mathbb{F}_q; T)$, as in Example 5.3.3, the α_i determine all the numbers $\#C(\mathbb{F}_{q^n})$:

$$\#C(\mathbb{F}_{q^n}) = q^n + 1 - \sum_{i=1}^{2g}\alpha_i^n, \qquad \text{so} \qquad \left|\#C(\mathbb{F}_{q^n}) - (q^n+1)\right| \leqslant 2g\, q^{n/2}.$$

For $C = \mathbb{P}^1$, of genus 0, Example 5.3.3 gives the theorem with $P = 1$.

REMARK 5.3.6 (A brief and incomplete history). F. K. Schmidt proved rationality and the functional equation for curves. Hasse proved the Riemann hypothesis for elliptic curves in the 1930s, and Weil proved it for all curves, and for abelian varieties, using intersection theory on surfaces [Wei49].
Weil's conjectures were guided by the Lefschetz fixed-point theorem. If $f\colon M \to M$ is a reasonable map of a compact manifold of dimension d, then the number of fixed points of f, counted with multiplicity, is

$$\sum_{i=0}^{d}(-1)^i \operatorname{tr}\left(f^* \mid H^i(M, \mathbb{Q})\right).$$

The set $C(\mathbb{F}_{q^n})$ is exactly the fixed-point set of the n-th power of the Frobenius morphism on $C(\overline{\mathbb{F}}_q)$. If C comes from a curve over $\mathbb{C}$, then H^0 and H^2 of the Riemann surface have rank 1 and H^1 has rank $2g$. So one expects cohomology groups that behave like these but exist in characteristic p, with F acting by 1 on H^0, by q on H^2, and with eigenvalues $\alpha_1, \dots, \alpha_{2g}$ on H^1. This gives the shape of Theorem 5.3.5.
Weil stated his conjectures for varieties of any dimension. Dwork proved rationality in 1960 using p-adic analysis [Dwo60]. The cohomology theory Weil was looking for is étale cohomology, developed by M. Artin, Grothendieck and others: its Grothendieck–Lefschetz trace formula proves every part of the conjectures except the Riemann hypothesis, which Deligne proved in 1973 [Del74]. See [Oor14].

5.3.3. Elliptic curves. We now prove Theorem 5.3.5 for elliptic curves. The result is Theorem 5.3.9. Let E be an elliptic curve over $\mathbb{F}_q$, let $\mathsf{F} \in \operatorname{End}(E)$ be its q-th power Frobenius endomorphism (Section 5.2), let $\ell \neq p$ be a prime, and let F_ℓ be the map induced by F on $T_\ell(E)$ (Section 4.5). Then $\deg \mathsf{F} = q$ by Proposition 3.2.12, and F^n is the q^n-th power Frobenius endomorphism. So the discussion after Corollary 4.4.4, applied to E over $\mathbb{F}_{q^n}$, gives

$$\#E(\mathbb{F}_{q^n}) = \deg(1 - \mathsf{F}^n). \tag{5.3.4}$$

DEFINITION 5.3.7. The trace of Frobenius of $E/\mathbb{F}_q$ is $a \coloneqq q + 1 - \#E(\mathbb{F}_q)$.

The name comes from Proposition 4.6.7 and (5.3.4): for every $\ell \neq p$,

$$\operatorname{tr} \mathsf{F}_\ell = 1 + \deg \mathsf{F} - \deg(1 - \mathsf{F}) = 1 + q - \#E(\mathbb{F}_q) = a,$$
$$\det \mathsf{F}_\ell = \deg \mathsf{F} = q$$

[Sil09, Remark V.2.6].

THEOREM 5.3.8. *Let $E/\mathbb{F}_q$ be an elliptic curve with trace of Frobenius a.*

(1) The characteristic polynomial of F_ℓ is $T^2 - aT + q$. Its complex roots α and β are complex conjugates with $|\alpha| = |\beta| = \sqrt{q}$, and for every $n \geqslant 1$,

$$\#E(\mathbb{F}_{q^n}) = q^n + 1 - \alpha^n - \beta^n. \tag{5.3.5}$$

(2) $\mathsf{F}^2 - a\mathsf{F} + q = 0$ in $\operatorname{End}(E)$.

PROOF. By the computation above, $\det(T - \mathsf{F}_\ell) = T^2 - \operatorname{tr}(\mathsf{F}_\ell)T + \det \mathsf{F}_\ell = T^2 - aT + q$.

For part 1, let $m/n \in \mathbb{Q}$. By Proposition 4.6.7 and Corollary 4.3.11,

$$\det\left(\frac{m}{n} - \mathsf{F}_\ell\right) = \frac{\det(m - n\mathsf{F}_\ell)}{n^2} = \frac{\deg(m - n\mathsf{F})}{n^2} \geqslant 0.$$

So the real quadratic polynomial $T^2 - aT + q$ is non-negative on $\mathbb{Q}$, hence on $\mathbb{R}$, and its discriminant satisfies $a^2 - 4q \leqslant 0$. Its roots are therefore complex conjugates or a repeated real root, so in either case $|\alpha|^2 = \alpha\beta = q$. (The inequality $a^2 \leqslant 4q$ is Hasse's bound, Theorem 5.2.1 with $r = 1$.)

Viewing α and β in an algebraic closure of $\mathbb{Q}_\ell$, we can triangularize F_ℓ with α and β on the diagonal. Then $(\mathsf{F}^n)_\ell = \mathsf{F}_\ell^n$ is triangular with α^n and β^n on the diagonal. By (5.3.4) and Proposition 4.6.7,

$$\#E(\mathbb{F}_{q^n}) = \deg(1 - \mathsf{F}^n) = \det(1 - \mathsf{F}_\ell^n) = (1 - \alpha^n)(1 - \beta^n) = 1 - \alpha^n - \beta^n + q^n.$$

Since $\alpha^n + \beta^n$ is a polynomial in $a = \alpha + \beta$ and $q = \alpha\beta$ with integer coefficients, it does not matter whether α and β are read in $\mathbb{C}$ or in $\overline{\mathbb{Q}}_\ell$.

For part 2, let $\psi := \mathsf{F}^2 - a\mathsf{F} + q \in \operatorname{End}(E)$. By the Cayley–Hamilton theorem, F_ℓ satisfies its characteristic polynomial, so $\psi_\ell = \mathsf{F}_\ell^2 - a\mathsf{F}_\ell + q = 0$. By Theorem 4.5.6, $\psi = 0$ [Sil09, Theorem V.2.3.1]. □

Theorem 5.3.8 recovers Hasse's bound over every extension at once: by (5.3.5),

$$|\#E(\mathbb{F}_{q^n}) - (q^n + 1)| = |\alpha^n + \beta^n| \leqslant |\alpha|^n + |\beta|^n = 2q^{n/2},$$

which is Theorem 5.2.1 with $r = n$.

THEOREM 5.3.9 (Weil conjectures for elliptic curves). *Let $E/\mathbb{F}_q$ be an elliptic curve with trace of Frobenius a, and let α and β be as in Theorem 5.3.8. Then*

$$Z(E/\mathbb{F}_q; T) = \frac{\det(1 - \mathsf{F}_\ell T)}{(1 - T)(1 - qT)} = \frac{1 - aT + qT^2}{(1 - T)(1 - qT)}, \tag{5.3.6}$$
$$1 - aT + qT^2 = (1 - \alpha T)(1 - \beta T), \qquad |\alpha| = |\beta| = \sqrt{q},$$

and

$$Z\left(E/\mathbb{F}_q; \frac{1}{qT}\right) = Z(E/\mathbb{F}_q; T).$$

PROOF. For a 2×2 matrix M, $\det(1 - M) = 1 - \operatorname{tr} M + \det M$. By (5.3.4) and Proposition 4.6.7, and since $\det \mathsf{F}_\ell^n = q^n$,

$$\#E(\mathbb{F}_{q^n}) = \deg(1 - \mathsf{F}^n) = \det(1 - \mathsf{F}_\ell^n) = 1 - \operatorname{tr}(\mathsf{F}_\ell^n) + q^n.$$

So, working in $\mathbb{Q}_\ell[[T]]$ and applying Lemma 5.3.4 to 1, q and F_ℓ,

$$Z(E/\mathbb{F}_q; T) = \exp\left(\sum_{n \geqslant 1} (1 + q^n)\frac{T^n}{n}\right) \exp\left(-\sum_{n \geqslant 1} \operatorname{tr}(\mathsf{F}_\ell^n)\frac{T^n}{n}\right) = \frac{\det(1 - \mathsf{F}_\ell T)}{(1 - T)(1 - qT)}.$$

By Theorem 5.3.8, $\det(1-\mathsf{F}_\ell T) = T^2\det(T^{-1}-\mathsf{F}_\ell) = 1-aT+qT^2 = (1-\alpha T)(1-\beta T)$, and $|\alpha| = |\beta| = \sqrt{q}$. For the functional equation, substitute $T \mapsto 1/(qT)$ and multiply numerator and denominator by qT^2:

$$\frac{1-a/(qT)+1/(qT^2)}{(1-1/(qT))(1-1/T)} = \frac{qT^2-aT+1}{(qT-1)(T-1)} = Z(E/\mathbb{F}_q;T)$$

[Sil09, Theorem V.2.4]. □

So E satisfies Theorem 5.3.5 with $g=1$ and $P(T) = 1-aT+qT^2 = \det(1-\mathsf{F}_\ell T)$.

COROLLARY 5.3.10. *Let $E/\mathbb{F}_q$ be an elliptic curve with trace of Frobenius a. Put $a_n := q^n+1-\#E(\mathbb{F}_{q^n})$ for $n \geqslant 1$, and $a_0 := 2$. Then $a_1 = a$ and*

$$a_{n+2} = a\,a_{n+1} - q\,a_n \qquad \text{for all } n \geqslant 0.$$

In particular, the single number $\#E(\mathbb{F}_q)$ determines $\#E(\mathbb{F}_{q^n})$ for every $n \geqslant 1$.

PROOF. By (5.3.5), $a_n = \alpha^n + \beta^n$ for $n \geqslant 1$, and also for $n = 0$. Since α is a root of $T^2 - aT + q$, $\alpha^{n+2} = a\alpha^{n+1} - q\alpha^n$, and likewise for β. Add the two [Sil09, Exercise 5.13]. □

For instance $a_2 = a^2 - 2q$, so $\#E(\mathbb{F}_{q^2}) = q^2 + 1 - a^2 + 2q$.

REMARK 5.3.11. To see why part 3 of Theorem 5.3.5 is called the Riemann hypothesis, set $T = q^{-s}$ as in Section 5.3.1:

$$\zeta_E(s) := Z(E/\mathbb{F}_q;q^{-s}) = \frac{1-aq^{-s}+q^{1-2s}}{(1-q^{-s})(1-q^{1-s})}.$$

The functional equation becomes $\zeta_E(s) = \zeta_E(1-s)$, and if $\zeta_E(s) = 0$, then $q^s \in \{\alpha,\beta\}$, so $|q^s| = \sqrt{q}$, that is, $\operatorname{Re} s = 1/2$ [Sil09, Remark V.2.5].

EXAMPLE 5.3.12. Let $E\colon y^2+y = x^3-x$ over $\mathbb{F}_2$, the reduction modulo 2 of the curve with LMFDB label 37.a1, which has good reduction away from 37 [LMFDB]. For $x = 0$ and $x = 1$ the equation reads $y^2+y = 0$, which has the two solutions $y = 0, 1$. With the point at infinity, $\#E(\mathbb{F}_2) = 5$, so $a = 2+1-5 = -2$ and

$$Z(E/\mathbb{F}_2;T) = \frac{1+2T+2T^2}{(1-T)(1-2T)}.$$

The roots of T^2+2T+2 are $\alpha,\beta = -1\pm i$, of absolute value $\sqrt{2}$. Since $\alpha^2 = -2i$, $\alpha^3 = 2+2i$ and $\alpha^4 = -4$, (5.3.5) predicts

$$\#E(\mathbb{F}_4) = 5, \qquad \#E(\mathbb{F}_8) = 9-4 = 5, \qquad \#E(\mathbb{F}_{16}) = 17+8 = 25.$$

Magma agrees [BCP97]. It writes the denominator of the zeta function expanded.

```
E := EllipticCurve([GF(2) | 0, 0, 1, -1, 0]);
TraceOfFrobenius(E);                        // -2
[#BaseChange(E, GF(2^n)) : n in [1..4]];    // [ 5, 5, 5, 25 ]
Z := ZetaFunction(E); K<T> := Parent(Z); Z;
// (2*T^2 + 2*T + 1)/(2*T^2 - 3*T + 1)
```

EXAMPLE 5.3.13. Let $E\colon y^2 = x^3+x$ over $\mathbb{F}_3$. The values of x^3+x at $x = 0,1,2$ are $0,2,1$, and 2 is not a square in $\mathbb{F}_3$, so $E(\mathbb{F}_3) = \{O,(0,0),(2,1),(2,2)\}$. Hence $a = 0$,

$$Z(E/\mathbb{F}_3;T) = \frac{1+3T^2}{(1-T)(1-3T)},$$

$\alpha,\beta = \pm i\sqrt{3}$, and $\#E(\mathbb{F}_{3^n}) = 3^n+1$ for n odd and $3^n+1-2(-3)^{n/2}$ for n even. Part 2 of Theorem 5.3.8 says that $\mathsf{F}^2 = [-3]$: applying the 9-th power map to the coordinates of a point of $E(\overline{\mathbb{F}}_3)$ is the same as multiplying it by -3. We check this on a random point of $E(\mathbb{F}_{3^5})$.

```
E := EllipticCurve([GF(3) | 1, 0]);
```

```
TraceOfFrobenius(E);                        // 0
[#BaseChange(E, GF(3^n)) : n in [1..4]];    // [ 4, 16, 28, 64 ]
E5 := BaseChange(E, GF(3^5));
P := Random(E5);
E5![P[1]^9, P[2]^9, P[3]^9] eq -3*P;        // true
```

By Corollary 5.5.7, an elliptic curve over $\mathbb{F}_q$ is supersingular if and only if $a \equiv 0 \pmod{p}$. So this curve is supersingular.

EXAMPLE 5.3.14. Let E be 37.a1 again, now reduced modulo the prime $q = 1013$: $E\colon y^2 + y = x^3 + 1012x$ over $\mathbb{F}_{1013}$. Magma computes $\#E(\mathbb{F}_{1013}) = 978$, so $a = 36$ and

$$Z(E/\mathbb{F}_{1013}; T) = \frac{1 - 36T + 1013T^2}{(1-T)(1-1013T)}.$$

The roots of $T^2 - 36T + 1013$ are $18 \pm \sqrt{689}\,i \approx 18 \pm 26.2488i$, of absolute value $\sqrt{1013} \approx 31.8277$. Then Corollary 5.3.10 predicts $\#E(\mathbb{F}_{1013^2}) = q^2 + 1 - (a^2 - 2q) = 1026900$.

```
q := 1013;
E := EllipticCurve([GF(q) | 0, 0, 1, -1, 0]);
#E, TraceOfFrobenius(E);                    // 978 36
#BaseChange(E, GF(q^2));                    // 1026900
Z := ZetaFunction(E); K<T> := Parent(Z); Z;
// (1013*T^2 - 36*T + 1)/(1013*T^2 - 1014*T + 1)
```

PROPOSITION 5.3.15. *Let E_1 and E_2 be elliptic curves over $\mathbb{F}_q$. The following are equivalent:*
(1) E_1 and E_2 are isogenous over $\mathbb{F}_q$.
(2) $\#E_1(\mathbb{F}_q) = \#E_2(\mathbb{F}_q)$.
(3) $Z(E_1/\mathbb{F}_q; T) = Z(E_2/\mathbb{F}_q; T)$.
(4) E_1 and E_2 have the same number of closed points of each degree.

PROOF. (See [Sil09, Exercise 5.4].) Let F_1 and F_2 be the q-th power Frobenius endomorphisms of E_1 and E_2. Parts 2, 3 and 4 of Proposition 5.3.15 are equivalent by Theorem 5.3.9 (the zeta function depends only on q and a) and by parts 1 and 3 of Proposition 5.3.2.

Suppose that $\varphi\colon E_1 \to E_2$ is a non-zero isogeny over $\mathbb{F}_q$. Its coordinate functions have coefficients in $\mathbb{F}_q$, which the q-th power map fixes, so $\varphi \circ \mathsf{F}_1 = \mathsf{F}_2 \circ \varphi$, and hence $\varphi \circ (1 - \mathsf{F}_1) = (1 - \mathsf{F}_2) \circ \varphi$. Taking degrees and cancelling $\deg \varphi \neq 0$, (5.3.4) gives $\#E_1(\mathbb{F}_q) = \#E_2(\mathbb{F}_q)$.

Conversely, suppose that $\#E_1(\mathbb{F}_q) = \#E_2(\mathbb{F}_q)$, so that $\mathsf{F}_{1,\ell}$ and $\mathsf{F}_{2,\ell}$ have the same characteristic polynomial $f(T) = T^2 - aT + q$. By Tate's theorem, mentioned after Theorem 4.5.6, the map $\mathrm{Hom}(E_1, E_2) \otimes \mathbb{Z}_\ell \to \mathrm{Hom}_{\mathrm{Gal}_{\mathbb{F}_q}}(T_\ell(E_1), T_\ell(E_2))$ is an isomorphism [Sil09, Theorem III.7.7(a)]. Since σ acts on $E_i(\overline{\mathbb{F}}_q)$ as F_i and topologically generates $\mathrm{Gal}_{\mathbb{F}_q}$, a $\mathbb{Z}_\ell$-linear map is $\mathrm{Gal}_{\mathbb{F}_q}$-equivariant if and only if it intertwines $\mathsf{F}_{1,\ell}$ and $\mathsf{F}_{2,\ell}$. It suffices to find a non-zero such map, since then $\mathrm{Hom}(E_1, E_2) \otimes \mathbb{Z}_\ell \neq 0$, and $\mathrm{Hom}(E_1, E_2)$ is a free $\mathbb{Z}$-module (Theorem 4.5.6), so it is non-zero.

- If $a^2 \neq 4q$, then $\mathsf{F}_{i,\ell}$ is not a scalar, so $V_i := T_\ell(E_i) \otimes \mathbb{Q}_\ell$ has a basis $v, \mathsf{F}_{i,\ell}v$ and $V_i \cong \mathbb{Q}_\ell[T]/(f)$, with T acting by $\mathsf{F}_{i,\ell}$. An isomorphism $V_1 \cong V_2$ of $\mathbb{Q}_\ell[T]$-modules, multiplied by a large power of ℓ, maps $T_\ell(E_1)$ into $T_\ell(E_2)$.
- If $a^2 = 4q$, then $f(T) = (T - a/2)^2$ with $a/2 \in \mathbb{Z}$, and by Proposition 4.6.7, $\deg(\mathsf{F}_i - [a/2]) = \det(\mathsf{F}_{i,\ell} - a/2) = f(a/2) = 0$. So $\mathsf{F}_i = [a/2]$, and every $\mathbb{Z}_\ell$-linear map intertwines the two.

□

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5.4. The endomorphism ring

In this section and the next, k is a field of characteristic $p > 0$, not necessarily finite, and E is an elliptic curve over k. We use the Frobenius isogeny $\mathsf{F}^r\colon E \to E^{(p^r)}$ and its Verschiebung $\mathsf{V}^r = \widehat{\mathsf{F}^r}$ of Definition 4.3.12.

Over $\mathbb{F}_q$ with $q = p^e$ the two readings agree: $E^{(p^e)} = E$, and the q-th power Frobenius endomorphism is F^e, with dual V^e.

Corollary 4.5.1 left two possibilities for $E[p^r](\overline{k})$, and Theorem 4.7.8 three possibilities for $\mathrm{End}(E_{\overline{k}})$. The following theorem of Deuring says that these are not independent: the p-torsion decides the endomorphism ring.

THEOREM 5.4.1 (Supersingular and ordinary elliptic curves). *Let k be a field of characteristic p, and let E be an elliptic curve over k. The following conditions are equivalent.*

(1) $E[p^r](\overline{k}) = 0$ for some (all) $r \geqslant 1$.
(2) V^r is purely inseparable for some (all) $r \geqslant 1$.
(3) $[p]\colon E \to E$ is purely inseparable and $j(E) \in \mathbb{F}_{p^2}$.
(4) $\mathrm{End}(E_{\overline{k}})$ is an order in a quaternion algebra.
(5) The formal group $\hat{E}$ has height 2.

If they hold, E is supersingular. Otherwise E is ordinary, and then

$$E[p^r](\overline{k}) \cong \mathbb{Z}/p^r\mathbb{Z} \qquad \text{for all } r \geqslant 1;$$

if moreover $j(E) \in \overline{\mathbb{F}}_p$, then $\mathrm{End}(E_{\overline{k}})$ is an order in an imaginary quadratic field.

PROOF. This is [Sil09, Theorem V.3.1], due to Deuring [Deu41]. Two of the conditions are already familiar. For conditions 1 and 2, the proof of Corollary 4.5.1 gives $\#E[p^r](\overline{k}) = \deg_s[p^r] = \deg_s \mathsf{V}^r$, because F^r is purely inseparable and $[p^r] = \mathsf{V}^r \circ \mathsf{F}^r$. So the p^r-torsion vanishes exactly when V^r has separable degree 1. Condition 4 is case 3 of Theorem 4.7.8, and the ordinary case is case 2. Compare Remark 4.7.12. □

By condition 3 of Theorem 5.4.1 there are only finitely many supersingular elliptic curves over $\overline{\mathbb{F}}_p$ up to isomorphism, since each has j-invariant in the finite field $\mathbb{F}_{p^2}$. Counting them is the subject of Theorem 5.5.10.

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5.5. The Cartier operator and the Hasse invariant

None of the conditions of Theorem 5.4.1 is easy to check on a given Weierstrass equation. The Cartier operator turns the dichotomy into the vanishing of a single coefficient.

SITUATION 5.5.1. In this section k is a **perfect** field of characteristic $p > 0$, not necessarily finite, so that $a \mapsto a^p$ is an automorphism of k and we may write $a^{1/p}$ for its inverse.

Perfectness is also what makes the definition below work at every closed point: it guarantees $dt \neq 0$ for a uniformizer t, which fails over an imperfect field (Warning 3.4.5).

DEFINITION 5.5.2. In Situation 5.5.1, let C be a nice curve of genus $g \geqslant 1$ over k, let P be a closed point of C and let t be a uniformizer at P. The functions $1, t, \dots, t^{p-1}$ form a p-basis of $\mathcal{O}_{C,P}$: every differential regular at P is uniquely

$$\omega = \left(\sum_{j=0}^{p-1} f_j^p\, t^j\right) dt, \qquad f_j \in \mathcal{O}_{C,P}.$$

The Cartier operator is

$$\mathcal{C}\colon \Omega_{k(C)/k} \to \Omega_{k(C)/k}, \qquad \mathcal{C}(\omega) \coloneqq f_{p-1}\, dt.$$

The value $\mathcal{C}(\omega)$ does not depend on the uniformizer, the local definitions agree on overlaps, and the result extends from differentials regular at P to all of $\Omega_{k(C)/k}$. See [AH17, Section 2.2], which follows [Ser58, Section 10]. The operator is due to Cartier [Car57]. There is a second description, the one used in computations: if

$x \in k(C)$ is a separating element, so that dx is a basis of $\Omega_{k(C)/k}$ (Proposition 3.4.2), then every differential is uniquely

$$\omega = \left(\sum_{j=0}^{p-1} g_j^p \, x^j \right) dx, \qquad g_j \in k(C), \qquad \text{and} \qquad \mathcal{C}(\omega) = g_{p-1}\, dx. \tag{5.5.1}$$

PROPOSITION 5.5.3. *In Situation 5.5.1, let C be a nice curve of genus $g \geqslant 1$ over k.*

(1) $\mathcal{C}(\omega_1 + \omega_2) = \mathcal{C}(\omega_1) + \mathcal{C}(\omega_2)$.

(2) $\mathcal{C}(f^p \omega) = f\, \mathcal{C}(\omega)$ for $f \in k(C)$. In particular $\mathcal{C}(c\,\omega) = c^{1/p}\, \mathcal{C}(\omega)$ for $c \in k$, so $\mathcal{C}$ is semilinear for the inverse of the p-th power map on k.

(3) $\mathcal{C}(df) = 0$ for $f \in k(C)$.

(4) If ω is regular, then so is $\mathcal{C}(\omega)$. So $\mathcal{C}$ restricts to an operator on the space of regular differentials of C, which has dimension g.

PROOF. Parts 1 to 3 are immediate from (5.5.1), and part 4 is the statement that $\mathcal{C}$ acts on $H^0(C, \Omega_{C/k})$. See [AH17, Section 2.2] and [Ser58, Section 10]. Dividing by a fixed non-zero regular differential identifies the space of regular differentials with $\mathcal{L}(K_C)$, whose dimension is the genus (Definition 3.5.3). □

Note that the semilinearity in part 2 is for the inverse of the p-th power map, not for the Frobenius automorphism $\sigma\colon x \mapsto x^q$ of Section 5.1. The two agree only when $q = p$.

For an elliptic curve the space of regular differentials is one-dimensional, spanned by the invariant differential ω, so $\mathcal{C}$ acts on it by a single scalar. That scalar is the Hasse invariant.

DEFINITION 5.5.4. Let q be a power of $p \geqslant 3$ and let E be an elliptic curve over $\mathbb{F}_q$, given by $y^2 = f(x)$ with $f \in \mathbb{F}_q[x]$ a cubic with distinct roots in $\overline{\mathbb{F}}_q$ (Lemma 4.1.4), with invariant differential $\omega = dx/(2y)$. Write

$$f(x)^{(p-1)/2} = \sum_{m \geqslant 0} c_m x^m.$$

The Hasse invariant of the equation is $A := c_{p-1}$, the coefficient of x^{p-1} in $f(x)^{(p-1)/2}$.

The Hasse invariant is exactly the scalar by which the Cartier operator acts. Indeed, using $y^2 = f(x)$,

$$\frac{dx}{y} = \frac{(y^2)^{(p-1)/2}}{y^p}\, dx = y^{-p} f(x)^{(p-1)/2}\, dx = y^{-p} \left(\sum_{m \geqslant 0} c_m x^m \right) dx,$$

so part 2 of Proposition 5.5.3 and (5.5.1) give

$$\mathcal{C}\left(\frac{dx}{y}\right) = \frac{1}{y}\, \mathcal{C}\left(\sum_{m \geqslant 0} c_m x^m\, dx \right) = A^{1/p}\, \frac{dx}{y}. \tag{5.5.2}$$

Only the terms with $m \equiv p-1 \pmod p$ survive, and $f^{(p-1)/2}$ has degree $3(p-1)/2 < 2p-1$, so $m = p-1$ is the only one. Since $2 \in \mathbb{F}_p$ is fixed by $c \mapsto c^{1/p}$, the same scalar appears for $\omega = dx/(2y)$: $\mathcal{C}(\omega) = A^{1/p}\, \omega$. This is the genus-1 case of the computation in [AH17, Section 3].

REMARK 5.5.5. Three warnings. First, A depends on the equation and not only on E: the change of coordinates $(x, y) \mapsto (u^2 x, u^3 y)$ of (4.1.3) replaces A by $u^{-(p-1)} A$. What is intrinsic is whether A vanishes, equivalently whether $\mathcal{C} = 0$ on the regular differentials of E. Second, the definition asks for $p \geqslant 3$, although the Cartier operator itself does not. Over $\overline{\mathbb{F}}_2$ the only supersingular elliptic curve is $y^2 + y = x^3$ [Sil09, Section V.4]. Third, Silverman [Sil09] uses "Hasse invariant" for the binary invariant of Theorem 5.4.1, which is 0 for a supersingular curve and 1 for an ordinary one. Here it is the scalar A attached to an equation.

The proof that A detects supersingularity counts points with the quadratic character.

LEMMA 5.5.6. *Let q be a power of $p \geqslant 3$, let $\chi\colon \mathbb{F}_q^\times \to \{\pm 1\}$ be the unique character of order 2, extended by $\chi(0) \coloneqq 0$, and let $E/\mathbb{F}_q$ be given by $y^2 = f(x)$ as in Definition 5.5.4. Then*

$$\#E(\mathbb{F}_q) = 1 + q + \sum_{x \in \mathbb{F}_q} \chi(f(x)). \tag{5.5.3}$$

PROOF. For a fixed $x \in \mathbb{F}_q$, the number of $y \in \mathbb{F}_q$ with $y^2 = f(x)$ is 2 if $f(x)$ is a non-zero square, 1 if $f(x) = 0$ and 0 otherwise, that is, $1 + \chi(f(x))$. Summing over the q values of x and adding the point at infinity gives (5.5.3) [Sil09, Application V.1.3]. □

COROLLARY 5.5.7. *Let E be an elliptic curve over $\mathbb{F}_q$, $q = p^e$, with trace of Frobenius a (Definition 5.3.7). Then E is supersingular if and only if $a \equiv 0 \pmod{p}$.*

PROOF. Write $\mathsf{F} \in \operatorname{End}(E)$ for the q-th power Frobenius endomorphism. It is the e-fold composite of the p-th power Frobenius, so its dual is V^e in the notation of Definition 4.3.12. By part 2 of Theorem 5.3.8 and $\widehat{\mathsf{F}} \circ \mathsf{F} = [q]$,

$$\left(\mathsf{F} - [a] + \widehat{\mathsf{F}}\right) \circ \mathsf{F} = \mathsf{F}^2 - a\mathsf{F} + q = 0,$$

and $\operatorname{End}(E)$ has no zero divisors (Proposition 4.3.7) with $\mathsf{F} \neq 0$, so $\mathsf{V}^e = \widehat{\mathsf{F}} = [a] - \mathsf{F}$. By Corollary 4.4.4, $[a] - \mathsf{F}$ is separable if and only if $p \nmid a$. Finally $\deg_s \mathsf{V}^e = \#E[p^e](\overline{k}) \in \{1, p^e\}$ by part 2 of Corollary 4.5.1, so V^e is inseparable if and only if it is purely inseparable, and part 2 of Theorem 5.4.1, with $r = e$, finishes the proof [Sil09, Exercise 5.10]. □

THEOREM 5.5.8. *Let q be a power of $p \geqslant 3$ and let E be an elliptic curve over $\mathbb{F}_q$, given by $y^2 = f(x)$ as in Definition 5.5.4, with Hasse invariant A. For $r \geqslant 1$, let A_{p^r} be the coefficient of x^{p^r-1} in $f(x)^{(p^r-1)/2}$, so that $A_p = A$. The following are equivalent.*

(1) E is supersingular.
(2) $A_q = 0$.
(3) $A = 0$.
(4) $\mathcal{C} = 0$ on the regular differentials of E.

PROOF. This is [Sil09, Theorem V.4.1(a)]. The argument is Manin's [Man61].

Since $\mathbb{F}_q^\times$ is cyclic of order $q-1$, we have $\chi(z) = z^{(q-1)/2}$ in $\mathbb{F}_q$ for every $z \in \mathbb{F}_q$, including $z = 0$. So Lemma 5.5.6 reads

$$\#E(\mathbb{F}_q) = 1 + \sum_{x \in \mathbb{F}_q} f(x)^{(q-1)/2} \qquad \text{in } \mathbb{F}_q,$$

using $q = 0$ in $\mathbb{F}_q$. Again because $\mathbb{F}_q^\times$ is cyclic, $\sum_{x \in \mathbb{F}_q} x^n$ is -1 when $n > 0$ and $q - 1 \mid n$, and 0 otherwise. The polynomial $f(x)^{(q-1)/2}$ has degree $3(q-1)/2 < 2(q-1)$, so the only surviving term is the one with $n = q - 1$, and $\#E(\mathbb{F}_q) = 1 - A_q$ in $\mathbb{F}_q$. Comparing with $\#E(\mathbb{F}_q) = q + 1 - a$ gives $a \equiv A_q \pmod{p}$, so conditions 1 and 2 are equivalent by Corollary 5.5.7.

For conditions 2 and 3, write

$$f(x)^{(p^{r+1}-1)/2} = \left(f(x)^{(p^r-1)/2}\right)^p f(x)^{(p-1)/2}.$$

A term of the right-hand side contributing to $x^{p^{r+1}-1}$ is x^{pn} times x^m with $pn + m = p^{r+1} - 1$, so $m \equiv p - 1 \pmod{p}$. Since $m \leqslant 3(p-1)/2 < 2p - 1$, the only possibility is $m = p - 1$, and then $n = p^r - 1$. Hence $A_{p^{r+1}} = A_{p^r}^p A$, and by induction $A_q = 0$ if and only if $A = 0$.

For conditions 3 and 4, the regular differentials of E are the $\mathbb{F}_q$-multiples of ω, and $\mathcal{C}(\omega) = A^{1/p}\omega$ by (5.5.2). □

EXAMPLE 5.5.9. Let $E\colon y^2 = x^3 - x$, and write $m \coloneqq (p-1)/2$. Then $f^{(p-1)/2} = x^m(x^2-1)^m$, whose exponents are all congruent to m modulo 2. So the coefficient A of $x^{p-1} = x^{2m}$ vanishes unless m is even, that is,

unless $p \equiv 1 \pmod 4$. And when $p \equiv 1 \pmod 4$ the one surviving term gives $A = (-1)^{m/2}\binom{m}{m/2}$, which is non-zero in $\mathbb{F}_p$ because $\binom{m}{m/2}$ is a quotient of factorials of integers smaller than p. Hence E is supersingular exactly when $p \equiv 3 \pmod 4$. For $p = 11$ we get $A = 0$, and for $p = 13$, $A = (-1)^3\binom{6}{3} = -20 = 6$ in $\mathbb{F}_{13}$. Magma agrees [BCP97], and lists the supersingular j-invariants of each characteristic. Note $1728 = 1$ in $\mathbb{F}_{11}$.

```
for p in [11, 13] do
  P<x> := PolynomialRing(GF(p));
  f := x^3 - x;
  E := EllipticCurve([GF(p) | -1, 0]);
  A := Coefficient(f^((p-1) div 2), p-1);
  print p, A, IsSupersingular(E);
  print [ j : j in GF(p^2) | IsSupersingular(EllipticCurveFromjInvariant(j)) ];
end for;
// 11 0 true
// [ 0, 1 ]
// 13 6 false
// [ 5 ]
```

Both lists are as short as part 3 of Theorem 5.4.1 promises, and they satisfy the mass formula of Theorem 5.5.10: in characteristic 11 the two supersingular curves have $j = 0$ and $j = 1728$, so $\#\operatorname{Aut}(E)$ is 6 and 4 by Theorem 4.7.11, and $1/6 + 1/4 = 5/12 = (11-1)/24$; in characteristic 13 the only one has $j = 5$, with $\#\operatorname{Aut}(E) = 2$ and $1/2 = (13-1)/24$.

We close with a count of the supersingular curves, weighted by their automorphisms.

THEOREM 5.5.10 (Mass formula). *Let p be a prime. Then*

$$\sum_E \frac{1}{\#\operatorname{Aut}(E)} = \frac{p-1}{24}, \tag{5.5.4}$$

where E runs over the isomorphism classes of supersingular elliptic curves over $\overline{\mathbb{F}}_p$.

PROOF. Due to Deuring [Deu41]. See [Voi21, (42.3.9)] and [Sil09, Exercise 5.9]. Remark 5.5.11 explains what a proof needs. □

REMARK 5.5.11. The mass formula does not follow from Theorems 5.4.1 and 5.5.8 alone: those say which curves are supersingular, not how many there are. Two routes to (5.5.4) are worth recording.

The first is analytic, and is the one that explains the denominator. By part condition 4 of Theorem 5.4.1 the endomorphism ring of a supersingular curve is an order in a quaternion algebra, and Deuring's correspondence [Voi21, Corollary 42.3.7] makes this precise: the supersingular elliptic curves over $\overline{\mathbb{F}}_p$, up to isomorphism, are in bijection with the class set of a maximal order $\mathcal{O}$ in the definite quaternion algebra over $\mathbb{Q}$ ramified exactly at p and ∞, and under the bijection $\operatorname{Aut}(E)$ is the unit group of the corresponding order. The count on the quaternionic side is the Eichler mass formula [Voi21, Theorem 25.1.1]: for a maximal order in a definite quaternion algebra over $\mathbb{Q}$ of discriminant D,

$$\sum_{[J]} \frac{1}{w_J} = \frac{\varphi(D)}{12}, \qquad w_J := \#\left(\mathcal{O}_L(J)^\times/\{\pm 1\}\right),$$

which for $D = p$ gives $(p-1)/12$. The quotient by $\{\pm 1\}$ accounts for the factor 2 between that and (5.5.4). The Eichler mass formula is proved by analytic means, from the analytic class number formula and the zeta function of the order [Voi21, Main Theorem 26.1.5 and Section 26.2]. Over $\mathbb{Q}$ it goes back to Hey and to Brandt.

The second route is a count. The supersingular j-invariants in characteristic p are the roots of an explicit polynomial, and there are $\lfloor p/12 \rfloor + \varepsilon$ of them, with $\varepsilon \in \{0, 1, 2\}$ depending on p mod 12 [Sil09, Theorem V.4.1]. This is due to Igusa [Igu58]. Weighting each by $\#\operatorname{Aut}(E)$, which is 2, 4 or 6 according as $j(E) \neq 0, 1728$,

$j(E) = 1728$ or $j(E) = 0$ when $p \geqslant 5$ (Theorem 4.7.11), and checking $p = 2, 3$ by hand, gives (5.5.4) case by case.

CHAPTER 6

Elliptic curves over local fields

Sources. This chapter follows [Sil09, Chapter VII]. For the reduction types we also follow [Poo08, Section 6] and [Mil20, Section II.3]. Silverman's Section VII.4 is merged here with its converse, Section VII.7. The last section, on the Kodaira–Néron classification and Tate's algorithm, follows [Sil94, Chapter IV], [Liu02, Section 10.2] and [Sil09, Section C.15], with [Mil20, Section II.7] for the motivation.

A number field is a finite extension of $\mathbb{Q}$. A global function field is a finite extension of $\mathbb{F}_p(t)$ for some prime p or, equivalently, it is the function field $k(C)$ of a geometrically integral curve C defined over a finite field $\mathbb{F}_q$. We say that k is a global field if it is either a number field or a global function field.

For k a global field, we let $\mathcal{P}_k$ denote the set of places of k: equivalence classes of non-trivial absolute values on k. For $v \in \mathcal{P}_k$, we denote by k_v the completion of k at v. A local field is the completion of a global field at a place.

Remark 6.0.1. A place v need not come from a discrete valuation.

- If k is a number field, the archimedean places correspond to the real embeddings and the pairs of complex conjugate embeddings of k, and for them k_v is $\mathbb{R}$ or $\mathbb{C}$. There v is an absolute value but not a valuation, and k_v has no ring of integers in the sense below.
- The non-archimedean places of a number field correspond to the non-zero prime ideals $\mathfrak{p}$ of its ring of integers. A global function field has only non-archimedean places, and they correspond to the closed points of the nice curve with function field k.

In the non-archimedean case we write $v\colon k_v^\times \to \mathbb{Z}$ for the normalized discrete valuation of the place, extended by $v(0) = \infty$. Then k_v is complete for v and its residue field is finite.

Situation 6.0.2.

- k is a global field.
- $v \in \mathcal{P}_k$ is a non-archimedean place, identified with its normalized discrete valuation $v\colon k_v^\times \to \mathbb{Z}$.
- k_v is the local field, the completion of k at v.
- $\mathcal{O}_v \coloneqq \{x \in k_v : v(x) \geqslant 0\}$ is the ring of integers of k_v.
- $\mathcal{O}_v^\times \coloneqq \{x \in k_v : v(x) = 0\}$ is the unit group of $\mathcal{O}_v$.
- $\mathfrak{m}_v \coloneqq \{x \in k_v : v(x) > 0\}$ is the maximal ideal of $\mathcal{O}_v$.
- π_v is a uniformizer for $\mathfrak{m}_v$, that is, a generator $\mathfrak{m}_v = \pi_v \mathcal{O}_v$. Since v is normalized, $v(\pi_v) = 1$.
- $\kappa(v) \coloneqq \mathcal{O}_v/\mathfrak{m}_v$ is the residue field, with $q \coloneqq \#\kappa(v)$ elements and characteristic $p \coloneqq \operatorname{char} \kappa(v)$.

The residue field $\kappa(v)$ is finite, hence perfect. This is not an extra hypothesis. In particular p is positive. The characteristic of k is arbitrary: it is 0 or p.

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6.1. Minimal Weierstrass equations

We are in Situation 6.0.2. An elliptic curve E over k_v has a Weierstrass equation

$$y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$$

by part 1 of Theorem 4.1.6, which we encode by $(a_1, a_2, a_3, a_4, a_6) \in k_v^5$.

Given $u \in k_v^\times$, substituting $u^{-2}x$ for x and $u^{-3}y$ for y and multiplying by u^6 yields another Weierstrass equation for E with coefficients

$$(ua_1, u^2a_2, u^3a_3, u^4a_4, u^6a_6).$$

This is the change of variables (4.1.3) with parameters $(u^{-1}, 0, 0, 0)$, and (4.1.4) gives the same coefficients.

So, choosing $u \in k_v^\times$ with

$$v(u^i a_i) = v(u^i) + v(a_i) = i\,v(u) + v(a_i) \geqslant 0, \quad \iff \quad v(u) \geqslant -v(a_i)/i,$$

for each $i \in \{1,2,3,4,6\}$ with $a_i \neq 0$, for instance $u = \pi_v^n$ with n large, ensures that

$$(ua_1, u^2a_2, u^3a_3, u^4a_4, u^6a_6) \in \mathcal{O}_v^5.$$

In particular, $v(\Delta(ua_1, \ldots, u^6a_6)) \geqslant 0$, since the discriminant is a polynomial with integer coefficients in the coefficients of the equation (Definition 4.1.2).

So integrality costs nothing, and by itself it says little. The same substitution with $u = \pi_v$ keeps the coefficients integral, multiplies Δ by π_v^{12}, and pushes the equation modulo π_v towards $y^2 = x^3$: a curve can look singular modulo π_v only because its equation was scaled too far (Example 6.2.3). The useful integral equations are the least scaled ones, in the same way that the useful homogeneous coordinates of a k_v-point are those in $\mathcal{O}_v$ not all in $\mathfrak{m}_v$. Minimizing $v(\Delta)$ makes this precise, and Proposition 6.1.2 shows that what is left is a change of variables over $\mathcal{O}_v$, which still makes sense modulo π_v.

Definition 6.1.1. A Weierstrass equation $(a_1, a_2, a_3, a_4, a_6) \in k_v^5$ for E is called v-minimal if its coefficients lie in $\mathcal{O}_v$ and $v(\Delta(a_1, \ldots, a_6))$ is minimal among all Weierstrass equations for E with coefficients in $\mathcal{O}_v$. This minimal value of $v(\Delta)$ is the valuation of the minimal discriminant of E at v.

The following is [Sil09, Proposition VII.1.3]. It uses the transformation formulas (4.1.4) of Proposition 4.1.7.

Proposition 6.1.2. *Let E be an elliptic curve over k_v.*

(1) E has a v-minimal Weierstrass equation.

(2) If a Weierstrass equation for E has coefficients in $\mathcal{O}_v$, then any change of variables (4.1.3) taking it to a v-minimal Weierstrass equation has $u, r, s, t \in \mathcal{O}_v$.

(3) Any two v-minimal Weierstrass equations for E are related by a change of variables (4.1.3) with $u \in \mathcal{O}_v^\times$ and $r, s, t \in \mathcal{O}_v$.

(4) The invariant differential ω of a v-minimal Weierstrass equation for E is unique up to multiplication by an element of $\mathcal{O}_v^\times$.

Proof. For part 1, the discussion before Definition 6.1.1 gives a Weierstrass equation for E with coefficients in $\mathcal{O}_v$. For every such equation $v(\Delta)$ lies in $\mathbb{Z}_{\geqslant 0}$, because v is discrete and $\Delta \in \mathcal{O}_v$, so the minimum is attained.

For part 2, let the unprimed equation have coefficients in $\mathcal{O}_v$ and let the primed one, obtained by (4.1.3), be v-minimal. By minimality $v(\Delta') \leqslant v(\Delta)$, and $u^{12}\Delta' = \Delta$ by Proposition 4.1.7, so $12\,v(u) = v(\Delta) - v(\Delta') \geqslant 0$ and $u \in \mathcal{O}_v$. All the a_i, a_i', b_i and b_i' lie in $\mathcal{O}_v$, since the b_i are polynomials with integer coefficients in the a_i. By the b_6 and b_8 lines of (4.1.4), r is a root of

$$P_6(T) \coloneqq 4T^3 + b_2T^2 + 2b_4T + b_6 - u^6b_6',$$
$$P_8(T) \coloneqq 3T^4 + b_2T^3 + 3b_4T^2 + 3b_6T + b_8 - u^8b_8',$$

hence of the monic polynomial

$$T\,P_6(T) - P_8(T) = T^4 - b_4T^2 - (2b_6 + u^6b_6')\,T - (b_8 - u^8b_8')$$

with coefficients in $\mathcal{O}_v$. So r is integral over $\mathcal{O}_v$, and $r \in \mathcal{O}_v$ because a discrete valuation ring is integrally closed. Next, the a_2 line of (4.1.4) says that s is a root of the monic polynomial

$$T^2 + a_1T - (a_2 + 3r - u^2a_2'),$$

whose coefficients are now in $\mathcal{O}_v$, so $s \in \mathcal{O}_v$. Finally, the a_6 line says that t is a root of the monic polynomial

$$T^2 + (a_3 + ra_1)\,T - (a_6 + ra_4 + r^2a_2 + r^3 - u^6a_6'),$$

so $t \in \mathcal{O}_v$.

For part 3, two Weierstrass equations for E are related by some change of variables (4.1.3) with $u \in k_v^\times$ and $r, s, t \in k_v$, by part 2 of Theorem 4.1.6. If both are v-minimal, the first has coefficients in $\mathcal{O}_v$, so $u, r, s, t \in \mathcal{O}_v$ by part 2. Moreover $v(\Delta) = v(\Delta')$, so $v(u) = 0$ and $u \in \mathcal{O}_v^\times$.

Part 4 follows from part 3, since $\omega = u^{-1}\omega'$ by Proposition 4.1.7. □

DRAFT
13 Oct 2026

6.2. Reduction modulo π

We are in Situation 6.0.2. The scheme $\operatorname{Spec} \mathcal{O}_v$ has two prime ideals: the generic point (0), with residue field k_v, and the special point $\mathfrak{m}_v$, with residue field $\kappa(v)$, which is its only closed point. The following definitions are [Poo17, Definitions 3.2.6 and 3.2.8].

DEFINITION 6.2.1. Let X be a k_v-scheme. An integral model of X is an $\mathcal{O}_v$-scheme $\mathcal{X}$ together with an isomorphism of k_v-schemes $\mathcal{X}_{k_v} := \mathcal{X} \times_{\operatorname{Spec} \mathcal{O}_v} \operatorname{Spec} k_v \to X$. The k_v-scheme $\mathcal{X}_{k_v}$ is the generic fiber of $\mathcal{X}$, and the $\kappa(v)$-scheme

$$\mathcal{X}_\pi := \mathcal{X} \times_{\operatorname{Spec} \mathcal{O}_v} \operatorname{Spec} \kappa(v)$$

is its special fiber.

The subscript π is only notation: the special fiber depends on $\mathfrak{m}_v$, not on the choice of the uniformizer π_v. As it stands, the definition allows useless models: X itself, viewed as an $\mathcal{O}_v$-scheme, is an integral model of X with empty special fiber. The models we use are the following.

Let E be an elliptic curve over k_v. The previous section produced Weierstrass equations for E with coefficients in $\mathcal{O}_v$, and each of them defines an integral model of E [Poo17, Section 5.7.6].

DEFINITION 6.2.2. Let $(a_1, a_2, a_3, a_4, a_6) \in \mathcal{O}_v^5$ be a Weierstrass equation for E. Its Weierstrass model is the closed subscheme $\mathcal{W} \subseteq \mathbb{P}^2_{\mathcal{O}_v}$ cut out by

$$Y^2Z + a_1XYZ + a_3YZ^2 = X^3 + a_2X^2Z + a_4XZ^2 + a_6Z^3.$$

If the equation is v-minimal (Definition 6.1.1), then $\mathcal{W}$ is a minimal Weierstrass model of E.

The generic fiber of $\mathcal{W}$ is the curve in $\mathbb{P}^2_{k_v}$ cut out by the same equation, which is E, so $\mathcal{W}$ is an integral model of E. It is proper over $\mathcal{O}_v$, being closed in $\mathbb{P}^2_{\mathcal{O}_v}$. Write $a \mapsto \bar{a}$ for the reduction $\mathcal{O}_v \to \kappa(v)$. The special fiber of $\mathcal{W}$ is the Weierstrass cubic

$$\mathcal{W}_\pi \colon Y^2Z + \bar{a}_1XYZ + \bar{a}_3YZ^2 = X^3 + \bar{a}_2X^2Z + \bar{a}_4XZ^2 + \bar{a}_6Z^3$$

over $\kappa(v)$. Its discriminant is $\bar{\Delta}$, since Δ is a polynomial with integer coefficients in the a_i, so by Proposition 4.1.5 the cubic $\mathcal{W}_\pi$ is smooth if $v(\Delta) = 0$ and singular if $v(\Delta) > 0$.

We would like to define the reduction of E modulo π to be the special fiber of such a model. Since E has many integral models, there are two ways to make this a definition:

- show that $\mathcal{X}_\pi \cong \mathcal{X}'_\pi$ for any two integral models $\mathcal{X}$ and $\mathcal{X}'$ of E; or
- find a canonical integral model for every elliptic curve over k_v, and define the reduction modulo π in terms of it.

The next example shows that the first way is not an option, even among Weierstrass models.

EXAMPLE 6.2.3. Let $k = \mathbb{Q}$ and let $v = p$ be a prime with $p \geqslant 5$, so that $k_v = \mathbb{Q}_p$, $\mathcal{O}_v = \mathbb{Z}_p$, $\pi_v = p$ and $\kappa(v) = \mathbb{F}_p$. The generic point of $\operatorname{Spec} \mathbb{Z}_p$ is (0) and its special point is $p\mathbb{Z}_p$.
The elliptic curves

$$E \colon Y^2Z = X^3 + Z^3 \qquad \text{and} \qquad E' \colon Y^2Z = X^3 + p^6Z^3$$

in $\mathbb{P}^2_{\mathbb{Q}_p}$ are isomorphic over $\mathbb{Q}_p$. Indeed, the linear automorphism $(X : Y : Z) \mapsto (p^2X : p^3Y : Z)$ of $\mathbb{P}^2_{\mathbb{Q}_p}$ pulls back the equation of E' to p^6 times the equation of E,

$$(p^3Y)^2Z - (p^2X)^3 - p^6Z^3 = p^6\,(Y^2Z - X^3 - Z^3),$$

so it maps E isomorphically onto E'. It is the substitution of Section 6.1 with $u = p$.
Since both equations have coefficients in $\mathbb{Z}_p$, the Weierstrass models

$$\mathcal{W} \colon Y^2Z = X^3 + Z^3 \qquad \text{and} \qquad \mathcal{W}' \colon Y^2Z = X^3 + p^6Z^3$$

in $\mathbb{P}^2_{\mathbb{Z}_p}$ have generic fibers $\mathcal{W}_{\mathbb{Q}_p} = E$ and $\mathcal{W}'_{\mathbb{Q}_p} = E' \cong E$: both are integral models of E. Their special fibers are

$$\mathcal{W}_p \colon Y^2Z = X^3 + Z^3 \qquad \text{and} \qquad \mathcal{W}'_p \colon Y^2Z = X^3$$

over $\mathbb{F}_p$. The equation of $\mathcal{W}$ has discriminant $\Delta = -2^4 3^3$, a unit in $\mathbb{Z}_p$ since $p \geqslant 5$, so $\mathcal{W}_p$ is an elliptic curve over $\mathbb{F}_p$. The equation of $\mathcal{W}'$ has discriminant $p^{12}\Delta$, and $\mathcal{W}'_p$ has a cusp at the rational point $(0:0:1)$. So $\mathcal{W}_p \ncong \mathcal{W}'_p$, although $\mathcal{W}_{\mathbb{Q}_p} \cong \mathcal{W}'_{\mathbb{Q}_p}$. See Figure 6.2.1. The first equation is p-minimal, and the second is not.

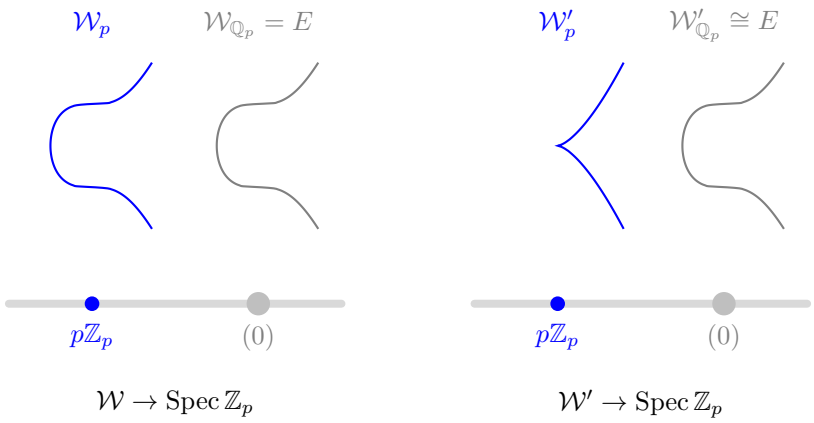


FIGURE 6.2.1. The models $\mathcal{W}$ and $\mathcal{W}'$ of Example 6.2.3 over $\operatorname{Spec} \mathbb{Z}_p$: the same generic fiber E over (0), and non-isomorphic special fibers over $p\mathbb{Z}_p$.

So we take the second way. By part 3 of Proposition 6.1.2, a minimal Weierstrass model of E is unique up to a change of variables with $u \in \mathcal{O}_v^\times$ and $r, s, t \in \mathcal{O}_v$, and we follow [Sil09, Section VII.2] in defining the reduction by it.

DEFINITION 6.2.4. Let E be an elliptic curve over k_v, and let $\mathcal{W}$ be a minimal Weierstrass model of E. The reduction of E modulo π is the special fiber

$$E_\pi \coloneqq \mathcal{W}_\pi,$$

a Weierstrass cubic over $\kappa(v)$. We write E_π^{ns} for its nonsingular part, as in Proposition 4.2.11. It is E_π if E_π is smooth.

REMARK 6.2.5. The reduction E_π is well defined up to isomorphism over $\kappa(v)$. Two minimal Weierstrass equations for E are related by a change of variables (4.1.3) with $u \in \mathcal{O}_v^\times$ and $r, s, t \in \mathcal{O}_v$ (part 3 of Proposition 6.1.2). The formulas (4.1.4) express each $u^i a'_i$ as a polynomial with integer coefficients in the a_j, r, s and t, so they remain true after reduction, and the change of variables with parameters $\bar{u} \neq 0$, $\bar{r}$, $\bar{s}$, $\bar{t}$ takes the reduction of one equation to the reduction of the other.

Luckily, there is also a canonical integral model with good functorial properties: the Néron model. The following is [Poo17, Definition 3.5.82], for an elliptic curve.

DEFINITION 6.2.6. Let E be an elliptic curve over k_v. A Néron model of E is a smooth $\mathcal{O}_v$-scheme $\mathcal{E}$ together with an isomorphism $\mathcal{E}_{k_v} \to E$, such that for every smooth $\mathcal{O}_v$-scheme $\mathcal{T}$ the map $\mathcal{E}(\mathcal{T}) \to E(\mathcal{T}_{k_v})$, restriction to the generic fiber, is bijective.

THEOREM 6.2.7 (Néron). *Let E be an elliptic curve over k_v.*

(1) E has a Néron model $\mathcal{E}$. It is of finite type over $\mathcal{O}_v$, unique up to unique isomorphism, and a group scheme over $\mathcal{O}_v$.

(2) Restriction to the generic fiber is a bijection $\mathcal{E}(\mathcal{O}_v) \to E(k_v)$.

(3) *Let $\mathcal{W}$ be a minimal Weierstrass model of E, and $\mathcal{E}^0 \subseteq \mathcal{E}$ the open subscheme obtained by removing the non-identity components of the special fiber $\mathcal{E}_\pi$. Then the smooth locus $\mathcal{W}^{\text{smooth}}$ of $\mathcal{W} \to \operatorname{Spec} \mathcal{O}_v$ is isomorphic to $\mathcal{E}^0$.*
(4) *If E has a Weierstrass equation with coefficients in $\mathcal{O}_v$ and $v(\Delta) = 0$, then its Weierstrass model $\mathcal{W}$ is smooth and proper over $\mathcal{O}_v$, and $\mathcal{W} \cong \mathcal{E}$.*

PROOF. Existence and finite type are [Poo17, Theorem 5.7.25]. Uniqueness follows from Yoneda's lemma in the category of smooth $\mathcal{O}_v$-schemes, and the group structure of E makes $\mathcal{E}$ a group object there [Poo17, Section 3.5.16]. Part 2 is the definition with $\mathcal{T} = \operatorname{Spec} \mathcal{O}_v$. Part 3 is [Poo17, Theorem 5.7.30]. For part 4, the equation is v-minimal because $v(\Delta) \geqslant 0$ for every integral equation, and then $\mathcal{W}$ is an abelian scheme over $\mathcal{O}_v$, in particular smooth, hence regular [Poo17, Section 5.7.6]. So $\mathcal{W}$ is already the minimal regular proper model of E constructed there, and it is its own smooth locus, so $\mathcal{W} \cong \mathcal{E}$ by the first isomorphism of [Poo17, Theorem 5.7.30]. □

Part 3 of Theorem 6.2.7 is how the Néron model is constructed from a minimal Weierstrass model [Poo17, Section 5.7.6]. Blowing up the non-regular closed points of $\mathcal{W}$ repeatedly yields a regular proper integral model of E, in fact the minimal one. Removing the closed subset of its special fiber where it is not smooth over $\mathcal{O}_v$, which may contain entire components (those of multiplicity greater than 1), yields $\mathcal{E}$. Removing only the singular point of E_π from $\mathcal{W}$, if there is one, yields $\mathcal{E}^0$.

So the special fiber $\mathcal{E}_\pi$ of the Néron model and the reduction E_π of Definition 6.2.4 agree when E_π is smooth, by part 4 of Theorem 6.2.7, but not in general, despite the notation. When E_π is singular, E_π is proper and not smooth, while $\mathcal{E}_\pi$ is smooth but not proper, since its identity component is the affine curve E_π^{ns} (Section 6.3), and possibly not connected (Theorem 6.3.9). What they share is

$$E_\pi^{\text{ns}} \cong \mathcal{W}_\pi^{\text{smooth}} \cong \mathcal{E}_\pi^0,$$

the identity component of $\mathcal{E}_\pi$, by part 3 of Theorem 6.2.7.

We now reduce k_v-points. A k_v-point P of $\mathbb{P}^2$ has homogeneous coordinates $P = (X_0 : Y_0 : Z_0)$ with $X_0, Y_0, Z_0 \in \mathcal{O}_v$, not all in $\mathfrak{m}_v$: scale by a power of π_v. They are unique up to multiplication by $\mathcal{O}_v^\times$, so $\bar{P} \coloneqq (\bar{X}_0 : \bar{Y}_0 : \bar{Z}_0)$ is a well-defined $\kappa(v)$-point of $\mathbb{P}^2$. If $P \in E(k_v)$, with $E \subseteq \mathbb{P}^2_{k_v}$ cut out by the equation of $\mathcal{W}$, then $\bar{P}$ satisfies the reduced equation, so $\bar{P} \in E_\pi(\kappa(v))$. This is the reduction map

$$E(k_v) \to E_\pi(\kappa(v)), \qquad P \mapsto \bar{P}.$$

In terms of models, P extends uniquely to an $\mathcal{O}_v$-point of the proper $\mathcal{O}_v$-scheme $\mathcal{W}$ [Poo17, Theorem 3.2.12], namely $\operatorname{Spec} \mathcal{O}_v \to \mathbb{P}^2_{\mathcal{O}_v}$ given by $(X_0 : Y_0 : Z_0)$, and $\bar{P}$ is its restriction to the special point. Let $O = (0 : 1 : 0)$. Following [Sil09, Section VII.2], we define the subsets

$$E_0(k_v) \coloneqq \{P \in E(k_v) : \bar{P} \in E_\pi^{\text{ns}}(\kappa(v))\},$$
$$E_1(k_v) \coloneqq \{P \in E(k_v) : \bar{P} = \bar{O}\}$$

of k_v-points with nonsingular reduction, and the kernel of reduction. Since $\bar{O} = (0 : 1 : 0)$ is a smooth $\kappa(v)$-point of E_π by Proposition 4.1.5, $E_1(k_v) \subseteq E_0(k_v)$. Neither set depends on the choice of minimal Weierstrass equation: the change of variables of Remark 6.2.5 has coefficients in $\mathcal{O}_v$, so it commutes with reduction of k_v-points, and its reduction maps nonsingular $\kappa(v)$-points of one reduced cubic to nonsingular $\kappa(v)$-points of the other. An $\mathcal{O}_v$-point of $\mathcal{W}$ lies in $\mathcal{W}^{\text{smooth}}$ exactly when its restriction to the special point does, so $E_0(k_v) = \mathcal{W}^{\text{smooth}}(\mathcal{O}_v)$, which is $\mathcal{E}^0(\mathcal{O}_v)$ by Theorem 6.2.7.

The following is [Sil09, Proposition VII.2.1].

PROPOSITION 6.2.8. *Let E be an elliptic curve over k_v. Then $E_0(k_v)$ is a subgroup of $E(k_v)$, and there is an exact sequence of abelian groups*

$$0 \to E_1(k_v) \to E_0(k_v) \to E_\pi^{\text{ns}}(\kappa(v)) \to 0,$$

where the map on the right is the reduction map $P \mapsto \bar{P}$.

PROOF. Fix a minimal Weierstrass equation for E, let F be its cubic form, with coefficients in $\mathcal{O}_v$, and let $f(x,y) \coloneqq F(x,y,1)$. Then $\bar{F}$ is the cubic form of E_π. Neither E nor E_π contains a line: the line at infinity meets each of them only at O, respectively $\bar{O}$ (Definition 4.1.2), and an affine line $y = \alpha x + \beta$ or $x = \gamma$ on the curve would make $f(x, \alpha x + \beta)$, of degree 3 in x, or $f(\gamma, y)$, monic of degree 2 in y, vanish identically.

The reduction map is surjective. Let $Q \in E_\pi^{\mathrm{ns}}(\kappa(v))$. If $Q = \bar{O}$, then Q is the reduction of O. Otherwise $Q = (\alpha, \beta)$ is affine, and $\partial\bar{f}/\partial x$ or $\partial\bar{f}/\partial y$ is nonzero at Q, since Q is nonsingular. Suppose that $(\partial\bar{f}/\partial y)(\alpha,\beta) \neq 0$, choose $x_0 \in \mathcal{O}_v$ with $\bar{x}_0 = \alpha$, and let $g(T) \coloneqq f(x_0, T) \in \mathcal{O}_v[T]$. Then $\bar{g}(\beta) = 0$ and $\bar{g}'(\beta) \neq 0$. The $\mathcal{O}_v$-scheme $\operatorname{Spec} \mathcal{O}_v[T, g'(T)^{-1}]/(g(T))$ is étale, and β is one of its $\kappa(v)$-points, so by Hensel's lemma [Poo17, Theorem 3.5.63], which uses that $\mathcal{O}_v$ is complete, there is $y_0 \in \mathcal{O}_v$ with $g(y_0) = 0$ and $\bar{y}_0 = \beta$. Then $P \coloneqq (x_0 : y_0 : 1) \in E(k_v)$ and $\bar{P} = Q$, so $P \in E_0(k_v)$. If instead $(\partial\bar{f}/\partial x)(\alpha,\beta) \neq 0$, exchange the roles of the two coordinates: lift β to $y_0 \in \mathcal{O}_v$ and apply Hensel's lemma to $f(T, y_0)$.

Lines and reduction. Let L be a line in $\mathbb{P}^2$ defined over k_v. Write it as $aX + bY + cZ = 0$ with $a, b, c \in \mathcal{O}_v$, not all in $\mathfrak{m}_v$. Then the linear form $\mathcal{O}_v^3 \to \mathcal{O}_v$ given by (a,b,c) is surjective, so its kernel M is a direct summand of $\mathcal{O}_v^3$, free of rank 2. Let w, w' be a basis of M. The map sending $(s : t)$ to $sw + tw'$, from $\mathbb{P}^1$ to L, is an isomorphism over k_v, and since M is a direct summand, $\bar{w}, \bar{w}'$ is a basis of the kernel of $(\bar{a}, \bar{b}, \bar{c})$, so its reduction is an isomorphism $\mathbb{P}^1 \to \bar{L}$ over $\kappa(v)$, compatible with the reduction of k_v-points. The binary cubic

$$G(s,t) \coloneqq F(sw + tw') \in \mathcal{O}_v[s,t]$$

is the restriction of F to L. It is nonzero, because E contains no line. We count the k_v-points of $L \cap E$ with multiplicity as the roots of G in $\mathbb{P}^1(k_v)$. Suppose that G has three roots $(s_i : t_i) \in \mathbb{P}^1(k_v)$, with $s_i, t_i \in \mathcal{O}_v$ not both in $\mathfrak{m}_v$, corresponding to $P_1, P_2, P_3 \in E(k_v)$. Then

$$G = c \prod_{i=1}^{3} (t_i s - s_i t)$$

for some $c \in k_v^\times$. By Gauss's lemma the product of the three linear forms has a coefficient in $\mathcal{O}_v^\times$, so $c \in \mathcal{O}_v$, and $\bar{G} = \bar{c} \prod_i (\bar{t}_i s - \bar{s}_i t)$. On the other hand $\bar{G}(s,t) = \bar{F}(s\bar{w} + t\bar{w}')$ is the restriction of $\bar{F}$ to $\bar{L}$, which is nonzero because E_π contains no line. So $\bar{c} \neq 0$, and $\bar{L}$ meets E_π in $\bar{P}_1, \bar{P}_2, \bar{P}_3$, counted with multiplicity.

The group law. Let $P_1, P_2 \in E_0(k_v)$, let L be the line through them (the tangent line to E at P_1 if $P_1 = P_2$), and let P_3 be the third k_v-point of $L \cap E$, counted with multiplicity, so that $P_3 \in E(k_v)$ and $P_1 + P_2 + P_3 = O$ by part 1 of Proposition 4.2.4. By the previous step $\bar{L}$ meets E_π in $\bar{P}_1, \bar{P}_2, \bar{P}_3$. If E_π has a singular point S, every line through S meets E_π at S with multiplicity at least 2, since all partial derivatives of $\bar{F}$ vanish at S. As $\bar{P}_1, \bar{P}_2 \neq S$, if $\bar{P}_3 = S$ the factorization of $\bar{G}$ would give S multiplicity exactly 1, a contradiction. So $\bar{P}_3 \in E_\pi^{\mathrm{ns}}(\kappa(v))$, that is, $P_3 \in E_0(k_v)$, and

$$\bar{P}_1 + \bar{P}_2 + \bar{P}_3 = \bar{O}$$

by the chord-and-tangent construction of the group law on E_π^{ns} (Proposition 4.2.11, or part 1 of Proposition 4.2.4 if E_π is smooth).

Taking $P_2 = O$, which lies in $E_0(k_v)$, gives $P_3 = -P_1 \in E_0(k_v)$ and $\overline{-P_1} = -\bar{P}_1$. In general, then, $P_1 + P_2 = -P_3 \in E_0(k_v)$ and $\overline{P_1 + P_2} = -\bar{P}_3 = \bar{P}_1 + \bar{P}_2$. So $E_0(k_v)$ is a subgroup of $E(k_v)$, and reduction is a homomorphism $E_0(k_v) \to E_\pi^{\mathrm{ns}}(\kappa(v))$. Its kernel is $E_1(k_v)$ by definition, and it is surjective by the first step. □

The following is [Sil09, Proposition VII.3.1], the local input to the weak Mordell–Weil theorem.

PROPOSITION 6.2.9. *Let E be an elliptic curve over k_v, and let $m \geqslant 1$ be an integer with $p \nmid m$.*

(1) The kernel of reduction $E_1(k_v)$ contains no k_v-point killed by m other than O: $E_1(k_v) \cap E[m](k_v) = \{O\}$.

(2) If E_π is smooth, then the reduction map $E[m](k_v) \to E_\pi(\kappa(v))$ is injective.

PROOF. For part 1, fix a minimal Weierstrass equation for E. The group $E_1(k_v)$ is isomorphic to the group $\hat{E}(\mathfrak{m}_v)$ attached to the formal group $\hat{E}$ of E [Sil09, Proposition VII.2.2], and every element of finite

order of $\hat{E}(\mathfrak{m}_v)$ has order a power of p [Sil09, Proposition IV.3.2(b)]. Since $p \nmid m$, the only k_v-point of $E_1(k_v)$ killed by m is O.

For part 2, if E_π is smooth, then $E_\pi^{\mathrm{ns}} = E_\pi$ and $E_0(k_v) = E(k_v)$, so by Proposition 6.2.8 reduction is a homomorphism $E(k_v) \to E_\pi(\kappa(v))$ with kernel $E_1(k_v)$. Its restriction to $E[m](k_v)$ has kernel $E_1(k_v) \cap E[m](k_v) = \{O\}$ by part 1. □

The next result, due to Cassels, bounds the denominators of the coordinates of a torsion k_v-point. It is [Sil09, Theorem VII.3.4]. Unlike Proposition 6.2.9, the Weierstrass equation need not be v-minimal.

THEOREM 6.2.10. *Let E be an elliptic curve over k_v, given by a Weierstrass equation*

$$y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$$

with all $a_i \in \mathcal{O}_v$, not necessarily v-minimal. Let $P \in E(k_v)$ be a k_v-point of exact order $m \geqslant 2$.

(1) If m is not a power of p, then $x(P), y(P) \in \mathcal{O}_v$.

(2) If $\operatorname{char} k = 0$ and $m = p^n$, then

$$\pi_v^{2r} x(P),\ \pi_v^{3r} y(P) \in \mathcal{O}_v, \qquad \text{where} \qquad r := \left\lfloor \frac{v(p)}{p^n - p^{n-1}} \right\rfloor.$$

In part 2, where $\operatorname{char} k = 0$, $v(p)$ is the ramification index of k_v over $\mathbb{Q}_p$. It is 1 for $k_v = \mathbb{Q}_p$.

PROOF. Write $x := x(P)$ and $y := y(P)$. Since $m \geqslant 2$, $P \neq O$ is affine.

Any integral equation. If $v(x) \geqslant 0$, then y is a root of the monic polynomial

$$T^2 + (a_1x + a_3)\,T - (x^3 + a_2x^2 + a_4x + a_6) \in \mathcal{O}_v[T],$$

so $y \in \mathcal{O}_v$, because a discrete valuation ring is integrally closed. Then both parts hold, since $r \geqslant 0$. Suppose instead that $v(x) < 0$. The right side of the Weierstrass equation has valuation exactly $3v(x)$. If $v(y) \geqslant v(x)$, the left side would have valuation at least $\min(2v(y), v(x)+v(y), v(y)) \geqslant 2v(x) > 3v(x)$. So $v(y) < v(x) < 0$, the left side has valuation exactly $2v(y)$, and $2v(y) = 3v(x)$:

$$v(x) = -2s, \qquad v(y) = -3s \qquad \text{for an integer } s \geqslant 1.$$

Reduction to a v-minimal equation. By part 2 of Proposition 6.1.2, a change of variables (4.1.3),

$$x = u^2x' + r_0, \qquad y = u^3y' + s_0u^2x' + t_0,$$

with $u, r_0, s_0, t_0 \in \mathcal{O}_v$, takes the equation to a v-minimal one, with coordinates x', y'. Suppose that $v(x) < 0$. Then $v(x'(P)) < 0$, for otherwise $x \in \mathcal{O}_v$. So by the previous step, applied to the v-minimal equation, $v(y'(P)) < v(x'(P)) < 0$, and therefore

$$v(x) \geqslant v(x'(P)), \qquad v(y) \geqslant v(y'(P)).$$

Hence both parts for the v-minimal equation imply them for the given one, and we may assume that the equation is v-minimal.

The v-minimal case. Suppose that $v(x) < 0$, with s as above. Then $P = (\pi_v^{3s}x : \pi_v^{3s}y : \pi_v^{3s})$ with $\pi_v^{3s}y \in \mathcal{O}_v^\times$ and the other two coordinates in $\mathfrak{m}_v$, so $\bar{P} = (0 : 1 : 0) = \bar{O}$ and $P \in E_1(k_v)$. Under the isomorphism $E_1(k_v) \cong \hat{E}(\mathfrak{m}_v)$ [Sil09, Proposition VII.2.2], P corresponds to $z := -x/y$, of exact order m, with $v(z) = -2s + 3s = s$.

For part 1: every element of finite order of $\hat{E}(\mathfrak{m}_v)$ has order a power of p [Sil09, Proposition IV.3.2(b)], which contradicts the hypothesis on m. So $v(x) \geqslant 0$, and the first step gives $x, y \in \mathcal{O}_v$.

For part 2: since z has exact order p^n in $\hat{E}(\mathfrak{m}_v)$, [Sil09, Theorem IV.6.1] gives $s = v(z) \leqslant v(p)/(p^n - p^{n-1})$, so $s \leqslant r$, as s is an integer. Hence $v(\pi_v^{2r}x) = 2(r - s) \geqslant 0$ and $v(\pi_v^{3r}y) = 3(r - s) \geqslant 0$. □

The following is [Sil09, Application VII.3.5].

EXAMPLE 6.2.11. Let $k = \mathbb{Q}$, let E be an elliptic curve over $\mathbb{Q}$ given by a Weierstrass equation with coefficients in $\mathbb{Z}$, and let $P \in E(\mathbb{Q})$ be a rational point of exact order $m \geqslant 3$. Let p be a prime and apply Theorem 6.2.10 to $P \in E(\mathbb{Q}_p)$. If m is not a power of p, then $x(P), y(P) \in \mathbb{Z}_p$ by part 1. If $m = p^n$, then

$v(p) = 1$, and

$$r = \left\lfloor \frac{1}{p^n - p^{n-1}} \right\rfloor = 0$$

unless $p^n - p^{n-1} = 1$, that is, $p = 2$ and $n = 1$, which $m \geqslant 3$ excludes. So $x(P), y(P) \in \mathbb{Z}_p$ by part 2. Hence $x(P), y(P) \in \mathbb{Z}$: a rational torsion point of order at least 3 has integer coordinates.
This fails for rational points of order 2. The curve

$$E\colon y^2 + xy = x^3 + 4x + 1$$

has discriminant $-65^2 \neq 0$, and $P = (-1/4, 1/8) \in E(\mathbb{Q})$: both sides equal $-1/64$. Since $-P = (x(P), -y(P) - x(P)) = P$, the rational point P has order 2. At $p = 2$, with $n = 1$, the theorem gives $r = 1$ and says only that $4x(P), 8y(P) \in \mathbb{Z}_2$, which is attained.

6.3. Good and bad reduction

We are in Situation 6.0.2. By Proposition 4.1.5 and Remark 4.1.11, the reduction E_π of an elliptic curve E over k_v (Definition 6.2.4) is smooth, or has a node, or has a cusp. Since $\kappa(v)$ is perfect, its singular point is a rational point and the tangent line at a cusp is defined over $\kappa(v)$, so Proposition 4.2.11 applies to E_π with no further hypothesis. The following is [Sil09, Section VII.5]. Compare [Poo08, Section 6.4] and [Mil20, Section II.3].

Definition 6.3.1. Let E be an elliptic curve over k_v, with reduction E_π.

(1) E has good reduction if E_π is smooth, hence an elliptic curve over $\kappa(v)$.
(2) E has multiplicative reduction if E_π has a node. The reduction is split if the slopes of the tangent lines at the node lie in $\kappa(v)$, and nonsplit otherwise.
(3) E has additive reduction if E_π has a cusp.

In the last two cases E has bad reduction. It has semistable reduction if its reduction is good or multiplicative.

The reduction type does not depend on the choice of minimal Weierstrass equation, since E_π is well defined up to isomorphism (Remark 6.2.5).

Proposition 6.3.2. *Let E be an elliptic curve over k_v, given by a minimal Weierstrass equation with discriminant Δ and invariants c_4, c_6.*

(1) E has good reduction if and only if $v(\Delta) = 0$.
(2) E has multiplicative reduction if and only if $v(\Delta) > 0$ and $v(c_4) = 0$. Then $E_\pi^{\mathrm{ns}} \cong \mathbb{G}_m$ if the reduction is split, and $E_\pi^{\mathrm{ns}} \cong \mathbb{G}_m^{(d)}$, with d as in (4.2.3) for the singular normal form of E_π and $L_d = \mathbb{F}_{q^2}$, if it is nonsplit.
(3) E has additive reduction if and only if $v(\Delta) > 0$ and $v(c_4) > 0$. Then $E_\pi^{\mathrm{ns}} \cong \mathbb{G}_a$.

If $\operatorname{char}\kappa(v) \neq 2$ and the reduction is multiplicative, it is split if and only if $-\bar{c}_6$ is a square in $\kappa(v)$.

Proof. The reduced equation has discriminant $\bar{\Delta}$ and invariants $\bar{c}_4$, $\bar{c}_6$, because these are polynomials with integer coefficients in the a_i. So the three cases follow from Proposition 4.1.5 and Remark 4.1.11, and the descriptions of E_π^{ns} from Proposition 4.2.11 over $\kappa(v)$. In the nonsplit case L_d is a quadratic extension of $\kappa(v)$, which is $\mathbb{F}_{q^2}$. For the last statement, move the node to $(0,0)$ as in (4.2.2), a change of variables with $u = 1$, which does not change c_6. There $b_4 = b_6 = 0$, so $c_6 = -b_2^3 + 36b_2b_4 - 216b_6 = -b_2^3$, and $-c_6 = b_2^3$ is a square exactly when $b_2 = d$ is, that is, when $L_d \cong \kappa(v) \times \kappa(v)$. Another minimal equation changes c_6 by u^{-6} with $u \in \mathcal{O}_v^\times$, a square. □

Counting $\kappa(v)$-points with Example 2.1.8 and part 4 of Proposition 2.1.9, and adding the singular point, gives the following table [Mil20, Section II.3]. Here $a \coloneqq q + 1 - \#E_\pi(\kappa(v))$, as in Definition 5.3.7 when E_π is smooth.

reduction	singularity	E_π^{ns}	$\#E_\pi^{\mathrm{ns}}(\kappa(v))$	a
good	none	E_π	$q+1-a$	$\lvert a\rvert \leqslant 2\sqrt{q}$
split multiplicative	node	$\mathbb{G}_{\mathrm{m}}$	$q-1$	1
nonsplit multiplicative	node	$\mathbb{G}_{\mathrm{m}}^{(d)}$	$q+1$	-1
additive	cusp	$\mathbb{G}_{\mathrm{a}}$	q	0

These values of a at the places of bad reduction are the ones that enter the L-function of an elliptic curve over a number field (Chapter 13).

REMARK 6.3.3. By part 3 of Theorem 6.2.7, E_π^{ns} is the identity component $\mathcal{E}_\pi^0$ of the special fiber of the Néron model. So the reduction is good, multiplicative or additive according as $\mathcal{E}_\pi^0$ is an elliptic curve, a torus (Definition 2.1.10), split exactly when the reduction is, or $\mathbb{G}_{\mathrm{a}}$. This is the form of the definition that generalizes to abelian varieties.

A Weierstrass equation with coefficients in $\mathcal{O}_v$ and $v(\Delta) < 12$, or with $v(c_4) = 0$, is v-minimal. Indeed, another integral equation has discriminant $u^{-12}\Delta$ and invariant $u^{-4}c_4$ (Proposition 4.1.7), both integral, so $v(u) \leqslant 0$ in either case, and its discriminant has valuation at least $v(\Delta)$. The following is [Sil09, Example VII.5.2].

EXAMPLE 6.3.4. Let $k = \mathbb{Q}$ and $v = p \geqslant 5$. Each of the following equations has $v(\Delta) < 12$, so it is p-minimal.

(1) $E\colon y^2 = x^3 + px^2 + 1$ has $\Delta = -16(4p^3 + 27)$, a unit: good reduction.
(2) $E\colon y^2 = x^3 + x^2 + p$ has $\Delta = -16p(27p + 4)$ and $c_4 = 16$: multiplicative reduction. The reduction $y^2 = x^2(x + 1)$ has tangent lines $y = \pm x$ at the node, so it is split. Accordingly $-c_6 = 64 + 864p$ is a square modulo p.
(3) Let $\varepsilon \in \mathbb{Z}$ be a non-square modulo p. Then $E\colon y^2 = x^3 + \varepsilon x^2 + p$ has $\Delta = -16p(4\varepsilon^3 + 27p)$ and $c_4 = 16\varepsilon^2$, and its reduction $y^2 = x^2(x + \varepsilon)$ has tangent slopes $\pm\sqrt{\varepsilon} \notin \mathbb{F}_p$: nonsplit multiplicative reduction.
(4) $E\colon y^2 = x^3 + p$ has $\Delta = -432p^2$ and $c_4 = 0$: additive reduction. Over $K' \coloneqq \mathbb{Q}_p(\varpi)$ with $\varpi^6 = p$, the substitution $x = \varpi^2 x'$, $y = \varpi^3 y'$ gives $y'^2 = x'^3 + 1$, with discriminant -432, a unit: E acquires good reduction over K'.

The last example shows that the reduction type can change under a finite extension. To make sense of this we need the reduction over a finite extension of k_v.

REMARK 6.3.5. Let K' be a finite extension of k_v. Then K' is again a local field: it is the completion of a finite extension of k at a place above v. We write v' for its normalized valuation, $\mathcal{O}'$ for its ring of integers, $\mathfrak{m}'$ for the maximal ideal, π' for a uniformizer and κ' for the residue field. The valuation v extends uniquely to K', and $v' = e\,v$ on k_v, where e is the ramification index of K'/k_v. The residue field κ' has degree f over $\kappa(v)$, and $ef = [K' : k_v]$. The extension is unramified if $e = 1$. In general one also asks that the residue extension $\kappa'/\kappa(v)$ be separable. Here it is an extension of finite fields, so this is automatic and we omit it. If K'/k_v is unramified, then $v' = v$ on k_v, π_v is also a uniformizer of K', and $[K' : k_v] = [\kappa' : \kappa(v)]$. Every finite extension of $\kappa(v)$ arises in this way from exactly one unramified extension of k_v (Section 6.4). The results of this chapter so far apply to $E_{K'}$, with its own minimal Weierstrass equation, reduction $(E_{K'})_{\pi'}$, and subgroups $E_0(K') \supseteq E_1(K')$.

DEFINITION 6.3.6. An elliptic curve E over k_v has potential good reduction if $E_{K'}$ has good reduction for some finite extension K' of k_v.

The following is [Sil09, Propositions VII.5.4 and VII.5.5].

PROPOSITION 6.3.7 (Semistable reduction theorem). *Let E be an elliptic curve over k_v, and K' a finite extension of k_v.*

(1) If K'/k_v is unramified, then a minimal Weierstrass equation for E is minimal for $E_{K'}$, and $E_{K'}$ has good, multiplicative or additive reduction according as E does.
(2) If E has good, respectively multiplicative, reduction, then a minimal Weierstrass equation for E is minimal for $E_{K'}$, and $E_{K'}$ has good, respectively multiplicative, reduction.

(3) There is a finite extension K'' of k_v such that $E_{K''}$ has good or split multiplicative reduction.

When a minimal Weierstrass equation for E stays minimal for $E_{K'}$, as in parts 1 and 2, the reduction $(E_{K'})_{\pi'}$ is the base change $(E_\pi)_{\kappa'}$: both are cut out by the same reduced equation. Whether a multiplicative reduction is split can therefore change: it is split over K' exactly when κ' contains the slopes of the tangent lines at the node. For instance, nonsplit multiplicative reduction becomes split over the unramified extension of degree 2.

PROOF. For part 2, fix a minimal Weierstrass equation for E, with Δ and c_4, and let a change of variables (4.1.3) take it to a minimal equation for $E_{K'}$, with Δ' and c_4'. By part 2 of Proposition 6.1.2 over K', the parameter u lies in $\mathcal{O}'$, and $\Delta' = u^{-12}\Delta$, $c_4' = u^{-4}c_4$ are integral, so

$$0 \leqslant v'(u) \leqslant \min\big(v'(\Delta)/12,\ v'(c_4)/4\big).$$

If $v(\Delta) = 0$ or $v(c_4) = 0$, then $v'(u) = 0$, so $v'(\Delta') = v'(\Delta)$: our equation, integral over K', is minimal for $E_{K'}$. Its $v'(\Delta)$ and $v'(c_4)$ are zero exactly when $v(\Delta)$ and $v(c_4)$ are, and Proposition 6.3.2 over K' gives the same type.

We prove part 1 only when $p \geqslant 5$. For $p = 2, 3$ it follows from Tate's algorithm (Corollary 6.5.13). Since 2 and 3 are units in $\mathcal{O}_v$, E has a minimal equation $y^2 = x^3 + Ax + B$, and $E_{K'}$ also has a minimal equation of this short form. It is obtained from ours by a change of variables $x = u^2x'$, $y = u^3y'$, the only ones preserving the short form, with $u \in \mathcal{O}'$ by part 2 of Proposition 6.1.2. Since K'/k_v is unramified, $v' = v$ on k_v (Remark 6.3.5), and $u_0 \coloneqq \pi_v^{v'(u)} \in k_v$ satisfies $u/u_0 \in \mathcal{O}'^\times$. The substitution $x = u_0^2x'$, $y = u_0^3y'$ gives $u_0^{-4}A = (u/u_0)^4u^{-4}A$ and $u_0^{-6}B$, both integral, over k_v, with discriminant of valuation $v(\Delta) - 12v'(u)$. By minimality over k_v, $v'(u) = 0$. So the equation is minimal over K', with the same $v(\Delta)$ and $v(c_4)$, and Proposition 6.3.2 gives the same type.

We prove part 3 only when $p \neq 2$. Over a finite extension K'' of k_v, E has a Legendre form $y^2 = x(x-1)(x-\lambda)$ with $\lambda \neq 0, 1$ [Sil09, Proposition III.1.7], for which

$$c_4 = 16(\lambda^2 - \lambda + 1), \qquad \Delta = 16\lambda^2(\lambda - 1)^2.$$

Let $\mathcal{O}''$ be the ring of integers of K'' and π'' a uniformizer. If $\lambda \in \mathcal{O}''$ and $\bar\lambda \neq 0, 1$, then Δ is a unit: good reduction. If $\lambda \in \mathcal{O}''$ and $\bar\lambda \in \{0, 1\}$, then c_4 is a unit and Δ is not: multiplicative reduction. If $\lambda \notin \mathcal{O}''$, let $r \geqslant 1$ with $\mu \coloneqq \pi''^r\lambda$ a unit, and adjoin a square root ϖ of π'', a separable extension since $p \neq 2$. The substitution $x = \pi''^{-r}x'$, $y = \varpi^{-3r}y'$ gives $y'^2 = x'(x' - \pi''^r)(x' - \mu)$, whose $c_4 = 16(\pi''^{2r} - \pi''^r\mu + \mu^2)$ is a unit and whose Δ is not: multiplicative reduction. In the multiplicative cases c_4 is a unit, so the equation is minimal over every finite extension (see before Example 6.3.4), and its reduction over an extension is the base change of the nodal cubic. The slopes of the tangent lines at the node generate an extension of degree at most 2 of the residue field, which comes from an unramified extension (Remark 6.3.5). Over it the reduction is split. □

PROPOSITION 6.3.8. *Let E be an elliptic curve over k_v. Then E has potential good reduction if and only if $j(E) \in \mathcal{O}_v$.*

PROOF. If $E_{K'}$ has good reduction, with a minimal equation over K' with Δ' a unit, then $j(E) = c_4'^3/\Delta'$ is integral in K', and it lies in k_v, so $j(E) \in \mathcal{O}_v$. We prove the converse only when $p \neq 2$. Take a Legendre form over a finite extension K'' of k_v, as in the proof of Proposition 6.3.7. Here K'' is separable over k_v, since it is obtained by adjoining the roots of the separable cubic $4x^3 + b_2x^2 + 2b_4x + b_6$ and square roots, and $p \neq 2$. Then

$$256(\lambda^2 - \lambda + 1)^3 - j\lambda^2(\lambda - 1)^2 = 0.$$

If j is integral, this polynomial in λ has integral coefficients and leading coefficient 256, a unit, so λ is integral. And reducing modulo π'' at $\bar\lambda = 0$ or 1 would give $256 = 0$. So $\bar\lambda \neq 0, 1$, and the Legendre form has good reduction. □

For instance, the curve of part 4 of Example 6.3.4 has $j = 0$, so potential good reduction, as we saw directly. The curve of part 2 has $v(j) = 3v(c_4) - v(\Delta) = -1$, and keeps multiplicative reduction over every finite extension.

Finally, the quotient $E(k_v)/E_0(k_v)$ of the filtration of Section 6.2 is finite. The first part below follows from Theorem 6.2.7: $E(k_v) = \mathcal{E}(\mathcal{O}_v)$ and $E_0(k_v) = \mathcal{E}^0(\mathcal{O}_v)$. Restricting an $\mathcal{O}_v$-point to the special point and taking its component gives a homomorphism from $\mathcal{E}(\mathcal{O}_v)$ to the group of components of $\mathcal{E}_\pi$ defined over $\kappa(v)$ (the $\kappa(v)$-points of its component group), finite because $\mathcal{E}$ is of finite type. Its kernel is $\mathcal{E}^0(\mathcal{O}_v)$: an $\mathcal{O}_v$-point whose special point lies in the open subscheme $\mathcal{E}^0$ factors through it, since every open subset of $\operatorname{Spec}\mathcal{O}_v$ containing the special point is everything. The second part comes from the classification of the special fibers of Néron models, Theorem 6.5.8. It is Corollary 6.5.9.

THEOREM 6.3.9 (Kodaira, Néron). *Let E be an elliptic curve over k_v, and Δ the discriminant of a minimal Weierstrass equation.*

(1) $E(k_v)/E_0(k_v)$ is finite.

(2) If E has split multiplicative reduction, then $E(k_v)/E_0(k_v)$ is cyclic of order $v(\Delta)$. Otherwise it has order at most 4.

PROOF. See [Sil09, Theorem VII.6.1 and Corollary VII.6.2]. □

DRAFT
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6.4. Inertia and the criterion of Néron–Ogg–Shafarevich

SITUATION 6.4.1. In addition to Situation 6.0.2:

- $\kappa(v)$ is identified with $\mathbb{F}_q$.
- k_v^{sep} is a separable closure of k_v, and $\mathrm{Gal}_{k_v} \coloneqq \mathrm{Gal}(k_v^{\mathrm{sep}}/k_v)$. We use k_v^{sep} rather than an algebraic closure because k_v is imperfect when $\operatorname{char} k = p$.
- v extends uniquely to k_v^{sep} (Remark 6.3.5). The residue field of its valuation ring is an algebraic closure $\overline{\mathbb{F}}_q$ of $\mathbb{F}_q$: since $\mathbb{F}_q$ is perfect, each of its finite extensions is separable, and is the residue field of an unramified extension of k_v inside k_v^{sep} (see below).
- $k_v^{\mathrm{unr}} \subseteq k_v^{\mathrm{sep}}$ is the maximal unramified extension of k_v, the union of the finite unramified extensions.
- $I_v \coloneqq \mathrm{Gal}(k_v^{\mathrm{sep}}/k_v^{\mathrm{unr}})$ is the inertia group. When $k = \mathbb{Q}$ and $v = p$, we write I_p for I_v.

Good reduction has a characterization in terms of the Galois action on torsion points alone, without Weierstrass equations. For instance, it shows at once that good reduction is an isogeny invariant (Corollary 6.4.8). We first review the inertia group. The unramified extensions of k_v correspond to the extensions of its residue field: for each $n \geqslant 1$ there is exactly one unramified extension of degree n inside k_v^{sep}, its residue field is $\mathbb{F}_{q^n}$, and it is Galois over k_v with group $\mathrm{Gal}(\mathbb{F}_{q^n}/\mathbb{F}_q)$, generated by the lift of the Frobenius automorphism $\sigma\colon \alpha \mapsto \alpha^q$. So there is an exact sequence

$$1 \to I_v \to \mathrm{Gal}_{k_v} \to \mathrm{Gal}_{\mathbb{F}_q} \to 1, \tag{6.4.1}$$

where the map on the right is the action on $\overline{\mathbb{F}}_q$: every $\tau \in \mathrm{Gal}_{k_v}$ preserves the valuation, hence acts on the residue field $\overline{\mathbb{F}}_q$. In other words, I_v is the subgroup of Gal_{k_v} acting trivially on $\overline{\mathbb{F}}_q$ [Sil09, Section VII.4]. The group $\mathrm{Gal}_{\mathbb{F}_q}$ is topologically generated by σ, and $\mathrm{Gal}(k_v^{\mathrm{unr}}/k_v) \cong \mathrm{Gal}_{\mathbb{F}_q}$.

REMARK 6.4.2. The inertia group has a largest pro-p subgroup P_v, the wild inertia group, and the tame quotient is $I_v/P_v \cong \prod_{\ell \neq p} \mathbb{Z}_\ell(1)$, with $\mathrm{Gal}_{\mathbb{F}_q}$ acting through the cyclotomic character. We will not need this.

DEFINITION 6.4.3. A set Σ with an action of Gal_{k_v} is unramified if I_v acts trivially on Σ, and ramified otherwise.

REMARK 6.4.4. By (6.4.1), Σ is unramified exactly when the action of Gal_{k_v} factors through $\mathrm{Gal}_{k_v}/I_v \cong \mathrm{Gal}_{\mathbb{F}_q}$: it is then determined by the action of a single element, any lift of σ, and it can be read off on the residue field. The name comes from fields. If Σ is finite and the action is continuous, that is, every element of Σ has an open stabilizer, as for $E[m](k_v^{\mathrm{sep}})$, then the kernel of the action is an open normal subgroup of Gal_{k_v}, and its fixed field L, the field cut out by Σ, is a finite Galois extension of k_v. Then Σ is unramified

exactly when $L \subseteq k_v^{\mathrm{unr}}$, that is, when L/k_v is unramified. For instance, the set of roots in k_v^{sep} of a separable polynomial $f \in k_v[x]$ is unramified exactly when the splitting field of f is unramified over k_v.
There is also a geometric picture. Think of $\operatorname{Spec} \mathcal{O}_v$ as a small disc, with the special point as its center and $\operatorname{Spec} k_v$ as the punctured disc. Finite Gal_{k_v}-sets play the role of finite covering spaces of the punctured disc, and I_v the role of the loop around the center: a covering extends across the center exactly when this loop acts trivially on its fibers. A family of elliptic curves over the whole disc has no monodromy, by Ehresmann's theorem, so good reduction, which extends E to a smooth family over $\operatorname{Spec} \mathcal{O}_v$, should force $E[m](k_v^{\mathrm{sep}})$ to be unramified. This is Proposition 6.4.6, and Theorem 6.4.7 is its converse. In the language of schemes, a finite étale k_v-scheme X has $X(k_v^{\mathrm{sep}})$ unramified exactly when X extends to a finite étale $\mathcal{O}_v$-scheme.

For an elliptic curve E over k_v and m with $\operatorname{char} k \nmid m$, this applies to $E[m](k_v^{\mathrm{sep}})$ (part 1 of Corollary 4.5.1): $E[m](k_v^{\mathrm{sep}})$ is unramified exactly when $I_v \subseteq \ker \bar{\rho}_{E,m}$, that is, when $E[m](k_v^{\mathrm{sep}}) \subseteq E(k_v^{\mathrm{unr}})$, or again when all the points of $E[m](k_v^{\mathrm{sep}})$ are defined over one finite unramified extension of k_v. Likewise $T_\ell(E)$ (Definition 4.5.4) is unramified exactly when every $E[\ell^n](k_v^{\mathrm{sep}})$ is.

Example 6.4.5.

(1) If $p \nmid m$, then $\mu_m(k_v^{\mathrm{sep}})$ is unramified. The polynomial $x^m - 1$ is separable over $\mathbb{F}_q$, so its m roots in $\overline{\mathbb{F}}_q$ lie in a finite field $\mathbb{F}_{q^n}$. By Hensel's lemma [Poo17, Theorem 3.5.63] they lift to m distinct roots of $x^m - 1$, hence to all of them, in the unramified extension of degree n. In particular $T_\ell(\mathbb{G}_{\mathrm{m}})$ is unramified for $\ell \neq p$, and an element $\tau \in \mathrm{Gal}_{k_v}$ acts on it through $\chi_\ell(\tau)$, which depends only on the image of τ in $\mathrm{Gal}_{\mathbb{F}_q}$: $\chi_\ell(\tau) = q$ for every τ lifting σ, because reduction is injective on $\mu_{\ell^n}(k_v^{\mathrm{sep}})$ and $\overline{\tau(\zeta)} = \bar{\zeta}^q$.
(2) Let $k_v = \mathbb{Q}_p$ with p odd. Then $\mathbb{Q}_p(\mu_p)$ is generated by a root of the Eisenstein polynomial $((x+1)^p - 1)/x$, so it is totally ramified of degree $p-1$, and I_p acts on $\mu_p(\mathbb{Q}_p^{\mathrm{sep}})$ through all of $(\mathbb{Z}/p\mathbb{Z})^\times$: $\mu_p(\mathbb{Q}_p^{\mathrm{sep}})$ is ramified, and so is $T_p(\mathbb{G}_{\mathrm{m}})$. For $p = 2$, $\mu_2(\mathbb{Q}_2^{\mathrm{sep}}) = \{\pm 1\}$ is unramified, but $\mu_4(\mathbb{Q}_2^{\mathrm{sep}})$ generates $\mathbb{Q}_2(i)$, totally ramified because $-1 + i$ is a root of the Eisenstein polynomial $x^2 + 2x + 2$. So $\mu_4(\mathbb{Q}_2^{\mathrm{sep}})$ and $T_2(\mathbb{G}_{\mathrm{m}})$ are ramified.
(3) For $m \geqslant 2$, the extension $\mathbb{Q}_p(p^{1/m})$ is totally ramified of degree m, since $x^m - p$ is Eisenstein. It is not contained in $\mathbb{Q}_p^{\mathrm{unr}}$, so some element of I_p moves $p^{1/m}$.

The torsion of an elliptic curve with good reduction behaves like $\mu_m(k_v^{\mathrm{sep}})$ with $p \nmid m$. The hypothesis $p \nmid m$ cannot be dropped. Let m be prime to $\operatorname{char} k$. If $E[m](k_v^{\mathrm{sep}})$ is unramified, the field L it cuts out is unramified, and $\mu_m(k_v^{\mathrm{sep}}) \subseteq L$ by Corollary 4.6.5 over L. So $\mu_m(k_v^{\mathrm{sep}})$ is unramified too. For $k_v = \mathbb{Q}_p$ with p odd, part 2 of Example 6.4.5 therefore shows that $E[p](\mathbb{Q}_p^{\mathrm{sep}})$ is ramified, whatever the reduction of E. (Over a field containing $\mu_p(k_v^{\mathrm{sep}})$, such as $\mathbb{Q}_p(\mu_p)$, this argument says nothing.) The following is [Sil09, Proposition VII.4.1].

Proposition 6.4.6. *Let E be an elliptic curve over k_v with good reduction.*
(1) If $p \nmid m$, then $E[m](k_v^{\mathrm{sep}})$ is unramified.
(2) If $\ell \neq p$, then $T_\ell(E)$ is unramified.

Proof. Part 2 follows from part 1 with $m = \ell^n$. For part 1, note that $\operatorname{char} k \nmid m$, since $\operatorname{char} k$ is 0 or p. Fix a minimal Weierstrass equation for E (the proof used its Δ without fixing one). Its discriminant is a unit by part 1 of Proposition 6.3.2. Let $K' \subseteq k_v^{\mathrm{sep}}$ be the finite Galois extension of k_v generated by the coordinates of the points of $E[m](k_v^{\mathrm{sep}})$, with ring of integers $\mathcal{O}'$ and residue field κ'. By part 2 of Proposition 6.3.7, our equation is minimal for $E_{K'}$, so the reduction of $E_{K'}$ is the elliptic curve $(E_\pi)_{\kappa'}$. By part 2 of Proposition 6.2.9 over K', the reduction map $E[m](K') \to E_\pi(\kappa')$ is injective.

Let $\tau \in I_v$ and $P \in E[m](K')$. Since τ preserves the valuation, it maps $\mathcal{O}'$ to itself, and reduction commutes with it: $\overline{\tau(P)} = \bar{\tau}(\bar{P})$, where $\bar{\tau}$ is the action on κ'. As $\tau \in I_v$, $\bar{\tau}$ is the identity, so $\overline{\tau(P) - P} = \bar{P} - \bar{P} = \bar{O}$. Since $\tau(P) - P \in E[m](K')$, injectivity gives $\tau(P) = P$. □

The converse is the criterion of Néron–Ogg–Shafarevich [Sil09, Theorem VII.7.1]. Silverman proves it over k_v^{unr}, which is not complete and has an infinite residue field. We work instead over a finite unramified

extension, where the results of this chapter apply as they stand, and where the last step needs only that $\kappa'^{\times}$ and $\mathbb{G}_{\mathrm{m}}^{(d)}(\kappa')$ are cyclic, instead of the ℓ-torsion of $\overline{\mathbb{F}}_q^{\times}$ and $\overline{\mathbb{F}}_q$.

THEOREM 6.4.7 (Néron–Ogg–Shafarevich). *Let E be an elliptic curve over k_v. The following are equivalent.*
(1) E has good reduction.
(2) $E[m](k_v^{\mathrm{sep}})$ is unramified for every $m \geqslant 1$ with $p \nmid m$.
(3) $T_\ell(E)$ is unramified for some prime $\ell \neq p$.
(4) $E[m](k_v^{\mathrm{sep}})$ is unramified for infinitely many $m \geqslant 1$ with $p \nmid m$.

PROOF. Part 1 implies part 2 by Proposition 6.4.6, part 2 implies part 3 trivially, and part 3 implies part 4 with $m = \ell^n$.

Assume part 4, and let Δ be the discriminant of a minimal Weierstrass equation for E. Choose m with $p \nmid m$, $m > \max(4, v(\Delta))$ and $E[m](k_v^{\mathrm{sep}})$ unramified. The points of $E[m](k_v^{\mathrm{sep}})$ are defined over a finite unramified extension K' of k_v, with residue field κ', and $E[m](K') \cong (\mathbb{Z}/m\mathbb{Z})^2$ by Corollary 4.5.1.

The component group. Since $v' = v$ on k_v (Remark 6.3.5), our minimal equation stays integral over K', with the same $v(\Delta)$, so the minimal discriminant of $E_{K'}$ has valuation at most $v(\Delta)$. By Theorem 6.3.9 over K', $\#E(K')/E_0(K') \leqslant \max(4, v(\Delta)) < m$.

An ℓ-torsion subgroup of rank 2. Let $H \coloneqq E[m](K') \cap E_0(K')$. Then $(\mathbb{Z}/m\mathbb{Z})^2/H$ injects into $E(K')/E_0(K')$, so H has index less than m in $(\mathbb{Z}/m\mathbb{Z})^2$, and $\#H > m$. A subgroup of $(\mathbb{Z}/m\mathbb{Z})^2$ whose ℓ-parts are all cyclic is cyclic of order dividing m. So H contains a subgroup $H_\ell \cong (\mathbb{Z}/\ell\mathbb{Z})^2$ for some prime $\ell \mid m$. By part 1 of Proposition 6.2.9 over K', $E_1(K') \cap E[\ell](K') = \{O\}$, so the reduction map of Proposition 6.2.8 over K' is injective on H_ℓ:

$$(\mathbb{Z}/\ell\mathbb{Z})^2 \hookrightarrow (E_{K'})_{\pi'}^{\mathrm{ns}}(\kappa').$$

The reduction type. If $E_{K'}$ had bad reduction, then by Proposition 6.3.2 over K' the group $(E_{K'})_{\pi'}^{\mathrm{ns}}(\kappa')$ would be $\mathbb{G}_{\mathrm{a}}(\kappa') = \kappa'$, a p-group with $\ell \neq p$; or $\mathbb{G}_{\mathrm{m}}(\kappa') = \kappa'^{\times}$, cyclic; or $\mathbb{G}_{\mathrm{m}}^{(d)}(\kappa')$, cyclic by part 4 of Proposition 2.1.9. None of these contains $(\mathbb{Z}/\ell\mathbb{Z})^2$. So $E_{K'}$ has good reduction, and so does E, by part 1 of Proposition 6.3.7. When $p \in \{2, 3\}$, this last step is the only one that uses Tate's algorithm. □

COROLLARY 6.4.8. *Let E_1 and E_2 be elliptic curves over k_v that are isogenous over k_v. Then E_1 has good reduction if and only if E_2 does.*

PROOF. Let $\varphi\colon E_1 \to E_2$ be a non-zero isogeny over k_v, and let m be prime to p and to $\deg\varphi$. The kernel of φ on $E_1(k_v^{\mathrm{sep}})$ has order dividing $\deg\varphi$ (Theorem 4.3.8), so it meets $E_1[m](k_v^{\mathrm{sep}})$ trivially, and $\varphi\colon E_1[m](k_v^{\mathrm{sep}}) \to E_2[m](k_v^{\mathrm{sep}})$ is a Gal_{k_v}-equivariant bijection between sets of order m^2. So $E_1[m](k_v^{\mathrm{sep}})$ is unramified if and only if $E_2[m](k_v^{\mathrm{sep}})$ is, and Theorem 6.4.7 applies, with infinitely many such m. □

COROLLARY 6.4.9. *Let E be an elliptic curve over k_v, and $\ell \neq p$ a prime. Then E has potential good reduction if and only if I_v acts on $T_\ell(E)$ through a finite quotient.*

PROOF. Suppose that E has potential good reduction. Then $E_{K'}$ has good reduction for some finite extension $K' \subseteq k_v^{\mathrm{sep}}$: if $\mathrm{char}\, k = 0$ every finite extension is separable, and if $p \neq 2$ the proof of Proposition 6.3.8 gives good reduction over a finite separable extension. Enlarging K' by part 2 of Proposition 6.3.7, we may take it finite Galois over k_v. The inertia group of K' is $I_v \cap \mathrm{Gal}_{K'}$, and it acts trivially on $T_\ell(E)$ by Theorem 6.4.7 over K'. It has finite index in I_v. Conversely, suppose that I_v acts through I_v/J with J open. The fixed field of J is a finite extension of k_v^{unr}, so it is $K'k_v^{\mathrm{unr}}$ for some finite extension K' of k_v. The inertia group of K' is $\mathrm{Gal}(k_v^{\mathrm{sep}}/K'k_v^{\mathrm{unr}}) = J$, which acts trivially on $T_\ell(E)$. So $E_{K'}$ has good reduction by Theorem 6.4.7 over K'. □

This is [Sil09, Corollaries VII.7.2 and VII.7.3]. We end with two examples.

EXAMPLE 6.4.10. Let $k = \mathbb{Q}$ and $v = p$.

(1) Let $p \geqslant 5$ and $E\colon y^2 = x^3 + p$, with additive reduction (part 4 of Example 6.3.4). Its $\mathbb{Q}_p^{\mathrm{sep}}$-points of order 2 are $(-\zeta p^{1/3}, 0)$ for the three cube roots of unity ζ. By part 3 of Example 6.4.5, some $\tau \in I_p$ moves $p^{1/3}$, hence moves the $\mathbb{Q}_p^{\mathrm{sep}}$-point $(-p^{1/3}, 0)$. So $E[2](\mathbb{Q}_p^{\mathrm{sep}})$ is ramified, and so is $E[m](\mathbb{Q}_p^{\mathrm{sep}})$ for every even m, consistent with Theorem 6.4.7. Part 2 shows that $m = 2$ alone need not detect bad reduction. Since E acquires good reduction over $\mathbb{Q}_p(p^{1/6})$, inertia acts on $T_\ell(E)$ through a finite quotient for every $\ell \neq p$, as Corollary 6.4.9 predicts.

(2) Let p be odd and $E\colon y^2 = x(x-1)(x+p)$, the Legendre form with $\lambda = -p$. Then $\Delta = 16p^2(p+1)^2$ and $c_4 = 16(p^2+p+1)$, a unit, so E has multiplicative reduction. Its reduction $y^2 = x^2(x-1)$ has tangent slopes $\pm\sqrt{-1}$, so it is split exactly when $p \equiv 1 \pmod 4$. Yet $E[2](\mathbb{Q}_p^{\mathrm{sep}}) = \{O, (0,0), (1,0), (-p,0)\}$ consists of rational points, so $E[2](\mathbb{Q}_p^{\mathrm{sep}})$ is unramified. One value of m does not suffice in Theorem 6.4.7. (In Chapter 12 we will see that for this curve, which is a Tate curve or an unramified quadratic twist of one, $E[\ell](\mathbb{Q}_p^{\mathrm{sep}})$ with $\ell \neq 2, p$ prime is unramified only when $\ell \mid v(\Delta) = 2$, that is, never.)

DRAFT
13 Oct 2026

6.5. The Kodaira–Néron classification and Tate's algorithm

SITUATION 6.5.1. In addition to Situation 6.0.2:

- $\bar\kappa$ is an algebraic closure of $\kappa(v)$.
- E is an elliptic curve over k_v, with a minimal Weierstrass equation, its minimal Weierstrass model $\mathcal{W}$ (Definition 6.2.2) and its Néron model $\mathcal{E}$ (Definition 6.2.6).
- $\mathcal{C}$ is the minimal regular proper model of E (Definition 6.5.3), with special fiber $\mathcal{C}_\pi$.

The residue field $\kappa(v)$ is perfect. The classification over $\kappa(v)$ and Tate's algorithm use this. Over an imperfect residue field both need more care [Liu02, Remark 10.2.3].

The reduction type of Section 6.3 is read off from E_π, a plane cubic, and it is coarse. Let $k = \mathbb{Q}$ and $p \geqslant 5$. The curves

$$E\colon y^2 = x^3 + p \qquad \text{and} \qquad E'\colon y^2 = x^3 + p^2$$

have $v(\Delta) = 2$ and 4, so both equations are minimal, and both reduce to the cuspidal cubic $y^2 = x^3$: both have additive reduction. Yet $E(\mathbb{Q}_p)/E_0(\mathbb{Q}_p)$ is trivial, while $E'(\mathbb{Q}_p)/E'_0(\mathbb{Q}_p)$ has order 3 (Example 6.5.14). By the discussion before Theorem 6.3.9, this quotient is the group of components of the special fiber of the Néron model, and the Néron model is built from the minimal Weierstrass model by blowing up (Theorem 6.2.7 and the paragraph after it). What tells E and E' apart is that the minimal Weierstrass model of E is regular and that of E' is not.

This section describes the special fibers that arise. For elliptic surfaces over $\mathbb{C}$ they were classified by Kodaira, and Néron found the same list over discrete valuation rings. Tate's algorithm then finds the fiber of a given curve from a Weierstrass equation, by a short list of divisibility tests. We prove little: the aim is to see where the fibers come from, by computing blow-ups in examples, and to be able to run the algorithm.

6.5.1. Regular models. Recall that a Noetherian local ring $(A, \mathfrak{m})$ is regular if $\dim_{A/\mathfrak{m}} \mathfrak{m}/\mathfrak{m}^2 = \dim A$ (Definition 2.4.1), and that a Noetherian scheme is regular if and only if its local rings at closed points are [Liu02, Corollary 4.2.17]. The Weierstrass model $\mathcal{W}$ of an integral equation is an integral scheme of dimension 2, flat and projective over $\mathcal{O}_v$: it is a fibered surface [Liu02, Definition 8.3.1], fibered over the curve-like base $\operatorname{Spec}\mathcal{O}_v$ with two fibers, the generic fiber E and the special fiber $\mathcal{W}_\pi$. Liu calls the regular fibered surfaces arithmetic surfaces [Liu02, Definition 8.3.14], and his intersection theory is for those. Since $\mathcal{W}$ is proper over $\mathcal{O}_v$, its closed points all lie on $\mathcal{W}_\pi$, so its regularity is decided there.

LEMMA 6.5.2. *Let $\mathcal{W}$ be the Weierstrass model of a Weierstrass equation with coefficients in $\mathcal{O}_v$, and let*

$$f(x,y) := y^2 + a_1xy + a_3y - x^3 - a_2x^2 - a_4x - a_6.$$

(1) $\mathcal{W}$ is regular at every closed point at which the cubic $\mathcal{W}_\pi$ is nonsingular.

(2) If $\mathcal{W}_\pi$ has a singular point, it is a rational point. Lift it to $(\alpha, \beta) \in \mathcal{O}_v^2$. Then $\mathcal{W}$ is regular at this closed point if and only if $v(f(\alpha,\beta)) = 1$.

PROOF. Let P be a closed point of $\mathcal{W}$, and $A \coloneqq \mathcal{O}_{\mathcal{W},P}$, with maximal ideal $\mathfrak{m}$. The scheme $\mathcal{W}$ is integral and $\pi_v \neq 0$ in A, so $\dim A/\pi_v A = \dim A - 1$ [Liu02, Theorem 2.5.15], and $A/\pi_v A$ is the local ring of the curve $\mathcal{W}_\pi$ at P, of dimension 1. So $\dim A = 2$.

For part 1, $\mathcal{W}_\pi$ is regular at P, since $\kappa(v)$ is perfect (part 2 of Proposition 2.4.2): the maximal ideal of $A/\pi_v A$ is generated by one element $\bar{t}$. Then $\mathfrak{m} = (\pi_v, t)$ is generated by $\dim A = 2$ elements, and A is regular.

For part 2, the singular point is a rational point by Remark 4.1.11, since $\kappa(v)$ is perfect, and it is affine, since $\bar{O}$ is a nonsingular point of $\mathcal{W}_\pi$ (Proposition 4.1.5). So P corresponds to the maximal ideal $\mathfrak{m}_P = (\pi_v, x-\alpha, y-\beta)$ of $B \coloneqq \mathcal{O}_v[x,y]$, and $A = B_{\mathfrak{m}_P}/(f)$. The local ring $B_{\mathfrak{m}_P}$ is regular of dimension 3, with $\mathfrak{m}_P$ generated by three elements, so A is regular if and only if $f \notin \mathfrak{m}_P^2 B_{\mathfrak{m}_P}$ [Liu02, Corollary 4.2.12 and Example 4.2.13]. Expand

$$f = f(\alpha,\beta) + f_x(\alpha,\beta)\,(x-\alpha) + f_y(\alpha,\beta)\,(y-\beta) + (\text{terms in } \mathfrak{m}_P^2),$$

where f_x, f_y are the partial derivatives. Since P is a singular point of the reduced cubic, $f(\alpha,\beta)$, $f_x(\alpha,\beta)$ and $f_y(\alpha,\beta)$ lie in $\mathfrak{m}_v$, so modulo $\mathfrak{m}_P^2$ only $f(\alpha,\beta)$ survives, and it lies in $\mathfrak{m}_P^2$ if and only if $\pi_v^2 \mid f(\alpha,\beta)$, because $\pi_v, x-\alpha, y-\beta$ are a basis of $\mathfrak{m}_P/\mathfrak{m}_P^2$. □

The condition does not depend on the lift: changing α by an element of $\mathfrak{m}_v$ changes $f(\alpha,\beta)$ by $f_x(\alpha,\beta)$ times it, plus terms in $\mathfrak{m}_v^2$, and $f_x(\alpha,\beta) \in \mathfrak{m}_v$. For $y^2 = x^3 + p$ the singular point is $(0,0)$ and $f(0,0) = -p$: the model is regular, and in the local ring at the cusp $p = y^2 - x^3$ is not a new generator. For $y^2 = x^3 + p^2$, $f(0,0) = -p^2$ and the model is not regular at the cusp. The fibers are the same, and only the surface sees the difference. See Figure 6.5.1. This is the arithmetic analogue of Kodaira's family $y^2 = x^3 + t$ versus $y^2 = x^3 + t^2$ over $\mathbb{C}[t]$ [Mil20, Example II.7.1].

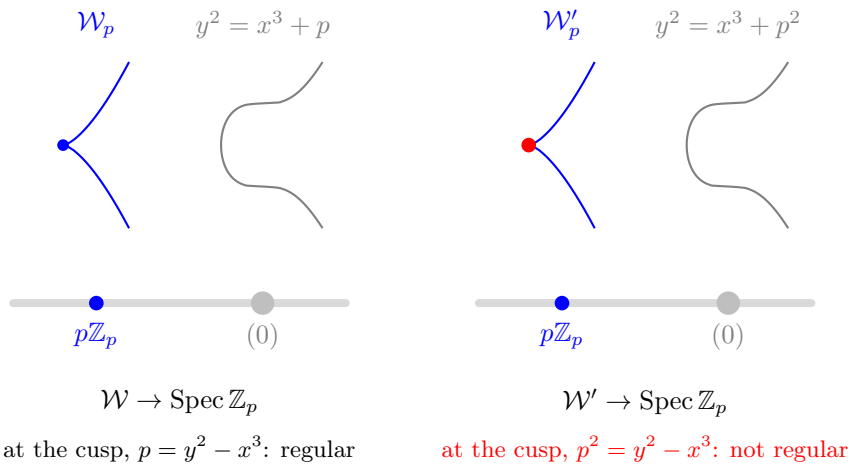


FIGURE 6.5.1. The Weierstrass models of $y^2 = x^3 + p$ and $y^2 = x^3 + p^2$ over $\operatorname{Spec} \mathbb{Z}_p$, with $p \geqslant 5$. The special fibers are the same cuspidal cubic; the first surface is regular at the cusp, and the second is not.

Every elliptic curve has a regular model, and among the regular ones there is a smallest. The following is [Liu02, Definition 9.3.12 and Proposition 10.1.8]. See also [Poo17, Section 9.3.1.6].

DEFINITION 6.5.3. A regular proper model of E is an integral model $\mathcal{X}$ of E (Definition 6.2.1) which is an integral regular scheme, flat and proper over $\mathcal{O}_v$. It is a minimal regular proper model if for every regular proper model $\mathcal{X}'$ of E, the identity of E extends to a morphism $\mathcal{X}' \to \mathcal{X}$ of $\mathcal{O}_v$-schemes.

A minimal regular proper model exists and is unique up to unique isomorphism [Liu02, Proposition 10.1.8]. For an elliptic curve it is constructed from the minimal Weierstrass model $\mathcal{W}$ by blowing up non-regular closed points until the result is regular [Poo17, Section 5.7.6], and its smooth locus is the Néron

model $\mathcal{E}$ [Liu02, Theorem 10.2.14]. So the Néron model, and the group $E(k_v)/E_0(k_v)$, are visible in the special fiber $\mathcal{C}_\pi$.

6.5.2. Blowing up, by hand. The blow-up of $\mathbb{A}^2_{\mathcal{O}_v}$ at the closed point (π_v, x, y) is covered by three affine charts, one for each generator of the ideal, in which that generator divides the other two [Liu02, Section 8.1]. In the chart of π_v, the coordinates are x_1, y_1 with $x = \pi_v x_1$ and $y = \pi_v y_1$. In the chart of x they are y_2, ϖ_2 with $y = xy_2$ and $\pi_v = x\varpi_2$. The chart of y is similar. The blow-up of $\mathcal{W}$ at that closed point is its strict transform: substitute, and divide f by the largest power of the chosen generator that divides it. The new special fiber is cut out by π_v, which now factors, and the exponents of its factors are the multiplicities of the components. The curve lying over the closed point that was blown up is the exceptional curve.

We work over $k_v = \mathbb{Q}_p$ with $p \geqslant 5$, and write p for π_v.

EXAMPLE 6.5.4. Let $E\colon y^2 = x^3 + x^2 + p^2$. Here $\Delta = -16p^2(27p^2 + 4)$ and $c_4 = 16$, so the reduction is multiplicative, with the node at $(0,0)$, and $v(f(0,0)) = 2$: the model is not regular there. In the chart $x = px_1$, $y = py_1$, dividing by p^2 gives

$$y_1^2 = x_1^2 + 1 + px_1^3,$$

with special fiber the conic $y_1^2 - x_1^2 = 1$: the exceptional curve Γ_1, smooth of multiplicity 1. Its two closed points at infinity are the directions $y_1 = \pm x_1$, the two tangent lines at the node, and that is where the strict transform Γ_0 of the nodal cubic meets it: blowing up separates the two branches through the node. Checking the other two charts with the criterion $f \notin \mathfrak{m}^2$ from the proof of Lemma 6.5.2, the blown-up surface is regular. So $\mathcal{C}_\pi$ consists of two smooth rational curves meeting transversally in two closed points, Figure 6.5.2: type I_2.

The same chart for $y^2 = x^3 + x^2 + p^n$ with $n \geqslant 3$ gives $y_1^2 = x_1^2 + p^{n-2} + px_1^3$. Now the exceptional curve is the pair of lines $y_1 = \pm x_1$, each meeting Γ_0 at infinity, and they cross at the origin, where the equation has the same shape with $n-2$ in place of n. So each blow-up adds two components, or one when it reaches $n = 2$, and lowers n by 2. In the end $\mathcal{C}_\pi$ is a cycle of n smooth rational curves: type I_n. For $n = 1$ the model is already regular, and $\mathcal{C}_\pi = E_\pi$ is the nodal cubic.

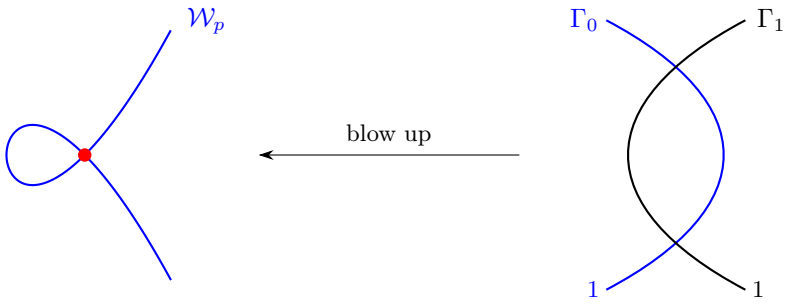


FIGURE 6.5.2. Blowing up the node of the special fiber of $y^2 = x^3 + x^2 + p^2$, where the surface is not regular, separates its two branches: type I_2. The strict transform Γ_0 of the nodal cubic is blue.

EXAMPLE 6.5.5. Blow up the cusp of the special fiber.

(1) Let $E\colon y^2 = x^3 + px$, with $\Delta = -64p^3$. At the cusp $f(0,0) = 0$, so the model is not regular there. In the chart $y = xy_2$, $p = x\varpi_2$, dividing by x^2 gives

$$y_2^2 = x + \varpi_2,$$

a regular surface, on which y_2 and ϖ_2 are local coordinates at the origin. The special fiber is $p = x\varpi_2 = (y_2^2 - \varpi_2)\,\varpi_2$: the strict transform $\Gamma_0 = \{\varpi_2 = 0\}$ of the cusp and the exceptional curve $\Gamma_1 = \{\varpi_2 = y_2^2\}$, both of multiplicity 1, tangent at one closed point. The other charts are regular, and this is type III.

(2) Let $E\colon y^2 = x^3 + p^2$, with $\Delta = -432p^4$. The same chart gives $y_2^2 = x + \varpi_2^2$, and the special fiber is $p = (y_2^2 - \varpi_2^2)\,\varpi_2$: three lines $\varpi_2 = 0$ and $y_2 = \pm\varpi_2$, through one closed point, pairwise transversal since $p \neq 2$. This is type IV.

See Figure 6.5.3. For $y^2 = x^3 + p$ the model is already regular, and $\mathcal{C}_\pi$ is the cuspidal cubic: type II.

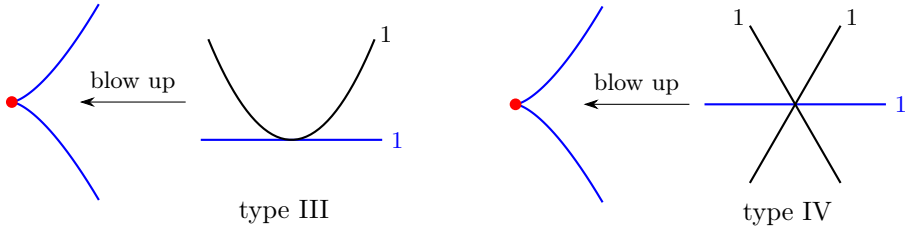


FIGURE 6.5.3. Blowing up the cusp of $y^2 = x^3 + px$ gives two curves tangent at one closed point, type III (left); for $y^2 = x^3 + p^2$ it gives three concurrent lines, type IV (right).

EXAMPLE 6.5.6. Let $E\colon y^2 = (x - ap)(x - bp)(x - cp)$ with $a, b, c \in \mathbb{Z}$ distinct modulo p, so that $v(\Delta) = 6$. In the chart $x = px_1$, $y = py_1$, dividing by p^2 gives

$$y_1^2 = p\,g(x_1), \qquad g(x_1) \coloneqq (x_1 - a)(x_1 - b)(x_1 - c).$$

Away from the roots of g, the function $p = y_1^2/g(x_1)$ vanishes to order 2 along the exceptional curve $y_1 = 0$: a line of multiplicity 2. At the three closed points $(p, y_1, x_1 - a)$, and likewise for b and c, the surface $y_1^2 = p\,(x_1 - a)\,u$, with u a unit, is not regular. Blowing up such a closed point, in the chart $p = (x_1 - a)p'$, $y_1 = (x_1 - a)y'$, gives the regular surface $y'^2 = p'u$, on which $p = (x_1 - a)\,y'^2/u$: a new exceptional curve $x_1 = a$ of multiplicity 1, crossing the double line $y' = 0$. With the strict transform of the cusp, which crosses the double line in the chart of x, the special fiber is four lines of multiplicity 1 crossing a line of multiplicity 2, Figure 6.5.4: type I_0^*. This is the first fiber that is not reduced.

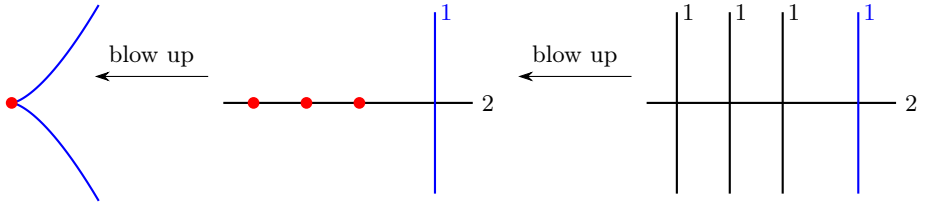


FIGURE 6.5.4. Type I_0^* in two blow-ups. The first produces a double line with three non-regular closed points (red), and blowing those up adds three simple lines. The strict transform of the cusp is blue.

6.5.3. The Kodaira–Néron classification. Let $\mathcal{X}$ be a regular proper model of E. Its special fiber is the divisor of the function π_v,

$$\mathcal{X}_\pi = \sum_i d_i \Gamma_i,$$

where the Γ_i are the irreducible components and $d_i \geqslant 1$ is the multiplicity of Γ_i: the valuation of π_v in the discrete valuation ring $\mathcal{O}_{\mathcal{X},\Gamma_i}$, the local ring at the generic point of Γ_i. A regular surface proper over $\mathcal{O}_v$ carries an intersection pairing between divisors supported on the special fiber [Liu02, Section 9.1.2]. For $i \neq j$, $\Gamma_i \cdot \Gamma_j \geqslant 0$ counts the closed points of $\Gamma_i \cap \Gamma_j$ with multiplicities and degrees. A transversal crossing at a rational point counts 1, the tangency of type III counts 2.

To see the fiber over $\bar{\kappa}$, let K' be a finite unramified extension of k_v, with residue field κ'. Then $\mathcal{C} \times_{\mathcal{O}_v} \mathcal{O}_{K'}$ is the minimal regular proper model of $E_{K'}$ [Liu02, Proposition 10.1.17], with special fiber $\mathcal{C}_\pi \times_{\kappa(v)} \kappa'$. Taking

κ' large enough, every component of $\mathcal{C}_{\bar\kappa} \coloneqq \mathcal{C}_\pi \times_{\kappa(v)} \bar\kappa$ and every crossing point is defined over κ'. So, replacing k_v by K', the components of $\mathcal{C}_\pi$ may be assumed geometrically irreducible with a rational point, and then they and their intersection numbers are those of $\mathcal{C}_{\bar\kappa}$. We use three facts.

PROPOSITION 6.5.7. *Let $\mathcal{X}$ be a regular proper model of E, with special fiber $\sum_i d_i\Gamma_i$.*
(1) $\Gamma_i \cdot \mathcal{X}_\pi = 0$ for every i, that is, $\sum_j d_j\, \Gamma_i \cdot \Gamma_j = 0$.
(2) The closure of the rational point O meets exactly one component, Γ_0, and $d_0 = 1$.
(3) If $\mathcal{X} = \mathcal{C}$, $\mathcal{C}_\pi$ has at least two components, and they are geometrically irreducible with a rational point, then every Γ_i is isomorphic to $\mathbb{P}^1_{\kappa(v)}$, and $\Gamma_i \cdot \Gamma_i = -2$.

PROOF. Part 1 holds because $\mathcal{X}_\pi$ is the divisor of the function π_v [Liu02, Proposition 9.1.21]. For part 2, O extends to a section $\sigma\colon \operatorname{Spec}\mathcal{O}_v \to \mathcal{X}$, since $\mathcal{X}$ is proper. Pulling back the divisor of π_v along σ gives $1 = v(\pi_v) = \sum_i d_i\,(\sigma\cdot\Gamma_i)$, with every $\sigma\cdot\Gamma_i$ a non-negative integer, positive when σ meets Γ_i. Part 3 is the adjunction formula: on the minimal model the relative canonical divisor is trivial [Liu02, Corollary 9.3.27], and a component with trivial canonical degree in a fiber with several components is a conic over its field of constants, which is $\kappa(v)$ here, with self-intersection -2 [Liu02, Proposition 9.4.8]. A geometrically irreducible conic over a perfect field is smooth [Liu02, Section 10.2.1], and a smooth conic with a rational point is $\mathbb{P}^1$. □

So over $\bar\kappa$ every component is a copy of $\mathbb{P}^1$ with self-intersection -2, unless $\mathcal{C}_{\bar\kappa}$ is irreducible. This is a strong constraint. Let $\Gamma_0, \dots, \Gamma_{m-1}$ be the components of $\mathcal{C}_{\bar\kappa}$, with $m \geqslant 2$, and form the symmetric matrix $A \coloneqq (-\Gamma_i\cdot\Gamma_j)_{i,j}$. It has 2 on the diagonal and entries $\leqslant 0$ off it, the special fiber is connected [Liu02, Corollary 8.3.6], and $A\,(d_0,\dots,d_{m-1})^{\mathsf{T}} = 0$ by part 1 of Proposition 6.5.7, with every $d_i > 0$. Encode A in the dual graph of the fiber: one vertex for each component, and $\Gamma_i\cdot\Gamma_j$ edges between Γ_i and Γ_j. A connected graph whose matrix of this form has a kernel vector with positive entries is one of the extended Dynkin diagrams $\tilde A_{m-1}$, $\tilde D_{m-1}$, $\tilde E_6$, $\tilde E_7$, $\tilde E_8$, and the kernel vector is unique up to scaling [Liu02, Section 10.2.1]. So the multiplicities d_i are forced, once $d_0 = 1$ on the component Γ_0 meeting O, by part 2 of Proposition 6.5.7. For instance, for $\tilde D_4$, with a central vertex joined to four others, the equations say $2d_{\text{end}} = d_{\text{center}}$ at each end and $2d_{\text{center}} = \sum d_{\text{end}}$ at the center: so $d = 1$ at the ends and $d = 2$ at the center, which is type I_0^*. Figure 6.5.5 shows the others.

The dual graph does not determine the fiber. Types I_2 and III both have dual graph $\tilde A_1$, two vertices joined by two edges: in I_2 the two curves cross at two closed points, and in III they are tangent at one. Types I_3 and IV both have $\tilde A_2$: a triangle of three crossing points, or three lines through one. And when $\mathcal{C}_{\bar\kappa}$ is irreducible it is a smooth, nodal or cuspidal cubic, types I_0, I_1 and II. With this caveat the list is complete, which is the theorem of Kodaira and Néron.

The Néron model $\mathcal{E}$ is the smooth locus of $\mathcal{C}$, so its special fiber is obtained from $\mathcal{C}_\pi$ by removing the components of multiplicity greater than 1 and all singular points of the reduced fiber, among them the crossing points. The identity component $\mathcal{E}_\pi^0$ is what remains of Γ_0, which is E_π^{ns} (Theorem 6.2.7).

THEOREM 6.5.8 (Kodaira, Néron). *Let $\mathcal{E}_\pi^0$ be the identity component of the special fiber of the Néron model, and*

$$\Phi \coloneqq \mathcal{E}_\pi/\mathcal{E}_\pi^0,$$

the component group of E, a finite étale group scheme over $\kappa(v)$.
(1) The special fiber $\mathcal{C}_{\bar\kappa}$ is one of the ten configurations of Figure 6.5.6, with the multiplicities shown there.
(2) $\Phi(\bar\kappa)$ is the group in the corresponding row. Its order is the number of components of multiplicity 1.
(3) $E(k_v)/E_0(k_v) \cong \Phi(\kappa(v))$, the subgroup of $\Phi(\bar\kappa)$ fixed by the Frobenius automorphism of $\bar\kappa$.
(4) If $p \geqslant 5$, then $v(\Delta)$ and $j(E)$ are as in the last two columns of Figure 6.5.6.

The configuration is the Kodaira type of E, written with Kodaira's symbols I_n, II, …, II^*, and the integer

$$c_v \coloneqq \#\Phi(\kappa(v)) = [E(k_v) : E_0(k_v)]$$

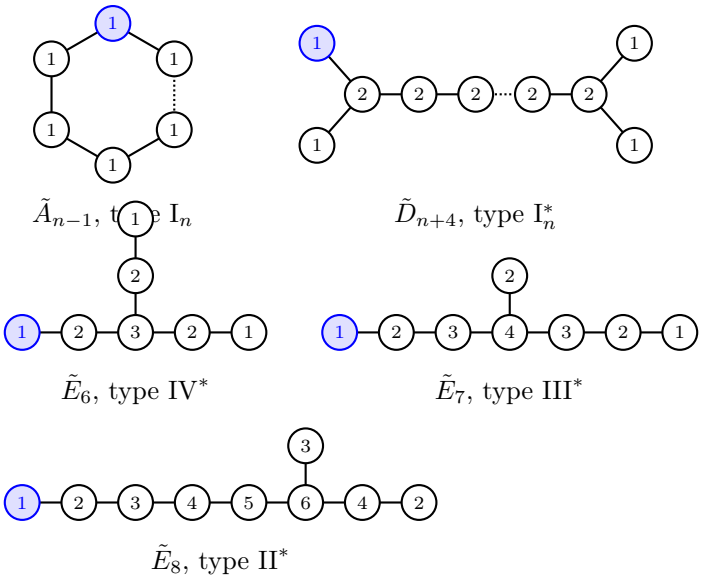


FIGURE 6.5.5. The dual graphs of the special fibers with at least two components over $\bar{\kappa}$, labeled by the multiplicities d_i. The component Γ_0 meeting O is blue. A cycle of two vertices joined by a double edge is $\tilde{A}_1$.

is the Tamagawa number of E at v.

PROOF. Parts 1, 2 and 4 are [Sil09, Theorem C.15.2 and Table 15.1], proved in [Sil94, Sections IV.8 and IV.9]. Part 1 is also [Liu02, Section 10.2.1]. The order of $\Phi(\bar{\kappa})$ counts the components of $\mathcal{E}_{\bar{\kappa}}$, which are the components of multiplicity 1 of $\mathcal{C}_{\bar{\kappa}}$ with their singular points removed. For part 3, $E(k_v) = \mathcal{E}(\mathcal{O}_v)$ and $E_0(k_v) = \mathcal{E}^0(\mathcal{O}_v)$ by Theorem 6.2.7, and reduction maps them onto $\mathcal{E}_\pi(\kappa(v))$ and $\mathcal{E}^0_\pi(\kappa(v))$ by Hensel's lemma [Poo17, Theorem 3.5.63], since both schemes are smooth over $\mathcal{O}_v$, with the same kernel. So $E(k_v)/E_0(k_v) \cong \mathcal{E}_\pi(\kappa(v))/\mathcal{E}^0_\pi(\kappa(v))$, and this is $\Phi(\kappa(v))$ because $H^1(\kappa(v), \mathcal{E}^0_\pi) = 0$, by Lang's theorem for the smooth connected group $\mathcal{E}^0_\pi$ over the finite field $\kappa(v)$ [Poo17, Theorem 5.12.19]. Compare [Sil09, Theorem C.15.1]. □

COROLLARY 6.5.9. *$c_v \leqslant 4$ unless E has multiplicative reduction, of type I_n with $n = v(\Delta)$. Then $c_v = n$ if the reduction is split, and $c_v = \#(\mathbb{Z}/n\mathbb{Z})[2]$, which is 1 or 2, if it is nonsplit.*

PROOF. For the additive types, $\#\Phi(\bar{\kappa}) \leqslant 4$ by the table. For type I_n, the components form an n-gon, and $\Phi(\bar{\kappa}) = \mathbb{Z}/n\mathbb{Z}$ numbers them around it, starting from Γ_0. Frobenius fixes Γ_0, and it preserves or reverses the direction around the cycle according as it fixes or swaps the two branches of E_π at the node, that is, according as the reduction is split or nonsplit. So it acts on $\mathbb{Z}/n\mathbb{Z}$ by 1 or by -1, with fixed subgroup $\mathbb{Z}/n\mathbb{Z}$ or $(\mathbb{Z}/n\mathbb{Z})[2]$. □

This is the second part of Theorem 6.3.9. Over $\kappa(v)$ itself the components of $\mathcal{C}_\pi$ need not be geometrically irreducible, nor the crossing points rational, and Liu refines Kodaira's symbols to record this: $\mathrm{I}_{n,2}$ for nonsplit multiplicative reduction, IV_2, $\mathrm{I}^*_{0,2}$, $\mathrm{I}^*_{0,3}$, $\mathrm{I}^*_{n,2}$ and IV^*_2 [Liu02, Section 10.2.1]. The Tamagawa number c_v is the number of Frobenius-fixed elements of $\Phi(\bar{\kappa})$, and Tate's algorithm below computes it.

REMARK 6.5.10. Let $p \geqslant 5$. A Weierstrass equation with coefficients in $\mathcal{O}_v$ is minimal if and only if $v(\Delta) < 12$ or $v(c_4) < 4$ [Sil09, Remark C.15.3]. The Kodaira type of E is then read off from $v(\Delta)$ and $v(j)$ by the table: if $v(j) \geqslant 0$, the values $v(\Delta) = 0, 2, 3, 4, 6, 8, 9, 10$ give $\mathrm{I}_0, \mathrm{II}, \mathrm{III}, \mathrm{IV}, \mathrm{I}_0^*, \mathrm{IV}^*, \mathrm{III}^*, \mathrm{II}^*$; if $v(j) = -n < 0$, the type is I_n or I_n^* according as $v(\Delta) = n$ or $n + 6$. Both $v(j) \geqslant 0$ and $v(j) < 0$ occur with $v(\Delta) = 8$: types IV^* and I_2^*.

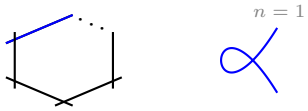
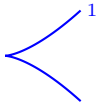
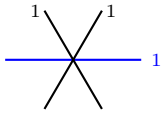
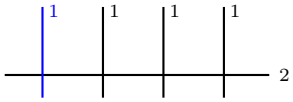
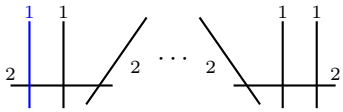
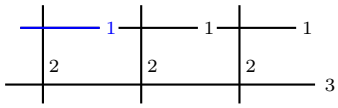
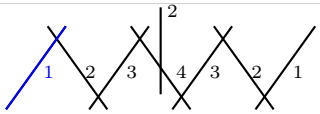
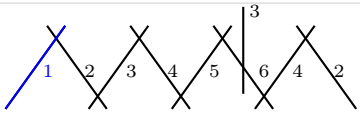
type	special fiber $\mathcal{C}_\pi$ over $\bar\kappa$	m	graph	$\Phi(\bar\kappa)$	$v(\Delta)$	j
I_0		1	–	0	0	$v(j)\geqslant 0$
I_n		n	$\tilde A_{n-1}$ $(n\geqslant 1)$	$\mathbb{Z}/n\mathbb{Z}$	n	$v(j)=-n$
II		1	$\tilde A_0$	0	2	$\bar\jmath=0$
III		2	$\tilde A_1$	$\mathbb{Z}/2\mathbb{Z}$	3	$\bar\jmath=1728$
IV		3	$\tilde A_2$	$\mathbb{Z}/3\mathbb{Z}$	4	$\bar\jmath=0$
I_0^*		5	$\tilde D_4$	$(\mathbb{Z}/2\mathbb{Z})^2$	6	$v(j)\geqslant 0$
I_n^*		$n+5$	$\tilde D_{n+4}$	$(\mathbb{Z}/2\mathbb{Z})^2$ if n is even $\mathbb{Z}/4\mathbb{Z}$ if n is odd	$n+6$	$v(j)=-n$
IV^*		7	$\tilde E_6$	$\mathbb{Z}/3\mathbb{Z}$	8	$\bar\jmath=0$
III^*		8	$\tilde E_7$	$\mathbb{Z}/2\mathbb{Z}$	9	$\bar\jmath=1728$
II^*		9	$\tilde E_8$	0	10	$\bar\jmath=0$

FIGURE 6.5.6. The Kodaira types. For each, the special fiber of the minimal regular proper model over $\bar\kappa$: every component is a smooth rational curve except in types I_0, I_1 and II, the numbers are the multiplicities, and the component Γ_0 meeting O is blue. Then the number m of components, the dual graph, the group $\Phi(\bar\kappa)$, and, valid only when $p\geqslant 5$, the valuation of the minimal discriminant and the behavior of j ($\bar\jmath$ is the reduction of $j\in\mathcal{O}_v$). After [Sil09, Table 15.1].

REMARK 6.5.11. Let $p \geqslant 5$, and let $E\colon y^2 = x^3 + Ax + B$ be minimal. Its quadratic twist by π_v, $E^{(\pi)}\colon y^2 = x^3 + \pi_v^2 Ax + \pi_v^3 B$, has Δ multiplied by π_v^6, c_4 by π_v^2, and the same j. By Remark 6.5.10, after rescaling by π_v^{-12} when the new equation is not minimal, twisting exchanges

$$\mathrm{I}_0 \leftrightarrow \mathrm{I}_0^*, \qquad \mathrm{I}_n \leftrightarrow \mathrm{I}_n^*, \qquad \mathrm{II} \leftrightarrow \mathrm{IV}^*, \qquad \mathrm{III} \leftrightarrow \mathrm{III}^*, \qquad \mathrm{IV} \leftrightarrow \mathrm{II}^*.$$

This explains the second half of the table: the starred types are the twists of the unstarred ones, and their $v(\Delta)$ is shifted by 6 modulo 12.

6.5.4. Tate's algorithm. When $p \geqslant 5$, Remark 6.5.10 finds the Kodaira type from $v(\Delta)$ and $v(j)$. When $p = 2$ or 3 this fails twice over. Minimality is no longer visible from $v(\Delta)$ and $v(c_4)$: $y^2 = x^3 + 16$ over $\mathbb{Q}_2$ has $v(\Delta) = 12$ and $c_4 = 0$, and it turns out to be far from minimal. And the type is not determined by $v(\Delta)$: over $\mathbb{Q}_3$, $y^2 + y = x^3$ has type II with $v(\Delta) = 3$, and over $\mathbb{Q}_2$, $y^2 = x^3 - x$ has type III with $v(\Delta) = 6$ (Example 6.5.16).

Tate's algorithm works for every residue field that is perfect. It takes a Weierstrass equation with coefficients in $\mathcal{O}_v$ and runs through a list of tests, each a divisibility condition on the coefficients or a question about the roots of a polynomial over $\kappa(v)$, changing variables along the way. It stops at the Kodaira type, together with c_v, or it finds that the equation is not minimal, scales it, and starts again. We state it as in [Sil94, Section IV.9], and Figure 6.5.7 draws it.

THEOREM 6.5.12 (Tate's algorithm). *Let E be given by a Weierstrass equation with coefficients $a_i \in \mathcal{O}_v$, with b_2, b_6, b_8 and Δ as in Definition 4.1.2. Write $a_{i,j} \coloneqq \pi_v^{-j} a_i$ whenever $\pi_v^j \mid a_i$. Go through the following items in order. The changes of variables they prescribe are of the form (4.1.3) with $u = 1$ and $r, s, t \in \mathcal{O}_v$, and every root of a polynomial over $\kappa(v)$ that they move is in $\kappa(v)$. The first test that holds gives the Kodaira type of E and c_v, and then the current equation is minimal.*

(1) If $\pi_v \nmid \Delta$, the type is I_0, and $c_v = 1$.

(2) Otherwise the reduced cubic is singular. Move its singular point to $(0,0)$, so that $\pi_v \mid a_3, a_4, a_6$. If $\pi_v \nmid b_2$, the type is I_n with $n = v(\Delta)$. And $c_v = n$ if $T^2 + a_1 T - a_2$ splits into linear factors over $\kappa(v)$, otherwise $c_v = 1$ if n is odd and $c_v = 2$ if n is even.

(3) If $\pi_v^2 \nmid a_6$, the type is II, and $c_v = 1$.

(4) If $\pi_v^3 \nmid b_8$, the type is III, and $c_v = 2$.

(5) If $\pi_v^3 \nmid b_6$, the type is IV. And $c_v = 3$ if $T^2 + a_{3,1} T - a_{6,2}$ splits over $\kappa(v)$, otherwise $c_v = 1$.

(6) Otherwise change variables so that $\pi_v \mid a_1, a_2$, $\pi_v^2 \mid a_3, a_4$ and $\pi_v^3 \mid a_6$, and let

$$P(T) \coloneqq T^3 + a_{2,1} T^2 + a_{4,2} T + a_{6,3}.$$

If its reduction has three distinct roots in $\bar{\kappa}$, the type is I_0^*, and $c_v = 1 + \#\{\text{roots in } \kappa(v)\}$.

(7) If it has a simple root and a double root, move the double root to $T = 0$, so that $\pi_v^2 \nmid a_2$, $\pi_v^3 \mid a_4$ and $\pi_v^4 \mid a_6$. The type is I_n^* for some $n \geqslant 1$, found as follows. For $n = 1, 2, \ldots$ in turn, consider the quadratic polynomial

$$Q_n \coloneqq \begin{cases} Y^2 + a_{3,(n+3)/2}\, Y - a_{6,n+3} & \text{if } n \text{ is odd,} \\ a_{2,1} X^2 + a_{4,(n+4)/2}\, X + a_{6,n+3} & \text{if } n \text{ is even.} \end{cases}$$

If its reduction has distinct roots in $\bar{\kappa}$, the type is I_n^*, and $c_v = 4$ if the roots lie in $\kappa(v)$, otherwise $c_v = 2$. If it has a double root, move it to 0, which makes $\pi_v^{(n+5)/2} \mid a_3$ and $\pi_v^{n+4} \mid a_6$ if n is odd, and $\pi_v^{(n+6)/2} \mid a_4$ and $\pi_v^{n+4} \mid a_6$ if n is even, and go on to $n+1$. This stops after finitely many steps.

(8) If P has a triple root, move it to $T = 0$, so that $\pi_v^2 \mid a_2$, $\pi_v^3 \mid a_4$ and $\pi_v^4 \mid a_6$. If the reduction of $Y^2 + a_{3,2} Y - a_{6,4}$ has distinct roots in $\bar{\kappa}$, the type is IV^*. And $c_v = 3$ if they lie in $\kappa(v)$, otherwise $c_v = 1$.

(9) Otherwise move its double root to 0, so that $\pi_v^3 \mid a_3$ and $\pi_v^5 \mid a_6$. If $\pi_v^4 \nmid a_4$, the type is III^*, and $c_v = 2$.

(10) If $\pi_v^6 \nmid a_6$, the type is II^*, and $c_v = 1$.

(11) Otherwise $\pi_v^i \mid a_i$ for every i, and the equation is not minimal. Replace each a_i by $\pi_v^{-i} a_i$, which is the substitution $x = \pi_v^2 x'$, $y = \pi_v^3 y'$ of Section 6.1, and start again.

In particular, an integral Weierstrass equation is minimal if and only if the algorithm, started on it, does not reach part 11.

PROOF. See [Sil94, Section IV.9], and Tate's original letter [Tat75]. □

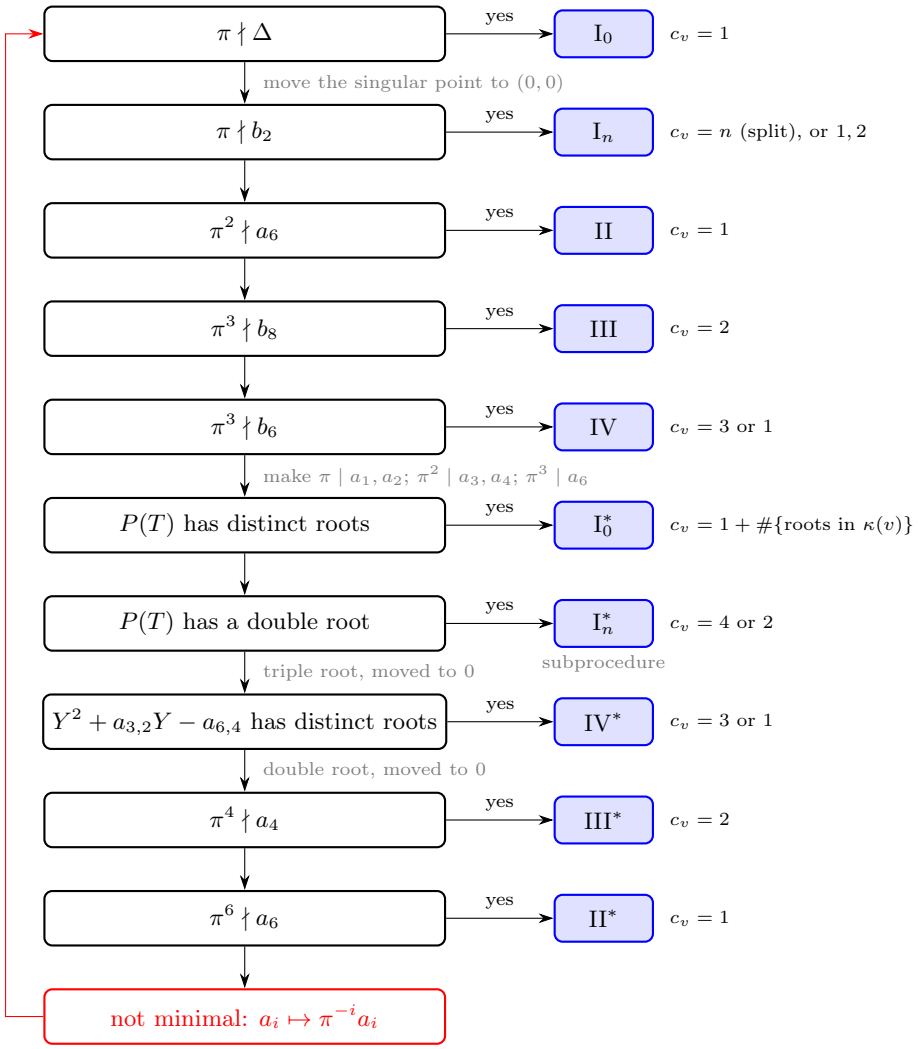


FIGURE 6.5.7. Tate's algorithm (Theorem 6.5.12). A downward arrow means that the test failed; the gray notes are the changes of variables made on the way. The type I_n^* calls the subprocedure of part 7, and the red box sends a non-minimal equation back to the start.

The roots that the algorithm moves are in $\kappa(v)$ because $\kappa(v)$ is perfect: the singular point of a cubic, a double or triple root of a cubic, and a double root of a quadratic over a perfect field are all rational. This is where perfectness is used.

Why does it work? Each test is a question about the model at hand, and Silverman's proof answers it with the blow-ups of the previous subsections, written for a general Weierstrass equation. Part 2 asks whether the singular point is a node or a cusp: after moving it to $(0,0)$, the quadratic part of f is $y^2 + a_1xy - a_2x^2$ modulo π_v, a product of two distinct lines exactly when $b_2 = a_1^2 + 4a_2$ is a unit. Part 3 is Lemma 6.5.2, since $f(0,0) = -a_6$: if $v(a_6) = 1$ the Weierstrass model is regular, $\mathcal{C} = \mathcal{W}$, and the fiber is the cuspidal cubic. Parts 4 and 5 look at the exceptional curve of the first blow-up, as in Example 6.5.5, whose computations

they generalize. Part 6 finds the double line of Example 6.5.6, with P recording where the surface fails to be regular on it. Each further part blows up again, and the polynomials Q_n record the new non-regular closed points.

Here is the payoff promised in the proof of Proposition 6.3.7.

COROLLARY 6.5.13. *Let K' be a finite unramified extension of k_v. A minimal Weierstrass equation for E is minimal for $E_{K'}$, and E and $E_{K'}$ have the same Kodaira type.*

PROOF. Run Tate's algorithm on the same equation over k_v and over K'. Since K'/k_v is unramified, π_v is a uniformizer of K', and the valuation of K' restricts to v. So every divisibility test has the same answer over both fields. The roots that the algorithm moves are in $\kappa(v)$, so the same changes of variables, with coefficients in $\mathcal{O}_v$, serve over both. And whether a polynomial over $\kappa(v)$ has distinct roots in $\bar{\kappa}$ does not depend on the field. So the algorithm takes the same path over k_v and over K', and stops at the same type. Over k_v it does not reach part 11, since the equation is minimal, so it does not over K' either. Alternatively, the minimal regular proper model is compatible with unramified base change [Liu02, Proposition 10.1.17]. □

Only c_v can change over K': the tests "splits over $\kappa(v)$" become tests over the larger residue field. For instance, a nonsplit I_n becomes split over the unramified quadratic extension, and c_v grows from 1 or 2 to n.

6.5.5. Examples. The examples below were checked with Magma's `LocalInformation` [BCP97], which runs Tate's algorithm.

EXAMPLE 6.5.14. Let $k = \mathbb{Q}$ and $p \geqslant 5$. The curves $y^2 = x^3 + p^k$, with $j = 0$, and $y^2 = x^3 + p^k x$, with $j = 1728$, have $v(\Delta) = 2k$ and $3k$, and $c_4 = 0$, respectively $v(c_4) = k$. For $k = 1, \dots, 5$, respectively $k = 1, 2, 3$, the equations are minimal, and Remark 6.5.10 gives the types in the following table. For $k = 6$, respectively $k = 4$, they are not minimal: the substitution $x = p^2 x'$, $y = p^3 y'$ gives $y'^2 = x'^3 + 1$, respectively $y'^2 = x'^3 + x'$, with good reduction.
The Tamagawa numbers come from Theorem 6.5.12. For $y^2 = x^3 + p^2$, part 5 looks at $T^2 - 1$, which splits, so $c_v = 3$: this is the curve E' from the start of the section. For $y^2 = x^3 + p^3$, part 6 looks at $P(T) = T^3 + 1$, whose roots in $\mathbb{F}_p$ are -1 and, when $p \equiv 1 \pmod 3$, two more. So $c_v = 4$ or 2. For $y^2 = x^3 + p^2 x$, $P(T) = T^3 + T$, and $c_v = 4$ or 2 according as $p \equiv 1$ or $3 \pmod 4$.

k	1	2	3	4	5
$y^2 = x^3 + p^k$	II, 1	IV, 3	I_0^*, 4 or 2	IV^*, 3	II^*, 1
$y^2 = x^3 + p^k x$	III, 2	I_0^*, 4 or 2	III^*, 2		

Each entry is the Kodaira type and c_v. Moving three places along the first row, or two along the second, is the quadratic twist by p, as in Remark 6.5.11.

EXAMPLE 6.5.15. Let $k = \mathbb{Q}$, $p \geqslant 5$ and $n \geqslant 1$. The curve $y^2 = x^3 + x^2 + p^n$ has split multiplicative reduction of type I_n, with $c_v = n$, by part 2 of Theorem 6.5.12: $T^2 - 1$ splits. Its quadratic twist by p is $y^2 = x^3 + px^2 + p^{n+3}$, with $v(\Delta) = n + 6$. Run Tate's algorithm on it. It is already in the shape of part 6, with $P(T) = T^3 + T^2 + p^n \equiv T^2(T + 1)$: a double root at 0, so we are in part 7. For $m < n$ the polynomial Q_m is Y^2 or X^2 modulo p, with a double root at 0 and nothing to move, and Q_n is $Y^2 - 1$ if n is odd and $X^2 + 1$ if n is even. So the type is I_n^*, with $c_v = 4$ if n is odd or $p \equiv 1 \pmod 4$, and $c_v = 2$ otherwise.

EXAMPLE 6.5.16. Three curves over $\mathbb{Q}$ at $p = 2$ and $p = 3$, where Remark 6.5.10 does not apply.

(1) $y^2 = x^3 + 16$ at $p = 2$, with $\Delta = -2^{12}3^3$. The reduction $y^2 = x^3$ is singular at $(0, 0)$, and $b_2 = 0$, $a_6 = 16$, $b_8 = 0$, $b_6 = 64$ are divisible by 2, 4, 8, 8: we reach part 6, with nothing to change, and $P(T) = T^3 + 2 \equiv T^3$, a triple root at 0. In part 8, $Y^2 + a_{3,2}Y - a_{6,4} = Y^2 - 1 \equiv (Y + 1)^2$ modulo 2, a double root at 1. Moving it to 0 with $y = y' + 4$ gives $y'^2 + 8y' = x^3$, so $a_3 = 8$ and $a_4 = a_6 = 0$: both tests of parts 9 and 10 fail, and the equation is not minimal. Scaling gives $y^2 + y = x^3$, with $\Delta = -27$: good reduction at 2.
(2) $y^2 + y = x^3$ at $p = 3$, with $\Delta = -27$. Modulo 3, $\partial f/\partial y = 2y + 1$ vanishes at $y = 1$, $\partial f/\partial x = -3x^2$ vanishes identically, and the equation forces $x^3 = 2$, so the singular point is $(2, 1)$. With $x = x' + 2$,

$y = y' + 1$ the equation becomes

$$y'^2 + 3y' = x'^3 + 6x'^2 + 12x' + 6,$$

with $b_2 = 24$, divisible by 3, and $a_6 = 6$, not divisible by 9: type II by part 3, with $v(\Delta) = 3$, not the 2 of the table.

(3) $y^2 = x^3 - x$ at $p = 2$, with $\Delta = 64$. The reduction $y^2 = x^3 + x$ is singular at $(1, 0)$, and $x = x' + 1$ gives $y^2 = x'^3 + 3x'^2 + 2x'$, with $b_2 = 12$, $a_6 = 0$ and $b_8 = -a_4^2 = -4$, not divisible by 8: type III by part 4, with $c_v = 2$ and $v(\Delta) = 6$.

REMARK 6.5.17. The exponent of the conductor of E at v is an integer $f_v \geqslant 0$ measuring the ramification of the Galois action on the torsion of E. It enters the L-function in Chapter 13. It is 0 for good reduction and 1 for multiplicative reduction, and if $p \geqslant 5$ it is 2 for additive reduction [Sil09, Table 15.1]. Ogg's formula

$$f_v = v(\Delta) + 1 - m,$$

with Δ minimal and m the number of components of $\mathcal{C}_{\bar{\kappa}}$, computes it in general. In Example 6.5.16 it gives $f_3 = 3 + 1 - 1 = 3$ for $y^2 + y = x^3$ and $f_2 = 6 + 1 - 2 = 5$ for $y^2 = x^3 - x$, so these curves have conductors 27 and 32.

CHAPTER 7

Galois cohomology

Sources. This chapter follows [Sil09, Appendix B] and [Sil09, Section VIII.2], with the left actions of [Poo02, Sections 2–6]. For the point of view on twists, torsors, Brauer groups and étale cohomology we follow [Poo17, Sections 1.3, 1.5, 4.4, 4.5, 5.12 and 6.2–6.6].

SITUATION 7.0.1.

- k is a field, not assumed perfect, with a separable closure k^{sep} and absolute Galois group $\mathrm{Gal}_k := \mathrm{Gal}(k^{\mathrm{sep}}/k)$. For an extension L of k inside k^{sep}, $\mathrm{Gal}_L := \mathrm{Gal}(k^{\mathrm{sep}}/L)$.
- G is a profinite group, often finite or Gal_k, with elements g, h. Elements of a Galois group, such as Gal_k or $\mathrm{Gal}(L/k)$, are σ, τ, γ. Where σ is introduced as a generator of a cyclic or procyclic group, such as a Frobenius automorphism, it keeps that meaning, and other Galois elements there are τ, γ.
- A is a discrete G-module, written additively, with the left action $a \mapsto ga$ (Definition 7.1.2).
- For a discrete Gal_k-module A we write $H^i(k, A) := H^i(\mathrm{Gal}_k, A)$. For a smooth commutative algebraic group A over k, such as $\mathbb{G}_{\mathrm{m}}$, μ_m or E, we abuse notation and write $H^i(k, A) := H^i(\mathrm{Gal}_k, A(k^{\mathrm{sep}}))$. Only here does A stand for $A(k^{\mathrm{sep}})$.
- E is an elliptic curve over k.
- $m \geqslant 1$ is an integer and $\mu_m := \mathbb{G}_{\mathrm{m}}[m]$, the kernel of $x \mapsto x^m$ on $\mathbb{G}_{\mathrm{m}}$. Unless stated otherwise, $\mathrm{char}\, k \nmid m$. Then $\mu_m(k^{\mathrm{sep}}) \cong \mathbb{Z}/m\mathbb{Z}$ and $E[m](k^{\mathrm{sep}}) \cong (\mathbb{Z}/m\mathbb{Z})^2$ as groups.
- For an abelian group B, $B/m := B/mB$ and $B[m]$ is its m-torsion subgroup. For a field L, $L^\times/L^{\times m}$ is $L^\times/m$, written multiplicatively.

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7.1. Profinite groups and discrete modules

We are in Situation 7.0.1. Galois cohomology studies the group Gal_k through the abelian groups on which it acts: the multiplicative group $(k^{\mathrm{sep}})^\times$, its roots of unity, the group $E(k^{\mathrm{sep}})$ and its torsion. Silverman [Sil09] assumes k perfect and works with an algebraic closure. Nothing in Silverman's Appendix B changes with k^{sep} in its place. Silverman also writes right actions, $a \mapsto a^\sigma$, where we write left actions, $a \mapsto ga$.

Restriction makes the finite groups $\mathrm{Gal}(L/k)$, for L running over the finite Galois extensions of k inside k^{sep}, an inverse system, and $\mathrm{Gal}_k = \varprojlim_L \mathrm{Gal}(L/k)$. So Gal_k is a profinite group: an inverse limit of finite groups, compact, Hausdorff and totally disconnected. Its topology is the Krull topology.

PROPOSITION 7.1.1 (Infinite Galois theory).

(1) The subgroups Gal_L, for L running over the finite Galois extensions of k inside k^{sep}, are open and normal in Gal_k, and they form a basis of open neighborhoods of 1.

(2) The maps $H \mapsto (k^{\mathrm{sep}})^H$ and $L \mapsto \mathrm{Gal}_L$ are inverse bijections between the closed subgroups of Gal_k and the extensions of k inside k^{sep}. The subgroup H is open if and only if $(k^{\mathrm{sep}})^H$ is finite over k, and then $[\mathrm{Gal}_k : H] = [(k^{\mathrm{sep}})^H : k]$. It is normal if and only if $(k^{\mathrm{sep}})^H$ is Galois over k, and then $\mathrm{Gal}((k^{\mathrm{sep}})^H/k) \cong \mathrm{Gal}_k/H$ by restriction. In particular, $(k^{\mathrm{sep}})^{\mathrm{Gal}_k} = k$.

PROOF. See [Sil09, Section B.2] and [Mil22, Theorem 7.13]. □

In every profinite group the open normal subgroups form a basis of open neighborhoods of 1, and an open subgroup has finite index, since its cosets form an open cover of a compact group. We met a first example in Chapter 5: by (5.1.1), $\mathrm{Gal}_{\mathbb{F}_q} \cong \widehat{\mathbb{Z}}$, and the Frobenius automorphism $\sigma\colon \alpha \mapsto \alpha^q$ generates a dense subgroup $\sigma^{\mathbb{Z}}$ of $\mathrm{Gal}_{\mathbb{F}_q}$. We say that σ is a topological generator.

DEFINITION 7.1.2. A discrete G-module is an abelian group A with a left action of G by group automorphisms, written $a \mapsto ga$, such that the action map $G \times A \to A$ is continuous for the discrete topology on A. Equivalently, the stabilizer of every $a \in A$ is open in G, that is, A is the union of the subgroups A^U of elements fixed by U, for U running over the open subgroups of G. A morphism of discrete G-modules is a homomorphism that commutes with the action. A G-group is defined in the same way, without the condition that the group be commutative, and with its law written multiplicatively.

EXAMPLE 7.1.3. The following are discrete Gal_k-modules, with the action of Gal_k on k^{sep}.

(1) The additive group $\mathbb{G}_{\mathrm{a}}(k^{\mathrm{sep}}) = k^{\mathrm{sep}}$, the multiplicative group $\mathbb{G}_{\mathrm{m}}(k^{\mathrm{sep}}) = (k^{\mathrm{sep}})^\times$ and its subgroup $\mu_m(k^{\mathrm{sep}})$. The stabilizer of $\alpha \in k^{\mathrm{sep}}$ contains $\mathrm{Gal}_{k(\alpha)}$, which is open because $k(\alpha)$ is finite over k (part 2 of Proposition 7.1.1).
(2) Every abelian group, such as $\mathbb{Z}$, $\mathbb{Z}/m\mathbb{Z}$, $\mathbb{Q}$ or $\mathbb{Q}/\mathbb{Z}$, with the trivial action.
(3) The group $A(k^{\mathrm{sep}})$ of k^{sep}-points of a commutative algebraic group A over k, such as $E(k^{\mathrm{sep}})$ and $E[m](k^{\mathrm{sep}})$. The coordinates of a k^{sep}-point lie in a finite extension of k, as in part 1. For any k-variety X, the same argument shows that Gal_k acts continuously on the set $X(k^{\mathrm{sep}})$, and $X(k) = X(k^{\mathrm{sep}})^{\mathrm{Gal}_k}$ by Proposition 2.1.17.
(4) The group $\mathrm{GL}_n(k^{\mathrm{sep}})$ is a Gal_k-group, acting on the entries of a matrix.

The two finite modules of the chapter are $\mu_m(k^{\mathrm{sep}})$ and $E[m](k^{\mathrm{sep}})$. Their actions are recorded by homomorphisms. Since $\mu_m(k^{\mathrm{sep}})$ is cyclic of order m, its automorphism group is $(\mathbb{Z}/m\mathbb{Z})^\times$, and the action is the cyclotomic character

$$\chi_m \colon \mathrm{Gal}_k \to (\mathbb{Z}/m\mathbb{Z})^\times, \qquad \sigma(\zeta) = \zeta^{\chi_m(\sigma)} \quad \text{for } \zeta \in \mu_m(k^{\mathrm{sep}}).$$

Its kernel is $\mathrm{Gal}_{k(\mu_m)}$, where $k(\mu_m)$ is the extension of k generated by $\mu_m(k^{\mathrm{sep}})$. Likewise, the action on $E[m](k^{\mathrm{sep}}) \cong (\mathbb{Z}/m\mathbb{Z})^2$ is the mod m representation $\bar\rho_{E,m} \colon \mathrm{Gal}_k \to \mathrm{GL}_2(\mathbb{Z}/m\mathbb{Z})$ of Chapter 4, after the choice of a basis. Its kernel is $\mathrm{Gal}_{k(E[m])}$, where $k(E[m])$ is the extension of k generated by the coordinates of the torsion points in $E[m](k^{\mathrm{sep}})$. Both kernels are open, and $k(\mu_m)$ and $k(E[m])$ are finite Galois extensions of k.

PROPOSITION 7.1.4. *Let E be an elliptic curve over k. Then $\det \bar\rho_{E,m} = \chi_m$. In particular $k(\mu_m) \subseteq k(E[m])$.*

PROOF. Let (S, T) be the chosen basis of $E[m](k^{\mathrm{sep}})$, and $\sigma \in \mathrm{Gal}_k$ with $\sigma(S) = aS + cT$ and $\sigma(T) = bS + dT$. The Weil pairing e_m is bilinear and alternating, by parts 1 and 2 of Proposition 4.6.3, so

$$e_m(\sigma(S), \sigma(T)) = e_m(S, T)^{ad} e_m(T, S)^{cb} = e_m(S, T)^{ad-bc}.$$

By part 4 of the same proposition, $e_m(\sigma(S), \sigma(T)) = \sigma(e_m(S, T))$, which is $e_m(S, T)^{\chi_m(\sigma)}$. The root of unity $e_m(S, T)$ has order m: if it had order d, then $e_m(S, [d]T) = e_m(S, T)^d = 1$ and $e_m(T, [d]T) = 1$, so $e_m(S', [d]T) = 1$ for every $S' \in E[m](k^{\mathrm{sep}})$, and $[d]T = O$ by part 3 of Proposition 4.6.3. So $d = m$. Hence $ad - bc \equiv \chi_m(\sigma) \pmod m$. If σ acts trivially on $E[m](k^{\mathrm{sep}})$, then $\chi_m(\sigma) = \det 1 = 1$. □

This is the first use of the Weil pairing in this chapter. It returns in Section 7.9.

EXAMPLE 7.1.5. Suppose $\mathrm{char}\, k \neq 2$ and $E \colon y^2 = f(x)$, with $f \in k[x]$ a separable cubic polynomial with roots $e_1, e_2, e_3 \in k^{\mathrm{sep}}$. Then $E[2](k^{\mathrm{sep}}) = \{O, (e_1, 0), (e_2, 0), (e_3, 0)\}$ (Example 4.2.9), so $k(E[2])$ is the splitting field of f. An element of Gal_k acts on $E[2](k^{\mathrm{sep}})$ by permuting the three non-zero elements as it permutes the roots, and $\mathrm{GL}_2(\mathbb{F}_2)$ is the group S_3 of all permutations of these three elements. So the image of $\bar\rho_{E,2}$ is the Galois group of f. For $y^2 = x^3 - 2$ over $\mathbb{Q}$ it is all of $\mathrm{GL}_2(\mathbb{F}_2)$, and $\mathbb{Q}(E[2]) = \mathbb{Q}(\sqrt[3]{2}, \sqrt{-3})$ has degree 6. Here χ_2 is trivial, and every matrix in $\mathrm{GL}_2(\mathbb{F}_2)$ has determinant 1, as Proposition 7.1.4 requires.

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7.2. Cohomology in low degrees

We are in Situation 7.0.1. Cohomology measures the failure of the functor $A \mapsto A^G$ to preserve surjections. We need the groups $H^i(G, A)$ only for $i \leqslant 2$, and we define them with cochains.

Definition 7.2.1. For $i = 0, 1, 2, 3$, the group $C^i(G, A)$ of i-cochains is the group of continuous maps $G^i \to A$, under addition of values, so that $C^0(G, A) = A$. We write ξ_g, η_{g_1,g_2} for the values of a 1-cochain ξ and a 2-cochain η. The coboundary maps $d^i \colon C^i(G, A) \to C^{i+1}(G, A)$, for $i \leqslant 2$, are

$$(d^0 a)_g \coloneqq ga - a,$$
$$(d^1 \xi)_{g_1,g_2} \coloneqq g_1 \xi_{g_2} - \xi_{g_1 g_2} + \xi_{g_1},$$
$$(d^2 \eta)_{g_1,g_2,g_3} \coloneqq g_1 \eta_{g_2,g_3} - \eta_{g_1 g_2,g_3} + \eta_{g_1,g_2 g_3} - \eta_{g_1,g_2}.$$

A direct computation gives $d^1 \circ d^0 = 0$ and $d^2 \circ d^1 = 0$. The group of i-cocycles is $Z^i(G, A) \coloneqq \ker d^i$, the group of i-coboundaries is $B^i(G, A) \coloneqq \operatorname{im} d^{i-1}$ (with $B^0 \coloneqq 0$), and the i-th cohomology group is

$$H^i(G, A) \coloneqq Z^i(G, A)/B^i(G, A), \qquad i = 0, 1, 2.$$

Two cocycles with the same class are cohomologous.

Unwinding the definition: $H^0(G, A) = A^G$, the subgroup of the a with $ga = a$ for every $g \in G$. A left 1-cocycle, or simply 1-cocycle, is a continuous map $\xi \colon G \to A$ with

$$\xi_{gh} = \xi_g + g\xi_h \qquad \text{for all } g, h \in G, \tag{7.2.1}$$

and a left 1-coboundary, or simply 1-coboundary, is a map $g \mapsto ga - a$, for some $a \in A$. Every coboundary is continuous, since the stabilizer of a is open. Taking $g = h = 1$ in (7.2.1) gives $\xi_1 = 0$. If G is finite, every map $G^i \to A$ is continuous. When the group law of A is written multiplicatively, as for $L^\times$ or $\mu_m(L)$, the conditions read $\xi_{gh} = \xi_g \cdot g(\xi_h)$ and $\xi_g = g(a)/a$.

Remark 7.2.2. Silverman's cocycles are right cocycles, written for the right action $a \mapsto a^\sigma$: $\xi_{\sigma\tau} = \xi_\sigma^\tau + \xi_\tau$, with coboundaries $\xi_\sigma = a^\sigma - a$ [Sil09, Section B.1]. Setting $ga \coloneqq a^{g^{-1}}$ turns a right action into a left one, and $\xi \mapsto (g \mapsto \xi_{g^{-1}})$ turns right cocycles into left ones, so the two definitions give isomorphic groups. The groups of Definition 7.2.1 are the first three of a sequence of functors $H^i(G, -)$, $i \geqslant 0$, that turn short exact sequences into long exact sequences [Poo02, Section 3], and that can also be defined as derived functors of $A \mapsto A^G$ [Poo17, Section 6.4]. We need only $i \leqslant 2$.

Proposition 7.2.3 (Trivial action). *If G acts trivially on A, then $H^0(G, A) = A$ and $H^1(G, A) = \operatorname{Hom}_{\text{conts}}(G, A)$, the group of continuous homomorphisms.*

Proof. With the trivial action, (7.2.1) says that ξ is a homomorphism, and every coboundary $g \mapsto a - a$ is 0 [Sil09, Remark B.2.2]. □

The first example of each of our two families:

- If $\mu_m(k^{\text{sep}}) \subseteq k$, then Gal_k acts trivially on $\mu_m(k^{\text{sep}})$, and $H^1(k, \mu_m) = \operatorname{Hom}_{\text{conts}}(\operatorname{Gal}_k, \mu_m(k^{\text{sep}}))$. The kernel of such a homomorphism χ is an open normal subgroup, the fixed field L of $\ker \chi$ is a cyclic extension of k of degree dividing m, and χ identifies $\operatorname{Gal}(L/k)$ with a subgroup of $\mu_m(k^{\text{sep}})$. We return to this in Corollary 7.4.7.
- If $E[m](k^{\text{sep}}) \subseteq E(k)$, then Gal_k acts trivially on $E[m](k^{\text{sep}})$, and
$$H^1(k, E[m]) = \operatorname{Hom}_{\text{conts}}(\operatorname{Gal}_k, E[m](k^{\text{sep}})) \cong \operatorname{Hom}_{\text{conts}}(\operatorname{Gal}_k, \mathbb{Z}/m\mathbb{Z})^2,$$
the isomorphism depending on the choice of a basis.

When the action is not trivial, cocycles are harder to find. For cyclic groups there is a formula.

Proposition 7.2.4.

(1) Let $G = \langle \sigma \rangle$ be cyclic of order n, and $N \coloneqq 1 + \sigma + \cdots + \sigma^{n-1}$, acting on A. Then $\xi \mapsto \xi_\sigma$ induces an isomorphism
$$H^1(G, A) \to \ker(N \colon A \to A)/(\sigma - 1)A.$$
(2) Let $G = \widehat{\mathbb{Z}}$, with topological generator σ, and suppose that A is a torsion group. Then $\xi \mapsto \xi_\sigma$ induces an isomorphism $H^1(G, A) \to A/(\sigma - 1)A$.

PROOF. Let ξ be a 1-cocycle and $x := \xi_\sigma$. By induction on i, using $\xi_{\sigma^{i+1}} = \xi_\sigma + \sigma\xi_{\sigma^i}$,

$$\xi_{\sigma^i} = (1 + \sigma + \cdots + \sigma^{i-1})x \qquad (i \geqslant 0), \tag{7.2.2}$$

with the empty sum for $i = 0$. So ξ is determined by x on the subgroup $\sigma^{\mathbb{Z}_{\geqslant 0}}$, which is all of G in part 1 and is dense in part 2. Since ξ is continuous and A is discrete, ξ is determined by x in both cases. The coboundary of a has $x = \sigma a - a$. It remains to find the image of $\xi \mapsto x$.

In part 1, $Nx = \xi_{\sigma^n} = \xi_1 = 0$. Conversely, for $x \in \ker N$, define ξ_{σ^i} by (7.2.2). Replacing i by $i + n$ adds $\sigma^i Nx = 0$, so ξ is well defined on G, and it is a cocycle, because

$$\xi_{\sigma^{i+j}} = (1 + \cdots + \sigma^{i-1})x + \sigma^i(1 + \cdots + \sigma^{j-1})x = \xi_{\sigma^i} + \sigma^i\xi_{\sigma^j}.$$

In part 2, let $x \in A$. Its stabilizer is open, so it contains $r\widehat{\mathbb{Z}}$ for some $r \geqslant 1$, and x has finite order e. Define ξ_{σ^i} by (7.2.2) for $i \geqslant 0$. For $n := re$,

$$\xi_{\sigma^{i+n}} - \xi_{\sigma^i} = \sigma^i(1 + \sigma + \cdots + \sigma^{n-1})x = \sigma^i e(1 + \sigma + \cdots + \sigma^{r-1})x = 0,$$

since $\sigma^r x = x$. So ξ only depends on i mod n, and it extends to a continuous map on $\widehat{\mathbb{Z}}$, through $\widehat{\mathbb{Z}} \to \mathbb{Z}/n\mathbb{Z}$. The cocycle condition holds on the dense subset $\sigma^{\mathbb{Z}_{\geqslant 0}}$ by the computation above, hence everywhere by continuity. □

EXAMPLE 7.2.5.

(1) Let $G = \{1, \sigma\}$ act on $\mathbb{Z}$ by $\sigma a = -a$. Then $N = 1 + \sigma = 0$, so $H^1(G, \mathbb{Z}) = \mathbb{Z}/2\mathbb{Z}$ and $H^0(G, \mathbb{Z}) = 0$. With the trivial action instead, $H^0(G, \mathbb{Z}) = \mathbb{Z}$ and $H^1(G, \mathbb{Z}) = \mathrm{Hom}(G, \mathbb{Z}) = 0$.

(2) Let $G = \mathrm{Gal}(\mathbb{C}/\mathbb{R}) = \{1, c\}$. In multiplicative notation $Nz = z\bar{z}$, and $(c-1)\mathbb{C}^\times$ is the set of quotients $\bar{z}/z$. Every w with $w\bar{w} = 1$ is such a quotient: if $w \neq -1$, take $z := 1 + \bar{w}$, and $-1 = \bar{i}/i$. So $H^1(G, \mathbb{C}^\times) = 1$, an instance of Hilbert's Theorem 90 below. On $\mu_m(\mathbb{C})$, c acts by inversion, so $N = 1$ and $(c-1)\mu_m(\mathbb{C}) = \mu_m(\mathbb{C})^2$, and $H^1(G, \mu_m(\mathbb{C}))$ has order $\gcd(2, m)$. This is also the order of $\mathbb{R}^\times/\mathbb{R}^{\times m}$, as Corollary 7.4.6 will explain.

For a group A that is not commutative, half of the theory survives. It is needed for the general linear group in Hilbert 90 and for automorphism groups in Section 7.6.

DEFINITION 7.2.6. Let A be a G-group, written multiplicatively. Then $H^0(G, A) := A^G$. A 1-cocycle is a continuous map $\xi\colon G \to A$ with $\xi_{gh} = \xi_g \cdot g\xi_h$ for all $g, h \in G$. Two 1-cocycles ξ, ξ' are cohomologous if there is $b \in A$ with $\xi'_g = b^{-1}\,\xi_g\, gb$ for every $g \in G$. This is an equivalence relation, and the set of classes is $H^1(G, A)$. It is a pointed set: a set with a distinguished element, the class of the trivial cocycle $g \mapsto 1$. Its elements are the cocycles of the form $g \mapsto b^{-1}\, gb$.

If A is commutative, Definitions 7.2.1 and 7.2.6 agree: $b^{-1}\,\xi_g\, gb$ is ξ_g times the coboundary of b. In general $H^1(G, A)$ is not a group [Sil09, Section B.3]. For an algebraic group A over k, not necessarily commutative, we still write $H^1(k, A) := H^1(\mathrm{Gal}_k, A(k^{\mathrm{sep}}))$.

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7.3. Functoriality, inflation and restriction

We are in Situation 7.0.1. A morphism $f\colon A \to B$ of discrete G-modules induces homomorphisms $f_*\colon H^i(G, A) \to H^i(G, B)$, by composing cochains with f. The main tool for computing with them is the long exact sequence.

PROPOSITION 7.3.1. *Let $0 \to A \xrightarrow{\iota} B \xrightarrow{\psi} C \to 0$ be an exact sequence of discrete G-modules. There is an exact sequence*

$$0 \to A^G \to B^G \to C^G \xrightarrow{\delta} H^1(G, A) \to H^1(G, B) \to H^1(G, C)$$
$$\xrightarrow{\delta} H^2(G, A) \to H^2(G, B) \to H^2(G, C),$$

where the unlabeled maps are ι_* and ψ_*. The connecting homomorphisms are as follows. For $c \in C^G$, choose $b \in B$ with $\psi(b) = c$. Then

$$\delta(c) = \text{the class of } g \mapsto gb - b. \tag{7.3.1}$$

For a 1-cocycle γ with values in C, choose a continuous $\beta\colon G \to B$ with $\psi \circ \beta = \gamma$. Then $\delta([\gamma])$ is the class of $d^1\beta$, which takes values in A.

PROOF. We check the part through $H^1(G, B)$, which is what we use most. The cochain $g \mapsto gb - b$ is $d^0 b$, a cocycle with values in B. Its values lie in A, since $\psi(gb - b) = gc - c = 0$, so it is a cocycle with values in A. Another choice $b + a$, with $a \in A$, adds the coboundary $d^0 a$. So δ is well defined, and it is a homomorphism.

Exactness at C^G: $\delta(c) = 0$ means that $gb - b = ga - a$ for some $a \in A$ and every g, that is, $b - a \in B^G$, and $\psi(b - a) = c$. Exactness at $H^1(G, A)$: the class of $d^0 b$ is 0 in $H^1(G, B)$. Conversely, if a cocycle ξ with values in A is $d^0 b$ for some $b \in B$, then $g\psi(b) - \psi(b) = \psi(\xi_g) = 0$, so $\psi(b) \in C^G$, and $\delta(\psi(b)) = [\xi]$. Exactness at $H^1(G, B)$: $\psi \circ \iota = 0$. Conversely, if $\psi \circ \xi = d^0 c$ for some $c \in C$, choose b with $\psi(b) = c$. Then $\xi - d^0 b$ is a cocycle with values in A, with the same class as ξ in $H^1(G, B)$. The rest is the same kind of diagram chase [Poo02, Theorem 1]. The lift β exists because γ takes finitely many values and is constant on the cosets of an open normal subgroup, as in the proof of Proposition 7.3.6 below. □

Students should compute with (7.3.1): Sections 7.4 and 7.5 apply it to the two Kummer sequences.

REMARK 7.3.2. For an exact sequence $1 \to A \to B \to C \to 1$ of G-groups, with A normal in B, the same formula (7.3.1), written $g \mapsto b^{-1} gb$, defines $\delta\colon C^G \to H^1(G, A)$, and

$$1 \to A^G \to B^G \to C^G \xrightarrow{\delta} H^1(G, A) \to H^1(G, B) \to H^1(G, C)$$

is an exact sequence of pointed sets: at each term, the elements mapping to the distinguished element are the image of the previous map. If A is central in B, then A is commutative, and the sequence continues with $H^1(G, C) \to H^2(G, A)$. For example, $1 \to (k^{\mathrm{sep}})^\times \to \mathrm{GL}_n(k^{\mathrm{sep}}) \to \mathrm{PGL}_n(k^{\mathrm{sep}}) \to 1$ gives a map $H^1(k, \mathrm{PGL}_n) \to H^2(k, \mathbb{G}_{\mathrm{m}})$ [Poo17, Section 1.5.5].

Next we change the group. Let $H \subseteq G$ be a closed subgroup. Then A is also a discrete H-module, and if H is normal, then A^H is a discrete G/H-module.

DEFINITION 7.3.3. The restriction $\mathrm{Res}\colon H^i(G, A) \to H^i(H, A)$ maps the class of a cocycle ξ to the class of its restriction to H^i. If H is normal, the inflation $\mathrm{Inf}\colon H^i(G/H, A^H) \to H^i(G, A)$ maps the class of ξ to the class of the composition $G^i \to (G/H)^i \xrightarrow{\xi} A^H \subseteq A$. A cocycle of G of this form is inflated from G/H [Sil09, Section B.1].

For $i = 0$, restriction is the inclusion $A^G \subseteq A^H$, and inflation is the identity of A^G. For a field extension L of k inside k^{sep}, we write $\mathrm{Res}\colon H^i(k, A) \to H^i(L, A)$ for the restriction to Gal_L.

REMARK 7.3.4. If H is open, there is also a corestriction $\mathrm{Cor}\colon H^1(H, A) \to H^1(G, A)$, with $\mathrm{Cor} \circ \mathrm{Res} = [G : H]$ [Mil20, Section IV.1]. For $i = 0$ it is the norm $a \mapsto \sum_t ta$, for t running over representatives of G/H. We do not use it: the consequences we need are proved directly in Corollary 7.3.7.

The group G acts on $H^1(H, A)$ for H normal, through

$$(g \cdot \xi)_h \coloneqq g\, \xi_{g^{-1}hg} \qquad (g \in G,\ h \in H).$$

This maps cocycles to cocycles and coboundaries to coboundaries: $g \cdot d^0 a = d^0(ga)$.

PROPOSITION 7.3.5 (Inflation–restriction). *Let H be a closed normal subgroup of G. Then H acts trivially on $H^1(H, A)$, and the sequence*

$$0 \to H^1(G/H, A^H) \xrightarrow{\mathrm{Inf}} H^1(G, A) \xrightarrow{\mathrm{Res}} H^1(H, A)^{G/H}$$

is exact.

PROOF. Let ξ be a cocycle on G, or on H if $g \in H$. Then $\xi_{g^{-1}hg} = \xi_{g^{-1}} + g^{-1}\xi_h + g^{-1}h\xi_g$, and $0 = \xi_{gg^{-1}} = \xi_g + g\xi_{g^{-1}}$, so

$$(g \cdot \xi)_h = \xi_h + h\xi_g - \xi_g.$$

So $g \cdot \xi$ and ξ differ by the coboundary of ξ_g on H. For $g \in H$ this shows that H acts trivially, and for ξ defined on G it shows that Res takes values in the invariants.

A cocycle inflated from G/H is constant on H, equal to $\xi_1 = 0$, so $\mathrm{Res} \circ \mathrm{Inf} = 0$. Inflation is injective: if $\xi\colon G/H \to A^H$ inflates to $g \mapsto ga - a$, then for $h \in H$, $ha - a = \xi_1 = 0$, so $a \in A^H$ and ξ is already a coboundary on G/H. Finally let ξ be a cocycle on G whose restriction to H is $h \mapsto ha - a$. Subtracting the coboundary of a, we may assume $\xi_h = 0$ for $h \in H$. Then $\xi_{gh} = \xi_g + g\xi_h = \xi_g$, so ξ is constant on the cosets gH, and $h\xi_g = \xi_h + h\xi_g = \xi_{hg} = \xi_{g(g^{-1}hg)} = \xi_g$, so $\xi_g \in A^H$. Hence ξ is inflated from a continuous cocycle on G/H [Sil09, Proposition B.1.3 and Exercise B.3]. □

The cohomology of a profinite group is the union of the cohomology of its finite quotients. For $U \subseteq V$ open normal subgroups of G, inflation $H^i(G/V, A^V) \to H^i(G/U, A^U)$ makes these groups a direct system.

PROPOSITION 7.3.6. *For $i = 0, 1, 2$, inflation induces an isomorphism*

$$\varinjlim_U H^i(G/U, A^U) \to H^i(G, A),$$

where U runs over the open normal subgroups of G. For $G = \mathrm{Gal}_k$, the groups G/U are the $\mathrm{Gal}(L/k)$, for L finite Galois over k inside k^{sep}, and $A^U = A^{\mathrm{Gal}_L}$. The same holds for H^1 of a G-group.

PROOF. Let $f\colon G^i \to A$ be continuous. Since G^i is compact and A discrete, f takes finitely many values, and its fibers are open. Each element of G^i has a neighborhood xU_x^i, for an open normal subgroup U_x, on which f is constant. Finitely many of these cover G^i. Let U be the intersection of the corresponding U_x and of the stabilizers of the finitely many values of f, shrunk to an open normal subgroup. Then f is constant on the cosets of U^i and takes values in A^U, so it is inflated from a cochain $(G/U)^i \to A^U$. Inflation commutes with the coboundary maps, and $G^i \to (G/U)^i$ is surjective, so f is a cocycle exactly when the cochain on G/U is one. So the map is surjective. If an inflated cocycle is the coboundary of a cochain on G, that cochain is itself inflated from some G/U' with $U' \subseteq U$, and the cocycle is a coboundary on G/U'. So the map is injective [Sil09, Exercise B.4]. For Gal_k, see part 1 of Proposition 7.1.1. □

COROLLARY 7.3.7. *Let $i \in \{1, 2\}$.*

(1) If G is finite of order n, then $nH^i(G, A) = 0$.
(2) The group $H^i(G, A)$ is torsion.
(3) If $mA = 0$, then $mH^i(G, A) = 0$.
(4) If L is a finite Galois extension of k inside k^{sep}, then $\ker(\mathrm{Res}\colon H^1(k, A) \to H^1(L, A))$ is killed by $[L : k]$.

PROOF. For part 1 with $i = 1$, let ξ be a cocycle and $a \coloneqq \sum_{h \in G} \xi_h$. As h runs over G, so does gh, so summing (7.2.1) over h gives $a = n\xi_g + ga$. Hence $n\xi = d^0(-a)$. For $i = 2$, let ξ be a 2-cocycle and $\eta_g \coloneqq \sum_{h \in G} \xi_{g,h}$. Summing $d^2\xi = 0$ over $g_3 \in G$ gives $g_1\eta_{g_2} - \eta_{g_1 g_2} + \eta_{g_1} - n\xi_{g_1, g_2} = 0$, so $n\xi = d^1\eta$. Part 2 follows from part 1 and Proposition 7.3.6. Part 3 holds already for cochains. For part 4, the kernel is the image of $H^1(\mathrm{Gal}(L/k), A^{\mathrm{Gal}_L})$ by Proposition 7.3.5, and part 1 applies. □

EXAMPLE 7.3.8. Let $L \coloneqq k(E[m])$. Then Gal_L acts trivially on $E[m](k^{\mathrm{sep}})$, and Proposition 7.3.5 and Proposition 7.2.3 give an exact sequence

$$0 \to H^1(\mathrm{Gal}(L/k), E[m](k^{\mathrm{sep}})) \to H^1(k, E[m]) \to \mathrm{Hom}_{\mathrm{conts}}(\mathrm{Gal}_L, E[m](k^{\mathrm{sep}}))^{\mathrm{Gal}(L/k)}.$$

The group on the left is finite, as $\mathrm{Gal}(L/k)$ and $E[m](k^{\mathrm{sep}})$ are. So, up to a finite group, $H^1(k, E[m])$ is a group of homomorphisms, that is, of abelian extensions of L. This is how Theorem 7.8.6 will be proved.

7.4. Hilbert 90 and Kummer theory

We are in Situation 7.0.1. This section computes the cohomology of the multiplicative group and of μ_m, the first half of the chapter's main dictionary. The second half, for E and $E[m]$, is Section 7.5. For a finite Galois extension L/k with group G, the norm and the trace of $a \in L$ are $N_{L/k}(a) \coloneqq \prod_{g\in G} ga$ and $\mathrm{Tr}_{L/k}(a) \coloneqq \sum_{g\in G} ga$, both in k.

THEOREM 7.4.1 (Dedekind). *Let F be a field and H a group. Distinct homomorphisms $\chi_1, \dots, \chi_r \colon H \to F^\times$ are linearly independent over F: if $a_1\chi_1 + \cdots + a_r\chi_r = 0$ as a function $H \to F$, with $a_i \in F$, then $a_1 = \cdots = a_r = 0$.*

PROOF. See [Mil22, Theorem 5.14]. □

THEOREM 7.4.2 (Hilbert 90). *Let L/k be a finite Galois extension with group G, inside k^{sep}.*
(1) $H^1(G, L^\times) = 1$, and $H^1(k, \mathbb{G}_m) = 1$.
(2) $H^1(G, L) = 0$, and $H^1(k, \mathbb{G}_a) = 0$.
(3) For every $n \geqslant 1$, $H^1(G, \mathrm{GL}_n(L))$ and $H^1(k, \mathrm{GL}_n)$ have one element.

PROOF. In each part, the statement over k follows from the one for finite extensions by Proposition 7.3.6, since $L^\times = ((k^{\mathrm{sep}})^\times)^{\mathrm{Gal}_L}$, and likewise for L and $\mathrm{GL}_n(L)$.

For part 1, let $\xi \colon G \to L^\times$ be a cocycle. The elements of G restrict to distinct homomorphisms $L^\times \to L^\times$, and the ξ_τ are non-zero, so by Theorem 7.4.1 there is $c \in L^\times$ with

$$b \coloneqq \sum_{\tau\in G} \xi_\tau \cdot \tau(c) \neq 0.$$

The cocycle condition gives $\sigma(\xi_\tau) = \xi_\sigma^{-1}\xi_{\sigma\tau}$, so

$$\sigma(b) = \sum_{\tau\in G} \sigma(\xi_\tau)\cdot(\sigma\tau)(c) = \xi_\sigma^{-1} \sum_{\tau\in G} \xi_{\sigma\tau}\cdot(\sigma\tau)(c) = \xi_\sigma^{-1} b,$$

since $\sigma\tau$ runs over G with τ. So $\xi_\sigma = b/\sigma(b) = \sigma(b^{-1})/b^{-1}$, the coboundary of b^{-1}.

For part 2, Theorem 7.4.1 with all coefficients 1 gives $c \in L$ with $t \coloneqq \mathrm{Tr}_{L/k}(c) \neq 0$. For a cocycle $\xi \colon G \to L$, let $b \coloneqq t^{-1}\sum_\tau \xi_\tau \cdot \tau(c)$. Then $\sigma(\xi_\tau) = \xi_{\sigma\tau} - \xi_\sigma$, so

$$\sigma(b) = t^{-1} \sum_{\tau\in G} (\xi_{\sigma\tau} - \xi_\sigma)\cdot(\sigma\tau)(c) = b - \xi_\sigma,$$

and ξ is the coboundary of $-b$.

For part 3, let $\xi \colon G \to \mathrm{GL}_n(L)$ be a cocycle, and let W be L^n with the action $\sigma * w \coloneqq \xi_\sigma \cdot \sigma(w)$, where $\sigma(w)$ is the action on coordinates. The cocycle condition says exactly that this is an action, and it is semilinear: $\sigma * (\lambda w) = (\sigma(\lambda))(\sigma * w)$ for $\lambda \in L$. By Galois descent for vector spaces, every n-dimensional L-vector space with a semilinear action of G is isomorphic to L^n with the action on coordinates [Poo17, Corollary 1.3.12]. So there is $\varphi \in \mathrm{GL}_n(L)$ with $\varphi(\sigma(w)) = \sigma * \varphi(w) = \xi_\sigma \cdot (\sigma(\varphi))(\sigma(w))$ for all w, that is, $\xi_\sigma = \varphi\,(\sigma(\varphi))^{-1}$. With $b \coloneqq \varphi^{-1}$ this is $\xi_\sigma = b^{-1}\,\sigma(b)$, so ξ is cohomologous to the trivial cocycle [Poo17, Proposition 1.3.15]. □

Part 1 was already used in the proof of Proposition 2.1.17. Hilbert stated it for cyclic extensions.

COROLLARY 7.4.3. *Let L/k be a finite cyclic extension, with $\mathrm{Gal}(L/k) = \langle\sigma\rangle$. If $a \in L^\times$ has $N_{L/k}(a) = 1$, then $a = \sigma(b)/b$ for some $b \in L^\times$.*

PROOF. In multiplicative notation, part 1 of Proposition 7.2.4 identifies $H^1(\mathrm{Gal}(L/k), L^\times)$ with the group of the a of norm 1, modulo the quotients $\sigma(b)/b$. It is trivial by part 1 of Theorem 7.4.2. □

EXAMPLE 7.4.4. Let (x, y) be a rational point of the circle $x^2 + y^2 = 1$ over $\mathbb{Q}$. The element $a \coloneqq x + iy$ of $\mathbb{Q}(i)$ has norm $x^2 + y^2 = 1$, so by Corollary 7.4.3, with σ complex conjugation, $a = \sigma(\beta)/\beta$ for some

$\beta = u - iv \in \mathbb{Q}(i)^\times$. Then

$$x + iy = \frac{u+iv}{u-iv} = \frac{u^2-v^2}{u^2+v^2} + i\,\frac{2uv}{u^2+v^2},$$

which is the usual parametrization of the rational points of the circle.

DEFINITION 7.4.5. The Brauer group of k is $\mathrm{Br}(k) \coloneqq H^2(k, \mathbb{G}_\mathrm{m}) = H^2(\mathrm{Gal}_k, (k^\mathrm{sep})^\times)$.

This is the cohomological definition. The group $\mathrm{Br}(k)$ is also the group of similarity classes of central simple algebras over k [Poo17, Theorem 1.5.12]. For instance, the class of a quaternion algebra (Definition 4.7.1) is an element of $\mathrm{Br}(k)[2]$, and it is zero exactly when the algebra is split (Section 7.9). If k is finite, then $\mathrm{Br}(k) = 0$ [Poo17, Theorem 1.5.32].

COROLLARY 7.4.6 (Kummer sequence).

(1) The sequence of discrete Gal_k*-modules*

$$1 \to \mu_m(k^\mathrm{sep}) \to (k^\mathrm{sep})^\times \xrightarrow{x \mapsto x^m} (k^\mathrm{sep})^\times \to 1 \tag{7.4.1}$$

is exact.

(2) Its connecting homomorphism induces an isomorphism

$$\delta\colon k^\times/k^{\times m} \to H^1(k, \mu_m), \qquad \delta(a) = \textit{the class of } \sigma \mapsto \sigma(\alpha)/\alpha,$$

for any $\alpha \in k^\mathrm{sep}$ *with* $\alpha^m = a$.

(3) There is an isomorphism $H^2(k, \mu_m) \cong \mathrm{Br}(k)[m]$.

PROOF. For $a \in (k^\mathrm{sep})^\times$, the polynomial $x^m - a$ is separable: its derivative mx^{m-1} vanishes only at 0, since $m \neq 0$ in k, and 0 is not a root. So it has m distinct roots, all in k^sep. So $x \mapsto x^m$ is surjective on $(k^\mathrm{sep})^\times$, with kernel $\mu_m(k^\mathrm{sep})$ of order m, and it commutes with the action of Gal_k. This is part 1. Proposition 7.3.1 and part 1 of Theorem 7.4.2 give exact sequences

$$k^\times \xrightarrow{m} k^\times \xrightarrow{\delta} H^1(k, \mu_m) \to H^1(k, \mathbb{G}_\mathrm{m}) = 1,$$

$$1 = H^1(k, \mathbb{G}_\mathrm{m}) \to H^2(k, \mu_m) \to \mathrm{Br}(k) \xrightarrow{m} \mathrm{Br}(k),$$

which are parts 2 and 3. The formula for δ is (7.3.1), as α lifts a [Sil09, Proposition VIII.2.2]. □

The proof shows more: injectivity of δ on $k^\times/k^{\times m}$ is elementary, and only its surjectivity uses Hilbert 90. If $\mu_m(k^\mathrm{sep}) \subseteq k$, then $\delta(a)$ is the homomorphism $\sigma \mapsto \sigma(\alpha)/\alpha$, and this is Kummer theory.

COROLLARY 7.4.7 (Kummer theory). *Suppose that* $\mu_m(k^\mathrm{sep}) \subseteq k$.

(1) The map δ *is an isomorphism* $k^\times/k^{\times m} \to \mathrm{Hom}_\mathrm{conts}(\mathrm{Gal}_k, \mu_m(k^\mathrm{sep}))$. *For* $a \in k^\times$ *and* $\alpha^m = a$, $\ker \delta(a) = \mathrm{Gal}_{k(\alpha)}$, *and* $k(\alpha)/k$ *is cyclic of degree the order of* a *in* $k^\times/k^{\times m}$.

(2) The map $\Delta \mapsto k(\Delta^{1/m})$, *the extension of* k *inside* k^sep *generated by the* m*-th roots of the elements of* Δ, *is a bijection from the subgroups* $k^{\times m} \subseteq \Delta \subseteq k^\times$ *with* $\Delta/k^{\times m}$ *finite to the finite abelian extensions of* k *inside* k^sep *of exponent dividing* m. *Its inverse is* $L \mapsto k^\times \cap L^{\times m}$, *and* $[k(\Delta^{1/m}) : k] = (\Delta : k^{\times m})$.

PROOF. We use that a finite abelian group B of exponent dividing m has as many homomorphisms to the cyclic group $\mu_m(k^\mathrm{sep})$ as elements [Mil22, discussion before Theorem 5.30].

Part 1 is part 2 of Corollary 7.4.6 and Proposition 7.2.3. The kernel of $\chi \coloneqq \delta(a)$ is the set of σ fixing α, which is $\mathrm{Gal}_{k(\alpha)}$. It is normal, so $k(\alpha)/k$ is Galois with group $\mathrm{Gal}_k / \ker \chi \cong \chi(\mathrm{Gal}_k)$, a subgroup of the cyclic group $\mu_m(k^\mathrm{sep})$, of order the order of χ, which is the order of a.

For part 2, write $X \coloneqq \mathrm{Hom}_\mathrm{conts}(\mathrm{Gal}_k, \mu_m(k^\mathrm{sep}))$. To a finite subgroup $\Phi \subseteq X$ attach the fixed field L_Φ of $\bigcap_{\chi\in\Phi} \ker \chi$, an open normal subgroup. To a finite abelian extension L of exponent dividing m attach $\Phi_L \coloneqq \{\chi \in X : \chi(\mathrm{Gal}_L) = 1\}$, which is $\mathrm{Hom}(\mathrm{Gal}(L/k), \mu_m(k^\mathrm{sep}))$, of order $[L : k]$. The pairing $\mathrm{Gal}(L_\Phi/k) \times \Phi \to \mu_m(k^\mathrm{sep})$ has trivial left kernel, so $\mathrm{Gal}(L_\Phi/k)$ embeds in $\mathrm{Hom}(\Phi, \mu_m(k^\mathrm{sep}))$. It is therefore abelian of exponent dividing m, and $[L_\Phi : k] \leqslant \#\Phi$. Since $\Phi \subseteq \Phi_{L_\Phi}$,

$$[L_\Phi : k] \leqslant \#\Phi \leqslant \#\Phi_{L_\Phi} = [L_\Phi : k].$$

So $\Phi = \Phi_{L_\Phi}$ and $[L_\Phi : k] = \#\Phi$. Conversely, $L_{\Phi_L} \subseteq L$ and $[L_{\Phi_L} : k] = \#\Phi_L = [L : k]$, so $L_{\Phi_L} = L$. Thus $\Phi \mapsto L_\Phi$ and $L \mapsto \Phi_L$ are inverse bijections. Now transport them along δ. By part 1, for $\Phi = \delta(\Delta/k^{\times m})$, $\bigcap_{a\in\Delta} \ker \delta(a)$ is $\mathrm{Gal}_{k(\Delta^{1/m})}$, so $L_\Phi = k(\Delta^{1/m})$. And $\delta(a) \in \Phi_L$ if and only if $a \in L^{\times m}$, by Corollary 7.4.6 over L, since the restriction of $\delta(a)$ to Gal_L is given by the same formula. □

Example 7.4.8.

(1) Let $m = 2$ and $\mathrm{char}\, k \neq 2$. Then $\mu_2(k^{\mathrm{sep}}) = \{\pm 1\} \subseteq k$, and $d \mapsto k(\sqrt{d})$ is a bijection from the non-trivial elements of $k^\times/k^{\times 2}$ to the quadratic extensions of k inside k^{sep}. For $k = \mathbb{Q}$, the quadratic fields are the $\mathbb{Q}(\sqrt{d})$ with $d \neq 1$ a squarefree integer, each obtained once. The homomorphism $\delta(d)$ is the quadratic character $\chi_d\colon \sigma \mapsto \sigma(\sqrt{d})/\sqrt{d}$.

(2) Without the roots of unity, part 1 of Corollary 7.4.7 fails: $2 \notin \mathbb{Q}^{\times 3}$, but $\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}$ is not Galois. The cocycle $\sigma \mapsto \sigma(\sqrt[3]{2})/\sqrt[3]{2}$ is not a homomorphism, because complex conjugation acts non-trivially on $\mu_3(\overline{\mathbb{Q}})$.

The same argument with the trivial Gal_k-modules $0 \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{Q}/\mathbb{Z} \to 0$ computes $H^2(k, \mathbb{Z})$. It is needed for cyclic algebras in Section 7.9 and for the Tate curve in Chapter 12.

Proposition 7.4.9. *The connecting homomorphism of* $0 \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{Q}/\mathbb{Z} \to 0$ *is an isomorphism*

$$\mathrm{Hom}_{\mathrm{conts}}(\mathrm{Gal}_k, \mathbb{Q}/\mathbb{Z}) = H^1(k, \mathbb{Q}/\mathbb{Z}) \to H^2(k, \mathbb{Z}).$$

Proof. Multiplication by $n \geqslant 1$ is an automorphism of $\mathbb{Q}$, hence of $H^i(k, \mathbb{Q})$, which is torsion for $i = 1, 2$ by part 2 of Corollary 7.3.7. So $H^1(k, \mathbb{Q}) = H^2(k, \mathbb{Q}) = 0$, and Proposition 7.3.1 gives the isomorphism. The equality is Proposition 7.2.3. □

Remark 7.4.10. Suppose $\mathrm{char}\, k = p > 0$ and $m = p$. Then $\mu_p(k^{\mathrm{sep}}) = \{1\}$, so $H^1(k, \mu_p) = 1$. This agrees with $k^\times/k^{\times p}$ when k is perfect, but not when k is imperfect: then $x \mapsto x^p$ is not surjective on $(k^{\mathrm{sep}})^\times$, and the Kummer sequence (7.4.1) is not exact. For cyclic extensions of degree p, its replacement is the Artin–Schreier sequence

$$0 \to \mathbb{Z}/p\mathbb{Z} \to k^{\mathrm{sep}} \xrightarrow{x\mapsto x^p - x} k^{\mathrm{sep}} \to 0,$$

exact because $x^p - x - a$ has derivative -1, hence is separable. With part 2 of Theorem 7.4.2 in place of part 1, it gives an isomorphism $k/\wp(k) \to H^1(k, \mathbb{Z}/p\mathbb{Z}) = \mathrm{Hom}_{\mathrm{conts}}(\mathrm{Gal}_k, \mathbb{Z}/p\mathbb{Z})$, where $\wp(k) := \{c^p - c : c \in k\}$, mapping a to $\sigma \mapsto \sigma(\alpha) - \alpha$ for a root α of $x^p - x - a$. So the cyclic extensions of k of degree p are the $k(\alpha)$ with $a \notin \wp(k)$. For $p = 2$ these are the quadratic algebras $k[t]/(t^2 + t + d)$ of Proposition 2.1.9 that are fields. The group $k^\times/k^{\times p}$ does not disappear: it is H^1 of the group scheme μ_p for the fppf topology (Section 7.10).

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7.5. The Kummer sequence of an elliptic curve

We are in Situation 7.0.1. The multiplication map $[m]$ on E plays the role of $x \mapsto x^m$ on $\mathbb{G}_{\mathrm{m}}$. The arithmetic use of the resulting sequence is the subject of Chapters 8 and 10.

Proposition 7.5.1.

(1) The sequence of discrete Gal_k*-modules*

$$0 \to E[m](k^{\mathrm{sep}}) \to E(k^{\mathrm{sep}}) \xrightarrow{[m]} E(k^{\mathrm{sep}}) \to 0$$

is exact.

(2) There is an exact sequence

$$0 \to E(k)/m \xrightarrow{\delta} H^1(k, E[m]) \to H^1(k, E)[m] \to 0, \tag{7.5.1}$$

where $\delta(P)$ *is the class of* $\sigma \mapsto \sigma(Q) - Q$*, for any* $Q \in E(k^{\mathrm{sep}})$ *with* $[m]Q = P$.

Proof. By Corollary 4.4.3, $[m]$ is a separable isogeny. It is surjective on $E(\overline{k})$ by Proposition 3.2.2. For $P \in E(k^{\mathrm{sep}})$, the fiber $[m]^{-1}(P)$, base changed to k^{sep}, is étale over k^{sep}, for the same reason as the fiber over O in the discussion after Theorem 4.3.8. So its $\overline{k}$-points lie in $E(k^{\mathrm{sep}})$, and $[m]$ is surjective on $E(k^{\mathrm{sep}})$.

Its kernel is $E[m](k^{\mathrm{sep}})$, and it commutes with the action of Gal_k. This is part 1. Proposition 7.3.1 gives an exact sequence

$$E(k) \xrightarrow{[m]} E(k) \xrightarrow{\delta} H^1(k, E[m]) \to H^1(k, E) \xrightarrow{[m]} H^1(k, E),$$

from which (7.5.1) is extracted, and δ is given by (7.3.1) [Sil09, Section VIII.2]. □

We call δ the Kummer map of E. Unlike $H^1(k, \mathbb{G}_{\mathrm{m}})$, the group $H^1(k, E)$ is not zero in general, and the image of δ is a subgroup of $H^1(k, E[m])$ that is hard to determine. The group $H^1(k, E)$ classifies the torsors under E of Section 7.10, studied in Chapter 10. Over a finite field, however, $H^1(\mathbb{F}_q, E) = 0$ by Lang's theorem [Poo17, Theorem 5.12.19], and the two families behave alike.

The same argument applies to a non-zero separable isogeny $\varphi\colon E_1 \to E_2$ over k. We write $E_1[\varphi]$ for its kernel, a finite group scheme, so that $E_1[\varphi](k^{\mathrm{sep}}) = \ker\varphi$ (the discussion after Theorem 4.3.8). Then $0 \to E_1[\varphi](k^{\mathrm{sep}}) \to E_1(k^{\mathrm{sep}}) \to E_2(k^{\mathrm{sep}}) \to 0$ is exact, and gives

$$0 \to E_2(k)/\varphi(E_1(k)) \xrightarrow{\delta} H^1(k, E_1[\varphi]) \to H^1(k, E_1)[\varphi] \to 0,$$

where $H^1(k, E_1)[\varphi]$ is the kernel of $\varphi_*\colon H^1(k, E_1) \to H^1(k, E_2)$.

EXAMPLE 7.5.2. Suppose $\mathrm{char}\, k \neq 2$ and $\deg\varphi = 2$, as in Example 4.3.13. Then $E_1[\varphi](k^{\mathrm{sep}}) = \{O, T\}$, with T fixed by Gal_k, so $E_1[\varphi](k^{\mathrm{sep}}) \cong \mathbb{Z}/2\mathbb{Z} \cong \mu_2(k^{\mathrm{sep}})$ as discrete Gal_k-modules. By Corollary 7.4.6, $H^1(k, E_1[\varphi]) \cong H^1(k, \mu_2) \cong k^\times/k^{\times 2}$. So the Kummer map of φ is a homomorphism $E_2(k)/\varphi(E_1(k)) \to k^\times/k^{\times 2}$. It is the Kummer map of $\mathbb{G}_{\mathrm{m}}$ in disguise. Its explicit formula is in Chapter 10.

REMARK 7.5.3. The two Kummer sequences side by side, with $\mathrm{char}\, k \nmid m$:

	multiplicative group	elliptic curve
torsion	$\mu_m(k^{\mathrm{sep}}) \cong \mathbb{Z}/m\mathbb{Z}$	$E[m](k^{\mathrm{sep}}) \cong (\mathbb{Z}/m\mathbb{Z})^2$
Galois action	χ_m	$\bar\rho_{E,m}$, with $\det = \chi_m$
H^1 of the group	$H^1(k, \mathbb{G}_{\mathrm{m}}) = 1$ (Hilbert 90)	$H^1(k, E)$, not 0 in general
Kummer map	$k^\times/m \cong H^1(k, \mu_m)$	$E(k)/m \hookrightarrow H^1(k, E[m])$
torsors	all trivial	genus one curves with Jacobian E (Chapter 10)

Over a number field k, the group $k^\times/m$ is infinite, while $E(k)/m$ is finite (Chapter 8). The difference comes from the local fields, and Sections 7.7 and 7.8 prepare it.

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7.6. Non-abelian H^1 and twists

We are in Situation 7.0.1. Varieties that become isomorphic over k^{sep} are classified by a non-abelian H^1. This section gives the general principle and its simplest examples. The quartic, sextic and cubic twists of elliptic curves, and the torsors under E, are in Chapter 10.

Let X and Y be k-varieties. The group Gal_k acts on the morphisms $\varphi\colon X_{k^{\mathrm{sep}}} \to Y_{k^{\mathrm{sep}}}$ by ${}^\sigma\varphi := \sigma\circ\varphi\circ\sigma^{-1}$, where σ acts on k^{sep}-points through its action on k^{sep}. So $({}^\sigma\varphi)(\sigma(P)) = \sigma(\varphi(P))$ for $P \in X(k^{\mathrm{sep}})$. A morphism is defined over k exactly when it is fixed by Gal_k [Poo17, Section 4.4], and every φ is defined over a finite extension of k, so this makes $\mathrm{Aut}(X_{k^{\mathrm{sep}}})$ a Gal_k-group.

DEFINITION 7.6.1. A twist of a k-variety X is a k-variety Y such that $Y_{k^{\mathrm{sep}}} \cong X_{k^{\mathrm{sep}}}$. We write $\mathrm{Twist}(X/k)$ for the set of k-isomorphism classes of twists of X, a pointed set with distinguished element the class of X.

THEOREM 7.6.2 (Twisting principle). *Let X be a quasi-projective k-variety. For a twist Y of X, choose an isomorphism $\varphi\colon X_{k^{\mathrm{sep}}} \to Y_{k^{\mathrm{sep}}}$. Then $\xi_\sigma := \varphi^{-1}\circ{}^\sigma\varphi$ is a 1-cocycle with values in $\mathrm{Aut}(X_{k^{\mathrm{sep}}})$, and $Y \mapsto [\xi]$ is a well defined bijection of pointed sets*

$$\mathrm{Twist}(X/k) \to H^1(k, \mathrm{Aut}(X_{k^{\mathrm{sep}}})).$$

PROOF. Since $\sigma(\varphi^{-1}\circ{}^\tau\varphi) = ({}^\sigma\varphi)^{-1}\circ{}^{\sigma\tau}\varphi$,

$$\xi_{\sigma\tau} = \varphi^{-1}\circ{}^{\sigma\tau}\varphi = (\varphi^{-1}\circ{}^\sigma\varphi)\circ({}^\sigma\varphi)^{-1}\circ{}^{\sigma\tau}\varphi = \xi_\sigma\cdot{}^\sigma\xi_\tau.$$

If φ is defined over a finite Galois extension L, then $\xi_{\sigma\tau} = \xi_\sigma$ for $\tau \in \mathrm{Gal}_L$, so ξ is continuous. Another choice $\varphi \circ \alpha$, with $\alpha \in \mathrm{Aut}(X_{k^{\mathrm{sep}}})$, replaces ξ_σ by $\alpha^{-1}\, \xi_\sigma\, {}^\sigma\alpha$, a cohomologous cocycle, and composing φ with a k-isomorphism $Y \to Y'$ does not change ξ. So the map is well defined, and it maps X to the trivial class.

Injectivity: suppose that Y and Y', with φ and φ', give cohomologous cocycles. Changing φ by an automorphism of $X_{k^{\mathrm{sep}}}$, we may assume that the cocycles are equal: $\varphi^{-1} \circ {}^\sigma\varphi = \varphi'^{-1} \circ {}^\sigma\varphi'$ for all σ. Then $\psi \coloneqq \varphi' \circ \varphi^{-1} \colon Y_{k^{\mathrm{sep}}} \to Y'_{k^{\mathrm{sep}}}$ satisfies ${}^\sigma\psi = \psi$ for all σ, so it is defined over k, and $Y \cong Y'$. Surjectivity is Galois descent: a cocycle defines a descent datum on X_L for a finite Galois extension L, and quasi-projectivity makes it effective. Its proof generalizes the descent for vector spaces used for part 3 of Theorem 7.4.2. See [Poo17, Corollary 4.4.6 and Theorem 4.5.2] and, for curves, [Sil09, Theorem X.2.2]. □

The same proof applies to varieties with extra structure, with Aut replaced by the automorphisms preserving that structure. For an algebraic group, these are the automorphisms of algebraic groups. For an elliptic curve E, they are the automorphisms of $E_{k^{\mathrm{sep}}}$ fixing O, the group $\mathrm{Aut}(E_{k^{\mathrm{sep}}})$ of Chapter 4. A twist of E in this sense is an elliptic curve E' over k with $E'_{k^{\mathrm{sep}}} \cong E_{k^{\mathrm{sep}}}$ as elliptic curves [Sil09, Proposition X.5.3], [Poo02, Section 5].

Example 7.6.3. A morphism $\mathbb{G}_{\mathrm{m}} \to \mathbb{G}_{\mathrm{m}}$ over a field is a unit of the ring $k^{\mathrm{sep}}[t, t^{-1}]$, so it has the form $t \mapsto ct^n$, and it is a homomorphism exactly when $c = 1$. So the automorphisms of $\mathbb{G}_{\mathrm{m}}$ as an algebraic group, over any field, are $t \mapsto t^{\pm 1}$, and Gal_k acts trivially on this group of order 2. By the twisting principle and Proposition 7.2.3, the twists of $\mathbb{G}_{\mathrm{m}}$ as an algebraic group are classified by

$$H^1(k, \mathbb{Z}/2\mathbb{Z}) = \mathrm{Hom}_{\mathrm{conts}}(\mathrm{Gal}_k, \mathbb{Z}/2\mathbb{Z}),$$

that is, by the separable quadratic extensions of k, together with the trivial class. If $\mathrm{char}\, k \neq 2$, this group is $k^\times/k^{\times 2}$ by Corollary 7.4.6, and in characteristic 2 it is $k/\wp(k)$ by Remark 7.4.10.

These twists are the twisted multiplicative groups $\mathbb{G}_{\mathrm{m}}^{(d)}$ of Example 2.1.8. Suppose $\mathrm{char}\, k \neq 2$ and d is not a square, so that $L_d = k(\sqrt{d})$. The homomorphism $\psi \colon \mathbb{G}_{\mathrm{m}}^{(d)} \to \mathbb{G}_{\mathrm{m}}$, $(x, y) \mapsto x + \sqrt{d}\, y$, of the proof of Proposition 2.1.9 is an isomorphism over L_d. For $(x, y) \in \mathbb{G}_{\mathrm{m}}^{(d)}(k^{\mathrm{sep}})$ and $\sigma \in \mathrm{Gal}_k$,

$$({}^\sigma\psi)(x, y) = x + \sigma(\sqrt{d})\, y = \psi(x, y)^{\chi_d(\sigma)},$$

since $x - \sqrt{d}\, y = (x + \sqrt{d}\, y)^{-1}$ on $\mathbb{G}_{\mathrm{m}}^{(d)}$, where χ_d is the quadratic character of Example 7.4.8. So for $\varphi \coloneqq \psi^{-1}$, the cocycle $\varphi^{-1} \circ {}^\sigma\varphi$ is $t \mapsto t^{\chi_d(\sigma)}$, the class $\chi_d = \delta(d)$. The twisting principle therefore says that $\mathbb{G}_{\mathrm{m}}^{(d)} \cong \mathbb{G}_{\mathrm{m}}^{(d')}$ exactly when $d/d' \in k^{\times 2}$, and that $\mathbb{G}_{\mathrm{m}}^{(d)} \not\cong \mathbb{G}_{\mathrm{m}}$ when $d \notin k^{\times 2}$. These are parts 2 and 3 of Proposition 2.1.9, and every twist of $\mathbb{G}_{\mathrm{m}}$ as an algebraic group is of this form. As a variety, $\mathbb{G}_{\mathrm{m}}$ has the automorphisms $t \mapsto ct^{\pm 1}$ as well, and its twists as a variety also include the affine curves $x^2 - dy^2 = c$, for $c \in k^\times$.

Example 7.6.4. Let E be an elliptic curve over k with $j(E) \neq 0, 1728$. By Theorem 4.7.11, $\mathrm{Aut}(E_{k^{\mathrm{sep}}}) = \{\pm 1\}$, with the trivial action of Gal_k, since $[-1]$ is defined over k. So the twists of E as an elliptic curve are classified by $\mathrm{Hom}_{\mathrm{conts}}(\mathrm{Gal}_k, \{\pm 1\})$. Suppose $\mathrm{char}\, k \neq 2, 3$, so that E has a short Weierstrass model. Then this group is $H^1(k, \mu_2) \cong k^\times/k^{\times 2}$, and the class of d is the quadratic twist E^d of Chapter 4. Indeed, for $E \colon y^2 = x^3 + Ax + B$, the isomorphism $\varphi \colon E \to E^d$, $(x, y) \mapsto (dx, d^{3/2}y)$, satisfies

$$({}^\sigma\varphi)(x, y) = \big(dx, \sigma(d^{3/2})\, y\big) = \big(dx, \chi_d(\sigma)\, d^{3/2} y\big) = \varphi([\chi_d(\sigma)](x, y)),$$

as $\sigma(d^{3/2}) = d \cdot \sigma(\sqrt{d})$. So $\varphi^{-1} \circ {}^\sigma\varphi = [\chi_d(\sigma)]$, the class $\delta(d)$. This explains the computation before Example 4.1.9: E^d and $E^{d'}$ are isomorphic over k exactly when $d/d' \in k^{\times 2}$, and every twist of E is one of them. For $j(E) = 0$ or 1728, and in characteristic 2 and 3, $\mathrm{Aut}(E_{k^{\mathrm{sep}}})$ is larger, and so is the set of twists (Chapter 10).

Example 7.6.5. The automorphism group of $\mathbb{P}^1_{k^{\mathrm{sep}}}$ is $\mathrm{PGL}_2(k^{\mathrm{sep}})$, so the twists of $\mathbb{P}^1$ are classified by $H^1(k, \mathrm{PGL}_2)$. They are the nice curves of genus 0 over k [Poo17, Example 4.5.8], that is, the smooth plane conics. The conic C_2 of Example 2.3.6 is a non-trivial twist of $\mathbb{P}^1_{\mathbb{Q}}$, trivialized over $\mathbb{Q}(\sqrt{3})$. The map $H^1(k, \mathrm{PGL}_2) \to \mathrm{Br}(k)$ of Remark 7.3.2 sends a conic to the class of a quaternion algebra [Poo17, Section 4.5], an element of $\mathrm{Br}(k)[2]$ (Section 7.9). A conic is isomorphic to $\mathbb{P}^1$ if and only if it has a rational point [Poo17, Proposition 4.5.10].

7.7. Local fields: unramified classes

SITUATION 7.7.1. In addition to Situation 7.0.1, we use the notation of Situation 6.4.1: k_v is the completion of a global field at a non-archimedean place v, with normalized valuation v, ring of integers $\mathcal{O}_v$, uniformizer π_v and residue field $\mathbb{F}_q$ of characteristic p, and $I_v \subseteq \mathrm{Gal}_{k_v}$ is the inertia group.

- E is an elliptic curve over k_v.
- $\sigma \in \mathrm{Gal}(k_v^{\mathrm{unr}}/k_v)$ is the Frobenius automorphism, which corresponds to $\alpha \mapsto \alpha^q$ under $\mathrm{Gal}(k_v^{\mathrm{unr}}/k_v) \cong \mathrm{Gal}_{\mathbb{F}_q} \cong \widehat{\mathbb{Z}}$, and is a topological generator. It is not the Frobenius isogeny F.
- For a discrete Gal_{k_v}-module A, $H^i(k_v, A) \coloneqq H^i(\mathrm{Gal}_{k_v}, A)$, and likewise for algebraic groups, with k_v^{sep} in place of k^{sep}.

The inertia group is a closed normal subgroup of Gal_{k_v}, with quotient $\mathrm{Gal}_{k_v}/I_v \cong \mathrm{Gal}(k_v^{\mathrm{unr}}/k_v)$ by (6.4.1). A class is unramified when it comes from this quotient.

DEFINITION 7.7.2. Let A be a discrete Gal_{k_v}-module. A class $\xi \in H^1(k_v, A)$ is unramified if its restriction to $H^1(I_v, A)$ is trivial. The unramified classes form the subgroup

$$H^1_{\mathrm{unr}}(k_v, A) \coloneqq \ker\bigl(H^1(k_v, A) \xrightarrow{\mathrm{Res}} H^1(I_v, A)\bigr).$$

PROPOSITION 7.7.3. *Let A be a discrete Gal_{k_v}-module.*
(1) Inflation is an isomorphism $H^1(\mathrm{Gal}(k_v^{\mathrm{unr}}/k_v), A^{I_v}) \to H^1_{\mathrm{unr}}(k_v, A)$.
(2) If A is a torsion group, then $\xi \mapsto \xi_\sigma$ induces an isomorphism $H^1_{\mathrm{unr}}(k_v, A) \cong A^{I_v}/(\sigma - 1)A^{I_v}$.
(3) If A is finite, then $\#H^1_{\mathrm{unr}}(k_v, A) = \#A^{\mathrm{Gal}_{k_v}}$.

PROOF. Part 1 is Proposition 7.3.5 with $H = I_v$. Part 2 follows, by part 2 of Proposition 7.2.4 for the procyclic group $\mathrm{Gal}(k_v^{\mathrm{unr}}/k_v)$. For part 3, let $B \coloneqq A^{I_v}$, a finite group. The kernel of $\sigma - 1$ on B is the set of elements fixed by σ, hence by the closure of $\sigma^{\mathbb{Z}}$, which is all of Gal_{k_v}/I_v. So it is $A^{\mathrm{Gal}_{k_v}}$, and $\#B/(\sigma - 1)B = \#\ker(\sigma - 1) = \#A^{\mathrm{Gal}_{k_v}}$. □

For the multiplicative group, unramified classes are those of units.

PROPOSITION 7.7.4. *Suppose $p \nmid m$, and let $a \in k_v^\times$. The class $\delta(a) \in H^1(k_v, \mu_m)$ is unramified if and only if $v(a) \equiv 0 \pmod m$. So δ induces an isomorphism $\mathcal{O}_v^\times/\mathcal{O}_v^{\times m} \to H^1_{\mathrm{unr}}(k_v, \mu_m)$.*

PROOF. Let $\alpha \in k_v^{\mathrm{sep}}$ with $\alpha^m = a$. The restriction of $\delta(a)$ to I_v is the cocycle $\tau \mapsto \tau(\alpha)/\alpha$, with values in $\mu_m(k_v^{\mathrm{sep}})$, on which I_v acts trivially by part 1 of Example 6.4.5. So it is a homomorphism, and it is trivial if and only if I_v fixes α, that is, $\alpha \in k_v^{\mathrm{unr}}$. If $v(a) = mn$, then $a = \pi_v^{mn}u$ with $u \in \mathcal{O}_v^\times$. The polynomial $x^m - \bar{u}$ is separable over $\mathbb{F}_q$, so its m roots lie in a finite field $\mathbb{F}_{q^r}$, and by Hensel's lemma [Poo17, Theorem 3.5.63] they lift to the m roots of $x^m - u$ in the unramified extension of degree r. One of them is α/π_v^n, so $\alpha \in k_v^{\mathrm{unr}}$. Conversely, the valuation of k_v^{unr} has value group $\mathbb{Z}$, so if $\alpha \in k_v^{\mathrm{unr}}$, then $v(a) = mv(\alpha) \equiv 0 \pmod m$. The elements a with $v(a) \equiv 0 \pmod m$ are those of $\mathcal{O}_v^\times k_v^{\times m}$, and $\mathcal{O}_v^\times \cap k_v^{\times m} = \mathcal{O}_v^{\times m}$. □

As a check of part 3 of Proposition 7.7.3: by Hensel's lemma, $\mathcal{O}_v^\times/\mathcal{O}_v^{\times m} \cong \mathbb{F}_q^\times/\mathbb{F}_q^{\times m}$, of order $\gcd(m, q-1) = \#\mu_m(\mathbb{F}_q) = \#\mu_m(k_v)$. For $m \geqslant 2$, the class of π_v is ramified: $k_v(\pi_v^{1/m})$ is totally ramified, as in part 3 of Example 6.4.5.

For an elliptic curve the definition applies to $E[m]$. If E has good reduction and $p \nmid m$, then $E[m](k_v^{\mathrm{sep}})$ is unramified by Proposition 6.4.6, so

$$H^1_{\mathrm{unr}}(k_v, E[m]) \cong E[m](k_v^{\mathrm{sep}})/(\sigma - 1)E[m](k_v^{\mathrm{sep}}),$$

of order $\#E(k_v)[m]$. Which classes come from $E(k_v)$ under the Kummer map is the subject of Chapter 8.

7.8. Global fields: classes unramified outside S

SITUATION 7.8.1. In addition to Situation 7.0.1:

- k is a global field, with set of places $\mathcal{P}_k$ and completions k_v.
- For each $v \in \mathcal{P}_k$, we fix a k-embedding $k^{\mathrm{sep}} \to k_v^{\mathrm{sep}}$. Restriction to k^{sep} is then an injection $\mathrm{Gal}_{k_v} \to \mathrm{Gal}_k$, whose image is a decomposition group at v. Another embedding changes it by conjugation.
- For a discrete Gal_k-module A, $\mathrm{res}_v \colon H^1(k, A) \to H^1(k_v, A)$ is the restriction along $\mathrm{Gal}_{k_v} \to \mathrm{Gal}_k$. For a commutative algebraic group A over k, it is composed with $A(k^{\mathrm{sep}}) \to A(k_v^{\mathrm{sep}})$.
- $S \subseteq \mathcal{P}_k$ is a finite non-empty set of places containing the archimedean ones and the non-archimedean v with $v(m) > 0$, so that m is a unit outside S.
- $\mathcal{O}_S := \{a \in k : v(a) \geqslant 0 \text{ for all non-archimedean } v \notin S\}$ is the ring of S-integers, with unit group $\mathcal{O}_S^\times$ and ideal class group $\mathrm{Cl}(\mathcal{O}_S)$.

A class $\xi \in H^1(k, A)$ is unramified at a non-archimedean place v if $\mathrm{res}_v(\xi)$ is unramified in the sense of Definition 7.7.2. This does not depend on the embedding: conjugating I_v by $\gamma \in \mathrm{Gal}_k$ replaces the restriction of ξ by $\gamma \cdot \xi$, which is cohomologous to it by the proof of Proposition 7.3.5 [Sil09, Section VIII.2 and Exercise B.6].

DEFINITION 7.8.2. Let A be a discrete Gal_k-module, or a commutative algebraic group over k, and $T \subseteq \mathcal{P}_k$ a finite set of places. The group of classes unramified outside T is

$$H^1(k, A; T) := \{\xi \in H^1(k, A) : \xi \text{ is unramified at every non-archimedean } v \notin T\}.$$

Also

$$k(T, m) := \{a \in k^\times / k^{\times m} : v(a) \equiv 0 \pmod{m} \text{ for all non-archimedean } v \notin T\}.$$

The Kummer classes unramified outside S can be computed exactly.

PROPOSITION 7.8.3.
(1) The isomorphism $\delta \colon k^\times / k^{\times m} \to H^1(k, \mu_m)$ maps $k(S, m)$ onto $H^1(k, \mu_m; S)$.
(2) There is an exact sequence

$$1 \to \mathcal{O}_S^\times / \mathcal{O}_S^{\times m} \to k(S, m) \to \mathrm{Cl}(\mathcal{O}_S)[m] \to 0,$$

where the map on the right sends a to the class of the fractional ideal $\mathfrak{a}$ with $a\mathcal{O}_S = \mathfrak{a}^m$.
(3) If $\mathrm{Cl}(\mathcal{O}_S)$ is finite and $\mathcal{O}_S^\times$ is finitely generated, then $k(S, m)$ and $H^1(k, \mu_m; S)$ are finite.

PROOF. For part 1, $\mathrm{res}_v \delta(a) = \delta(a)$ computed over k_v, by the formula of Corollary 7.4.6. For non-archimedean $v \notin S$ the residue characteristic does not divide m, and Proposition 7.7.4 says that $\delta(a)$ is unramified at v exactly when $v(a) \equiv 0 \pmod{m}$.

For part 2, the prime ideals of the Dedekind domain $\mathcal{O}_S$ are those of the non-archimedean places outside S. If a represents an element of $k(S, m)$, the exponents of its fractional ideal $a\mathcal{O}_S$ are divisible by m, so it is $\mathfrak{a}^m$ for a unique fractional ideal $\mathfrak{a}$. Replacing a by ac^m replaces $\mathfrak{a}$ by $c\mathfrak{a}$, so $a \mapsto [\mathfrak{a}]$ is a well defined homomorphism, with values in $\mathrm{Cl}(\mathcal{O}_S)[m]$. It is surjective: if $\mathfrak{a}^m = a\mathcal{O}_S$, then a represents an element of $k(S, m)$ mapping to $[\mathfrak{a}]$. If $\mathfrak{a} = c\mathcal{O}_S$ is principal, then $a/c^m \in \mathcal{O}_S^\times$, so the kernel is the image of $\mathcal{O}_S^\times$. Finally, if $u \in \mathcal{O}_S^\times$ is c^m with $c \in k^\times$, then $mv(c) = v(u) = 0$ for $v \notin S$, so $c \in \mathcal{O}_S^\times$. Part 3 follows: $\mathcal{O}_S^\times / \mathcal{O}_S^{\times m}$ is a finitely generated group killed by m. □

The proofs of parts 2 and 3 use only that S is finite, non-empty and contains the archimedean places. So these two parts hold for such S even when m is not a unit outside S.

THEOREM 7.8.4. *If k is a number field, then $\mathrm{Cl}(\mathcal{O}_S)$ is finite and $\mathcal{O}_S^\times$ is finitely generated (Dirichlet's S-unit theorem). The same holds for a global function field, since S is non-empty.*

PROOF. For number fields, see [Sil09, proof of Proposition VIII.1.6], which quotes them. □

EXAMPLE 7.8.5. Let $k = \mathbb{Q}$, $m = 2$ and $S = \{\infty, 2\}$. Then $\mathcal{O}_S = \mathbb{Z}[1/2]$, a principal ideal domain with unit group $\{\pm 2^n : n \in \mathbb{Z}\}$. So $\mathbb{Q}(S,2) = \langle -1, 2\rangle$, of order 4, and its non-trivial elements correspond to $\mathbb{Q}(i)$, $\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(\sqrt{-2})$, the quadratic fields unramified outside 2 and ∞. More generally, for a finite set S of primes together with ∞, $\mathbb{Q}(S,2)$ is generated by -1 and the primes in S, and has order $2^{\#S}$. For $k = \mathbb{Q}(\sqrt{-5})$, $m = 2$ and S the archimedean place, which is allowed in parts 2 and 3 of Proposition 7.8.3 by the remark after it, $\mathcal{O}_S^\times/\mathcal{O}_S^{\times 2} = \{\pm 1\}$, and $\mathrm{Cl}(\mathcal{O}_S)$ has order 2, generated by $\mathfrak{p} := (2, 1+\sqrt{-5})$ with $\mathfrak{p}^2 = (2)$. So $k(S,2) = \langle -1, 2\rangle$ has order 4, although 2 is not a unit.

The next theorem is the cohomological form of the finiteness that drives the weak Mordell–Weil theorem.

THEOREM 7.8.6. *Suppose k is a number field. Let A be a finite discrete Gal_k-module with $mA = 0$, and $T \subseteq \mathcal{P}_k$ a finite set of places. Then $H^1(k, A; T)$ is finite.*

PROOF. Enlarging T enlarges $H^1(k, A; T)$, so we may assume that T satisfies the conditions on S in Situation 7.8.1. Let L be the fixed field of the kernels of the actions of Gal_k on A and on $\mu_m(k^{\mathrm{sep}})$, a finite Galois extension of k, and T_L the set of places of L above T. By Proposition 7.3.5, the kernel of Res: $H^1(k, A) \to H^1(L, A)$ is $H^1(\mathrm{Gal}(L/k), A)$, which is finite. The restriction maps $H^1(k, A; T)$ into $H^1(L, A; T_L)$: for compatible embeddings, the inertia group of L at a place w above v is contained in I_v. So it suffices to show that $H^1(L, A; T_L)$ is finite.

Over L, the action on A is trivial, so $H^1(L, A; T_L)$ is the group of continuous homomorphisms $\mathrm{Gal}_L \to A$ that are trivial on the inertia groups at the places outside T_L (Proposition 7.2.3). The finite abelian group A is killed by m, so it embeds in $(\mathbb{Z}/m\mathbb{Z})^r$ for some r, and this embeds $H^1(L, A; T_L)$ into $H^1(L, \mathbb{Z}/m\mathbb{Z}; T_L)^r$. As $\mu_m(k^{\mathrm{sep}}) \subseteq L$, the choice of a primitive m-th root of unity identifies $\mathbb{Z}/m\mathbb{Z}$ with $\mu_m(k^{\mathrm{sep}})$ as Gal_L-modules, and $H^1(L, \mu_m; T_L) \cong L(T_L, m)$ is finite by Proposition 7.8.3 and Theorem 7.8.4 over L [Sil09, Lemma X.4.3]. □

For $A = E[m](k^{\mathrm{sep}})$ this says that $H^1(k, E[m]; T)$ is finite. Chapter 8 shows that the image of the Kummer map $E(k)/m \to H^1(k, E[m])$ lies in such a group, for T containing the places of bad reduction. The same theorem for μ_m, with the infinite group $k^\times/k^{\times m}$ in place of $E(k)/m$, only says that the elements of $k^\times/k^{\times m}$ unramified outside T form a finite group.

REMARK 7.8.7. One might hope that an element of $k^\times$ that is an m-th power in every completion k_v is an m-th power in k, that is, that $H^1(k, \mu_m) \to \prod_v H^1(k_v, \mu_m)$ is injective. This is usually true, but not always. Let $k = \mathbb{Q}(\sqrt{7})$ and $m = 8$. Since

$$x^8 - 16 = (x^2 - 2)(x^2 + 2)(x^2 - 2x + 2)(x^2 + 2x + 2),$$

and the last two factors have roots $\pm 1 \pm i$, the number 16 is an 8-th power in a field of characteristic 0 exactly when one of 2, -2, -1 is a square there. The field $\mathbb{Q}(\sqrt{7})$ contains none of $\sqrt{2}$, $\sqrt{-2}$, i, so 16 is not an 8-th power in k. But it is one in every completion.

- At the two real places, $2 > 0$ is a square.
- At a place over an odd prime $\ell \neq 7$, the residue field contains $\mathbb{F}_\ell$, in which $\left(\frac{-1}{\ell}\right)\left(\frac{2}{\ell}\right) = \left(\frac{-2}{\ell}\right)$. So one of $-1, 2, -2$ is a square there, and it lifts by Hensel's lemma.
- At the place over 7, the residue field is $\mathbb{F}_7$, where $2 = 3^2$ is a square.
- At the place over 2, $-7 \equiv 1 \pmod 8$ is a square in $\mathbb{Q}_2$, so $\mathbb{Q}_2(\sqrt{7}) = \mathbb{Q}_2(\sqrt{-1})$, and -1 is a square.

Over $\mathbb{Q}$ itself, 16 is not an 8-th power in $\mathbb{Q}_2$, since $v_2(16) = 4$. This is Wang's counterexample to a theorem claimed by Grunwald. The Grunwald–Wang theorem says that such failures occur only for $8 \mid m$ and in a precise special case.

DRAFT
14 Oct 2026

7.9. Cup products and the Brauer group

We are in Situation 7.0.1. Silverman does not treat cup products. We define them in the degrees we need, prove the one property we use, and state the results on Brauer groups of local and global fields. Tate local duality and the Cassels–Tate pairing, which are built from them, are in Chapter 10.

Let A, B, C be discrete G-modules and $A \times B \to C$, $(a, b) \mapsto a \cdot b$, a bilinear map with $g(a \cdot b) = ga \cdot gb$.

Definition 7.9.1. The cup product $H^i(G,A)\times H^j(G,B)\to H^{i+j}(G,C)$, for $i,j\in\{0,1\}$, is induced by the following products of cocycles:

$$a\cup b\coloneqq a\cdot b,\qquad (a\cup\eta)_g\coloneqq a\cdot\eta_g,\qquad (\xi\cup b)_g\coloneqq \xi_g\cdot b,\qquad (\xi\cup\eta)_{g,h}\coloneqq \xi_g\cdot g\eta_h,$$

for $a\in A^G$, $b\in B^G$, and 1-cocycles ξ and η.

Proposition 7.9.2. *The cup product is well defined and bilinear. In degrees* 1 *and* 1*, let* $\cup'$ *be the cup product for the bilinear map* $B\times A\to C$, $(b,a)\mapsto a\cdot b$. *Then* $\xi\cup\eta=-(\eta\cup'\xi)$ *in* $H^2(G,C)$.

Proof. We treat degrees 1 and 1. The others are similar and shorter. Let $c\coloneqq\xi\cup\eta$. Using $\xi_{g_1g_2}=\xi_{g_1}+g_1\xi_{g_2}$,

$$(d^2c)_{g_1,g_2,g_3}=\xi_{g_1}\cdot g_1\big(\eta_{g_2g_3}-\eta_{g_2}-g_2\eta_{g_3}\big)=0.$$

If $\xi=d^0a$, then $c=d^1(g\mapsto a\cdot\eta_g)$, and if $\eta=d^0b$, then $c=d^1(g\mapsto-\xi_g\cdot gb)$, as one checks with the cocycle conditions. So the class of c depends only on the classes of ξ and η. Bilinearity is clear. Finally, let $f_g\coloneqq-\xi_g\cdot\eta_g$. Expanding $\xi_{gh}\cdot\eta_{gh}=(\xi_g+g\xi_h)\cdot(\eta_g+g\eta_h)$ gives

$$(d^1f)_{g,h}=\xi_g\cdot g\eta_h+g\xi_h\cdot\eta_g=(\xi\cup\eta)_{g,h}+(\eta\cup'\xi)_{g,h}.$$

□

Cup products are also compatible with restriction and inflation, since these act on cochains by composition.

The multiplicative group. Suppose $\mu_m(k^{\mathrm{sep}})\subseteq k$, and fix a primitive m-th root of unity ζ. The pairing $\mu_m(k^{\mathrm{sep}})\times\mu_m(k^{\mathrm{sep}})\to\mu_m(k^{\mathrm{sep}})$, $(\zeta^i,\zeta^j)\mapsto\zeta^{ij}$, is bilinear and Gal_k-equivariant, as Gal_k acts trivially. For $a,b\in k^\times$, let

$$(a,b)_\zeta\coloneqq\delta(a)\cup\delta(b)\in H^2(k,\mu_m)\cong\mathrm{Br}(k)[m],$$

by Corollary 7.4.6. The pairing is symmetric in its two arguments, so Proposition 7.9.2 gives $(b,a)_\zeta=-(a,b)_\zeta$. The class $(a,b)_\zeta$ is, up to sign, the class of the cyclic algebra built from a, b and ζ [Poo17, Section 1.5.7.4].

Example 7.9.3. Let $m=2$ and $\operatorname{char}k\neq2$, so that $\zeta=-1$ is canonical, and write $(a,b)\coloneqq(a,b)_{-1}\in\mathrm{Br}(k)[2]$. It is the class of the quaternion algebra $(a,b\mid k)$ of Definition 4.7.1: the sign does not matter in a group of exponent 2 [Poo17, Section 1.5.7.4 and Remark 1.5.24]. Since $\mathrm{Br}(k)$ is the group of similarity classes of central simple algebras [Poo17, Theorem 1.5.12], $(a,b)=0$ if and only if $(a,b\mid k)$ is split. By Proposition 4.7.5, this holds if and only if b is a norm from $k(\sqrt{a})$, that is, if and only if $ax^2+by^2=z^2$ has a non-trivial solution in k: if $b=z^2-ax^2$, take $y=1$, and conversely a solution with $y\neq0$ writes b as a norm, while one with $y=0$ makes a a square. For instance, the conic C_2 of Example 2.3.6 is $3X^2+3Y^2=(3Z)^2$, and it has no rational point, so $(3,3)\neq0$ in $\mathrm{Br}(\mathbb{Q})$.

Over local and global fields the Brauer group is known completely. These theorems are by-products of class field theory, and we only state them.

Theorem 7.9.4.

(1) *If* k_v *is a non-archimedean local field, there is an isomorphism* $\mathrm{inv}_v\colon\mathrm{Br}(k_v)\to\mathbb{Q}/\mathbb{Z}$. *Also* $\mathrm{Br}(\mathbb{R})\cong\frac{1}{2}\mathbb{Z}/\mathbb{Z}$ *and* $\mathrm{Br}(\mathbb{C})=0$.

(2) *If* k *is a global field, the sequence*

$$0\to\mathrm{Br}(k)\to\bigoplus_{v\in\mathcal{P}_k}\mathrm{Br}(k_v)\xrightarrow{\sum_v\mathrm{inv}_v}\mathbb{Q}/\mathbb{Z}\to0$$

is exact, where inv_v *is the inclusion of part 1 for archimedean* v.

Proof. See [Poo17, Theorems 1.5.34 and 1.5.36]. Part 2 is the theorem of Albert, Brauer, Hasse and Noether. □

For a local field k_v containing $\mu_m(k_v^{\mathrm{sep}})$, the Hilbert symbol of $a, b \in k_v^\times$ is the root of unity $\zeta^{m\,\mathrm{inv}_v((a,b)_\zeta)}$, also for archimedean v. For $m = 2$ it is 1 if $ax^2 + by^2 = z^2$ has a non-trivial solution in k_v, and -1 otherwise. If $\mu_m(k^{\mathrm{sep}}) \subseteq k$ and $a, b \in k^\times$, the class $(a, b) \in \mathrm{Br}(k)$ has $\sum_v \mathrm{inv}_v = 0$ by part 2 of Theorem 7.9.4, so the Hilbert symbols satisfy the product formula $\prod_v (a, b)_v = 1$. For $k = \mathbb{Q}$ and $m = 2$, the product formula is equivalent to quadratic reciprocity.

The elliptic curve. The Weil pairing $e_m \colon E[m](k^{\mathrm{sep}}) \times E[m](k^{\mathrm{sep}}) \to \mu_m(k^{\mathrm{sep}})$ is bilinear and Gal_k-equivariant (Proposition 4.6.3). In multiplicative notation for $\mu_m(k^{\mathrm{sep}})$, it gives a cup product

$$H^1(k, E[m]) \times H^1(k, E[m]) \to H^2(k, \mu_m) \cong \mathrm{Br}(k)[m].$$

PROPOSITION 7.9.5. *The cup product with the Weil pairing on $H^1(k, E[m])$ is symmetric.*

PROOF. Since e_m is alternating, $e_m(T, S) = e_m(S, T)^{-1}$ (part 2 of Proposition 4.6.3). So the cup product $\cup'$ of Proposition 7.9.2, for the pairing $(T, S) \mapsto e_m(S, T)$, is the inverse of the cup product for e_m itself. Proposition 7.9.2 then gives $\xi \cup \eta = (\eta \cup' \xi)^{-1} = \eta \cup \xi$. □

Combined with inv_v over a local field, this pairing is Tate local duality, and over a global field it underlies the Cassels–Tate pairing. Both are in Chapter 10.

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7.10. Étale cohomology and the geometric point of view

We are in Situation 7.0.1. This section is a survey, without proofs, of the geometric meaning of the groups of this chapter. It follows [Poo17, Sections 5.12 and 6.2–6.6].

Sheaves on a point. An étale k-scheme is a finite disjoint union of spectra of finite separable extensions of k. A sheaf on the small étale site of $\mathrm{Spec}\, k$ is determined by its value on $\mathrm{Spec}\, k^{\mathrm{sep}}$, as a direct limit, and this is an equivalence between sheaves of sets and discrete Gal_k-sets, and between abelian sheaves and discrete Gal_k-modules. Under it, $H^i_{\mathrm{et}}(\mathrm{Spec}\, k, F) \cong H^i(k, F(k^{\mathrm{sep}}))$ [Poo17, Theorem 6.4.6]. So $\mathrm{Spec}\, k$, which has a single point, behaves like a topological space whose only invariant is its fundamental group Gal_k: finite étale covers play the role of covering spaces, and Galois cohomology plays the role of the cohomology $H^i(\pi_1(X), M)$ of a space X with contractible universal cover.

DEFINITION 7.10.1. Let G be a smooth algebraic group over k. A *G-torsor* is a k-variety X with a right action $X \times G \to X$ such that $X_{k^{\mathrm{sep}}}$, with its $G_{k^{\mathrm{sep}}}$-action, is isomorphic to $G_{k^{\mathrm{sep}}}$ with the action by right translation. It is trivial if it is isomorphic to G with this action [Poo17, Definition 5.12.3].

PROPOSITION 7.10.2. *Let G be a smooth algebraic group over k.*

(1) For a G-torsor X and $x \in X(k^{\mathrm{sep}})$, there is a unique $\xi_\sigma \in G(k^{\mathrm{sep}})$ with $\sigma(x) = x\,\xi_\sigma$, and ξ is a 1-cocycle. Its class $[X] \in H^1(k, G)$ does not depend on x, and $X \mapsto [X]$ is a bijection from the isomorphism classes of G-torsors to $H^1(k, G)$.

(2) A G-torsor X is trivial if and only if $X(k)$ is non-empty, if and only if $[X]$ is the distinguished element.

PROOF. See [Poo17, Section 5.12.4, Remark 5.12.13 and Proposition 5.12.14]. The cocycle condition is the computation $x\,\xi_{\sigma\tau} = \sigma(\tau(x)) = \sigma(x\,\xi_\tau) = x\,\xi_\sigma\,\sigma(\xi_\tau)$. □

EXAMPLE 7.10.3.

(1) Every $\mathbb{G}_{\mathrm{m}}$-torsor is trivial, by part 1 of Theorem 7.4.2. This is Hilbert 90 restated.

(2) For $a \in k^\times$, the curve $X_a \colon x^m = a$ in $\mathbb{G}_{\mathrm{m}}$, with μ_m acting by multiplication, is a μ_m-torsor. For a root $\alpha \in X_a(k^{\mathrm{sep}})$, $\sigma(\alpha) = \alpha \cdot \sigma(\alpha)/\alpha$, so $[X_a] = \delta(a)$. By Corollary 7.4.6, every μ_m-torsor is some X_a.

(3) For $P \in E(k)$, the fiber $[m]^{-1}(P) \subseteq E$, with $E[m]$ acting by translation, is an $E[m]$-torsor. For Q in it, $\sigma(Q) = Q + (\sigma(Q) - Q)$, so its class is $\delta(P)$: the Kummer map of Proposition 7.5.1 sends P to this torsor, which is trivial exactly when $P \in mE(k)$.

Torsors under E itself are the genus one curves with Jacobian E [Poo17, Example 5.12.12], and they form the group $H^1(k, E)$, the Weil–Châtelet group of Chapter 10.

Twists without cocycles. Let T be a G-torsor and X a quasi-projective k-variety with a left action of G. The contracted product $T \times^G X$, the quotient of $T \times X$ by $(t, x) \mapsto (tg^{-1}, gx)$, is a twist of X, and its class is the image of $[T]$ under $H^1(k, G) \to H^1(k, \mathrm{Aut}(X_{k^{\mathrm{sep}}}))$ [Poo17, Section 5.12.5.2]. Conversely, when the automorphisms of X form an algebraic group, a twist Y gives the torsor of isomorphisms from X to Y. A cocycle is then nothing but a choice of a k^{sep}-point of this torsor, that is, of an isomorphism over k^{sep}, as in Theorem 7.6.2.

Over a base. Over a locally noetherian scheme T, torsors under an fppf group scheme G are trivialized fppf locally, and they are classified by $H^1_{\mathtt{fppf}}(T, G)$ [Poo17, Theorem 6.5.10]. When G is smooth, this is $H^1_{\mathrm{et}}(T, G)$ [Poo17, Remark 6.6.3]. Over $\operatorname{Spec} k$, "étale locally" means "over a finite separable extension", and this recovers Proposition 7.10.2. Hilbert 90 becomes $H^1_{\mathrm{et}}(T, \mathbb{G}_{\mathrm{m}}) \cong \operatorname{Pic} T$ [Poo17, Proposition 6.6.1], and at $T = \operatorname{Spec} k$ the Picard group is 0. The Brauer group of a scheme is $\operatorname{Br} T := H^2_{\mathrm{et}}(T, \mathbb{G}_{\mathrm{m}})$ [Poo17, Definition 6.6.4].

On $T = \operatorname{Spec} \mathcal{O}_S$ of Situation 7.8.1, where m is invertible, the Kummer sequence $1 \to \mu_m \to \mathbb{G}_{\mathrm{m}} \xrightarrow{m} \mathbb{G}_{\mathrm{m}} \to 1$ is exact for the étale topology, and the argument of Corollary 7.4.6 gives

$$1 \to \mathcal{O}_S^\times / \mathcal{O}_S^{\times m} \to H^1_{\mathrm{et}}(\operatorname{Spec} \mathcal{O}_S, \mu_m) \to \operatorname{Pic}(\mathcal{O}_S)[m] \to 0,$$

with $\operatorname{Pic}(\mathcal{O}_S) = \operatorname{Cl}(\mathcal{O}_S)$. This is the sequence of part 2 of Proposition 7.8.3: a class of $H^1(k, \mu_m)$ is unramified outside S exactly when its torsor extends to a μ_m-torsor over $\operatorname{Spec} \mathcal{O}_S$.

Characteristic p. If $\operatorname{char} k = p$, the group schemes μ_p and $E[p]$ are not étale, and Galois cohomology sees only their k^{sep}-points, such as $\mu_p(k^{\mathrm{sep}}) = \{1\}$. The Kummer sequence for $m = p$ is exact for the fppf topology, and $H^1_{\mathtt{fppf}}(\operatorname{Spec} k, \mu_p) \cong k^\times / k^{\times p}$ [Poo17, Remark 1.5.14]. This is the setting for p-descent in characteristic p.

CHAPTER 8

The Mordell–Weil theorem

Sources. This chapter follows [Sil09, Sections VIII.1–VIII.3, VIII.5 and VIII.9], with the point of view of [Poo02, Sections 8–10]. The counting of elements of bounded height follows [HS00, Sections B.4–B.6], and the heights on projective space [HS00, Sections B.1, B.2 and B.7].

The Mordell–Weil theorem says that the group $E(k)$ of rational points of an elliptic curve over a number field is finitely generated. Its proof has two steps of a very different nature. The first, the weak Mordell–Weil theorem, says that $E(k)/mE(k)$ is finite for every $m \geqslant 2$, and the second step needs it for a single m. It is an algebraic statement, and it is proved with the Galois cohomology of Chapter 7 and the local fields of Chapter 6. The second is a theory of heights. This chapter starts with the first step, first in a case where everything is explicit.

Situation 8.0.1.

- k is a number field, with set of places $\mathcal{P}_k$ and completions k_v. For a non-archimedean place v we use the notation of Situations 6.0.2 and 6.4.1: the ring of integers $\mathcal{O}_v$, the residue field $\kappa(v)$ of characteristic p, and the inertia group I_v.
- Since $\operatorname{char} k = 0$, separable closures are algebraic closures. We write $\overline{k}$ and $\overline{k_v}$ for the k^{sep} and k_v^{sep} of Chapters 6 and 7, and $\mathrm{Gal}_k \coloneqq \mathrm{Gal}(\overline{k}/k)$. For an extension F of k inside $\overline{k}$, $\mathrm{Gal}_F \coloneqq \mathrm{Gal}(\overline{k}/F)$.
- Elements of Galois groups are σ, τ, γ. For a $\overline{k}$-point Q of a variety over k, $\sigma(Q)$ is obtained by applying σ to the coordinates of Q.
- As in Situation 7.8.1, for each $v \in \mathcal{P}_k$ we fix a k-embedding $\overline{k} \to \overline{k_v}$. Restriction to $\overline{k}$ then identifies Gal_{k_v} with a decomposition group in Gal_k, and I_v with a subgroup of it.
- E is an elliptic curve over k, and $m \geqslant 2$ is an integer. For an abelian group B, $B/m \coloneqq B/mB$ and $B[m]$ is its m-torsion subgroup.
- For a non-archimedean place v, E has good (respectively bad) reduction at v if E_{k_v} has good (respectively bad) reduction (Definition 6.3.1).

Extensions of k are compared with inertia in the same way as cohomology classes in Section 7.8.

Definition 8.0.2. Let F be a Galois extension of k inside $\overline{k}$, possibly infinite, and let v be a non-archimedean place of k. Then F is unramified at v if $I_v \subseteq \mathrm{Gal}_F$, that is, if I_v fixes F. For a finite set $T \subseteq \mathcal{P}_k$, F is unramified outside T if it is unramified at every non-archimedean $v \notin T$.

Another embedding $\overline{k} \to \overline{k_v}$ replaces I_v by a conjugate $\gamma I_v \gamma^{-1}$, with $\gamma \in \mathrm{Gal}_k$, and Gal_F is normal, so the condition does not depend on the embedding. A compositum of Galois extensions unramified at v is unramified at v, since $\mathrm{Gal}_{FF'} = \mathrm{Gal}_F \cap \mathrm{Gal}_{F'}$, and so is a Galois subextension $F'' \subseteq F$, since $\mathrm{Gal}_F \subseteq \mathrm{Gal}_{F''}$. Let $\chi\colon \mathrm{Gal}_k \to A$ be a continuous homomorphism to a finite abelian group, a class of $H^1(k, A)$ for the trivial action (Proposition 7.2.3). The fixed field of $\ker \chi$ is unramified at v exactly when χ is trivial on I_v, that is, when the class χ is unramified at v in the sense of Section 7.8. For a quadratic field this is the familiar criterion: if p is an odd prime and $d \in \mathbb{Q}^\times$, then $\mathbb{Q}(\sqrt{d})$ is unramified at p if and only if $v_p(d)$ is even, by Proposition 7.7.4 applied to the class $\delta(d)\colon \sigma \mapsto \sigma(\sqrt{d})/\sqrt{d}$.

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8.1. The case $m = 2$ over $\mathbb{Q}$ with rational 2-torsion

The proof of the weak Mordell–Weil theorem is easier to see in a special case, where the class group and the unit theorem disappear and the local input is elementary. This section follows [Sil09, Section VIII.1] for $m = 2$ and $k = \mathbb{Q}$.

Situation 8.1.1. In addition to Situation 8.0.1:

- $k = \mathbb{Q}$ and $m = 2$. For a prime p, $I_p \subseteq \mathrm{Gal}_{\mathbb{Q}_p} \subseteq \mathrm{Gal}_{\mathbb{Q}}$ is the inertia group at p.
- E is given by
$$y^2 = (x - e_1)(x - e_2)(x - e_3)$$
with distinct $e_1, e_2, e_3 \in \mathbb{Z}$. Its points of order two are $T_i := (e_i, 0)$, so $E[2](\overline{\mathbb{Q}}) = \{O, T_1, T_2, T_3\} \subseteq E(\mathbb{Q})$.
- $S := \{\infty, 2\} \cup \{p : p \mid (e_1 - e_2)(e_1 - e_3)(e_2 - e_3)\}$, a finite set of places of $\mathbb{Q}$.
- L is the compositum inside $\overline{\mathbb{Q}}$ of the fields $\mathbb{Q}(Q)$, for the $\overline{\mathbb{Q}}$-points Q of E with $2Q \in E(\mathbb{Q})$. Silverman [Sil09] writes $L = \mathbb{Q}([2]^{-1}E(\mathbb{Q}))$.

The group law of E and the map [2] are given by rational functions with coefficients in $\mathbb{Q}$ (Algorithm 4.2.7), so every $\sigma \in \mathrm{Gal}_{\mathbb{Q}}$ acts on $E(\overline{\mathbb{Q}})$ by group automorphisms that commute with [2].

Definition 8.1.2. Let $P \in E(\mathbb{Q})$ and $\sigma \in \mathrm{Gal}_{\mathbb{Q}}$, and choose a $\overline{\mathbb{Q}}$-point Q of E with $2Q = P$. The Kummer pairing of E is the map
$$\kappa\colon E(\mathbb{Q}) \times \mathrm{Gal}_{\mathbb{Q}} \to E[2](\overline{\mathbb{Q}}), \qquad \kappa(P, \sigma) := \sigma(Q) - Q.$$

Such a $\overline{\mathbb{Q}}$-point Q exists, since [2] is surjective on $E(\overline{\mathbb{Q}})$ (part 1 of Proposition 7.5.1).

Proposition 8.1.3. *Let $P, P' \in E(\mathbb{Q})$ and $\sigma, \tau \in \mathrm{Gal}_{\mathbb{Q}}$.*

(1) $\kappa(P, \sigma)$ lies in $E[2](\overline{\mathbb{Q}})$ and does not depend on the choice of Q.
(2) $\kappa(P + P', \sigma) = \kappa(P, \sigma) + \kappa(P', \sigma)$ and $\kappa(P, \sigma\tau) = \kappa(P, \sigma) + \kappa(P, \tau)$.
(3) $\kappa(P, \sigma) = O$ for every $\sigma \in \mathrm{Gal}_{\mathbb{Q}}$ if and only if $P \in 2E(\mathbb{Q})$.
(4) $\kappa(P, \sigma) = O$ for every $P \in E(\mathbb{Q})$ if and only if $\sigma \in \mathrm{Gal}_L$.
(5) $L/\mathbb{Q}$ is Galois, $\mathrm{Gal}(L/\mathbb{Q})$ is abelian of exponent dividing 2, and κ induces a pairing
$$E(\mathbb{Q})/2 \times \mathrm{Gal}(L/\mathbb{Q}) \to E[2](\overline{\mathbb{Q}})$$
for which $E(\mathbb{Q})/2 \to \mathrm{Hom}_{\mathrm{conts}}(\mathrm{Gal}(L/\mathbb{Q}), E[2](\overline{\mathbb{Q}}))$ and $\mathrm{Gal}(L/\mathbb{Q}) \to \mathrm{Hom}(E(\mathbb{Q})/2, E[2](\overline{\mathbb{Q}}))$ are injective. In particular $E(\mathbb{Q})/2$ is finite if and only if $L/\mathbb{Q}$ is finite.

Proof. For part 1, $2(\sigma(Q) - Q) = \sigma(2Q) - 2Q = \sigma(P) - P = O$, as P is a rational point. Another choice of Q is $Q + T$ with $T \in E[2](\overline{\mathbb{Q}})$, and $\sigma(T) = T$ because T is a rational point. So $\sigma(Q+T) - (Q+T) = \sigma(Q) - Q$.

For part 2, if $2Q = P$ and $2Q' = P'$, then $2(Q + Q') = P + P'$, and $\sigma(Q + Q') - (Q + Q') = (\sigma(Q) - Q) + (\sigma(Q') - Q')$. Next,
$$(\sigma\tau)(Q) - Q = \big(\sigma(Q) - Q\big) + \sigma\big(\tau(Q) - Q\big) = \kappa(P, \sigma) + \sigma\big(\kappa(P, \tau)\big),$$
and $\kappa(P, \tau)$ is a point of order dividing two, hence rational, hence fixed by σ. This and part 1 are the two places where we use that $E[2](\overline{\mathbb{Q}}) \subseteq E(\mathbb{Q})$.

For part 3, if $P = 2Q$ with $Q \in E(\mathbb{Q})$, then $\sigma(Q) = Q$ for every σ. Conversely, if $\sigma(Q) = Q$ for every σ, then the coordinates of Q lie in $\overline{\mathbb{Q}}^{\mathrm{Gal}_{\mathbb{Q}}} = \mathbb{Q}$ (Proposition 7.1.1), so $P = 2Q \in 2E(\mathbb{Q})$.

For part 4, if $\sigma \in \mathrm{Gal}_L$ and $2Q \in E(\mathbb{Q})$, then the coordinates of Q lie in L, so $\sigma(Q) = Q$. Conversely, if $\kappa(\cdot, \sigma) = O$, then σ fixes every $\overline{\mathbb{Q}}$-point Q with $2Q \in E(\mathbb{Q})$, hence every field $\mathbb{Q}(Q)$, hence their compositum L.

For part 5, by part 2 the map $\sigma \mapsto \kappa(\cdot, \sigma)$ is a homomorphism from $\mathrm{Gal}_{\mathbb{Q}}$ to the abelian group $\mathrm{Hom}(E(\mathbb{Q}), E[2](\overline{\mathbb{Q}}))$, which is killed by 2. Its kernel is Gal_L by part 4. So Gal_L is normal, $L/\mathbb{Q}$ is Galois by Proposition 7.1.1, and $\mathrm{Gal}(L/\mathbb{Q}) \cong \mathrm{Gal}_{\mathbb{Q}} / \mathrm{Gal}_L$ embeds in $\mathrm{Hom}(E(\mathbb{Q}), E[2](\overline{\mathbb{Q}}))$. It is therefore abelian of exponent dividing 2. The pairing factors through $E(\mathbb{Q})/2$ and $\mathrm{Gal}(L/\mathbb{Q})$, and parts 3 and 4 say that the two induced maps are injective. Each $\kappa(P, \cdot)$ is continuous, being constant on the cosets of $\mathrm{Gal}_{\mathbb{Q}(Q)}$. If $E(\mathbb{Q})/2$ is finite, $\mathrm{Gal}(L/\mathbb{Q})$ embeds in a finite group, and if $L/\mathbb{Q}$ is finite, $E(\mathbb{Q})/2$ does. □

In the language of Chapter 7, $\kappa(P, \cdot)$ is the cocycle of Proposition 7.5.1 representing the Kummer class $\delta(P)$. As $\mathrm{Gal}_{\mathbb{Q}}$ acts trivially on $E[2](\overline{\mathbb{Q}})$, cocycles are homomorphisms (Proposition 7.2.3), and part 3 is the injectivity of δ on $E(\mathbb{Q})/2$.

REMARK 8.1.4. Part 5 of Proposition 8.1.3 turns the finiteness of $E(\mathbb{Q})/2$ into the finiteness of the field L. This should be surprising. The same construction for the multiplicative group, with $\sqrt{a}$ in place of a half of P, is the pairing

$$\mathbb{Q}^\times/\mathbb{Q}^{\times 2} \times \mathrm{Gal}_\mathbb{Q} \to \mu_2(\overline{\mathbb{Q}}), \qquad (a,\sigma) \mapsto \sigma(\sqrt{a})/\sqrt{a},$$

which is perfect in the same sense (part 1 of Corollary 7.4.7, and [Sil09, Remark VIII.1.2.1]). There the analogue of L is $\mathbb{Q}(\sqrt{a} : a \in \mathbb{Q}^\times)$, the compositum of all quadratic fields, which is an infinite extension of $\mathbb{Q}$: the classes of -1 and of the primes are independent in $\mathbb{Q}^\times/\mathbb{Q}^{\times 2}$, by unique factorization. For E, by contrast, $E(\mathbb{Q})$ may be infinite, and the coordinates of the multiples of a rational point of infinite order involve more and more primes (Example 4.2.8). One would expect the halves of these rational points to generate more and more quadratic fields, as $\sqrt{p}$ does for each prime p. They do not, and Proposition 8.1.5 explains why.

PROPOSITION 8.1.5. *Let $p \notin S$ be a prime, $Q \in E(\overline{\mathbb{Q}})$ a $\overline{\mathbb{Q}}$-point with $2Q \in E(\mathbb{Q})$, and $\tau \in I_p$. Then $\tau(Q) = Q$. Hence $L/\mathbb{Q}$ is unramified outside S.*

PROOF. Through the embedding $\overline{\mathbb{Q}} \to \overline{\mathbb{Q}_p}$, Q is a $\overline{\mathbb{Q}_p}$-point of E. Let K' be the finite Galois extension of $\mathbb{Q}_p$ generated by the coordinates of Q and their conjugates, with ring of integers $\mathcal{O}'$ and residue field κ'. The discriminant of the equation of E is

$$\Delta = 16(e_1-e_2)^2(e_1-e_3)^2(e_2-e_3)^2,$$

and $p \nmid \Delta$ because $p \notin S$. So the equation is minimal at p and E has good reduction at p (part 1 of Proposition 6.3.2). Let E_p be its reduction modulo p, the E_π of Definition 6.2.4 with $\pi = p$, an elliptic curve over $\mathbb{F}_p$. By part 2 of Proposition 6.3.7, the equation is minimal for $E_{K'}$, and the reduction of $E_{K'}$ is the elliptic curve $(E_p)_{\kappa'}$. By Proposition 6.2.8 over K', reduction $R \mapsto \bar{R}$ is a homomorphism $E(K') \to E_p(\kappa')$.

The element τ maps K' to itself and preserves the valuation, so it maps $\mathcal{O}'$ to itself, and $\overline{\tau(R)} = \bar{\tau}(\bar{R})$, where $\bar{\tau}$ is the action of τ on κ'. As $\tau \in I_p$, $\bar{\tau}$ is the identity, so

$$\overline{\tau(Q) - Q} = \bar{\tau}(\bar{Q}) - \bar{Q} = \bar{O}.$$

By part 1 of Proposition 8.1.3, $\tau(Q) - Q \in E[2](\overline{\mathbb{Q}}) = \{O, T_1, T_2, T_3\}$. The reductions $\bar{T}_i = (\bar{e}_i, 0)$ are affine, so they differ from $\bar{O} = (0:1:0)$, and they are distinct because $p \nmid e_i - e_j$ for $i \neq j$. So reduction is injective on $E[2](\overline{\mathbb{Q}})$, and $\tau(Q) = Q$. Thus I_p fixes each field $\mathbb{Q}(Q)$, hence their compositum L. □

REMARK 8.1.6. Let us see where the same argument fails for the multiplicative group. Let p be an odd prime, $a \in \mathbb{Q}^\times$, $\alpha := \sqrt{a}$ and $\tau \in I_p$. Then $\tau(\alpha)/\alpha \in \mu_2(\overline{\mathbb{Q}})$, and $\mu_2(\overline{\mathbb{Q}}) = \{\pm 1\}$ injects into $\mathbb{F}_p^\times$ under reduction. If $v_p(a) = 0$, then α is a unit, it reduces to an element of $\overline{\mathbb{F}}_p^\times$, the reduction of $\tau(\alpha)/\alpha$ is 1, and τ fixes α. But if $v_p(a)$ is odd, α reduces to 0, which is not in $\mathbb{G}_\mathrm{m}(\overline{\mathbb{F}}_p)$. The argument has nothing to say, and indeed $\mathbb{Q}_p(\sqrt{p})$ is ramified (part 3 of Example 6.4.5). For E, every $\overline{\mathbb{Q}_p}$-point reduces to a point of $E_p(\overline{\mathbb{F}}_p)$, because E is projective. No rational point of E escapes the group under reduction, as p escapes $\mathbb{G}_\mathrm{m}$. We return to this in Remark 8.3.5.

The proposition has an elementary shadow. Let $P = (x_0, y_0) \in E(\mathbb{Q})$, and let $p \notin S$. If $v_p(x_0) < 0$, then $2v_p(y_0) = 3v_p(x_0)$, so $v_p(x_0 - e_i) = v_p(x_0)$ is even. If $v_p(x_0) \geqslant 0$, then at most one of the three factors $x_0 - e_i$ is divisible by p, as their differences $e_j - e_i$ are prime to p. That factor has the valuation of y_0^2, which is even. So in all cases $v_p(x_0 - e_i)$ is even when $x_0 \neq e_i$: the class of $x_0 - e_i$ in $\mathbb{Q}^\times/\mathbb{Q}^{\times 2}$ lies in the finite group $\mathbb{Q}(S,2)$ of Example 7.8.5, however large the numerator and denominator of x_0. The map $P \mapsto (x_0 - e_1, x_0 - e_2)$, corrected at T_1 and T_2, is an explicit form of the Kummer map $E(\mathbb{Q})/2 \to \mathbb{Q}(S,2)^2$ [Sil09, Proposition X.1.4], and it is the starting point of the descent in Chapter 10.

THEOREM 8.1.7. *In Situation 8.1.1, $[L:\mathbb{Q}] \leqslant 2^{\#S}$ and $\#E(\mathbb{Q})/2E(\mathbb{Q}) \leqslant 4^{\#S}$.*

PROOF. We do not know yet that L is finite. Let $F \subseteq L$ be a finite extension of $\mathbb{Q}$. As $\mathrm{Gal}(L/\mathbb{Q})$ is abelian of exponent dividing 2 (part 5 of Proposition 8.1.3), $F/\mathbb{Q}$ is Galois, abelian of exponent dividing 2. Since $\mu_2(\overline{\mathbb{Q}}) \subseteq \mathbb{Q}$, part 2 of Corollary 7.4.7 gives $F = \mathbb{Q}(\Delta^{1/2})$ for a subgroup $\mathbb{Q}^{\times 2} \subseteq \Delta \subseteq \mathbb{Q}^\times$ with $[F:\mathbb{Q}] = (\Delta : \mathbb{Q}^{\times 2})$. For $d \in \Delta$, the field $\mathbb{Q}(\sqrt{d}) \subseteq L$ is unramified outside S by Proposition 8.1.5, so the

class $\delta(d)$ of Corollary 7.4.6 lies in $H^1(\mathbb{Q}, \mu_2; S)$, and the class of d lies in $\mathbb{Q}(S,2)$ by part 1 of Proposition 7.8.3. So $\Delta/\mathbb{Q}^{\times 2} \subseteq \mathbb{Q}(S,2)$, a group of order $2^{\#S}$ (Example 7.8.5), and $[F:\mathbb{Q}] \leqslant 2^{\#S}$.

Choose such an F_0 of maximal degree. For any other F, the compositum F_0F is a finite extension inside L, of degree at most $[F_0:\mathbb{Q}]$, so $F \subseteq F_0$. As L is the union of its finite subextensions, $L = F_0$ and $[L:\mathbb{Q}] \leqslant 2^{\#S}$. So $\mathrm{Gal}(L/\mathbb{Q}) \cong (\mathbb{Z}/2\mathbb{Z})^t$ with $t \leqslant \#S$, and by part 5 of Proposition 8.1.3,

$$\#E(\mathbb{Q})/2E(\mathbb{Q}) \leqslant \#\operatorname{Hom}\big((\mathbb{Z}/2\mathbb{Z})^t, (\mathbb{Z}/2\mathbb{Z})^2\big) = 4^t \leqslant 4^{\#S}.$$

□

Granting the Mordell–Weil theorem, $E(\mathbb{Q}) \cong \mathbb{Z}^r \times E(\mathbb{Q})_{\mathrm{tors}}$ for some $r \geqslant 0$, and $E(\mathbb{Q})_{\mathrm{tors}}/2$ has the order of $E(\mathbb{Q})_{\mathrm{tors}}[2] = E[2](\overline{\mathbb{Q}})$, which is 4. So $\#E(\mathbb{Q})/2E(\mathbb{Q}) = 2^{r+2}$, and Theorem 8.1.7 bounds the rank: $r \leqslant 2\#S - 2$.

EXAMPLE 8.1.8. Let $E\colon y^2 = x^3 - x$, with $(e_1,e_2,e_3) = (0,1,-1)$. The differences $e_i - e_j$ are ± 1 and ± 2, so $S = \{\infty, 2\}$, $\mathbb{Q}(S,2) = \langle -1, 2\rangle$, $L \subseteq \mathbb{Q}(i, \sqrt{2})$, and $\#E(\mathbb{Q})/2E(\mathbb{Q}) \leqslant 16$. The halves of $T_1 = (0,0)$ are $(i, \pm(1-i))$ and $(-i, \pm(1+i))$, defined over $\mathbb{Q}(i)$. The halves of T_2 and T_3 generate $\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(\sqrt{2}, \sqrt{-2})$. So $L = \mathbb{Q}(i, \sqrt{2}) = \mathbb{Q}(\zeta_8)$, ramified only at 2, and the bound on $[L:\mathbb{Q}]$ is attained. Here $E(\mathbb{Q}) = E[2](\overline{\mathbb{Q}})$, so $\#E(\mathbb{Q})/2E(\mathbb{Q}) = 4$, and only 4 of the 16 homomorphisms $\mathrm{Gal}(L/\mathbb{Q}) \to E[2](\overline{\mathbb{Q}})$ come from rational points.

EXAMPLE 8.1.9. Let $E\colon y^2 = x^3 - 25x$, with $(e_1,e_2,e_3) = (0,5,-5)$. The differences $e_i - e_j$ are ± 5 and ± 10, so $S = \{\infty, 2, 5\}$, $\mathbb{Q}(S,2) = \langle -1, 2, 5\rangle$ and $L \subseteq \mathbb{Q}(i, \sqrt{2}, \sqrt{5})$. Magma finds that $E(\mathbb{Q})$ is generated by T_1, T_2 and $P := (-4, 6)$, of infinite order, so $\#E(\mathbb{Q})/2E(\mathbb{Q}) = 8$, against the bound 64. The multiples of P have

$$x(2P) = \Big(\frac{41}{12}\Big)^2, \qquad x(3P) = -\frac{2439844}{5094049} = -\Big(\frac{2\cdot 11\cdot 71}{37\cdot 61}\Big)^2,$$
$$x(4P) = \frac{11183412793921}{2234116132416} = \Big(\frac{3344161}{2^3\cdot 3\cdot 7^2\cdot 31\cdot 41}\Big)^2,$$

and $x(3P) - 5 = -(3^2\cdot 587/2257)^2$, $x(3P) + 5 = (4799/2257)^2$. The primes in these coordinates grow, but each appears to an even power, as Remark 8.1.6 predicts. So the classes of $x(nP) - e_i$ never leave $\langle -1, 2, 5\rangle$. For instance, $Q := (2+i, 7i-1)$ satisfies $2Q = P$, and the halves of $3P$ are the $Q+P+T$ with $T \in E[2](\overline{\mathbb{Q}})$, all defined over $\mathbb{Q}(i)$. Magma also finds that the halves of T_1 generate $\mathbb{Q}(i, \sqrt{5})$, those of T_2 generate $\mathbb{Q}(\sqrt{2}, \sqrt{5})$, and $L = \mathbb{Q}(i, \sqrt{2}, \sqrt{5})$, of degree $8 = 2^{\#S}$.

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14 Oct 2026

8.2. The weak Mordell–Weil theorem

We are in Situation 8.0.1. This section is [Sil09, Section VIII.1]: the argument of Section 8.1 for any m and any number field.

THEOREM 8.2.1 (Weak Mordell–Weil). *The group $E(k)/mE(k)$ is finite.*

We call Theorem 8.2.1 the weak Mordell–Weil theorem. The first step reduces it to the case $E[m](\overline{k}) \subseteq E(k)$, which made the Kummer pairing bilinear in Section 8.1.

LEMMA 8.2.2. *Let K be a finite Galois extension of k inside $\overline{k}$. If $E(K)/mE(K)$ is finite, then $E(k)/mE(k)$ is finite.*

PROOF. Let Φ be the kernel of $E(k)/m \to E(K)/m$, so that $0 \to \Phi \to E(k)/m \to E(K)/m$ is exact, and it suffices to show that Φ is finite. Let $\delta_k\colon E(k)/m \to H^1(k, E[m])$ and $\delta_K\colon E(K)/m \to H^1(K, E[m])$ be the Kummer maps, injective by Proposition 7.5.1. For $P \in E(k)$, the cocycle $\sigma \mapsto \sigma(Q) - Q$ restricted to Gal_K represents $\delta_K(P)$, so $\mathrm{Res}\circ\delta_k = \delta_K$ on $E(k)/m$. Hence δ_k maps Φ into the kernel of $\mathrm{Res}\colon H^1(k, E[m]) \to H^1(K, E[m])$. By Proposition 7.3.5, with $G = \mathrm{Gal}_k$ and $H = \mathrm{Gal}_K$, this kernel is the image of $H^1(\mathrm{Gal}(K/k), E[m](K))$. The cocycles defining this group are maps between the finite sets $\mathrm{Gal}(K/k)$ and $E[m](K)$, so it is finite, and so is Φ [Sil09, Lemma VIII.1.1.1 and Section VIII.2]. □

The field $K \coloneqq k(E[m])$ is a finite Galois extension of k, a number field, with $E[m](\overline{k}) \subseteq E(K)$ (the discussion after Example 7.1.3). By Lemma 8.2.2, it suffices to prove Theorem 8.2.1 when $E[m](\overline{k}) \subseteq E(k)$. Then Definition 8.1.2 generalizes word for word. For $P \in E(k)$ and $\sigma \in \mathrm{Gal}_k$, choose a $\overline{k}$-point Q with $mQ = P$, and let

$$\kappa\colon E(k) \times \mathrm{Gal}_k \to E[m](\overline{k}), \qquad \kappa(P, \sigma) \coloneqq \sigma(Q) - Q,$$

and let $L \coloneqq k([m]^{-1}E(k))$ be the compositum inside $\overline{k}$ of the fields $k(Q)$, for the $\overline{k}$-points Q of E with $mQ \in E(k)$.

PROPOSITION 8.2.3. *Suppose that $E[m](\overline{k}) \subseteq E(k)$. Then κ is well defined and bilinear, its left kernel is $mE(k)$, and its right kernel is Gal_L. So L/k is Galois, $\mathrm{Gal}(L/k)$ is abelian of exponent dividing m, and κ induces a pairing*

$$E(k)/m \times \mathrm{Gal}(L/k) \to E[m](\overline{k})$$

for which both induced maps are injective. In particular $E(k)/m$ is finite if and only if L/k is finite.

PROOF. The proof of Proposition 8.1.3 applies with 2 replaced by m and $\mathbb{Q}$ by k [Sil09, Proposition VIII.1.2]. □

Let $S \subseteq \mathcal{P}_k$ be the set of the archimedean places, the non-archimedean places v with $v(m) > 0$, and the places of bad reduction of E.

REMARK 8.2.4. The set S is finite [Sil09, Remark VIII.1.3]. Fix a Weierstrass equation for E over k, with coefficients $a_1, \dots, a_6$ and discriminant $\Delta \neq 0$. A non-zero element of k has valuation 0 at all but finitely many places, so for all but finitely many non-archimedean v, $v(a_i) \geqslant 0$ for every i and $v(\Delta) = 0$. At such v the equation is minimal, and E has good reduction by part 1 of Proposition 6.3.2. Only finitely many places divide m, and there are finitely many archimedean places. A single equation need not be minimal at every place at once.

The local input is the following lemma. It is the proof of Proposition 8.1.5, with the injectivity of reduction on torsion taken from Chapter 6, and it needs no hypothesis on $E[m](\overline{k})$.

LEMMA 8.2.5. *Let v be a non-archimedean place of k with $v(m) = 0$, and E' an elliptic curve over k_v with good reduction. Let $Q \in E'(\overline{k_v})$ be a $\overline{k_v}$-point with $mQ \in E'(k_v)$. Then $\tau(Q) = Q$ for every $\tau \in I_v$, that is, $Q \in E'(k_v^{\mathrm{unr}})$.*

PROOF. Let K' be the finite Galois extension of k_v generated by the coordinates of Q and their conjugates, with residue field κ'. Fix a minimal Weierstrass equation for E'. By part 2 of Proposition 6.3.7, it is minimal for $E'_{K'}$, whose reduction is $(E'_\pi)_{\kappa'}$, an elliptic curve. As in the proof of Proposition 8.1.5, reduction $E'(K') \to E'_\pi(\kappa')$ is a homomorphism commuting with τ, and $\tau(Q) - Q$ reduces to $\bar{O}$. Also $m(\tau(Q) - Q) = \tau(mQ) - mQ = O$, as mQ is a k_v-point. The residue characteristic p does not divide m, since $v(m) = 0$, so reduction is injective on $E'[m](K')$ by part 2 of Proposition 6.2.9 over K'. Hence $\tau(Q) = Q$. A $\overline{k_v}$-point fixed by I_v has its coordinates in the fixed field k_v^{unr} of I_v. □

PROPOSITION 8.2.6. *Let $Q \in E(\overline{k})$ be a $\overline{k}$-point with $mQ \in E(k)$, and $v \notin S$ a non-archimedean place. Then I_v fixes Q. Hence, if $E[m](\overline{k}) \subseteq E(k)$, then L/k is abelian of exponent dividing m and unramified outside S.*

PROOF. Apply Lemma 8.2.5 to $E' = E_{k_v}$, through the embedding $\overline{k} \to \overline{k_v}$: $v \notin S$ means that $v(m) = 0$ and that E has good reduction at v. Then I_v fixes each $k(Q)$, hence L. The rest is Proposition 8.2.3 [Sil09, Proposition VIII.1.5]. □

It remains to show that an extension like L is finite. In Section 8.1 this was an explicit computation with quadratic fields. In general it is Theorem 7.8.6, which packages Kummer theory, the finiteness of the class group and Dirichlet's S-unit theorem.

PROPOSITION 8.2.7. *Let $T \subseteq \mathcal{P}_k$ be a finite set of places, and F an abelian extension of k inside $\overline{k}$, of exponent dividing m and unramified outside T. Then F/k is finite.*

PROOF. Let $\chi\colon \mathrm{Gal}(F/k) \to \mathbb{Z}/m\mathbb{Z}$ be a continuous homomorphism, and $\tilde{\chi}$ its composite with $\mathrm{Gal}_k \to \mathrm{Gal}(F/k)$. It is trivial on I_v for every non-archimedean $v \notin T$, since $I_v \subseteq \mathrm{Gal}_F$. So $\tilde{\chi}$ is a class of $H^1(k, \mathbb{Z}/m\mathbb{Z}; T)$, for the trivial action (Proposition 7.2.3), and $\chi \mapsto \tilde{\chi}$ is injective. By Theorem 7.8.6, there are only finitely many such χ, say $\chi_1, \dots, \chi_n$. They separate the elements of $\mathrm{Gal}(F/k)$: if $\sigma \neq 1$, some open normal subgroup U does not contain σ, and the finite abelian group $\mathrm{Gal}(F/k)/U$ of exponent dividing m has a homomorphism to $\mathbb{Z}/m\mathbb{Z}$ that is non-zero on the image of σ. So $(\chi_1, \dots, \chi_n)\colon \mathrm{Gal}(F/k) \to (\mathbb{Z}/m\mathbb{Z})^n$ is injective, and F/k is finite. □

Silverman proves Proposition 8.2.7 directly [Sil09, Proposition VIII.1.6]: over $k(\mu_m)$, such an F is generated by m-th roots of the elements of a group like $k(T, m)$, which is finite by the class group and the S-unit theorem. That is the proof of Theorem 7.8.6, and it is what the proof of Theorem 8.1.7 did by hand. We can now prove Theorem 8.2.1.

PROOF. By Lemma 8.2.2 with $K = k(E[m])$ we may assume that $E[m](\overline{k}) \subseteq E(k)$. The set S is finite by Remark 8.2.4. By Proposition 8.2.6, L/k is abelian of exponent dividing m and unramified outside S, so it is finite by Proposition 8.2.7. By Proposition 8.2.3, $E(k)/m$ embeds in the finite group $\mathrm{Hom}(\mathrm{Gal}(L/k), E[m](\overline{k}))$. □

REMARK 8.2.8. Instead of Proposition 8.2.7, one can use the theorem of Hermite and Minkowski: a number field has only finitely many extensions of bounded degree unramified outside a finite set of places [Sil09, Remark VIII.1.7]. Here each $k(Q)$ has degree at most m^2 over k, since the conjugates of Q are among the $Q + T$ with $T \in E[m](\overline{k})$, so there are only finitely many fields $k(Q)$.

REMARK 8.2.9. Everything in the proof is effective except one step [Sil09, Remark VIII.1.8]. The field L, or rather the largest abelian extension of exponent m unramified outside S, can be computed, and with it the finite group $\mathrm{Hom}(\mathrm{Gal}(L/k), E[m](\overline{k}))$ into which $E(k)/m$ embeds. No algorithm is known to decide which of its elements come from rational points. This is the descent problem of Chapter 10.

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8.3. The Kummer pairing via cohomology

We are in Situation 8.0.1. This section is [Sil09, Section VIII.2]: it restates the previous one with the Kummer map $\delta\colon E(k)/m \to H^1(k, E[m])$ of Proposition 7.5.1, and adds the local picture that Chapter 10 needs. With $mQ = P$, the class $\delta(P)$ is represented by the cocycle $\sigma \mapsto \sigma(Q) - Q$, which is $\kappa(P, \cdot)$. If $E[m](\overline{k}) \subseteq E(k)$, then Gal_k acts trivially on $E[m](\overline{k})$, so $H^1(k, E[m]) = \mathrm{Hom}_{\mathrm{conts}}(\mathrm{Gal}_k, E[m](\overline{k}))$ (Proposition 7.2.3), $\delta(P) = \kappa(P, \cdot)$, and the injectivity of δ is the statement about the left kernel in Proposition 8.2.3. The field L is the fixed field of the intersection of the kernels of the $\delta(P)$. In general $\delta(P)$ is only a cocycle, and L plays no role. The ramification condition survives.

PROPOSITION 8.3.1. *The image of $\delta\colon E(k)/m \to H^1(k, E[m])$ lies in $H^1(k, E[m]; S)$.*

PROOF. Let $P \in E(k)$, $Q \in E(\overline{k})$ with $mQ = P$, and $v \notin S$ non-archimedean. The restriction of $\mathrm{res}_v\, \delta(P)$ to I_v is the cocycle $\tau \mapsto \tau(Q) - Q$, which is 0 by Proposition 8.2.6 [Sil09, Proposition VIII.2.1]. □

This gives a second proof of Theorem 8.2.1, without Lemma 8.2.2 and without the field L: δ embeds $E(k)/m$ in $H^1(k, E[m]; S)$, which is finite by Theorem 7.8.6 with $A = E[m](\overline{k})$. The reduction to $E[m](\overline{k}) \subseteq E(k)$ has not disappeared: it is the inflation–restriction step in the proof of Theorem 7.8.6.

8.3.1. The Kummer image at a good place. The Kummer map exists over k_v too, by Proposition 7.5.1 over k_v:

$$\delta_v\colon E(k_v)/m \to H^1(k_v, E[m]),$$

and $\mathrm{res}_v \circ \delta = \delta_v$ on $E(k)$, with the same cocycle. At the places outside S its image is exactly the group of unramified classes of Definition 7.7.2. We first count $E(k_v)/m$.

PROPOSITION 8.3.2. *Let v be a non-archimedean place of k with $v(m) = 0$, and E' an elliptic curve over k_v. Then $\#E'(k_v)/mE'(k_v) = \#E'(k_v)[m]$, and both are finite.*

PROOF. Fix a minimal Weierstrass equation for E'. The kernel of reduction $E'_1(k_v)$ is isomorphic to the group $\hat{E}'(\mathfrak{m}_v)$ of the formal group of E' [Sil09, Proposition VII.2.2]. Since $m \in \mathcal{O}_v^\times$, multiplication by m is an isomorphism of the formal group [Sil09, Proposition IV.2.3(b)], so it is bijective on $\hat{E}'(\mathfrak{m}_v)$, and $E'_1(k_v)$ is uniquely m-divisible. The quotient $B \coloneqq E'(k_v)/E'_1(k_v)$ is finite, since $E'(k_v)/E'_0(k_v)$ is finite by part 1 of Theorem 6.3.9, and $E'_0(k_v)/E'_1(k_v) \cong (E'_\pi)^{\mathrm{ns}}(\kappa(v))$ is finite by Proposition 6.2.8. The snake lemma for multiplication by m on $0 \to E'_1(k_v) \to E'(k_v) \to B \to 0$ gives an exact sequence

$$0 \to E'_1(k_v)[m] \to E'(k_v)[m] \to B[m] \to E'_1(k_v)/m \to E'(k_v)/m \to B/m \to 0,$$

whose terms $E'_1(k_v)[m]$ and $E'_1(k_v)/m$ are 0. So $\#E'(k_v)[m] = \#B[m]$ and $\#E'(k_v)/m = \#B/m$. For a finite abelian group B, $\#B[m] = \#B/m$, as both are the order of the kernel and of the cokernel of $m\colon B \to B$. □

REMARK 8.3.3. When $v(m) > 0$, the formal group contributes: $E'(k_v)$ contains a subgroup of finite index isomorphic to $\mathcal{O}_v$ [Sil09, Proposition VII.6.3]. This subgroup has no m-torsion, and $\#\mathcal{O}_v/m\mathcal{O}_v = q^{v(m)}$. The snake lemma for it, as in the proof of Proposition 8.3.2, gives

$$\#E'(k_v)/mE'(k_v) = q^{v(m)}\,\#E'(k_v)[m],$$

which Chapter 10 needs.

PROPOSITION 8.3.4. *Let v be a non-archimedean place of k with $v(m) = 0$, and E' an elliptic curve over k_v with good reduction. Then $\delta_v(E'(k_v)/m) = H^1_{\mathrm{unr}}(k_v, E'[m])$.*

PROOF. Let $P \in E'(k_v)$ and $Q \in E'(\overline{k_v})$ with $mQ = P$. The restriction of $\delta_v(P)$ to I_v is $\tau \mapsto \tau(Q) - Q$, which is 0 by Lemma 8.2.5. So the image lies in $H^1_{\mathrm{unr}}(k_v, E'[m])$. The map δ_v is injective, and by Proposition 8.3.2 and part 3 of Proposition 7.7.3, applied to the finite module $E'[m](\overline{k_v})$,

$$\#E'(k_v)/m = \#E'(k_v)[m] = \#E'[m](\overline{k_v})^{\mathrm{Gal}_{k_v}} = \#H^1_{\mathrm{unr}}(k_v, E'[m]).$$

So the image is all of $H^1_{\mathrm{unr}}(k_v, E'[m])$. □

Good reduction enters only through the containment. For bad reduction the image still has $\#E'(k_v)[m]$ elements, but they need not be unramified classes.

REMARK 8.3.5. We can now say why E differs from $\mathbb{G}_{\mathrm{m}}$, the question of Remark 8.1.4. For $\mathbb{G}_{\mathrm{m}}$ the Kummer map $\delta\colon k^\times/k^{\times m} \to H^1(k, \mu_m)$ is an isomorphism (Corollary 7.4.6), and its image is not unramified outside any finite set: if $v(m) = 0$ and $a \in k^\times$ has $v(a) = 1$, then $\delta(a)$ is ramified at v (Proposition 7.7.4). Locally, the unramified classes of $H^1(k_v, \mu_m)$ are the image of the units $\mathcal{O}_v^\times$, and the class of π_v is the difference between $k_v^\times$ and $\mathcal{O}_v^\times$.
For E there is no such difference. The group $\mathbb{G}_{\mathrm{m}}(k_v) = k_v^\times$ is larger than the group $\mathbb{G}_{\mathrm{m}}(\mathcal{O}_v) = \mathcal{O}_v^\times$ of $\mathcal{O}_v$-points of the model $\operatorname{Spec}\mathcal{O}_v[t, t^{-1}]$. But E is proper, so every k_v-point of E extends to an $\mathcal{O}_v$-point of a Weierstrass model, as in Section 6.2: $E(k_v)$ consists of integral points. By Proposition 8.3.4, at a good place $v \notin S$ the classes of $E(k_v)$ are exactly the unramified ones, the analogue of $\mathcal{O}_v^\times$, and nothing like π_v exists. So the analogue of $E(k)$ in this proof is not $k^\times$ but the group $\mathcal{O}_S^\times$ of S-units, the $a \in k^\times$ with $v(a) = 0$ at every non-archimedean $v \notin S$. For it, weak Mordell–Weil holds: $\mathcal{O}_S^\times/\mathcal{O}_S^{\times m}$ is finite, and lands in $k(S, m)$ (Proposition 7.8.3 and Theorem 7.8.4). Integral points return in Chapter 9. The descent from $\mathcal{O}_S^\times/\mathcal{O}_S^{\times m}$ to finite generation, which works for $\mathcal{O}_S^\times$ and $E(k)$ alike, is Remark 8.4.19.

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8.4. Heights on abelian groups

This section is [Sil09, Section VIII.3], with the parts of [Sil09, Section VIII.9] and [HS00, Sections B.4–B.6] that do not depend on the curve. The weak Mordell–Weil theorem says that $E(k)/m$ is finite. A height

turns this into finite generation, and once the group is finitely generated, the same height counts its elements. Neither step uses the curve, so this section works with an abstract abelian group.

SITUATION 8.4.1. In this section, independently of Situation 8.0.1:

- A is an abelian group, written additively, with elements P, Q, R, and torsion subgroup A_{tors}. We write $A_{\mathbb{R}} \coloneqq A \otimes_{\mathbb{Z}} \mathbb{R}$.
- $m \geqslant 2$ is an integer, and $A/m \coloneqq A/mA$.

8.4.1. Heights and finite generation.

DEFINITION 8.4.2. Let X be a set. A height on X is a function $h\colon X \to \mathbb{R}$. Two heights h and h' on X are equivalent, written $h = h' + O(1)$, if $|h - h'|$ is bounded on X. A height h has the Northcott property if the set $\{x \in X : h(x) \leqslant C\}$ is finite for every $C \in \mathbb{R}$.

An inequality $f \leqslant g + c$ that holds for one constant c holds for every larger one. So the constants that bound heights in this section are taken non-negative, and a lower bound is written $-c$ with $c \geqslant 0$.

The Northcott property depends only on the equivalence class: if $|h - h'| \leqslant c$ on X, with $c \geqslant 0$, then $\{h' \leqslant C\} \subseteq \{h \leqslant C + c\}$. A height with the Northcott property is bounded below. Indeed, if X is non-empty and $x_0 \in X$, the set $\{h \leqslant h(x_0)\}$ is finite and non-empty, and its minimum is a lower bound for h.

DEFINITION 8.4.3. A height h on A is quasi-quadratic if

$$h(P+Q) + h(P-Q) = 2h(P) + 2h(Q) + O(1), \tag{8.4.1}$$

that is, if the difference of the two sides is bounded on $A \times A$. It is quadratic if the two sides are equal for all $P, Q \in A$ (the parallelogram law). The pairing of a quadratic height q is

$$\langle P, Q \rangle_q \coloneqq q(P+Q) - q(P) - q(Q).$$

LEMMA 8.4.4. *Let h be a quasi-quadratic height on A.*

(1) For every $n \in \mathbb{Z}$, $h(nP) = n^2 h(P) + O(1)$, with a bound depending on n but not on P.

(2) If h is bounded below, then for every $Q \in A$ there is a constant $C(Q) \geqslant 0$ with $h(P+Q) \leqslant 2h(P) + C(Q)$ for all $P \in A$.

PROOF. Let $c \geqslant 0$ bound the difference of the two sides of (8.4.1). With $P = 0$, it gives $h(-Q) = h(Q) + O(1)$, so it suffices to prove the first part for $n \geqslant 0$, by induction. It holds for $n = 0$ and $n = 1$. If it holds for $n-1$ and n, then (8.4.1) for the pair (nP, P) gives

$$\begin{aligned} h((n+1)P) &= 2h(nP) + 2h(P) - h((n-1)P) + O(1) \\ &= \bigl(2n^2 + 2 - (n-1)^2\bigr) h(P) + O(1) = (n+1)^2 h(P) + O(1). \end{aligned}$$

For the second part, let $c' \geqslant 0$ with $h \geqslant -c'$, and $C(Q) \coloneqq 2h(Q) + 2c' + c$, which is at least $c \geqslant 0$. Then

$$h(P+Q) \leqslant 2h(P) + 2h(Q) - h(P-Q) + c \leqslant 2h(P) + 2h(Q) + c' + c \leqslant 2h(P) + C(Q).$$

□

THEOREM 8.4.5. *Let h be a height on A with the Northcott property. Suppose that there are real numbers $0 < a < b$ with the following properties.*

(1) For every $Q \in A$ there is a constant $C_1(Q) \geqslant 0$ such that $h(P+Q) \leqslant a\,h(P) + C_1(Q)$ for all $P \in A$.

(2) There is a constant $C_2 \geqslant 0$ such that $h(mP) \geqslant b\,h(P) - C_2$ for all $P \in A$.

If A/m is finite, then A is finitely generated. More precisely, let $Q_1, \dots, Q_s$ represent the cosets of mA in A, and $C \coloneqq \max_i C_1(-Q_i) + C_2$, so that $C \geqslant 0$. Then A is generated by $Q_1, \dots, Q_s$ and the finitely many $P \in A$ with $h(P) \leqslant 1 + C/(b-a)$.

PROOF. Let $P \in A$ and $P_0 \coloneqq P$. For $j \geqslant 1$, choose an index i_j and an element $P_j \in A$ with $P_{j-1} = mP_j + Q_{i_j}$. By the two hypotheses,

$$b\,h(P_j) \leqslant h(mP_j) + C_2 = h(P_{j-1} - Q_{i_j}) + C_2 \leqslant a\,h(P_{j-1}) + C.$$

Let $\rho \coloneqq a/b$, so $0 < \rho < 1$. Then $h(P_j) \leqslant \rho\, h(P_{j-1}) + C/b$, and, since $\rho > 0$, induction on n gives

$$h(P_n) \leqslant \rho^n h(P) + \frac{C}{b}\bigl(1 + \rho + \cdots + \rho^{n-1}\bigr) \leqslant \rho^n h(P) + \frac{C}{b-a},$$

where the last step uses $C \geqslant 0$. Choose n with $\rho^n h(P) \leqslant 1$. Then $h(P_n) \leqslant 1 + C/(b-a)$, and $P = m^n P_n + \sum_{j=1}^{n} m^{j-1} Q_{i_j}$. The set of P_n that can occur is finite by the Northcott property. This is the proof of [Sil09, Theorem VIII.3.1], which is the case $a = 2$ and $b = m^2$. □

By Lemma 8.4.4, a quasi-quadratic height with the Northcott property satisfies the hypotheses of Theorem 8.4.5 with $a = 2$ and $b = m^2$, for every $m \geqslant 2$. The case $a = 1$ and $b = m$ is the multiplicative group, in Remark 8.4.19.

REMARK 8.4.6. The proof is effective given three inputs: representatives $Q_1, \dots, Q_s$ of A/m, the constants $C_1(-Q_i)$ and C_2, and the list of elements of A of bounded height [Sil09, Remark VIII.3.2]. For the heights on $E(k)$, the constants and the list can be computed [Sil09, Exercise 8.18]. The representatives are the step that no known algorithm guarantees, as in Remark 8.2.9.

EXAMPLE 8.4.7. No hypothesis of Theorem 8.4.5 can be dropped.

(1) Let $A = \mathbb{Q}$, as an additive group, and $h(x) \coloneqq \log \max(|u|, |w|)$ for $x = u/w$ in lowest terms. It has the Northcott property, since there are at most $(2e^C + 1)^2$ such pairs (u, w) with $h(x) \leqslant C$. From $u/w + u'/w' = (uw' + u'w)/(ww')$ we get $h(x+y) \leqslant h(x) + h(y) + \log 2$, so part 1 of Theorem 8.4.5 holds with $a = 1$, and $\mathbb{Q}/m = 0$. But $\mathbb{Q}$ is not finitely generated, since a finitely generated subgroup of $\mathbb{Q}$ has bounded denominators. Here part 2 of Theorem 8.4.5 fails for every $b > 1$: $h(mx) \leqslant h(x) + \log m$.
(2) Let $A = \bigoplus_{i \geqslant 1} \mathbb{Z}$ and $h(x) \coloneqq \sum_i i\, x_i^2$. Each term satisfies the parallelogram law, so h is quadratic. It has the Northcott property: $h(x) \leqslant C$ forces $x_i = 0$ for $i > C$ and $|x_i| \leqslant \sqrt{C}$. But $A/2$ is infinite, and A is not finitely generated.
(3) Conversely, let A be finitely generated, and choose an isomorphism $A \cong A_{\mathrm{tors}} \times \mathbb{Z}^r$, with $r \geqslant 0$. For $P \in A$, let $x_1(P), \dots, x_r(P) \in \mathbb{Z}$ be the coordinates of the image of P in $\mathbb{Z}^r$, and $h(P) \coloneqq x_1(P)^2 + \cdots + x_r(P)^2$. Each x_i is a homomorphism $A \to \mathbb{Z}$, so each x_i^2 satisfies the parallelogram law, and h is quadratic. The set $\{h \leqslant C\}$ is the preimage of finitely many elements of $\mathbb{Z}^r$, and each preimage is a coset of the finite group A_{tors}. So h has the Northcott property.

So an abelian group A is finitely generated if and only if $A/2$ is finite and A has a quasi-quadratic height with the Northcott property.

8.4.2. The canonical height.

LEMMA 8.4.8. *Let q be a quadratic height on A. Then $q(0) = 0$, $q(-P) = q(P)$, the pairing $\langle\ ,\ \rangle_q$ is symmetric and bilinear, $\langle P, P\rangle_q = 2q(P)$, and $q(nP) = n^2 q(P)$ for every $n \in \mathbb{Z}$.*

PROOF. The parallelogram law with $P = Q = 0$ gives $2q(0) = 4q(0)$, and with $P = 0$ it gives $q(Q) + q(-Q) = 2q(Q)$. The pairing is symmetric by definition. Apply the parallelogram law to the pairs $(P+R, Q)$, $(P, R-Q)$, $(P+Q, R)$ and (R, Q):

$$\begin{aligned}
q(P+Q+R) + q(P+R-Q) &= 2q(P+R) + 2q(Q),\\
q(P+R-Q) + q(P+Q-R) &= 2q(P) + 2q(R-Q),\\
q(P+Q+R) + q(P+Q-R) &= 2q(P+Q) + 2q(R),\\
q(R+Q) + q(R-Q) &= 2q(R) + 2q(Q).
\end{aligned}$$

The first minus the second plus the third, minus twice the fourth, is twice

$$q(P+Q+R) - q(P+Q) - q(P+R) - q(Q+R) + q(P) + q(Q) + q(R) = 0,$$

which says that $\langle P+Q, R\rangle_q = \langle P, R\rangle_q + \langle Q, R\rangle_q$ [HS00, Lemma B.5.2]. The parallelogram law with $Q = P$ gives $q(2P) = 4q(P)$, so $\langle P, P\rangle_q = q(2P) - 2q(P) = 2q(P)$. By bilinearity, $q(nP) = \frac{1}{2}\langle nP, nP\rangle_q = n^2 q(P)$. □

PROPOSITION 8.4.9. *Let h be a quasi-quadratic height on A. For every $P \in A$ the limit*

$$\hat{h}(P) \coloneqq \lim_{n\to\infty} 4^{-n} h(2^n P)$$

exists, and $\hat{h}$ is the unique quadratic height on A with $\hat{h} = h + O(1)$.

PROOF. By part 1 of Lemma 8.4.4, there is a constant $c \geqslant 0$ with $|h(2P) - 4h(P)| \leqslant c$ for all P. Applied to $2^n P$ and divided by 4^{n+1}, it gives

$$\left|4^{-(n+1)} h(2^{n+1}P) - 4^{-n} h(2^n P)\right| \leqslant c\, 4^{-(n+1)}.$$

So the sequence is Cauchy, and summing from $n = 0$ gives $|\hat{h}(P) - h(P)| \leqslant c/3$. Let $c' \geqslant 0$ bound the difference of the two sides of (8.4.1). Applied to $2^n P$ and $2^n Q$ and divided by 4^n, the difference for $4^{-n} h(2^n \cdot)$ is at most $c'\, 4^{-n}$, and tends to 0. So $\hat{h}$ is quadratic. If q and q' are quadratic with $q - q'$ bounded, then $q - q'$ is quadratic, and $(q - q')(P) = n^{-2}(q - q')(nP)$ tends to 0 by Lemma 8.4.8. This is Tate's construction [Sil09, Proposition VIII.9.1 and Theorem VIII.9.3], [HS00, Theorem B.4.1]. □

We call $\hat{h}$ the canonical height of h. Equivalent quasi-quadratic heights have the same canonical height, by the uniqueness.

COROLLARY 8.4.10. *Let h be a quasi-quadratic height on A with the Northcott property.*
(1) The canonical height $\hat{h}$ has the Northcott property, and $\hat{h}(P) \geqslant 0$ for every $P \in A$.
(2) We have $\hat{h}(P) = 0$ if and only if $P \in A_{\mathrm{tors}}$, and A_{tors} is finite.
(3) For $P \in A$ and $T \in A_{\mathrm{tors}}$, $\langle P, T\rangle_{\hat{h}} = 0$ and $\hat{h}(P + T) = \hat{h}(P)$.

PROOF. The Northcott property passes to the equivalent height $\hat{h}$, so $\hat{h}$ is bounded below, say by $-c$ with $c \geqslant 0$. Then $\hat{h}(P) = n^{-2}\hat{h}(nP) \geqslant -c\, n^{-2}$ for every $n \geqslant 1$, so $\hat{h}(P) \geqslant 0$. If $nP = 0$ with $n \geqslant 1$, then $\hat{h}(P) = n^{-2}\hat{h}(0) = 0$. Conversely, if $\hat{h}(P) = 0$, then $\hat{h}(nP) = 0$ for every n, so the multiples of P lie in the finite set $\{\hat{h} \leqslant 0\}$, and P is torsion. That set contains A_{tors}. Finally, if $nT = 0$ with $n \geqslant 1$, then $n\langle P, T\rangle_{\hat{h}} = \langle P, nT\rangle_{\hat{h}} = 0$, and $\hat{h}(P + T) = \hat{h}(P) + \hat{h}(T) + \langle P, T\rangle_{\hat{h}} = \hat{h}(P)$. □

8.4.3. Lattices and Minkowski's theorem. Let V be a real vector space of finite dimension r, with a translation-invariant measure vol: in coordinates, a positive multiple of Lebesgue measure.

DEFINITION 8.4.11. A lattice in V is a subgroup Λ generated by a basis $\lambda_1, \ldots, \lambda_r$ of V. Its fundamental parallelepiped is $F \coloneqq \{t_1\lambda_1 + \cdots + t_r\lambda_r : 0 \leqslant t_i < 1\}$, and its covolume is $\mathrm{covol}(\Lambda) \coloneqq \mathrm{vol}(F)$.

The space V is the disjoint union of the translates $F + \lambda$, for $\lambda \in \Lambda$. Two bases of Λ differ by a matrix in $\mathrm{GL}_r(\mathbb{Z})$, of determinant ± 1, so the covolume does not depend on the basis.

THEOREM 8.4.12 (Minkowski). *Let Λ be a lattice in V, and $U \subseteq V$ a convex subset with $-U = U$. If $\mathrm{vol}(U) > 2^r\, \mathrm{covol}(\Lambda)$, then U contains a non-zero element of Λ.*

PROOF. Let $U' \coloneqq \frac{1}{2}U$, so $\mathrm{vol}(U') = 2^{-r}\, \mathrm{vol}(U) > \mathrm{vol}(F)$. The sets $U' \cap (F + \lambda)$, for $\lambda \in \Lambda$, partition U', and their translates $(U' \cap (F + \lambda)) - \lambda$ lie in F. Their volumes add up to $\mathrm{vol}(U') > \mathrm{vol}(F)$, so two of the translates meet: there are $\lambda \neq \lambda'$ in Λ and $x, x' \in U'$ with $x - \lambda = x' - \lambda'$. Then $2x$ and $-2x'$ lie in U, and so does their midpoint $x - x' = \lambda - \lambda'$, a non-zero element of Λ [HS00, Theorem B.5.4]. □

LEMMA 8.4.13. *Let Λ be a lattice in V, and q a quadratic form on V. Suppose that $q(\lambda) \neq 0$ for every non-zero $\lambda \in \Lambda$, and that $\{\lambda \in \Lambda : q(\lambda) \leqslant C\}$ is finite for every C. Then q is positive definite.*

PROOF. If $q(\lambda) < 0$ for some $\lambda \in \Lambda$, then the multiples $n\lambda$ satisfy $q(n\lambda) = n^2 q(\lambda) \leqslant 0$, and they are infinitely many. So q is positive on $\Lambda \setminus \{0\}$, and $\mu \coloneqq \min\{q(\lambda) : \lambda \in \Lambda, \lambda \neq 0\}$ exists, by the finiteness hypothesis, and is positive. Diagonalize q: there are coordinates $x_1, \ldots, x_r$ on V in which $q = x_1^2 + \cdots +$

$x_s^2 - x_{s+1}^2 - \cdots - x_{s+t}^2$, with $s+t \leqslant r$. Take vol to be Lebesgue measure in these coordinates. Suppose that $s < r$. For $\delta > 0$, the set

$$U_\delta := \Big\{ x \in V : \sum_{i \leqslant s} x_i^2 \leqslant \frac{\mu}{2} \text{ and } \sum_{i > s} x_i^2 \leqslant \delta \Big\}$$

is convex and symmetric, and its volume is a positive multiple of $\delta^{(r-s)/2}$, which tends to infinity. By Theorem 8.4.12, for δ large U_δ contains a non-zero $\lambda \in \Lambda$, and $q(\lambda) \leqslant \sum_{i \leqslant s} x_i(\lambda)^2 \leqslant \mu/2$, a contradiction. So $s = r$, and q is positive definite [Sil09, Lemma VIII.9.5]. □

REMARK 8.4.14. The finiteness hypothesis in Lemma 8.4.13 cannot be dropped [Sil09, Remark VIII.9.4 and Exercise 8.16]. On $V = \mathbb{R}^2$ with $\Lambda = \mathbb{Z}^2$, the form $q(x,y) := (x - \sqrt{2}\,y)^2$ is non-zero on $\Lambda \setminus \{0\}$, since $\sqrt{2}$ is irrational, but it vanishes on the line $x = \sqrt{2}\,y$. The positivity of $\hat{h}$ on A/A_{tors}, from Corollary 8.4.10, is the first hypothesis only. The point, due to Cassels, is that the second is needed to conclude that $\hat{h}$ is positive definite on $A_{\mathbb{R}}$.

8.4.4. Néron's counting theorem.

THEOREM 8.4.15. *Let h be a quasi-quadratic height on A with the Northcott property, and suppose that A/m is finite.*

(1) The group A is finitely generated, with A_{tors} finite, so $A \cong A_{\mathrm{tors}} \times \mathbb{Z}^r$ for some $r \geqslant 0$.
(2) The image Λ of A in $A_{\mathbb{R}}$ is a lattice, and the kernel of $A \to \Lambda$ is A_{tors}.
(3) There is a unique quadratic form q on $A_{\mathbb{R}}$ with $\hat{h}(P) = q(P \otimes 1)$ for all $P \in A$, and q is positive definite.

PROOF. The group A is finitely generated by Lemma 8.4.4 and Theorem 8.4.5, and A_{tors} is finite. Since $A_{\mathrm{tors}} \otimes \mathbb{R} = 0$, the group $A_{\mathbb{R}}$ is $(A/A_{\mathrm{tors}}) \otimes \mathbb{R} \cong \mathbb{R}^r$, in which $A/A_{\mathrm{tors}} \cong \mathbb{Z}^r$ is the standard lattice. By Lemma 8.4.8 and part 3 of Corollary 8.4.10, the pairing $\langle\ ,\ \rangle_{\hat{h}}$ is a symmetric bilinear form on A/A_{tors}. It extends uniquely to a symmetric bilinear form B on $A_{\mathbb{R}}$, and $q(v) := \frac{1}{2}B(v,v)$ satisfies $q(P \otimes 1) = \frac{1}{2}\langle P, P\rangle_{\hat{h}} = \hat{h}(P)$. By Corollary 8.4.10, q is non-zero on $\Lambda \setminus \{0\}$, and $\{\lambda \in \Lambda : q(\lambda) \leqslant C\}$ is the image of the finite set $\{\hat{h} \leqslant C\}$. So q is positive definite by Lemma 8.4.13 [Sil09, Proposition VIII.9.6]. □

DEFINITION 8.4.16. In Theorem 8.4.15, let $P_1, \ldots, P_r \in A$ map to a basis of Λ. The regulator of $\hat{h}$ is $\mathrm{Reg}(\hat{h}) := \det\big(\langle P_i, P_j\rangle_{\hat{h}}\big)_{i,j}$.

Another choice replaces the Gram matrix $G := (\langle P_i, P_j\rangle_{\hat{h}})$ by $U^{\mathsf{T}} G U$ with $U \in \mathrm{GL}_r(\mathbb{Z})$, so $\mathrm{Reg}(\hat{h})$ does not depend on the choice, and it is positive since q is positive definite. Let $\omega_r := \pi^{r/2}/\Gamma(r/2+1)$ be the volume of the unit ball in $\mathbb{R}^r$.

LEMMA 8.4.17. *Let Λ be a lattice in V, with $r \geqslant 1$, and $\|\ \|$ a norm on V, with unit ball $B := \{v : \|v\| \leqslant 1\}$. Then, as $T \to \infty$,*

$$\#\{\lambda \in \Lambda : \|\lambda\| \leqslant T\} = \frac{\mathrm{vol}(B)}{\mathrm{covol}(\Lambda)}\, T^r + O(T^{r-1}).$$

PROOF. Let F be a fundamental parallelepiped of Λ, and $D := \sup_{y \in F} \|y\|$, finite since F is bounded. Let $N(T)$ be the left side, and U_T the union of the $F + \lambda$ with $\|\lambda\| \leqslant T$. If $\|x\| \leqslant T - D$, write $x = y + \lambda$ with $y \in F$. Then $\|\lambda\| \leqslant \|x\| + \|y\| \leqslant T$, so $x \in U_T$. If $x = y + \lambda \in U_T$, then $\|x\| \leqslant T + D$. So $(T-D)B \subseteq U_T \subseteq (T+D)B$, and taking volumes, for $T \geqslant D$,

$$\mathrm{vol}(B)(T-D)^r \leqslant N(T)\,\mathrm{covol}(\Lambda) \leqslant \mathrm{vol}(B)(T+D)^r$$

[HS00, Lemma B.6.4]. □

THEOREM 8.4.18 (Néron). *Let h be a quasi-quadratic height on A with the Northcott property, suppose that A/m is finite, and let $r \geqslant 1$ be the rank of A. Then, as $T \to \infty$,*

$$\#\{P \in A : h(P) \leqslant T\} = \frac{\#A_{\mathrm{tors}}\, \omega_r\, 2^{r/2}}{\sqrt{\mathrm{Reg}(\hat{h})}}\, T^{r/2} + O\big(T^{(r-1)/2}\big). \tag{8.4.2}$$

PROOF. By Proposition 8.4.9 there is a constant $c \geqslant 0$ with $\{\hat{h} \leqslant T - c\} \subseteq \{h \leqslant T\} \subseteq \{\hat{h} \leqslant T + c\}$. By Theorem 8.4.15, the map $A \to \Lambda$ has fibers of size $\#A_{\mathrm{tors}}$, and $\hat{h}$ is the pullback of q, so $\#\{\hat{h} \leqslant T\} = \#A_{\mathrm{tors}} \cdot \#\{\lambda \in \Lambda : q(\lambda) \leqslant T\}$. Use the coordinates $t \mapsto \sum_i t_i(P_i \otimes 1)$ on $A_{\mathbb{R}}$, with Lebesgue measure, so that $\mathrm{covol}(\Lambda) = 1$ and $q(t) = \frac{1}{2} t^{\mathsf{T}} G t$. Apply Lemma 8.4.17 to the norm $\sqrt{q}$, and $\sqrt{T}$ in place of T. The substitution $t = \sqrt{2}\, G^{-1/2} u$ maps the unit ball onto $\{q \leqslant 1\}$, so $\mathrm{vol}\{q \leqslant 1\} = \omega_r 2^{r/2}/\sqrt{\det G}$. Finally $(T \pm c)^{r/2} = T^{r/2} + O(T^{r/2-1})$, which is within the error term [HS00, Theorem B.6.3]. □

Let $N(T)$ be the left side of (8.4.2). Then $r = \lim_{T\to\infty} 2 \log N(T)/\log T$: the rank can be read off from the growth of the number of elements of bounded height. Hindry and Silverman state Theorem 8.4.18 for an abelian variety, with the multiplicative height e^h, so that $T^{r/2}$ becomes $(\log T)^{r/2}$, and for a canonical height that may have a linear part. A quasi-quadratic height has none, since $h(-P) = h(P) + O(1)$.

8.4.5. The multiplicative group.

REMARK 8.4.19. Theorem 8.4.5 also applies to the multiplicative group, with a linear height in place of a quadratic one. Let k be a number field, $S \subseteq \mathcal{P}_k$ a finite set containing the archimedean places, and $\mathcal{O}_S^\times$ the group of S-units, as in Remark 8.3.5. In these notes $\mathcal{O}_S^\times$ is finitely generated by Dirichlet's S-unit theorem, which is only cited (Theorem 7.8.4), and the finiteness of $\mathcal{O}_S^\times/\mathcal{O}_S^{\times m}$ is deduced from it (Proposition 7.8.3). An independent proof goes as follows. Enlarging S enlarges $\mathcal{O}_S^\times$, and a subgroup of a finitely generated abelian group is finitely generated, so we may assume that S contains the places v with $v(m) > 0$.

(1) The group $\mathcal{O}_S^\times/\mathcal{O}_S^{\times m}$ is finite. Let $k' := k(\mu_m)$, and S' the set of places of k' above S. For $x \in \mathcal{O}_S^\times$, the extension $k'(x^{1/m})/k'$ has degree at most m and is unramified outside S', by Proposition 7.7.4. By Hermite's theorem there are only finitely many such extensions [HS00, Proposition C.1.6], [Sil09, Remark VIII.1.7], so their compositum F is finite over k'. By Corollary 7.4.6, the image of $\mathcal{O}_S^\times$ in $k'^\times/k'^{\times m}$ embeds in the finite group $\mathrm{Hom}(\mathrm{Gal}(F/k'), \mu_m(k'))$. The kernel of $k^\times/k^{\times m} \to k'^\times/k'^{\times m}$ is finite, as in Lemma 8.2.2, and $\mathcal{O}_S^\times \cap k^{\times m} = \mathcal{O}_S^{\times m}$, by valuations.

(2) For $x \in k^\times$, let $h(x) := h([x : 1])$ be the logarithmic height of Definition 8.5.3, with the absolute values $|\ |_v$ and local degrees n_v of Situation 8.5.1. It has the Northcott property on k, viewed inside $\mathbb{P}^1(k)$, by Theorem 8.5.13. By Proposition 8.5.12, it satisfies $h(xy) \leqslant h(x) + h(y)$ and $h(x^m) = m\, h(x)$.

(3) These are the hypotheses of Theorem 8.4.5 for $\mathcal{O}_S^\times$, written multiplicatively, with $a = 1$, $b = m$, $C_1(y) = h(y)$ and $C_2 = 0$. Here $C_1(y) \geqslant 0$, since every term of $h(y)$ is non-negative. So $\mathcal{O}_S^\times$ is finitely generated.

Hermite's theorem is itself proved with Minkowski's theorem [HS00, Theorem C.3.2], so this proof avoids Dirichlet's theorem but not the geometry of numbers. It does not give the rank, $\#S - 1$ [HS00, Theorem C.3.3].

The analogue of Theorem 8.4.18 is a count of S-units. For $x \in \mathcal{O}_S^\times$, we have $|x|_v = 1$ for $v \notin S$, and part 3 of Proposition 8.5.12 gives

$$h(x) = \frac{1}{2[k : \mathbb{Q}]} \sum_{v \in S} n_v \big|\log |x|_v\big|.$$

Under the logarithmic embedding $x \mapsto (n_v \log|x|_v)_{v\in S}$, whose kernel is the finite group of roots of unity of k and whose image is a lattice in the hyperplane $\sum_v y_v = 0$ by Dirichlet's theorem, h is a multiple of the ℓ^1 norm. It is a norm, not a quadratic form: $h(x^n) = |n|h(x)$. So Lemma 8.4.17 applies to h itself, and, if $\#S \geqslant 2$, the number of S-units with $h(x) \leqslant T$ is $c\, T^{\#S-1} + O(T^{\#S-2})$ for a constant $c > 0$, with exponent the rank instead of half the rank. The constant c involves the covolume of the image, the S-regulator, in place of $\mathrm{Reg}(\hat{h})$.

8.5. Heights on projective space

Theorem 8.4.5 needs a height on $E(k)$ with the Northcott property. It will come from a height on projective space, restricted to $E \subseteq \mathbb{P}^2$. This section constructs that height, following [Sil09, Section VIII.5] and [HS00, Sections B.1 and B.2]. A point of $\mathbb{P}^N(\overline{\mathbb{Q}})$ has large height when its coordinates are large at the archimedean places, or have large denominators at the non-archimedean ones. The main example is the projective line, and the multiplicative group inside it. There the minimal polynomial of an algebraic number computes its height, and it shows the valuations of the number at the non-archimedean places.

SITUATION 8.5.1. In this section we return to Situation 8.0.1. The curve E plays no role.

- As k is a number field, $\overline{k}$ is an algebraic closure of $\mathbb{Q}$, and we write $\overline{\mathbb{Q}}$ for it.
- $\mathcal{P}_k^\infty$ and $\mathcal{P}_k^0$ are the sets of archimedean and non-archimedean places of k.
- For $v \in \mathcal{P}_k$, $|\ |_v$ is the absolute value on k_v whose restriction to $\mathbb{Q}$ is the usual absolute value on $\mathbb{R}$ or the p-adic absolute value, with $|p|_p = p^{-1}$. We write $\mathbb{Q}_v$ for the completion of $\mathbb{Q}$ at this restriction, $\mathbb{R}$ or $\mathbb{Q}_p$, and $n_v := [k_v : \mathbb{Q}_v]$ for the local degree of v.
- For $v \in \mathcal{P}_k^0$, $q_v := \#\kappa(v)$, and $v\colon k_v^\times \to \mathbb{Z}$ is the normalized valuation, as in Situation 6.0.2.
- For a point $P = [x_0 : \cdots : x_N]$ of $\mathbb{P}^N(k)$ and $v \in \mathcal{P}_k$, $|P|_v := \max_i |x_i|_v$. It depends on the choice of coordinates.

The archimedean places correspond to the embeddings $\tau\colon k \to \mathbb{C}$, up to complex conjugation, with $|x|_v = |\tau(x)|$, and n_v is 1 or 2 as τ is real or not [HS00, Proposition B.1.3]. Let $v \in \mathcal{P}_k^0$ lie above p, with ramification index e and residue degree f over $\mathbb{Q}_p$, so that $n_v = ef$, $v(p) = e$ and $q_v = p^f$ (Remark 6.3.5). As $|\ |_v$ and v define the same place, $|x|_v = c^{v(x)}$ for a constant $0 < c < 1$, and $|p|_v = p^{-1}$ gives $c = p^{-1/e}$. So, for $x \in k_v^\times$,

$$|x|_v^{n_v} = q_v^{-v(x)}. \tag{8.5.1}$$

PROPOSITION 8.5.2.
(1) Let L be a finite extension of k, and $v \in \mathcal{P}_k$. Then $\sum_{w|v} n_w = [L : k]\, n_v$, the sum over the places $w \in \mathcal{P}_L$ above v.
(2) Let $x \in k^\times$. Then $\prod_{v \in \mathcal{P}_k} |x|_v^{n_v} = 1$.

These are the extension formula and the product formula [Sil09, (VIII.5.2) and (VIII.5.3)], [HS00, Propositions B.1.1 and B.1.2]. For $k = \mathbb{Q}$ and $x = \pm \prod_p p^{a_p}$, the product formula says $|x| \cdot \prod_p p^{-a_p} = 1$. Part 1, for $k = \mathbb{Q}$ and the archimedean place, gives $\sum_{v \in \mathcal{P}_L^\infty} n_v = [L : \mathbb{Q}]$.

8.5.1. The height of a point of projective space.

DEFINITION 8.5.3. The height relative to k of a point $P \in \mathbb{P}^N(k)$ is

$$H_k(P) := \prod_{v \in \mathcal{P}_k} |P|_v^{n_v}.$$

Let $P \in \mathbb{P}^N(\overline{\mathbb{Q}})$, and let $K \subseteq \overline{\mathbb{Q}}$ be a number field with $P \in \mathbb{P}^N(K)$. The height of P is $H(P) := H_K(P)^{1/[K:\mathbb{Q}]}$, and its logarithmic height is $h(P) := \log H(P)$.

The product is finite. Fix j with $x_j \neq 0$. At all but finitely many places, $|x_i|_v \leqslant 1$ for every i and $|x_j|_v = 1$, and there $|P|_v = 1$.

PROPOSITION 8.5.4. *Let $P \in \mathbb{P}^N(k)$.*
(1) The height $H_k(P)$ does not depend on the choice of coordinates of P.
(2) The height satisfies $H_k(P) \geqslant 1$.
(3) Let L be a finite extension of k. Then $H_L(P) = H_k(P)^{[L:k]}$.
In particular $H(P)$ does not depend on the choice of K, and $h(P) \geqslant 0$ for every $P \in \mathbb{P}^N(\overline{\mathbb{Q}})$.

PROOF. Other coordinates of P are $[\lambda x_0 : \cdots : \lambda x_N]$ with $\lambda \in k^\times$, and $|\lambda P|_v = |\lambda|_v |P|_v$. By part 2 of Proposition 8.5.2, the factors $|\lambda|_v^{n_v}$ multiply to 1. For part 2, choose coordinates with some $x_j = 1$. Then every factor $|P|_v^{n_v}$ is at least 1. For part 3, the coordinates lie in k, so $|P|_w = |P|_v$ for $w \mid v$, and part 1 of Proposition 8.5.2 gives

$$H_L(P) = \prod_{v \in \mathcal{P}_k} \prod_{w \mid v} |P|_v^{n_w} = \prod_{v \in \mathcal{P}_k} |P|_v^{[L:k] n_v} = H_k(P)^{[L:k]}.$$

Two number fields K, K' containing the coordinates are contained in their compositum KK', and part 3 for KK'/K and KK'/K' shows that $H_K(P)^{1/[K:\mathbb{Q}]} = H_{KK'}(P)^{1/[KK':\mathbb{Q}]} = H_{K'}(P)^{1/[K':\mathbb{Q}]}$ [Sil09, Proposition VIII.5.4]. □

So h is a height on $\mathbb{P}^N(\overline{\mathbb{Q}})$ in the sense of Definition 8.4.2, and $h \geqslant 0$. Over $\mathbb{Q}$ it has a familiar form.

EXAMPLE 8.5.5. Let $P \in \mathbb{P}^N(\mathbb{Q})$, and choose coordinates $x_0, \dots, x_N \in \mathbb{Z}$ with $\gcd(x_0, \dots, x_N) = 1$. At every prime p, $|x_i|_p \leqslant 1$ for every i, with equality for some i, so $|P|_p = 1$. Hence

$$H_{\mathbb{Q}}(P) = \max(|x_0|, \dots, |x_N|)$$

[Sil09, Example VIII.5.1 and Remark VIII.5.5]. There are at most $(2C+1)^{N+1}$ such rational points with $H_{\mathbb{Q}}(P) \leqslant C$. For $N = 1$ and $P = [u : w]$, this is the height of part 1 of Example 8.4.7, $h(u/w) = \log \max(|u|, |w|)$.

REMARK 8.5.6. Over a number field, the recipe of Example 8.5.5 may fail. Let $P = [x_0 : \cdots : x_N] \in \mathbb{P}^N(k)$, and $\mathfrak{b}$ the fractional ideal of $\mathcal{O}_k$ generated by $x_0, \dots, x_N$. For $v \in \mathcal{P}_k^0$, the exponent of the prime ideal of v in $\mathfrak{b}$ is $\min_i v(x_i)$, and by (8.5.1) $|P|_v^{n_v} = q_v^{-\min_i v(x_i)}$. So

$$H_k(P) = \frac{1}{\operatorname{Nm} \mathfrak{b}} \prod_{v \in \mathcal{P}_k^\infty} |P|_v^{n_v}$$

[Sil09, Exercise 8.9]. If $\mathfrak{b} = \lambda \mathcal{O}_k$ is principal, then the coordinates x_i/λ lie in $\mathcal{O}_k$ and generate the unit ideal, and only the archimedean places contribute, as over $\mathbb{Q}$. Otherwise no coordinates of P do: other coordinates generate $\lambda \mathfrak{b}$, which is not principal. For instance, in $k = \mathbb{Q}(\sqrt{-5})$, the ideal $\mathfrak{b} = (2, 1 + \sqrt{-5})$ has $\mathfrak{b}^2 = (2)$, so $\operatorname{Nm} \mathfrak{b} = 2$, and it is not principal, as $u^2 + 5w^2 = 2$ has no integer solution. The k-point $P = [2 : 1 + \sqrt{-5}]$ has $|P|_v^2 = \max(4, 6)$ at the complex place, and $H_k(P) = 6/2 = 3$.

8.5.2. The projective line and the multiplicative group. For $x \in \overline{\mathbb{Q}}$ we write $H(x) \coloneqq H([x : 1])$ and $h(x) \coloneqq \log H(x)$. The multiplicative group $\mathbb{G}_{\mathrm{m}} = \mathbb{P}^1 \setminus \{0, \infty\}$ has $\mathbb{G}_{\mathrm{m}}(\overline{\mathbb{Q}}) = \overline{\mathbb{Q}}^\times$, and on it h behaves like a height on an abelian group, with the roots of unity as torsion. If $x \in K$, then $|[x : 1]|_v = \max(1, |x|_v)$, so

$$H(x)^{[K:\mathbb{Q}]} = \prod_{v \in \mathcal{P}_K} \max(1, |x|_v)^{n_v},$$

and only the places where x is large contribute. For $K = \mathbb{Q}(x)$, the archimedean factors are $\max(1, |\tau(x)|)$, over the embeddings $\tau \colon K \to \mathbb{C}$, a complex place counting its two embeddings through $n_v = 2$. The non-archimedean factors are read from the minimal polynomial of x. The tool is Gauss's lemma.

LEMMA 8.5.7 (Gauss). *Let F be a field with a non-archimedean absolute value $|\ |$. For $g = \sum_i c_i T^i \in F[T]$, let $|g| \coloneqq \max_i |c_i|$, the Gauss norm of g. Then $|gg'| = |g||g'|$ for all $g, g' \in F[T]$. In particular, if $g = c \prod_{j=1}^d (T - \beta_j)$ with $c, \beta_j \in F$, then $|g| = |c| \prod_j \max(1, |\beta_j|)$.*

PROOF. We may assume that $g, g' \neq 0$. The norm $|g|$ is the absolute value of a coefficient of g. Dividing g and g' by such coefficients, we may assume that $|g| = |g'| = 1$. Let $\mathcal{O} \coloneqq \{c : |c| \leqslant 1\}$, with maximal ideal $\mathfrak{m} \coloneqq \{c : |c| < 1\}$. Then $g, g' \in \mathcal{O}[T]$, so $|gg'| \leqslant 1$ by the ultrametric inequality, and the reductions of g and g' in $(\mathcal{O}/\mathfrak{m})[T]$ are non-zero. As $(\mathcal{O}/\mathfrak{m})[T]$ is a domain, the reduction of gg' is non-zero, so some coefficient of gg' has absolute value 1. For the last claim, $|T - \beta| = \max(1, |\beta|)$ [HS00, proof of Proposition B.7.3]. □

To pass from $\mathbb{Q}(x)$ to its completions, we use the following description of the places of $\mathbb{Q}(x)$ above p.

PROPOSITION 8.5.8. *Let $x \in \overline{\mathbb{Q}}$, $K \coloneqq \mathbb{Q}(x)$, $g \in \mathbb{Q}[T]$ the monic minimal polynomial of x, and p a prime. Then $g = \prod_{v|p} g_v$ in $\mathbb{Q}_p[T]$, the product over the places $v \in \mathcal{P}_K$ above p, where the g_v are distinct, monic and irreducible over $\mathbb{Q}_p$, $\deg g_v = n_v$, and x is a root of g_v in K_v, with $K_v = \mathbb{Q}_p(x)$. Every root of g_v in a finite extension L of $\mathbb{Q}_p$ has absolute value $|x|_v$, for the absolute value of L extending $|\ |_p$.*

PROOF. The first part is the description of the extensions of $|\ |_p$ to K [Neu99, Proposition II.8.2]. For the second, let $\beta \in L$ be a root of g_v. As g_v is irreducible, there is a $\mathbb{Q}_p$-isomorphism $K_v = \mathbb{Q}_p(x) \to \mathbb{Q}_p(\beta)$ sending x to β. Through it, the absolute value of L restricts to an absolute value on K_v extending $|\ |_p$, which is $|\ |_v$ by uniqueness (Remark 6.3.5). So $|\beta| = |x|_v$. □

The valuations of the roots of a polynomial over a p-adic field are read from its coefficients. Let F be a finite extension of $\mathbb{Q}_p$, and $w\colon F^\times \to \mathbb{Q}$ the valuation with $w(p) = 1$, so that $|c| = p^{-w(c)}$. The Newton polygon of $g = \sum_{i=0}^d c_i T^i \in F[T]$, with $c_0 c_d \neq 0$, is the lower boundary of the convex hull of the points $(i, w(c_i))$ of $\mathbb{R}^2$, for the i with $c_i \neq 0$. It is the graph of a convex piecewise linear function on $[0, d]$, and consists of segments of distinct slopes.

PROPOSITION 8.5.9. *Let F and g be as above, and let L be a finite extension of F over which $g = c_d \prod_{j=1}^d (T - \beta_j)$ splits, with w extended to L. If the Newton polygon of g has a segment of slope s and horizontal length ℓ, then exactly ℓ of the roots β_j, counted with multiplicity, have $w(\beta_j) = -s$. Every root has this form for some segment.*

PROOF. Dividing g by c_d moves the points $(i, w(c_i))$ down by $w(c_d)$, so we may assume that $c_d = 1$. Order the roots so that $w(\beta_1) \leqslant \ldots \leqslant w(\beta_d)$, and let $r_i \coloneqq w(\beta_1) + \cdots + w(\beta_i)$, with $r_0 = 0$. Up to sign, c_{d-i} is the i-th elementary symmetric function of the roots, a sum of products of i distinct roots. Each product has valuation at least r_i, so $w(c_{d-i}) \geqslant r_i$. If $i \in \{0, d\}$ or $w(\beta_i) < w(\beta_{i+1})$, then $\beta_1 \cdots \beta_i$ is the only product of valuation r_i, so $w(c_{d-i}) = r_i$ by the ultrametric inequality. Let ρ be the piecewise linear function with $\rho(d - i) = r_i$. On $[d - i - 1, d - i]$ its slope is $-w(\beta_{i+1})$, which increases with $d - i$, so ρ is convex. Every point $(j, w(c_j))$ lies on or above the graph of ρ, and the vertices of the graph are among these points. So the graph of ρ is the Newton polygon. As ρ has slope $-w(\beta_{i+1})$ on $[d - i - 1, d - i]$, its segment of slope s has horizontal length the number of j with $w(\beta_j) = -s$. □

THEOREM 8.5.10. *Let $x \in \overline{\mathbb{Q}}^\times$, $K \coloneqq \mathbb{Q}(x)$ and $d \coloneqq [K : \mathbb{Q}]$. Let*

$$f = a_0 T^d + a_1 T^{d-1} + \cdots + a_d = a_0 (T - \alpha_1) \cdots (T - \alpha_d) \in \mathbb{Z}[T]$$

be the minimal polynomial of x, scaled so that its coefficients are coprime integers and $a_0 > 0$, with roots $\alpha_j \in \mathbb{C}$.

(1) For every prime p,

$$\prod_{v|p} \max(1, |x|_v)^{n_v} = |a_0|_p^{-1}, \qquad \prod_{v|p} |x|_v^{n_v} = |a_d/a_0|_p.$$

So the non-archimedean places contribute a_0 to $H_K([x : 1])$.

(2) The height of x is

$$H(x)^d = a_0 \prod_{j=1}^d \max(1, |\alpha_j|).$$

(3) Let p be a prime. If the Newton polygon of f over $\mathbb{Q}_p$ has a segment of slope s and horizontal length ℓ, then $\sum n_v = \ell$, the sum over the places $v \mid p$ of K with $|x|_v = p^s$.

(4) We have $x \in \overline{\mathbb{Z}}$ if and only if $a_0 = 1$, if and only if $|x|_v \leqslant 1$ for every $v \in \mathcal{P}_K^0$. In that case, x is a unit at every place above a prime p if and only if $p \nmid a_d$, and $a_d = (-1)^d \operatorname{Nm}_{K/\mathbb{Q}}(x)$.

PROOF. Let p be a prime, $g \coloneqq f/a_0$, and $g = \prod_{v|p} g_v$ as in Proposition 8.5.8. Let L be a finite extension of $\mathbb{Q}_p$ over which f splits. Each g_v is the product of n_v factors $T - \beta$ with $|\beta| = |x|_v$, so Lemma 8.5.7 over L gives $|g_v| = \max(1, |x|_v)^{n_v}$. The coefficients of f are coprime integers, so $|f| = 1$, and by Lemma 8.5.7 again

$$1 = |f| = |a_0|_p \prod_{v|p} |g_v| = |a_0|_p \prod_{v|p} \max(1, |x|_v)^{n_v}.$$

Likewise $|g_v(0)| = |x|_v^{n_v}$, and $g(0) = a_d/a_0$ is the product of the $g_v(0)$. The product of the $|a_0|_p^{-1}$ over all p is a_0, by unique factorization. This proves part 1.

The archimedean factors of $H_K([x:1])$ are $\prod_j \max(1, |\alpha_j|)$, as the α_j are the images $\tau(x)$ of x under the embeddings $\tau\colon K \to \mathbb{C}$. With part 1 this gives part 2. Part 3 is Proposition 8.5.9 for f over $\mathbb{Q}_p$: the roots of f in L are those of the g_v, and g_v has n_v of them, each with $|\beta| = |x|_v$.

By part 1, $a_0 = 1$ if and only if every factor $\max(1, |x|_v)^{n_v}$ with $v \in \mathcal{P}_K^0$ is 1, that is, $|x|_v \leqslant 1$ for every $v \in \mathcal{P}_K^0$. This holds if and only if x lies in $\mathcal{O}_K = \overline{\mathbb{Z}} \cap K$ [HS00, Section B.1, after Proposition B.1.3]. If $a_0 = 1$, each $|x|_v^{n_v}$ with $v \mid p$ is at most 1, and their product is $|a_d|_p$. So they are all 1 if and only if $p \nmid a_d$. Finally $a_d = (-1)^d \alpha_1 \cdots \alpha_d = (-1)^d \operatorname{Nm}_{K/\mathbb{Q}}(x)$. □

The right side of part 2 is the Mahler measure $M(f)$ of f [HS00, Section B.7, after Lemma B.7.3.1]. For an algebraic integer, the non-archimedean places contribute nothing to $H(x)$, and only the conjugates of x outside the unit disc count. The places where x is not a unit lie above the primes dividing the norm $\pm a_d$, and the Newton polygons of f at these primes give the values $|x|_v$. These values are the non-archimedean contribution to the height of $1/x \in \mathbb{G}_m(\overline{\mathbb{Q}})$. Its minimal polynomial is $\pm(a_d T^d + \cdots + a_1 T + 1)$, with leading coefficient $|a_d|$, so part 1 for $1/x$ gives $\prod_{v \in \mathcal{P}_K^0} \max(1, |x|_v^{-1})^{n_v} = |\operatorname{Nm}_{K/\mathbb{Q}}(x)|$.

EXAMPLE 8.5.11.

(1) Let $x \coloneqq 1 + 2i$, with $f = T^2 - 2T + 5$, so $x \in \overline{\mathbb{Z}}$ and $\operatorname{Nm}_{K/\mathbb{Q}}(x) = 5$. Over $\mathbb{Q}_5$, the Newton polygon joins $(0,1)$, $(1,0)$ and $(2,0)$, with slopes -1 and 0, each of length 1. So 5 has two places above it in $K = \mathbb{Q}(i)$, each with $n_v = 1$. At one, $|x|_v = 5^{-1}$, and at the other x is a unit. At every other prime x is a unit. The conjugates $1 \pm 2i$ have absolute value $\sqrt{5}$, so $H(x)^2 = 5$. The minimal polynomial of $1/x$ is $5T^2 - 2T + 1$, with roots of absolute value $1/\sqrt{5}$. Its Mahler measure is $5 \cdot 1 \cdot 1 = 5$ again, now entirely non-archimedean.

(2) Let $d \geqslant 1$ and $x \coloneqq 2^{1/d}$, the real root of $f = T^d - 2$, irreducible by Eisenstein's criterion. Over $\mathbb{Q}_2$ the Newton polygon is one segment, from $(0,1)$ to $(d,0)$, of slope $-1/d$ and length d. So $|x|_v = 2^{-1/d}$ at every place $v \mid 2$, and $\sum_{v|2} n_v = d$. The d conjugates have absolute value $2^{1/d}$, so $H(x)^d = 2$ and $h(x) = (\log 2)/d$.

(3) Let $x \coloneqq (3 + 4i)/5$. Both conjugates have absolute value 1, but $x \notin \overline{\mathbb{Z}}$: from $x + \bar{x} = 6/5$ and $x\bar{x} = 1$, $f = 5T^2 - 6T + 5$. Over $\mathbb{Q}_5$ the Newton polygon joins $(0,1)$, $(1,0)$ and $(2,1)$, with slopes -1 and 1. So $|x|_v = 5^{-1}$ at one place above 5 and $|x|_v = 5$ at the other, and $H(x)^2 = 5$, all from the leading coefficient.

PROPOSITION 8.5.12. *Let $x, y \in \overline{\mathbb{Q}}^\times$, $n \in \mathbb{Z}$, and K a number field containing x.*

(1) $h(x^n) = |n|\, h(x)$. In particular $h(1/x) = h(x)$.

(2) $h(xy) \leqslant h(x) + h(y)$, and $h(\zeta x) = h(x)$ for every root of unity ζ.

(3) We have

$$h(x) = \frac{1}{2[K:\mathbb{Q}]} \sum_{v \in \mathcal{P}_K} n_v \left|\log |x|_v\right|.$$

(4) For $x_1, \ldots, x_r \in \overline{\mathbb{Q}}$, $h(x_1 + \cdots + x_r) \leqslant \log r + h(x_1) + \cdots + h(x_r)$, with equality for $x_1 = \cdots = x_r = 1$.

PROOF. Let K contain all the elements involved, and $v \in \mathcal{P}_K$. For $n \geqslant 0$, $\max(1, |x^n|_v) = \max(1, |x|_v)^n$, so $h(x^n) = n\, h(x)$. Also $[1/x : 1] = [1 : x]$, and $|[1:x]|_v = |[x:1]|_v$, so $h(1/x) = h(x)$, which gives the case $n < 0$. This proves part 1. For part 2, if $a, b \geqslant 0$, then $\max(1, ab) \leqslant \max(1, a) \max(1, b)$. If $\zeta^n = 1$ with $n \geqslant 1$, then $|\zeta|_v = 1$ for every v, so $\max(1, |\zeta x|_v) = \max(1, |x|_v)$. For part 3, write $\log \max(1, a) = \frac{1}{2}(\log a + |\log a|)$ for $a > 0$, and sum over v: the terms $n_v \log |x|_v$ add up to 0 by part 2 of Proposition 8.5.2.

For part 4, let $\epsilon(v) \coloneqq 1$ if v is archimedean and $\epsilon(v) \coloneqq 0$ otherwise. The triangle and ultrametric inequalities read $|x_1 + \cdots + x_r|_v \leqslant r^{\epsilon(v)} \max_i |x_i|_v$, and $\max_i a_i \leqslant \prod_i \max(1, a_i)$ for $a_i \geqslant 0$. So

$$\max(1, |x_1 + \cdots + x_r|_v) \leqslant r^{\epsilon(v)} \prod_{i=1}^{r} \max(1, |x_i|_v).$$

Raise to the power n_v and multiply over v. The factors $r^{\epsilon(v) n_v}$ multiply to $r^{[K:\mathbb{Q}]}$, by part 1 of Proposition 8.5.2. For $x_i = 1$, both sides are $\log r$ [Sil09, Exercise 8.8]. □

The inequality of part 2 can be strict: $h(2 \cdot \frac{1}{2}) = 0$, while $h(2) + h(\frac{1}{2}) = 2 \log 2$. On the group $\mathbb{G}_{\mathrm{m}}(\overline{\mathbb{Q}})$, part 1 is an exact identity, not an estimate up to $O(1)$ as in Lemma 8.4.4. The height h is linear rather than quadratic, and it is already canonical: in the analogue of Proposition 8.4.9 for a linear height, $2^{-n} h(x^{2^n}) = h(x)$ for every n, with no limit to take. Part 3 is the formula of Remark 8.4.19 for S-units.

8.5.3. Northcott's theorem and Kronecker's theorem. For $P = [x_0 : \cdots : x_N] \in \mathbb{P}^N(\overline{\mathbb{Q}})$, let $\mathbb{Q}(P)$ be the field generated over $\mathbb{Q}$ by the ratios x_i/x_j with $x_j \neq 0$.

THEOREM 8.5.13 (Northcott). *Let $C, D \in \mathbb{R}$. The set of $P \in \mathbb{P}^N(\overline{\mathbb{Q}})$ with $H(P) \leqslant C$ and $[\mathbb{Q}(P) : \mathbb{Q}] \leqslant D$ is finite. In particular, h has the Northcott property on $\mathbb{P}^N(k)$.*

PROOF. Choose coordinates of P with some $x_j = 1$, so that $F \coloneqq \mathbb{Q}(P) = \mathbb{Q}(x_0, \ldots, x_N)$. Then $|P|_v \geqslant \max(1, |x_i|_v)$ for every i and every $v \in \mathcal{P}_F$, so $H(x_i) \leqslant H(P) \leqslant C$, and $[\mathbb{Q}(x_i) : \mathbb{Q}] \leqslant D$. So it suffices to show that the set of $x \in \overline{\mathbb{Q}}$ with $H(x) \leqslant C$ and $e \coloneqq [\mathbb{Q}(x) : \mathbb{Q}] \leqslant D$ is finite. For such an x, let $f = a_0 T^e + \cdots + a_e$ be as in Theorem 8.5.10. By part 2 of that theorem, $a_0 \leqslant H(x)^e$, and

$$|a_i| = a_0 \Big| \sum_{j_1 < \cdots < j_i} \alpha_{j_1} \cdots \alpha_{j_i} \Big| \leqslant \binom{e}{i} a_0 \prod_{j=1}^{e} \max(1, |\alpha_j|) = \binom{e}{i} H(x)^e,$$

since each product of i distinct roots is at most $\prod_j \max(1, |\alpha_j|)$ in absolute value. So the coefficients of f are integers bounded by $2^D \max(1, C)^D$. There are finitely many such f, each with at most D roots. For the last claim, $[\mathbb{Q}(P) : \mathbb{Q}] \leqslant [k : \mathbb{Q}]$ for $P \in \mathbb{P}^N(k)$, and $h(P) \leqslant C$ means $H(P) \leqslant e^C$ [Sil09, Theorem VIII.5.11], [HS00, Theorem B.2.3]. □

REMARK 8.5.14. Silverman bounds the coefficients through a statement for any polynomial $f = a_0(T - \alpha_1) \cdots (T - \alpha_d) \in \overline{\mathbb{Q}}[T]$, not only a minimal polynomial over $\mathbb{Q}$ [Sil09, Theorem VIII.5.9]:

$$2^{-d} \prod_{j=1}^{d} H(\alpha_j) \leqslant H([a_0 : \cdots : a_d]) \leqslant 2^{d-1} \prod_{j=1}^{d} H(\alpha_j).$$

Its proof is the computation in the proof of Theorem 8.5.10, place by place. At a non-archimedean place, Lemma 8.5.7 gives the exact equality $\max_i |a_i|_v = |a_0|_v \prod_j \max(1, |\alpha_j|_v)$. The constants $2^{\pm d}$ come from the archimedean places alone. For the minimal polynomial with coprime integer coefficients, part 2 of Theorem 8.5.10 gives the exact value $H(x)^d = M(f)$, and the bound in the proof above replaces the upper inequality.

THEOREM 8.5.15 (Kronecker). *Let $x \in \overline{\mathbb{Q}}^{\times}$. Then $h(x) = 0$ if and only if x is a root of unity. Equivalently, a non-zero $x \in \overline{\mathbb{Z}}$ whose conjugates all lie in the closed unit disc is a root of unity.*

PROOF. If ζ is a root of unity, then $|\zeta|_v = 1$ for every place, so $h(\zeta) = 0$. Conversely, let $h(x) = 0$, and $K \coloneqq \mathbb{Q}(x)$. By part 1 of Proposition 8.5.12, $h(x^n) = n\, h(x) = 0$ for every $n \geqslant 1$, and so the K-points $[x^n : 1]$ of $\mathbb{P}^1$ have height 1. By Theorem 8.5.13 there are finitely many of them, so $x^r = x^s$ for some $1 \leqslant r < s$, and $x^{s-r} = 1$ since $x \neq 0$. For the second form, the factors a_0 and $\max(1, |\alpha_j|)$ in part 2 of Theorem 8.5.10 are at least 1. So $h(x) = 0$ if and only if $a_0 = 1$ and every $|\alpha_j| \leqslant 1$, that is, by part 4, $x \in \overline{\mathbb{Z}}$ and its conjugates lie in the closed unit disc [Sil09, Exercise 8.6], [HS00, Corollary B.2.3.1]. □

The hypothesis $x \neq 0$ is needed, since $H(0) = H([0:1]) = 1$. So is the integrality: the number $(3+4i)/5$ of part 3 of Example 8.5.11 has both conjugates on the unit circle, and height $\sqrt{5}$. Kronecker's theorem is the analogue for $\mathbb{G}_{\mathrm{m}}$ of part 2 of Corollary 8.4.10: the zeros of the canonical height on $\mathbb{G}_{\mathrm{m}}(\overline{\mathbb{Q}})$ are exactly its torsion. As there, the finiteness of the torsion of $\mathbb{G}_{\mathrm{m}}(k) = k^\times$ follows from the Northcott property: the roots of unity of k are the $x \in k^\times$ with $h(x) = 0$, a finite set by Theorem 8.5.13.

REMARK 8.5.16. Kronecker's own argument needs no Northcott theorem. Let $x \in \overline{\mathbb{Z}}$ be non-zero, with conjugates $\alpha_1, \dots, \alpha_d$ in the closed unit disc. For $n \geqslant 1$, the polynomial $\prod_j (T - \alpha_j^n)$ has integer coefficients, being symmetric in the α_j, and its i-th coefficient is at most $\binom{d}{i}$ in absolute value. So there are finitely many such polynomials, the powers x^n take finitely many values, and x is a root of unity. The same bound makes the case of small degree a finite computation: of the 3159 monic integer polynomials of degree 4 with $|a_i| \leqslant \binom{4}{i}$, Magma finds 24 with non-zero constant term and all roots in the closed unit disc, and each is a product of cyclotomic polynomials.

REMARK 8.5.17. By part 2 of Example 8.5.11, $h(2^{1/d}) = (\log 2)/d$, so h takes arbitrarily small positive values on $\overline{\mathbb{Q}}^\times$, and Kronecker's theorem has no quantitative form with a uniform constant. Lehmer asked whether $[\mathbb{Q}(x) : \mathbb{Q}]\, h(x)$, which is $\log M(f)$ by part 2 of Theorem 8.5.10, is bounded below by a positive constant on the $x \in \overline{\mathbb{Q}}^\times$ that are not roots of unity. The question is open. The smallest value known is that of a root b of Lehmer's polynomial $T^{10} + T^9 - T^7 - T^6 - T^5 - T^4 - T^3 + T + 1$, irreducible and not cyclotomic, with $10\, h(b) = \log 1.17628\ldots \approx 0.1624$.

8.5.4. Galois conjugates, morphisms and embeddings.

PROPOSITION 8.5.18. *Let $P \in \mathbb{P}^N(\overline{\mathbb{Q}})$ and $\sigma \in \mathrm{Gal}_{\mathbb{Q}}$. Then $H(\sigma(P)) = H(P)$.*

PROOF. Let K be a number field with $P \in \mathbb{P}^N(K)$. Then σ restricts to an isomorphism $K \to \sigma(K)$. For $v \in \mathcal{P}_K$, the absolute value $|y|_{\sigma v} := |\sigma^{-1}(y)|_v$ on $\sigma(K)$ restricts to the same absolute value on $\mathbb{Q}$, and σ extends to an isomorphism $K_v \to \sigma(K)_{\sigma v}$, so $n_{\sigma v} = n_v$. So $v \mapsto \sigma v$ is a bijection $\mathcal{P}_K \to \mathcal{P}_{\sigma(K)}$, with $|\sigma(P)|_{\sigma v} = |P|_v$, and $H_{\sigma(K)}(\sigma(P)) = H_K(P)$. As $[\sigma(K) : \mathbb{Q}] = [K : \mathbb{Q}]$, this is the claim [Sil09, Theorem VIII.5.10]. □

For $x \in \overline{\mathbb{Q}}$, this is also visible in part 2 of Theorem 8.5.10: conjugates have the same minimal polynomial. A morphism of degree d from $\mathbb{P}^N$ to $\mathbb{P}^M$ is a map $F = [f_0 : \cdots : f_M]$, with $f_0, \dots, f_M \in \overline{\mathbb{Q}}[X_0, \dots, X_N]$ homogeneous of degree d and without common zero in $\overline{\mathbb{Q}}^{N+1}$ other than 0. It sends $P \in \mathbb{P}^N(\overline{\mathbb{Q}})$ to $F(P) := [f_0(P) : \cdots : f_M(P)]$, which is a point of $\mathbb{P}^M(\overline{\mathbb{Q}})$ since some $f_j(P) \neq 0$. It is a morphism $\mathbb{P}^N_{\overline{\mathbb{Q}}} \to \mathbb{P}^M_{\overline{\mathbb{Q}}}$ in the sense of Section 2.3.

THEOREM 8.5.19. *Let $F\colon \mathbb{P}^N \to \mathbb{P}^M$ be a morphism of degree d. There are constants $C_1, C_2 > 0$, depending on F, such that*

$$C_1 H(P)^d \leqslant H(F(P)) \leqslant C_2 H(P)^d \quad \text{for all } P \in \mathbb{P}^N(\overline{\mathbb{Q}}).$$

That is, $h \circ F = d\, h + O(1)$ on $\mathbb{P}^N(\overline{\mathbb{Q}})$.

PROOF. Let $P = [x_0 : \cdots : x_N] \in \mathbb{P}^N(\overline{\mathbb{Q}})$, and K a number field containing the x_i and the coefficients of the f_j, together with the coefficients of the polynomials g_{ij} below, which do not depend on P. For $v \in \mathcal{P}_K$, let $|F|_v$ be the largest $|c|_v$ over the coefficients c of the f_j, $|F(P)|_v := \max_j |f_j(P)|_v$, and $H(F)$ the height of the K-point of projective space whose coordinates are all these coefficients. Let $\epsilon(v)$ be as in the proof of Proposition 8.5.12. Each f_j has at most $c_1 := \binom{N+d}{N}$ monomials, so $|F(P)|_v \leqslant c_1^{\epsilon(v)} |F|_v |P|_v^d$. Raise to the power n_v, multiply over v, and take the $[K : \mathbb{Q}]$-th root: as in the proof of part 4 of Proposition 8.5.12, $H(F(P)) \leqslant c_1 H(F) H(P)^d$.

For the lower bound, the X_i vanish at the only common zero 0 of the f_j in $\overline{\mathbb{Q}}^{N+1}$. By the Nullstellensatz [Har77, Theorem I.1.3A], there are $e \geqslant 1$ and polynomials $g_{ij} \in \overline{\mathbb{Q}}[X_0, \dots, X_N]$ with $X_i^e = \sum_{j=0}^M g_{ij} f_j$ for

every i. Keeping the terms of degree e, we may assume that each g_{ij} is homogeneous of degree $e-d$. Let $|G|_v$ and $H(G)$ be defined from the coefficients of the g_{ij} as above, and $c_2 \coloneqq (M+1)\binom{N+e-d}{N}$. Then

$$|x_i|_v^e \leqslant (M+1)^{\epsilon(v)} \max_j |g_{ij}(P)|_v |f_j(P)|_v \leqslant c_2^{\epsilon(v)} |G|_v |P|_v^{e-d} |F(P)|_v.$$

Take the maximum over i, and divide by $|P|_v^{e-d}$:

$$|P|_v^d \leqslant c_2^{\epsilon(v)} |G|_v |F(P)|_v.$$

As before, $H(P)^d \leqslant c_2 H(G) H(F(P))$ [Sil09, Theorem VIII.5.6]. □

The constant C_1 is hard to make explicit, since it depends on the polynomials g_{ij} given by the Nullstellensatz [Sil09, Remark VIII.5.7]. For $d=1$, a matrix $A \in \mathrm{GL}_{N+1}(\overline{\mathbb{Q}})$ acting on $\mathbb{P}^N$ satisfies $h(AP) = h(P) + O(1)$ [Sil09, Corollary VIII.5.8]. The upper bound uses only the triangle inequality, but the lower bound fails without the hypothesis on common zeros [Sil09, Exercise 8.10]. For instance, $f_0 = X_0^2$ and $f_1 = X_0 X_1$ have the common zero $[0:1]$, and $[x^2 : x] = [x : 1]$ for $x \neq 0$, so the height of the image is $H(x)$, not of the order of $H(x)^2$.

The following two maps satisfy exact identities, with no constants.

Definition 8.5.20. Let $N, M \geqslant 0$ and $d \geqslant 1$.

(1) The Segre embedding is the map $S_{N,M} \colon \mathbb{P}^N \times \mathbb{P}^M \to \mathbb{P}^{NM+N+M}$,

$$S_{N,M}([x_0 : \cdots : x_N], [y_0 : \cdots : y_M]) \coloneqq [x_0 y_0 : x_0 y_1 : \cdots : x_i y_j : \cdots : x_N y_M],$$

with the $(N+1)(M+1)$ products in lexicographic order of (i,j) [Har77, Exercise I.2.14].

(2) Let $N_d \coloneqq \binom{N+d}{N} - 1$, and $m_0, \ldots, m_{N_d}$ the monomials of degree d in $X_0, \ldots, X_N$. The d-uple embedding is the morphism of degree d $\nu_d \colon \mathbb{P}^N \to \mathbb{P}^{N_d}$, $\nu_d(P) \coloneqq [m_0(P) : \cdots : m_{N_d}(P)]$ [Har77, Exercise I.2.12].

Both are well defined on $\overline{\mathbb{Q}}$-points: if some x_i and some y_j are non-zero, then so is $x_i y_j$, and the monomials include the X_i^d. Rescaling the coordinates of P and Q by λ and μ rescales $S_{N,M}(P,Q)$ by $\lambda\mu$, and $\nu_d(P)$ by λ^d.

Proposition 8.5.21. *Let $P = [x_0 : \cdots : x_N] \in \mathbb{P}^N(\overline{\mathbb{Q}})$, $Q \in \mathbb{P}^M(\overline{\mathbb{Q}})$, and $d \geqslant 1$.*

(1) $H(S_{N,M}(P,Q)) = H(P)\,H(Q)$.

(2) $H(\nu_d(P)) = H(P)^d = H([x_0^d : \cdots : x_N^d])$.

Proof. Let K be a number field containing the coordinates of P and Q, and $v \in \mathcal{P}_K$. The largest $|x_i y_j|_v = |x_i|_v |y_j|_v$ is attained at a pair (i,j) where both factors are largest, so $|S_{N,M}(P,Q)|_v = |P|_v |Q|_v$. Each monomial $m_l(P)$ is a product of d coordinates of P, so $|m_l(P)|_v \leqslant |P|_v^d$, with equality for $m_l = X_i^d$ when $|x_i|_v = |P|_v$. So $|\nu_d(P)|_v = |P|_v^d = \max_i |x_i^d|_v$. In each case, raise to the power n_v, multiply over v, and take the $[K:\mathbb{Q}]$-th root [HS00, Proposition B.2.4]. □

No triangle inequality enters, so these are equalities, while Theorem 8.5.19 only holds up to the constants C_1 and C_2. For $N=1$ and $P=[x:1]$, part 2 is the identity $H(x^d) = H(x)^d$ of part 1 of Proposition 8.5.12, and in logarithmic form part 1 reads $h(S_{N,M}(P,Q)) = h(P) + h(Q)$.

CHAPTER 9

Integral points

Sources. This chapter follows [Sil09, Chapter IX]. For Diophantine approximation see also [HS00, Part D].

SITUATION 9.0.1.

- Let k be a number field, with set of places $\mathcal{P}_k$ and completions k_v. As in Situation 8.5.1, for $v \in \mathcal{P}_k$ the absolute value $|\ |_v$ on k_v restricts to the usual absolute value on $\mathbb{R}$ or to the p-adic absolute value with $|p|_p = p^{-1}$, and $n_v := [k_v : \mathbb{Q}_v]$ is the local degree.
- We write $\overline{k} = \overline{\mathbb{Q}}$, and for each $v \in \mathcal{P}_k$ we fix a k-embedding $\overline{k} \to \overline{k_v}$, as in Situation 8.0.1. The absolute value $|\ |_v$ extends uniquely to $\overline{k_v}$ (for a non-archimedean v, by Remark 6.3.5, and for an archimedean v, $\overline{k_v} = \mathbb{C}$). Through the embedding, it defines $|\alpha|_v$ for $\alpha \in \overline{k}$.
- The height relative to k is H_k, as in Definition 8.5.3, and $H_k(x) := H_k([x : 1])$ for $x \in k$.

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9.1. Diophantine approximation

Diophantine approximation asks how well an irrational number can be approximated by rational numbers. For every rational number p/q the distance $|p/q - \sqrt{2}|$ is positive, and since $\mathbb{Q}$ is dense in $\mathbb{R}$, a good choice of p/q makes it as small as we like. The question is how small it can be made without making p and q large. Two elementary results set the scale. Throughout, a rational number p/q is written with $p, q \in \mathbb{Z}$ and $q \geqslant 1$.

PROPOSITION 9.1.1 (Dirichlet). *Let $\alpha \in \mathbb{R} \setminus \mathbb{Q}$. There are infinitely many rational numbers p/q, in lowest terms, with*

$$\left|\frac{p}{q} - \alpha\right| < \frac{1}{q^2}. \tag{9.1.1}$$

PROOF. Let $N \geqslant 1$ be an integer. For $t \in \mathbb{R}$, let $\{t\} := t - \lfloor t \rfloor \in [0, 1)$ be its fractional part. As α is irrational, the $N + 1$ numbers $\{q\alpha\}$ with $0 \leqslant q \leqslant N$ are distinct. The N intervals $[j/N, (j+1)/N)$, for $0 \leqslant j < N$, cover $[0, 1)$, so by the pigeonhole principle one of them contains $\{q_1\alpha\}$ and $\{q_2\alpha\}$ for some $0 \leqslant q_1 < q_2 \leqslant N$. Let $q := q_2 - q_1$ and $p := \lfloor q_2\alpha \rfloor - \lfloor q_1\alpha \rfloor$. Then $1 \leqslant q \leqslant N$ and $|q\alpha - p| = |\{q_2\alpha\} - \{q_1\alpha\}| < 1/N$, so

$$\left|\frac{p}{q} - \alpha\right| < \frac{1}{qN} \leqslant \frac{1}{q^2}.$$

Writing $p/q = p'/q'$ in lowest terms, $q' \leqslant q$, so p'/q' satisfies (9.1.1) too.

So there is at least one solution. Suppose there were only finitely many, $p_1/q_1, \ldots, p_r/q_r$. As α is irrational, $\delta := \min_i |p_i/q_i - \alpha| > 0$. The construction with $N > 1/\delta$ gives a rational number p/q satisfying (9.1.1) with $|p/q - \alpha| < 1/(qN) \leqslant 1/N < \delta$, so it is none of the p_i/q_i, a contradiction [Sil09, Proposition IX.1.2]. □

Proposition 9.1.1 has a picture. Around each rational number p/q in lowest terms, draw the disk in $\mathbb{C}$ of radius $1/q^2$. An irrational real number lies in infinitely many of these disks. A rational number lies in only finitely many. Indeed, let a/b be in lowest terms. For $p/q \neq a/b$,

$$\left|\frac{p}{q} - \frac{a}{b}\right| = \frac{|pb - aq|}{bq} \geqslant \frac{1}{bq},$$

and this is at least $1/q^2$ as soon as $q \geqslant b$. So a/b lies in its own disk and in disks with $q < b$ only. In Figure 9.1.1 the disks are translucent, so the shade darkens where many of them overlap. The cusps that point at the integers are made of the disks around $n \pm 1/q$: they come within $1/q - 1/q^2$ of n, and never reach it.

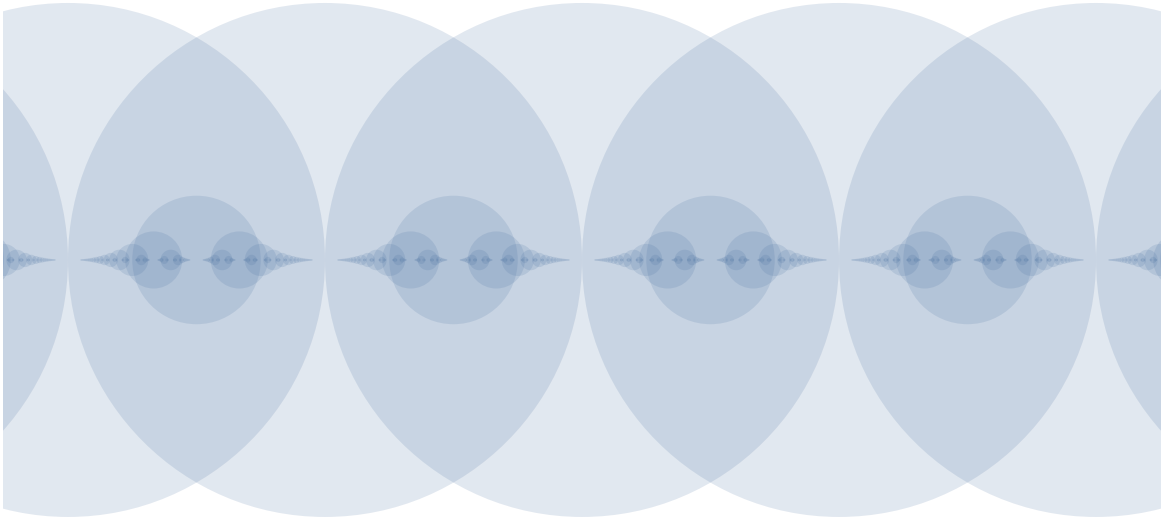


FIGURE 9.1.1. The disks of radius $1/q^2$ around the rational numbers p/q in lowest terms with $q \leqslant 20$, drawn translucent. Without the bound on q, every irrational real number lies in infinitely many of the disks (Proposition 9.1.1), and every rational number in finitely many. After a picture in [Sta23, at 11:02].

REMARK 9.1.2. Hurwitz showed that $1/q^2$ in (9.1.1) can be replaced by $1/(\sqrt{5}\,q^2)$, and that $\sqrt{5}$ cannot be replaced by any larger constant [Sil09, Remark IX.1.2.1].

Dirichlet's theorem says how well every irrational number can be approximated. For algebraic numbers there is a bound in the other direction.

PROPOSITION 9.1.3 (Liouville). *Let $\alpha \in \mathbb{C}$ be algebraic over $\mathbb{Q}$ of degree $d \geqslant 2$. There is a constant $c > 0$, depending on α, such that every rational number p/q satisfies*

$$\left|\frac{p}{q} - \alpha\right| \geqslant \frac{c}{q^d}. \tag{9.1.2}$$

PROOF. If $\alpha \notin \mathbb{R}$, then $|p/q - \alpha| \geqslant |\operatorname{Im}\alpha| \geqslant |\operatorname{Im}\alpha|/q^d$. Suppose that $\alpha \in \mathbb{R}$. Let $f = a_0T^d + a_1T^{d-1} + \cdots + a_d \in \mathbb{Z}[T]$ be a minimal polynomial of α, and let

$$c_1 \coloneqq \sup\{|f'(t)| : \alpha - 1 \leqslant t \leqslant \alpha + 1\}.$$

Here $c_1 > 0$, since f' is a nonzero polynomial. If $|p/q - \alpha| \geqslant 1$, then (9.1.2) holds with $c = 1$. Otherwise, by the mean value theorem,

$$\left|f\left(\frac{p}{q}\right)\right| = \left|f\left(\frac{p}{q}\right) - f(\alpha)\right| \leqslant c_1\left|\frac{p}{q} - \alpha\right|.$$

The number $q^d f(p/q) = a_0p^d + a_1p^{d-1}q + \cdots + a_dq^d$ is an integer. It is nonzero, since f is irreducible of degree $d \geqslant 2$ and so has no rational root. So $|f(p/q)| \geqslant q^{-d}$, and (9.1.2) holds with $c = 1/c_1$. So for a real α, $c \coloneqq \min(1, 1/c_1)$ works [Sil09, Proposition IX.1.3]. □

REMARK 9.1.4. The constant c in Proposition 9.1.3 is explicit in terms of α. Liouville used his theorem to write down the first transcendental numbers. Let $\alpha \coloneqq \sum_{n\geqslant 1} 10^{-n!}$, and for $N \geqslant 1$ let p_N/q_N be the partial

sum up to $n = N$, with $q_N := 10^{N!}$. Then

$$0 < \alpha - \frac{p_N}{q_N} < \frac{2}{10^{(N+1)!}} = \frac{2}{q_N^{N+1}}.$$

For every $d \geqslant 2$ and $c > 0$, this contradicts (9.1.2) once N is large. The decimal expansion of α is not eventually periodic, so α is not rational either. So α is transcendental [Sil09, Remark IX.1.3.1 and Exercise 9.2].

9.1.1. The approximation exponent. Dirichlet's theorem approximates every real number to within $1/q^2$, and Liouville's theorem says that an algebraic number of degree d cannot be approximated to within c/q^d. For quadratic irrationals the two exponents agree. For $d \geqslant 3$ we may ask for the best exponent. There is no reason to restrict the approximations to $\mathbb{Q}$, or to the usual absolute value, so we let them range over a number field k and measure the distance at a place of k. The size of q becomes the height $H_k(x)$.

Definition 9.1.5. Let $v \in \mathcal{P}_k$, and let $\tau\colon \mathbb{Z}_{\geqslant 1} \to \mathbb{R}_{>0}$ be a function. We say that k has approximation exponent τ at v if the following holds. For every $\alpha \in \overline{k}$, with $d := [k(\alpha) : k]$, and every real number $C > 0$, the set of $x \in k$ with

$$|x - \alpha|_v < C\, H_k(x)^{-\tau(d)} \tag{9.1.3}$$

is finite.

Remark 9.1.6.

(1) Equivalently, for every $\alpha \in \overline{k}$ and every $C > 0$, all but finitely many $x \in k$ satisfy

$$|x - \alpha|_v \geqslant C\, H_k(x)^{-\tau(d)}.$$

This is a lower bound for the distance from α to the elements of k, like Liouville's (9.1.2): it says that α cannot be approximated too well. A smaller τ is a stronger statement. If $\tau \leqslant \tau'$ and k has approximation exponent τ at v, then it also has approximation exponent τ' at v, since $H_k(x) \geqslant 1$ gives $H_k(x)^{-\tau'(d)} \leqslant H_k(x)^{-\tau(d)}$.

(2) The definition does not depend on the embedding $\overline{k} \to \overline{k_v}$ of Situation 9.0.1. Two k-embeddings ι, ι' have the same image, the algebraic closure of k in $\overline{k_v}$, so $\iota' = \iota \circ \sigma$ for some $\sigma \in \mathrm{Gal}_k$. Then $|\iota'(x - \alpha)| = |\iota(x - \sigma(\alpha))|$ for $x \in k$, and $\sigma(\alpha)$ has the same degree as α. Since α ranges over all of $\overline{k}$, the definition covers every extension of $|\ |_v$ to $k(\alpha)$. This is how [Sil09, Section IX.1] phrases it.

(3) For $k = \mathbb{Q}$ and $x = p/q$ in lowest terms, $[x : 1] = [p : q]$ has $|[p : q]|_\ell = 1$ at every prime ℓ, so $H_{\mathbb{Q}}(x) = \max(|p|, q)$.

Let ∞ be the archimedean place of $\mathbb{Q}$. In these terms, Liouville's theorem gives an approximation exponent of $\mathbb{Q}$ at ∞.

Corollary 9.1.7. *For every $\varepsilon > 0$, $\mathbb{Q}$ has approximation exponent $\tau(d) = d + \varepsilon$ at ∞.*

Proof. Let $\alpha \in \overline{\mathbb{Q}}$, with $d := [\mathbb{Q}(\alpha) : \mathbb{Q}]$, and let $C > 0$. Let $x = p/q$ in lowest terms satisfy (9.1.3). By part 3 of Remark 9.1.6, $H_{\mathbb{Q}}(x) = \max(|p|, q) \geqslant q$, so

$$|x - \alpha| < C\, H_{\mathbb{Q}}(x)^{-d-\varepsilon} \leqslant \frac{C}{q^{d+\varepsilon}}.$$

If $d \geqslant 2$, let c be the constant of Proposition 9.1.3. Then $c/q^d < C/q^{d+\varepsilon}$, that is, $q^\varepsilon < C/c$. If $d = 1$, then $\alpha = a/b$ in lowest terms, and for $x \neq \alpha$, $|x - \alpha| \geqslant 1/(bq)$ as in the discussion of Figure 9.1.1, so $q^\varepsilon < bC$. In both cases q is bounded, apart from $x = \alpha$ when $d = 1$. Then $|p| \leqslant q(|\alpha| + C)$ is bounded too, so there are finitely many such x. □

Remark 9.1.8. The ε in Corollary 9.1.7 cannot be removed for $d = 2$. More generally, Dirichlet's theorem shows that $\mathbb{Q}$ has no approximation exponent τ at ∞ with $\tau(d) \leqslant 2$ for some $d \geqslant 2$. Let $\alpha \in \mathbb{R}$ be algebraic of degree d, for example $\alpha = \sqrt[d]{2}$. By Proposition 9.1.1 there are infinitely many $x = p/q$ in lowest terms

with $|x-\alpha| < 1/q^2$. For them $|p| < q(|\alpha|+1)$, so $H_{\mathbb{Q}}(x) < (|\alpha|+1)q$, and

$$|x-\alpha| < \frac{1}{q^2} < \frac{(|\alpha|+1)^2}{H_{\mathbb{Q}}(x)^2} \leqslant (|\alpha|+1)^2 H_{\mathbb{Q}}(x)^{-\tau(d)}.$$

So (9.1.3) with $C = (|\alpha|+1)^2$ has infinitely many solutions.

REMARK 9.1.9. Corollary 9.1.7 also holds at every prime p of $\mathbb{Q}$, with the product formula in place of the mean value theorem. Let $\alpha \in \overline{\mathbb{Q}}$ have degree d, and let $f = a_0 T^d + \cdots + a_d \in \mathbb{Z}[T]$ be a minimal polynomial of α with coprime coefficients. Let $x = a/b \in \mathbb{Q}$ in lowest terms, and let $N \coloneqq b^d f(x) = \sum_i a_i a^{d-i} b^i$. This is an integer, and it is nonzero unless $x = \alpha$, since f is irreducible. Let $x \neq \alpha$, and let $A \coloneqq \sum_i |a_i|$. Then $|N| \leqslant A\, H_{\mathbb{Q}}(x)^d$. Since $|N|_\ell \leqslant 1$ for every prime ℓ, the product formula gives

$$|N|_p \geqslant \frac{1}{|N|} \geqslant \frac{1}{A\, H_{\mathbb{Q}}(x)^d}.$$

Now suppose that $|x-\alpha|_p \leqslant 1$, so that $|x|_p \leqslant B \coloneqq \max(1, |\alpha|_p)$. Write $f(X) - f(Y) = (X-Y)g(X,Y)$ with $g \in \mathbb{Z}[X,Y]$ of degree $d-1$. As $|\ |_p$ is non-archimedean, $|g(x,\alpha)|_p \leqslant B^{d-1}$, and

$$|N|_p \leqslant |f(x)|_p = |f(x) - f(\alpha)|_p \leqslant B^{d-1} |x-\alpha|_p.$$

So $|x-\alpha|_p \geqslant c\, H_{\mathbb{Q}}(x)^{-d}$ with $c \coloneqq 1/(AB^{d-1})$. Let $C > 0$ and $\varepsilon > 0$, and let x satisfy (9.1.3) with $\tau(d) = d + \varepsilon$. If $|x-\alpha|_p > 1$, then $H_{\mathbb{Q}}(x)^{d+\varepsilon} < C$. Otherwise $H_{\mathbb{Q}}(x)^{\varepsilon} < C/c$. In both cases the height of x is bounded, and by part 3 of Remark 9.1.6 there are finitely many such x.

Liouville's exponent was improved several times [Sil09, Section IX.1].

Year	Approximation exponent	Due to
1851	$\tau(d) = d + \varepsilon$	Liouville
1909	$\tau(d) = d/2 + 1 + \varepsilon$	Thue
1921	$\tau(d) = 2\sqrt{d} + \varepsilon$	Siegel
1947	$\tau(d) = \sqrt{2d} + \varepsilon$	Gelfond, Dyson
1955	$\tau(d) = 2 + \varepsilon$	Roth

By Remark 9.1.8, Roth's exponent is essentially the best possible. It is conjectured that ε can be replaced by some $\varepsilon(d)$ with $\varepsilon(d) \to 0$ as $d \to \infty$. Mahler handled several absolute values at once, and Schmidt the simultaneous approximation of several irrational numbers [Sil09, Section IX.1].

THEOREM 9.1.10 (Roth). *Let $v \in \mathcal{P}_k$ and $\varepsilon > 0$. Then k has approximation exponent $\tau(d) = 2 + \varepsilon$ at v.*

PROOF. Not proved in these notes. The ideas are elementary, but the details are long. See [Sil09, Theorem IX.1.4 and Section IX.8] for a sketch, and [HS00, Part D] for a proof. □

Theorems on Diophantine approximation give finiteness results for Diophantine equations.

EXAMPLE 9.1.11. Let $a \in \mathbb{Z}$. We show that the equation $x^3 - 2y^3 = a$ has finitely many solutions $(x,y) \in \mathbb{Z}^2$. If $y = 0$, then $x^3 = a$, which has at most one solution. Let $\beta \coloneqq \sqrt[3]{2} \in \mathbb{R}$, and let $\zeta \in \mathbb{C}$ be a primitive cube root of 1. For a solution with $y \neq 0$,

$$\left(\frac{x}{y} - \beta\right)\left(\frac{x}{y} - \zeta\beta\right)\left(\frac{x}{y} - \zeta^2\beta\right) = \frac{a}{y^3}.$$

For $t \in \mathbb{R}$, $|t - \zeta\beta|$ and $|t - \zeta^2\beta|$ are at least $|\mathrm{Im}(\zeta\beta)| = (\sqrt{3}/2)\beta$. So

$$\left|\frac{x}{y} - \beta\right| \leqslant \frac{c_a}{|y|^3}, \qquad c_a \coloneqq \frac{4|a|}{3\beta^2}.$$

If $|x/y - \beta| \geqslant 1$, then $|y|^3 \leqslant c_a$, and each of these finitely many y has at most one x. Otherwise $|x| < (\beta+1)|y|$. Since x/y need not be in lowest terms, part 3 of Remark 9.1.6 gives only $H_{\mathbb{Q}}(x/y) \leqslant \max(|x|,|y|) < (\beta+1)|y|$, but this suffices:

$$\left|\frac{x}{y} - \beta\right| \leqslant c_a(\beta+1)^3 H_{\mathbb{Q}}(x/y)^{-3} < C\, H_{\mathbb{Q}}(x/y)^{-5/2}, \qquad C \coloneqq c_a(\beta+1)^3 + 1.$$

As β has degree 3 over $\mathbb{Q}$, Theorem 9.1.10 with $\varepsilon = 1/2$ leaves finitely many values for $r := x/y$. Thue's exponent $\tau(3) = 5/2 + \varepsilon$ would do as well. Each value of r comes from at most one solution: $y^3(r^3 - 2) = a$ with $r^3 \neq 2$ determines y, and then $x = ry$. This is [Sil09, Example IX.1.5], with the details filled in.

REMARK 9.1.12. Theorem 9.1.10 says that a set is finite, but its proof gives no procedure to find the elements of the set. It is ineffective. So Example 9.1.11 gives no bound on $|x|$ and $|y|$ in terms of a, and the finiteness results that rely on Roth's theorem are ineffective too. Liouville's theorem, with its explicit constant, is effective, but too weak for these applications. Effective bounds come from a different method, lower bounds for linear forms in logarithms [Sil09, Remark IX.1.6 and Section IX.5].

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9.2. Distance functions and local heights

A Diophantine inequality $|x - \alpha|_v < C\, H_k(x)^{-\tau(d)}$ has two parts. The height $H_k(x)$ measures the arithmetic size of x, as in Chapter 8. The quantity $|x - \alpha|_v$ measures how close x is to α in the v-adic topology. This section defines a v-adic distance on a curve, following [Sil09, Section IX.2], compares it with an honest metric, and restates Theorem 9.1.10 in its terms. The distance turns out to be one place's share of a height: summed over all places, the distances to the poles of a function give its height. This is the theory of local heights.

SITUATION 9.2.1. We keep Situation 9.0.1.

- Fix a place $v \in \mathcal{P}_k$ and a nice curve C over k. Through $k \subseteq k_v$, $C(k) \subseteq C(k_v)$, and through the embedding $\overline{k} \to \overline{k_v}$, $C(\overline{k}) \subseteq C(\overline{k_v})$.
- The set $C(k_v)$ carries its v-adic topology (Lemma 9.2.4). The notation $P \to_v Q$ appears only below a limit, as in $\lim_{P \to_v Q}$ or $\liminf_{P \in C(k),\ P \to_v Q}$. There it means that $P \neq Q$ ranges over $C(k_v)$, or over the subset named below the limit, and tends to Q in this topology. In the text we say that P is v-adically close to Q.
- For functions f, g on a set X with values in $[0, \infty)$, we write $f \asymp g$ if there is a constant $c \geqslant 1$ with $c^{-1} g \leqslant f \leqslant c\, g$ on X.

9.2.1. The chordal distance. The honest metric comes from projective space. For a point $P = [x_0 : \cdots : x_N]$ of $\mathbb{P}^N(\overline{k_v})$ we write $\|P\|_v$ for the norm of its coordinates: $\max_i |x_i|_v$ if v is non-archimedean, written $|P|_v$ in Situation 8.5.1, and $\bigl(\sum_i |x_i|_v^2\bigr)^{1/2}$ if v is archimedean. For a second point $P' = [x'_0 : \cdots : x'_N]$ of $\mathbb{P}^N(\overline{k_v})$, we write $\|P \wedge P'\|_v$ for $\max_{i<j} |x_i x'_j - x_j x'_i|_v$ if v is non-archimedean, and for $\bigl(\sum_{i<j} |x_i x'_j - x_j x'_i|_v^2\bigr)^{1/2}$ if v is archimedean. Both depend on the choice of coordinates.

DEFINITION 9.2.2. The chordal distance between $P_1, P_2 \in \mathbb{P}^N(\overline{k_v})$ is

$$\delta_v(P_1, P_2) := \frac{\|P_1 \wedge P_2\|_v}{\|P_1\|_v \|P_2\|_v}.$$

Rescaling the coordinates of P_1 by λ multiplies the numerator and the denominator by $|\lambda|_v$, so δ_v does not depend on the coordinates.

PROPOSITION 9.2.3. *The chordal distance δ_v is a metric on $\mathbb{P}^N(\overline{k_v})$, with values in $[0,1]$. If v is non-archimedean, it is an ultrametric: $\delta_v(P_1, P_3) \leqslant \max(\delta_v(P_1, P_2), \delta_v(P_2, P_3))$.*

PROOF. The minors $x_i x'_j - x_j x'_i$ all vanish if and only if the coordinate vectors are proportional, so $\delta_v(P_1, P_2) = 0$ if and only if $P_1 = P_2$. Symmetry is clear.

Let v be non-archimedean. Choose coordinates x, y, w of P_1, P_2, P_3 of norm 1, by dividing by a coordinate of largest absolute value. Then each minor has absolute value at most 1, so $\delta_v \leqslant 1$. Choose l with $|y_l|_v = 1$. Expanding the right-hand side shows that

$$y_l(x_i w_j - x_j w_i) = (x_i y_l - x_l y_i) w_j - (x_j y_l - x_l y_j) w_i + x_l (y_i w_j - y_j w_i).$$

Since $|w_i|_v, |w_j|_v, |x_l|_v \leqslant 1$, this gives $|x_i w_j - x_j w_i|_v \leqslant \max(\|P_1 \wedge P_2\|_v, \|P_2 \wedge P_3\|_v)$, which is the ultrametric inequality.

Let v be archimedean, so that $\overline{k_v} = \mathbb{C}$. Choose coordinates $p, q \in \mathbb{C}^{N+1}$ of P_1, P_2 of norm 1, and let $\langle p, q\rangle := \sum_i p_i \bar{q}_i$. Lagrange's identity $\|p\|^2\|q\|^2 - |\langle p, q\rangle|^2 = \sum_{i<j} |p_i q_j - p_j q_i|^2$ gives $\delta_v(P_1, P_2)^2 = 1 - |\langle p, q\rangle|^2 \leqslant 1$. The Hermitian matrix $\pi_{P_1} := p\bar{p}^{\mathsf{T}}$, the orthogonal projection onto the line P_1, depends only on P_1, and for the Hilbert–Schmidt norm $\|A\|^2 := \operatorname{tr}(A\bar{A}^{\mathsf{T}})$,

$$\|\pi_{P_1} - \pi_{P_2}\|^2 = 2 - 2|\langle p, q\rangle|^2 = 2\,\delta_v(P_1, P_2)^2.$$

So $\sqrt{2}\,\delta_v$ is the distance of a norm, restricted to the matrices π_P, and satisfies the triangle inequality. □

For $N = 1$ and v archimedean, δ_v is the length of the chord between the images of P_1 and P_2 on the Riemann sphere of diameter 1, whence the name.

Lemma 9.2.4.

(1) *Let $Q = [1 : a_1 : \cdots : a_N] \in \mathbb{P}^N(k_v)$. There is a constant $c \geqslant 1$ such that every $P = [1 : x_1 : \cdots : x_N] \in \mathbb{P}^N(k_v)$ with $|x_j - a_j|_v \leqslant 1$ for all j satisfies*

$$c^{-1} \max_j |x_j - a_j|_v \leqslant \delta_v(P, Q) \leqslant c \max_j |x_j - a_j|_v. \tag{9.2.1}$$

So δ_v induces the topology of k_v^N on each standard affine chart, and $\mathbb{P}^N(k_v)$ is compact for it.

(2) *Let $C \subseteq \mathbb{P}^N$ be a closed embedding. Then $C(k_v)$ is a compact subset of $\mathbb{P}^N(k_v)$. Its topology does not depend on the embedding, and every $f \in k_v(C)$ defines a continuous map $C(k_v) \to \mathbb{P}^1(k_v)$.*

(3) *Let $Q \in C(k_v)$, and let $u \in k_v(C)$ be a uniformizer at Q. Then u maps an open neighborhood of Q in $C(k_v)$ homeomorphically onto an open neighborhood of 0 in k_v. In particular, Q is not an isolated point of $C(k_v)$.*

(4) *The set $C(k_v)$ is closed in $C(\overline{k_v})$ for δ_v.*

Proof. For part 1, the set of coordinate vectors of norm 1 is closed and bounded in k_v^{N+1}, hence compact, since k_v is locally compact. It maps onto $\mathbb{P}^N(k_v)$, continuously since $\|x \wedge y\|_v = \|x \wedge (y - x)\|_v \leqslant \|y - x\|_v$ for vectors of norm 1. So $\mathbb{P}^N(k_v)$ is compact. For P as in the statement, the norms $\|P\|_v$ lie between 1 and a constant depending on Q. The minors of the pair (P, Q) are $a_j - x_j$, and $x_i a_j - x_j a_i = a_j(x_i - a_i) - a_i(x_j - a_j)$. So $\|P \wedge Q\|_v$ is at least $\max_j |x_j - a_j|_v$ and at most a constant times it, which is (9.2.1). Let U be the set of $P = [1 : x_1 : \cdots : x_N]$ with $|x_j - a_j|_v < 1$ for all j. Its complement is the image of the closed set of coordinate vectors x of norm 1 with $|x_j - a_j x_0|_v \geqslant |x_0|_v$ for some j. So it is compact, it does not contain Q, and $\delta_v(\cdot, Q)$ is bounded below on it by a positive constant. Hence small δ_v-balls around Q lie in U, where (9.2.1) compares them with the polydiscs of k_v^N around $(a_1, \ldots, a_N)$. After permuting the coordinates, this holds for every Q.

So δ_v defines the analytic topology of [Poo17, Section 2.6.2] on $\mathbb{P}^N(k_v)$. Part 2 follows: $C(k_v)$ is closed, as the common zero set of polynomials, the analytic topology of $C(k_v)$ does not depend on the embedding, and morphisms are continuous for it [Poo17, Section 2.6.2]. By Proposition 3.2.1, f is a morphism $C_{k_v} \to \mathbb{P}^1$.

For part 3, the uniformizer u is regular on an open neighborhood U of Q, and $u\colon U \to \mathbb{A}^1$ is étale at Q, since $\operatorname{ord}_Q(u) = 1$ and the residue field of Q is k_v. Shrinking U, it is étale on U, and by [Poo17, Proposition 3.5.73] it is a local homeomorphism $U(k_v) \to k_v$. As k_v has no isolated points, neither does $C(k_v)$.

For part 4, $\mathbb{P}^N(k_v)$ is compact, hence closed in the metric space $\mathbb{P}^N(\overline{k_v})$, and $C(k_v) = C(\overline{k_v}) \cap \mathbb{P}^N(k_v)$. □

9.2.2. The v-adic distance on a curve. The chordal distance depends on an embedding into projective space. Silverman's distance uses a single function instead. By Riemann–Roch, a nice curve C of genus g over k_v has, for every $Q \in C(k_v)$ and $e \geqslant \max(2g, 1)$, a function $f \in k_v(C)$ with a pole of order e at Q and no other poles. Indeed, part 2 of Corollary 3.5.5 gives $\ell(eQ) - \ell((e-1)Q) = 1$. Then $t_Q := 1/f$ has a zero of order e at Q and no other zeros in $C(\overline{k_v})$ [Sil09, footnote in Section IX.2].

Definition 9.2.5. Let C be a nice curve over k_v, let $Q \in C(k_v)$, and let $t_Q \in k_v(C)$ have a zero of order $e \geqslant 1$ at Q and no other zeros in $C(\overline{k_v})$. The v-adic distance from $P \in C(k_v)$ to Q is

$$d_v(P, Q) := \min\bigl(|t_Q(P)|_v^{1/e}, 1\bigr),$$

with $d_v(P, Q) := 1$ if P is a pole of t_Q.

For a nice curve over k, we apply the definition to C_{k_v}. By part 2 of Lemma 9.2.4, $d_v(\cdot, Q)$ is continuous on $C(k_v)$, with values in $[0,1]$, and it vanishes only at Q. It depends on the choice of t_Q, but only up to bounded factors.

LEMMA 9.2.6. *Let d_v and d'_v be defined from two choices t_Q and t'_Q, with orders e and e' at Q. Then $d_v(\cdot, Q) \asymp d'_v(\cdot, Q)$ on $C(k_v)$.*

PROOF. The function $g := t'^{e}_Q / t^{e'}_Q$ has neither a zero nor a pole at Q. So there is an open neighborhood U of Q on which $|g|_v$ is bounded above and below by positive constants, and $|t_Q|_v, |t'_Q|_v < 1$. There $d'_v(P,Q)/d_v(P,Q) = |g(P)|_v^{1/(ee')}$ for $P \neq Q$. On the compact set $C(k_v) \setminus U$, both functions are continuous with values in $(0,1]$, so both are bounded below by a positive constant. □

The class of $d_v(\cdot, Q)$ under $\asymp$ contains an honest metric.

THEOREM 9.2.7. *Let $C \subseteq \mathbb{P}^N$ be a closed embedding of a nice curve over k_v, and let $Q \in C(k_v)$. Then $d_v(\cdot, Q) \asymp \delta_v(\cdot, Q)$ on $C(k_v)$.*

PROOF. Permuting the coordinates, write $Q = [1 : a_1 : \cdots : a_N]$ with $a_j \in k_v$, and let $x_j := X_j/X_0 \in k_v(C)$. The local ring $\mathcal{O}_{C,Q}$ is a quotient of the local ring of $\mathbb{P}^N_{k_v}$ at Q, whose maximal ideal is generated by the $x_j - a_j$. So the $x_j - a_j$ generate the maximal ideal of the discrete valuation ring $\mathcal{O}_{C,Q}$, and one of them, say $u := x_1 - a_1$, is a uniformizer. Then $x_j - a_j = u g_j$ with g_j regular at Q and $g_1 = 1$, and $t_Q = u^e h$ with h regular and non-zero at Q. On a small open neighborhood of Q, the $|g_j|_v$ are bounded, $|h|_v$ is bounded above and below by positive constants, and (9.2.1) applies. There

$$\delta_v(P,Q) \asymp \max_j |u(P) g_j(P)|_v \asymp |u(P)|_v \asymp |t_Q(P)|_v^{1/e} = d_v(P,Q).$$

Away from this neighborhood, both functions are continuous and positive on a compact set, as in the proof of Lemma 9.2.6. □

Theorem 9.2.7 makes [Sil09, Remark IX.2.2.1] precise. That remark defines a distance from several functions with no other common zero, such as the $x_j - a_j$, and [Sil09, Exercise 9.15] compares it with d_v. The constant in $\asymp$ depends on Q. A distance with constants uniform in both variables is a function on $C \times C$, and $-\log$ of it is a local height for the diagonal of $C \times C$ [Sil87].

REMARK 9.2.8. The v-adic distance is not a metric, and is not meant to be one. It is defined one Q at a time, after choosing t_Q, and $c\,t_Q$ is another choice for every $c \in k_v^\times$. So it makes no sense to ask whether $d_v(P,Q) = d_v(Q,P)$, or for the triangle inequality, until the choices are coordinated. Even natural choices fail. On $\mathbb{P}^1$, let $a \in k_v$ with $|a|_v > 1$, and take $t_a := x - a$ and $t_\infty := 1/x$. The rational point ∞ is a pole of t_a, so $d_v(\infty, a) = 1$, while $d_v(a, \infty) = \min(|1/a|_v, 1) = 1/|a|_v$.
What is well defined is the class of $d_v(\cdot, Q)$ under $\asymp$, by Lemma 9.2.6, and by Theorem 9.2.7 it contains the honest metric $\delta_v(\cdot, Q)$. The results of this chapter depend only on this class. They concern $\log d_v(P,Q)$ as P tends to Q, and replacing d_v by a function g with $g \asymp d_v$ changes $\log d_v$ by a bounded amount, which is negligible against $\log d_v \to -\infty$.

The two examples below take Q at infinity. The first one is the source of all heights.

EXAMPLE 9.2.9. Let $C = \mathbb{P}^1$, with affine coordinate x, and let $Q := \infty = [1:0]$. The function $t_\infty := 1/x$ has a simple zero at ∞ and no other zeros, so for $P = [x:1]$ with $x \in k_v$,

$$d_v(P, \infty) = \min\bigl(1, |x|_v^{-1}\bigr) = \frac{1}{\max(1, |x|_v)}.$$

The chordal distance is $\delta_v(P,\infty) = |x \cdot 0 - 1 \cdot 1|_v / (\|P\|_v \cdot 1)$. If v is non-archimedean, this is $1/\max(1,|x|_v) = d_v(P,\infty)$, so here $d_v(\cdot,\infty)$ is exactly the ultrametric distance to ∞. If v is archimedean, it is $1/\sqrt{1+|x|_v^2}$, and $1 \leqslant d_v(P,\infty)/\delta_v(P,\infty) \leqslant \sqrt{2}$.

Now let $x \in k$. The height of Definition 8.5.3, computed with the coordinates $[x : 1]$, is $H_k(x) = \prod_v \max(1, |x|_v)^{n_v}$. So

$$\log H_k(x) = \sum_{v \in \mathcal{P}_k} n_v\big(-\log d_v(x, \infty)\big). \tag{9.2.2}$$

The height of x measures how close x is to ∞, at all places at once. An element of k has large height when it is very close to ∞ at one place, or moderately close at many.

EXAMPLE 9.2.10. Let E be an elliptic curve over k_v, given by a Weierstrass equation, and let $Q := O$. The function x has a double pole at O and no other poles, so we may take $t_O := 1/x$, with $e = 2$:

$$d_v(P, O) = \min\big(1, |x(P)|_v^{-1/2}\big).$$

The uniformizer x/y at O is not allowed, since it also vanishes at some $\overline{k_v}$-point with $x = 0$. For $P_1, P_2 \in E(k_v)$, let $\hat{d}_v(P_1, P_2) := d_v(P_1 - P_2, O)$. It is a v-adic distance to P_2 in the sense of Definition 9.2.5, defined by the function $P \mapsto 1/x(P - P_2)$, which has a double zero at P_2 and no other zeros.
Let v be non-archimedean, and let the equation be minimal. A k_v-point P with $v(x(P)) < 0$ reduces to $\bar{O}$. So if $P \notin E_1(k_v)$, then $v(x(P)) \geqslant 0$, and $d_v(P, O) = 1$. If $P \in E_1(k_v) \setminus \{O\}$, then $3v(x(P)) = 2v(y(P)) = -6r$ for some $r \geqslant 1$ [Sil09, proof of Proposition VII.2.2]. Let $z := -x/y$. Then $|x(P)|_v^{-1/2} = |z(P)|_v$, so

$$d_v(P, O) = \begin{cases} |z(P)|_v, & P \in E_1(k_v), \\ 1, & P \notin E_1(k_v). \end{cases}$$

For non-archimedean v, $\hat{d}_v$ is a translation-invariant ultrametric on $E(k_v)$. Indeed, z identifies $E_1(k_v)$ with the group $\hat{E}(\mathfrak{m}_v)$ of the formal group of E, whose group law and inverse are power series $F(T_1, T_2) = T_1 + T_2 + \cdots$ and $i(T) = -T + \cdots$ with coefficients in $\mathcal{O}_v$ [Sil09, Section IV.1 and Proposition VII.2.2]. So $|z(P_1 + P_2)|_v \leqslant \max(|z(P_1)|_v, |z(P_2)|_v)$ and $|z(-P)|_v = |z(P)|_v$ on $E_1(k_v)$. If $P_1 - P_2$ or $P_2 - P_3$ is not in the subgroup $E_1(k_v)$, the ultrametric inequality reads $\hat{d}_v(P_1, P_3) \leqslant 1$. So on E, the class of every $d_v(\cdot, Q)$ contains a translation-invariant ultrametric, defined by the group law.
Let v be archimedean, and let the equation be $y^2 = x^3 + Ax + B$. By Theorem 1.7.1, $E(\mathbb{C}) \cong \mathbb{C}/\Lambda$ for a lattice Λ. Let $P \in E(k_v)$ correspond to $s + \Lambda$, so that $x(P) = \wp_\Lambda(s)$. The Laurent series $\wp_\Lambda(s) = s^{-2} + 3G_4(\Lambda)s^2 + \cdots$ [Sil09, Theorem VI.3.5] gives $|x(P)|_v^{-1/2} = |s|(1 + O(|s|^4))$ near $s = 0$. By the compactness argument of Lemma 9.2.6, $d_v(P, O) \asymp \min_{\omega \in \Lambda} |s - \omega|$ on $E(k_v)$. The right-hand side is the distance from s to Λ, so $\hat{d}_v$ is $\asymp$ to the flat metric of the torus $\mathbb{C}/\Lambda$.

The next result says that d_v measures the order of vanishing of every function at Q.

PROPOSITION 9.2.11. *Let $Q \in C(k_v)$, and let $F \in k_v(C)$ vanish at Q. Then*

$$\lim_{P \to_v Q} \frac{\log|F(P)|_v}{\log d_v(P, Q)} = \operatorname{ord}_Q(F).$$

In particular, the limit does not depend on the choice of t_Q.

PROOF. We follow [Sil09, Proposition IX.2.2]. Let $e := \operatorname{ord}_Q(t_Q)$ and $m := \operatorname{ord}_Q(F) \geqslant 1$. The function $g := F^e / t_Q^m$ has neither a zero nor a pole at Q, so $|g|_v$ is bounded above and below by positive constants on a neighborhood of Q. For $P \neq Q$ close to Q, $F(P) \neq 0$, $d_v(P, Q) = |t_Q(P)|_v^{1/e} < 1$, and

$$\log|F(P)|_v = m \log d_v(P, Q) + \tfrac{1}{e} \log|g(P)|_v.$$

Divide by $\log d_v(P, Q)$, which tends to $-\infty$ as P tends to Q. □

The next result is about the effect of a morphism on distances. It depends on the ramification, not on the degree.

PROPOSITION 9.2.12. *Let $\varphi\colon C_1 \to C_2$ be a non-constant morphism of nice curves over k_v, and let $Q \in C_1(k_v)$. Then*

$$\lim_{P \to_v Q} \frac{\log d_v(\varphi(P), \varphi(Q))}{\log d_v(P, Q)} = e_\varphi(Q).$$

PROOF. We follow [Sil09, Proposition IX.2.3]. Let $t_{\varphi(Q)} \in k_v(C_2)$ define $d_v(\cdot, \varphi(Q))$, with $e_2 \coloneqq \operatorname{ord}_{\varphi(Q)}(t_{\varphi(Q)})$. Writing $t_{\varphi(Q)}$ as the e_2-th power of a uniformizer times a unit at $\varphi(Q)$, Definition 3.2.7 gives $\operatorname{ord}_Q(\varphi^* t_{\varphi(Q)}) = e_\varphi(Q) e_2$. The morphism φ is continuous, by part 2 of Lemma 9.2.4, and its fibers are finite. So for $P \neq Q$ close to Q, $\varphi(P) \neq \varphi(Q)$ is close to $\varphi(Q)$, and

$$\log d_v(\varphi(P), \varphi(Q)) = \frac{1}{e_2} \log |(\varphi^* t_{\varphi(Q)})(P)|_v.$$

Apply Proposition 9.2.11 to $F \coloneqq \varphi^* t_{\varphi(Q)}$. □

Compare with heights. For a morphism $F\colon \mathbb{P}^N \to \mathbb{P}^M$ of degree d, $h \circ F = d\,h + O(1)$ by Theorem 8.5.19: heights see the degree. Proposition 9.2.12 says that distances see the ramification. For an unramified morphism of large degree, such as the multiplication $[m]$ on an elliptic curve over a field of characteristic 0, the log-ratio of distances tends to 1 while heights are multiplied by m^2. This contrast drives the proof of Siegel's theorem.

9.2.3. Local heights. Example 9.2.9 writes the height of $x \in k$ as a sum over the places of $-\log$ of a distance to ∞. Local heights generalize this to any divisor on any curve [HS00, Section B.8]. Intuitively, the local height of P for a divisor D is $-\log$ of the v-adic distance from P to D.

DEFINITION 9.2.13. Let C be a nice curve over k_v, and let $D = \sum_Q n_Q Q$ be a divisor on C. A local height for D at v is a continuous function $\lambda\colon C(k_v) \setminus \operatorname{supp} D \to \mathbb{R}$ such that, for each $Q \in C(k_v)$ with $n_Q \neq 0$ and a uniformizer u_Q at Q, the function $\lambda + n_Q \log |u_Q|_v$ is bounded on a punctured neighborhood of Q.

The closed points of the support of D that are not k_v-points impose no condition, since they are not in $C(k_v)$.

PROPOSITION 9.2.14. *Let $D = \sum_Q n_Q Q$ be a divisor on a nice curve C over k_v.*

(1) Two local heights for D differ by a bounded function.
(2) The function $-\sum_{Q \in C(k_v)} n_Q \log d_v(\cdot, Q)$ is a local height for D.
(3) Let $f \in k_v(C)^\times$. Then $-\log |f|_v$ is a local height for $\operatorname{div}(f)$.
(4) Let $f \in k_v(C)$ be non-constant, with pole divisor $D_\infty \coloneqq \sum_{\operatorname{ord}_Q(f) < 0} (-\operatorname{ord}_Q f) Q$. Then $\log \max(1, |f|_v)$ is a local height for D_∞.

PROOF. For part 1, the difference of two local heights is continuous on $C(k_v)$ minus the finitely many k_v-points of $\operatorname{supp} D$, and bounded on a punctured neighborhood of each of them. The complement of these neighborhoods is compact.

For part 2, each $-\log d_v(\cdot, Q)$ is continuous away from Q. Near Q, write $t_Q = u_Q^e h$ with h a unit at Q. Then $-\log d_v(\cdot, Q) + \log |u_Q|_v = -\frac{1}{e} \log |h|_v$ is bounded.

For part 3, near a k_v-point Q with $n \coloneqq \operatorname{ord}_Q(f)$, write $f = u_Q^n g$ with g a unit at Q. Then $-\log |f|_v + n \log |u_Q|_v = -\log |g|_v$ is bounded. For part 4, $\log \max(1, |f|_v)$ is continuous away from the poles, and equals $\log |f|_v$ near a pole. Conclude as for part 3. □

Parts 1 and 2 say that the local heights for D are the functions $-\sum n_Q \log d_v(\cdot, Q) + O(1)$. For instance, let $f \in k_v(C)$ be non-constant. The pole divisor of $1/f$ is the divisor of zeros of f, and $\log \max(1, |1/f|_v) = -\log \min(1, |f|_v)$. So part 4, applied to $1/f$, gives

$$\log \min(1, |f|_v) = \sum_{Q \in C(k_v),\ f(Q) = 0} \operatorname{ord}_Q(f) \log d_v(\cdot, Q) + O(1)$$

on $C(k_v)$ minus the zeros of f [Sil09, Exercise 9.16].

The local heights of a rational function add up to its height.

PROPOSITION 9.2.15. *Let C be a nice curve over k, let $f \in k(C)$ be non-constant, and let $P \in C(k)$ not be a pole of f. Then*

$$\log H_k(f(P)) = \sum_{v \in \mathcal{P}_k} n_v \log \max(1, |f(P)|_v).$$

If the only pole of f in $C(\overline{k})$ is a k-point Q, of order e, and $d_v(\cdot, Q)$ is defined by $t_Q := 1/f$ at every place v, then $\log H_k(f(P)) = e \sum_{v \in \mathcal{P}_k} n_v\big(-\log d_v(P, Q)\big)$.

PROOF. The first formula is Definition 8.5.3 for $[f(P) : 1]$. For the second, $t_Q = 1/f$ has a zero of order e at Q and no other zeros, and $-e \log d_v(P, Q) = -\log \min(|f(P)|_v^{-1}, 1) = \log \max(1, |f(P)|_v)$. □

For $C = \mathbb{P}^1$ and $f = x$, this is (9.2.2). Each term is a local height at v, by part 4 of Proposition 9.2.14. The sum is finite, since almost all terms vanish. For an arbitrary choice of local heights, the bounded errors at infinitely many places need not add up to a bounded error. The general theory controls them uniformly in v [HS00, Theorem B.8.1], but we only need the exact identity of Proposition 9.2.15.

REMARK 9.2.16. Let k' be a finite extension of k, and w a place of k' above v. The absolute values of Situation 9.0.1 are not normalized: $|\ |_w$ restricts to $|\ |_v$ on k_v. So a function $t_Q \in k_v(C)$ that defines $d_v(\cdot, Q)$ also defines $d_w(\cdot, Q)$, and the two agree on $C(k_v)$.

REMARK 9.2.17. A local height is defined only up to a bounded function, by part 1 of Proposition 9.2.14. On an elliptic curve E, the group law selects a canonical one, as Tate's limit of Proposition 8.4.9 selects the canonical height among equivalent quasi-quadratic heights. Néron's local height λ_v is the unique continuous function on $E(k_v) \setminus \{O\}$ such that $\lambda_v(P) - \frac{1}{2}\log|x(P)|_v$ has a limit as P tends to O, and that satisfies the duplication formula of [Sil09, Proposition C.18.1(iii)]. In particular it is a local height for the divisor O. Let $\hat{h}$ be the canonical height on $E(k)$ of [Sil09, Section VIII.9], not yet constructed in these notes. It satisfies $\hat{h}(P) = (2[k : \mathbb{Q}])^{-1} \log H_k(x(P)) + O(1)$, and it is the sum of the local heights: $\hat{h}(P) = [k : \mathbb{Q}]^{-1} \sum_v n_v \lambda_v(P)$ for $P \in E(k) \setminus \{O\}$ [Sil09, Theorem C.18.2], [HS00, Section B.9]. Proposition 9.2.15 with $f = x$ gives $\log H_k(x(P)) = 2 \sum_v n_v\big(-\log d_v(P, O)\big)$, so the functions $-\log d_v(\cdot, O)$ add up to $\hat{h}$ as well, up to a bounded function.
If v is non-archimedean and E has good reduction, with a minimal Weierstrass equation, then $\lambda_v(P) = \log \max(1, |x(P)|_v^{1/2}) = -\log d_v(P, O)$, with $t_O = 1/x$ as in Example 9.2.10 [Sil09, Theorem C.18.3(b)]. So in this case the choice of Example 9.2.10 is the canonical one. At a place of split multiplicative reduction, let $N := -v(j(E))$, and identify $E(k_v)/E_0(k_v)$ with $\mathbb{Z}/N\mathbb{Z}$. A k_v-point in the class of n, with $1 \leqslant n \leqslant N - 1$, has $\lambda_v = \frac{1}{2}B_2(n/N) \log|j(E)|_v$, with $B_2(T) := T^2 - T + \frac{1}{6}$ [Sil09, Theorem C.18.3(c)]. For $N = 2$ and $n = 1$, $B_2(1/2) = -1/12$ and $|j(E)|_v > 1$, so $\lambda_v < 0$ there. Since d_v and δ_v are at most 1, λ_v is then neither $-\log d_v(\cdot, O)$, for any choice of t_O, nor $-\log \delta_v(\cdot, O)$. Canonical does not mean metric.

9.2.4. Roth's theorem in terms of distances. Theorem 9.1.10 says how fast elements of k can approach an algebraic number. In terms of distances, it says how fast k-points of a curve can approach a $\overline{k}$-point.

COROLLARY 9.2.18. *Let C be a nice curve over k, let $f \in k(C)$ be non-constant, and let $Q \in C(\overline{k})$. Then*

$$\liminf_{P \in C(k),\ P \to_v Q} \frac{\log d_v(P, Q)}{\log H_k(f(P))} \geqslant -2.$$

If $Q \notin C(k_v)$, or if Q is not a limit of k-points, the lim inf *is taken to be* 0.

PROOF. We follow [Sil09, Corollary IX.2.4]. If $Q \notin C(k_v)$, then Q is not a limit of k_v-points, by part 4 of Lemma 9.2.4. So let $Q \in C(k_v)$, and let (P_i) be a sequence of distinct k-points, different from Q, tending to Q. We may replace f by $1/f$. Indeed f has finitely many zeros, and $H_k(1/a) = H_k(a)$ for $a \in k^\times$, since $[1/a : 1] = [1 : a]$ and permuting the coordinates does not change the height. So we may assume

that f is regular at Q. Let $\alpha := f(Q)$. It lies in $\overline{k}$, since $Q \in C(\overline{k})$, and in k_v, since $Q \in C(k_v)$. Let $e := \operatorname{ord}_Q(f - \alpha) \geqslant 1$. By Proposition 9.2.11,

$$\lim_{i\to\infty} \frac{\log|f(P_i) - \alpha|_v}{\log d_v(P_i, Q)} = e.$$

The fibers of f are finite. So $f(P_i) \neq \alpha$ for large i, each element of k is $f(P_i)$ for finitely many i, and $H_k(f(P_i)) \to \infty$ by Theorem 8.5.13. Let $\varepsilon > 0$ and $\tau := 2 + \varepsilon$. By Theorem 9.1.10, in the form of part 1 of Remark 9.1.6 with constant 1, $|f(P_i) - \alpha|_v \geqslant H_k(f(P_i))^{-\tau}$ for all large i. Then, for large i,

$$\frac{\log d_v(P_i, Q)}{\log H_k(f(P_i))} = \frac{\log d_v(P_i, Q)}{\log|f(P_i) - \alpha|_v} \cdot \frac{\log|f(P_i) - \alpha|_v}{\log H_k(f(P_i))},$$

where the first factor is positive and tends to $1/e$, and the second lies in $[-\tau, 0)$. So the $\liminf$ is at least $-\tau/e \geqslant -2 - \varepsilon$, for every $\varepsilon > 0$. □

Remark 9.2.19. Corollary 9.2.18 says nothing new for $Q \in C(k)$. Then $\alpha = f(Q) \in k$. For $w \in \mathcal{P}_k$, let $c_w := 2$ if w is archimedean and $c_w := 1$ otherwise, so that $\prod_w c_w^{n_w} = 2^{[k:\mathbb{Q}]}$. For $a \in k$ with $a \neq \alpha$, the product formula (part 2 of Proposition 8.5.2) gives

$$\begin{aligned} |a - \alpha|_v^{n_v} = \prod_{w\neq v} |a - \alpha|_w^{-n_w} &\geqslant \prod_w \bigl(c_w \max(1, |a|_w) \max(1, |\alpha|_w)\bigr)^{-n_w} \\ &= \frac{1}{2^{[k:\mathbb{Q}]} H_k(\alpha) H_k(a)}, \end{aligned}$$

So the proof gives a $\liminf$ of at least $-1/(e\, n_v) \geqslant -1$, which is better than -2. This is the argument for rational α in the proof of Corollary 9.1.7. The content of Corollary 9.2.18 is for $\overline{k}$-points that are not k-points.

Siegel's theorem, in the next section, improves the bound to 0 when C is an elliptic curve with infinitely many k-points. Let $S \subseteq \mathcal{P}_k$ be finite, and let P be a k-point with $|x(P)|_w \leqslant 1$ for all $w \notin S$. In Proposition 9.2.15 for $f = x$, only the places of S contribute, so P is v-adically very close to O for some $v \in S$. Writing $P = [m]P' + R$ with R in a finite set, Proposition 9.2.12 for the unramified morphism $P' \mapsto [m]P' + R$ keeps the log-ratio of distances near 1, while the height of P' is about m^{-2} times that of P [Sil09, Section IX.3].

CHAPTER 10

Twisting, Selmer, and Tate–Shafarevich

CHAPTER 11

Complex multiplication

CHAPTER 12

The Tate curve

CHAPTER 13

The L-function and BSD

CHAPTER 14

The Sato–Tate conjecture

CHAPTER 15

Introduction to modular curves

Glossary

absolute height: For $P \in \mathbb{P}^N(\overline{\mathbb{Q}})$, the number $H(P) = H_K(P)^{1/[K:\mathbb{Q}]}$ for any number field K with $P \in \mathbb{P}^N(K)$. p. 139

additive group: The k-variety $\mathbb{G}_{\mathrm{a}} = \operatorname{Spec} k[x] = \mathbb{A}^1_k$, with $\mathbb{G}_{\mathrm{a}}(R) = R$ under addition for every k-algebra R. p. 26

additive reduction: An elliptic curve over k_v has additive reduction if E_π has a cusp. Then $E_\pi^{\mathrm{ns}} \cong \mathbb{G}_{\mathrm{a}}$. p. 92

affine k-variety: A scheme over k isomorphic to $\operatorname{Spec} A$ for some k-algebra A of finite type. p. 25

affine n-space: The k-variety $\mathbb{A}^n_k = \operatorname{Spec} k[x_1, \ldots, x_n]$, with $\mathbb{A}^n(R) = R^n$ for every k-algebra R. p. 26

approximation exponent: For a number field k and a place v, a function τ on the positive integers such that, for every $\alpha \in \overline{k}$ of degree d over k and every $C > 0$, only finitely many $x \in k$ satisfy $|x - \alpha|_v < C H_k(x)^{-\tau(d)}$. p. 149

b-invariants: The polynomials b_2, b_4, b_6, b_8 in the coefficients of a Weierstrass equation, from which Δ is built. p. 45

bad reduction: Multiplicative or additive reduction. p. 92

base extension: For a k-variety X and a field $L \supseteq k$, the L-variety $X_L = X \times_{\operatorname{Spec} k} \operatorname{Spec} L$, defined by the same equations as X. p. 30

Brauer group: The group $\operatorname{Br}(k) := H^2(k, \mathbb{G}_{\mathrm{m}})$, equivalently the similarity classes of central simple algebras over k. p. 116

c-invariants: The polynomials $c_4 = b_2^2 - 24b_4$ and $c_6 = -b_2^3 + 36b_2b_4 - 216b_6$ of a Weierstrass equation. They satisfy $1728\Delta = c_4^3 - c_6^2$. p. 45

canonical divisor: The divisor of a non-zero differential on a nice curve. p. 38

canonical height: The quadratic height $\hat{h}(P) = \lim 4^{-n} h(2^n P)$ equivalent to a quasi-quadratic height h. p. 136

Cartier operator: The additive operator $\mathcal{C}$ on the differentials of a curve over a perfect field of characteristic p with $\mathcal{C}(f^p\omega) = f\mathcal{C}(\omega)$, given locally by $\mathcal{C}\bigl(\sum_j f_j^p t^j\, dt\bigr) = f_{p-1}\, dt$. p. 80

chart: A homeomorphism from an open subset of a topological space onto an open subset of $\mathbb{C}$. p. 11

chordal distance: For points P_1, P_2 of $\mathbb{P}^N(\overline{k_v})$, the number $\|P_1 \wedge P_2\|_v / (\|P_1\|_v \|P_2\|_v)$, built from the 2×2 minors of their coordinates. It is a metric, and an ultrametric at a non-archimedean place. p. 151

closed point: A point P of a k-variety whose residue field $k(P)$ is finite over k (for varieties, equivalently $\{P\}$ is closed). Its degree is $[k(P) : k]$. p. 33

coboundary: A left 1-coboundary: a map $G \to A$ of the form $g \mapsto ga - a$. p. 111

cocycle: A left 1-cocycle: a continuous map $\xi\colon G \to A$ with $\xi_{gh} = \xi_g + g\xi_h$. p. 111

cohomologous: Two cocycles with the same class in $H^1(G, A)$. p. 111

compatible charts: Two charts whose domains are disjoint, or whose transition maps are holomorphic. p. 11

completion: For a place v of a global field k, the completion k_v of k with respect to an absolute value in the class v. p. 85

complex structure: A family of pairwise compatible charts whose domains cover the space. p. 11

complex torus: The quotient $\mathbb{C}/\Lambda$ of $\mathbb{C}$ by a lattice Λ: a compact Riemann surface that is also an abelian group. p. 14

component group: The finite étale group scheme $\Phi = \mathcal{E}_\pi / \mathcal{E}_\pi^0$ of components of the special fiber of the Néron model. p. 102

connected: A topological space that is not the union of two disjoint non-empty open subsets. p. 11

connecting homomorphism: The map $\delta\colon C^G \to H^1(G, A)$ of a short exact sequence $0 \to A \to B \to C \to 0$, and its analogue in degree 1. p. 113

cubic twist: For $d \in k^\times$, the Selmer curve C_d, regarded as a twist of the Fermat cubic C_1. Its isomorphism class depends only on d in $k^\times/k^{\times 3}$. p. 72

cup product: The product $H^i(G,A) \times H^j(G,B) \to H^{i+j}(G,C)$ induced by a G-equivariant bilinear map $A \times B \to C$, given by $(\xi \cup \eta)_{g,h} = \xi_g \cdot g\eta_h$ in degrees 1 and 1. p. 123

curve: An integral variety of dimension 1. p. 33

cyclic isogeny: An isogeny of complex tori whose kernel is a cyclic group. p. 16

cyclotomic character: The action $\chi_m\colon \mathrm{Gal}_k \to (\mathbb{Z}/m\mathbb{Z})^\times$ of Gal_k on $\mu_m(k^{\mathrm{sep}})$, with $\sigma(\zeta) = \zeta^{\chi_m(\sigma)}$. p. 110

d-uple embedding: The map $\mathbb{P}^N \to \mathbb{P}^{N_d}$ given by all the monomials of degree d. p. 145

definite quaternion algebra: A quaternion algebra B over $\mathbb{Q}$ with $B \otimes_{\mathbb{Q}} \mathbb{R}$ isomorphic to Hamilton's quaternions. Here $(a,b \mid \mathbb{Q})$ is definite exactly when $a, b < 0$. p. 66

degree of a morphism: For a non-constant morphism of curves, the degree $[k(C_1) : \varphi^* k(C_2)]$ of the extension of function fields. p. 35

degree of an isogeny of complex tori: The order of its kernel. For $z \mapsto \alpha z$ from $\mathbb{C}/\Lambda_1$ to $\mathbb{C}/\Lambda_2$ it is the index $[\Lambda_2 : \alpha\Lambda_1]$. p. 17

differential: An element of $\Omega_{k(C)/k}$, the $k(C)$-vector space generated by the dx subject to the Leibniz rules and $da = 0$ for $a \in k$. p. 38

discrete module: An abelian group with a left action of a profinite group G whose stabilizers are open, that is, continuous for the discrete topology. p. 110

discriminant: The polynomial Δ in the coefficients of a Weierstrass equation that vanishes exactly when the cubic is singular; $\Delta = -16(4A^3 + 27B^2)$ for $y^2 = x^3 + Ax + B$. p. 45

divisor: A finite formal integer combination of closed points of a curve. p. 36

dual graph: For a special fiber with components Γ_i, the graph with one vertex for each component and $\Gamma_i \cdot \Gamma_j$ edges between Γ_i and Γ_j. p. 102

dual isogeny: For an isogeny φ of degree d, the unique isogeny $\hat{\varphi}$ with $\hat{\varphi} \circ \varphi = [d]$. p. 58

dual isogeny of complex tori: For an isogeny $z \mapsto \alpha z$ of complex tori of degree d, the isogeny $z \mapsto (d/\alpha)z$ in the opposite direction. Both composites are multiplication by d. p. 17

Eisenstein integers: The ring $\mathbb{Z}[\omega]$, where ω is a primitive cube root of 1. p. 14

Eisenstein series: For a lattice Λ and even $k > 2$, the sum $G_k(\Lambda)$ of ω^{-k} over the non-zero $\omega \in \Lambda$. p. 22

elliptic curve: A pair (E, O) of a nice curve E of genus 1 over k and a rational point $O \in E(k)$. p. 45

exceptional curve: The curve lying over the closed point that is blown up. p. 100

fibered surface: An integral scheme of dimension 2, flat and projective over $\mathcal{O}_v$, such as the Weierstrass model of an integral Weierstrass equation. p. 98

finite type: A scheme over k covered by finitely many open subschemes $\operatorname{Spec} A$, each with A a finitely generated k-algebra. p. 25

flex: A smooth $\overline{k}$-point of a plane cubic whose tangent line meets the cubic there with multiplicity 3. p. 46

free: An abelian group that has a $\mathbb{Z}$-basis. p. 13

Frobenius automorphism: For a finite field $\mathbb{F}_q$, the automorphism $x \mapsto x^q$ of $\overline{\mathbb{F}}_q$. It topologically generates $\mathrm{Gal}(\overline{\mathbb{F}}_q/\mathbb{F}_q)$. p. 73

function field: For an integral variety X, the local ring $k(X)$ at its generic point, which is a field. For $X = \operatorname{Spec} A$ it is $\operatorname{Frac} A$. p. 30

fundamental group: The group $\pi_1(X, x)$ of homotopy classes of loops in X based at x. p. 13

fundamental parallelogram: For a lattice with $\mathbb{Z}$-basis $\{\omega_1, \omega_2\}$, the half-open parallelogram $\{t_1\omega_1 + t_2\omega_2 : 0 \leqslant t_1, t_2 < 1\}$. p. 15

G-group: A group, not necessarily commutative, with a continuous left action of a profinite group G by group automorphisms. p. 110

Gauss norm: For a polynomial over a field with a non-archimedean absolute value, the largest absolute value of its coefficients. It is multiplicative (Gauss's lemma). p. 140

Gaussian integers: The ring $\mathbb{Z}[i]$. p. 13

generic fiber: For an $\mathcal{O}_v$-scheme $\mathcal{X}$, the k_v-scheme $\mathcal{X}_{k_v} = \mathcal{X} \times_{\operatorname{Spec} \mathcal{O}_v} \operatorname{Spec} k_v$. p. 87

generic point: For an integral scheme, the point whose closure is the whole scheme; for $\operatorname{Spec} \mathcal{O}_v$, the prime ideal (0), with residue field k_v. p. 87

genus: For a nice curve, $\ell(K_C)$, the dimension of the space of regular differentials. p. 40

geometrically integral: A k-variety X such that $X_{\overline{k}}$ is integral. p. 30

global field: A number field or a global function field. p. 85

global function field: A finite extension of $\mathbb{F}_p(t)$ for a prime p; equivalently, the function field of a geometrically integral curve over a finite field. p. 85

good reduction: An elliptic curve over k_v has good reduction if its reduction E_π is smooth, equivalently $v(\Delta) = 0$ for a minimal Weierstrass equation. p. 92

group cohomology: The groups $H^i(G, A) = Z^i(G, A)/B^i(G, A)$ of continuous cocycles modulo coboundaries. p. 111

group variety: A k-variety G with morphisms $m\colon G \times G \to G$, $i\colon G \to G$ and $e\colon \operatorname{Spec} k \to G$ satisfying the group axioms; equivalently, a group structure on every $G(R)$, natural in the k-algebra R. p. 26

Hasse invariant: For $p \geqslant 3$ and an elliptic curve $y^2 = f(x)$ over $\mathbb{F}_q$, the coefficient A of x^{p-1} in $f(x)^{(p-1)/2}$. It satisfies $\mathcal{C}(\omega) = A^{1/p}\omega$, and vanishes exactly for a supersingular curve. p. 81

Hausdorff: A topological space in which any two distinct points of the space have disjoint open neighborhoods. p. 11

height function: A function $h\colon X \to \mathbb{R}$ on a set X. Two heights are equivalent if their difference is bounded. p. 134

height relative to a number field: For $P \in \mathbb{P}^N(k)$, the number $H_k(P) = \prod_v \max_i |x_i|_v^{n_v}$, independent of the coordinates x_i by the product formula. p. 139

Hilbert symbol: For a local field k_v with $\mu_m(k_v^{\mathrm{sep}}) \subseteq k_v$, the root of unity attached to the local invariant of the cup product of the Kummer classes of a and b. p. 124

holomorphic differential: A meromorphic differential on a Riemann surface whose local expressions $f_\phi(z)\,dz$ are all holomorphic. p. 19

holomorphic map: A continuous map between Riemann surfaces that is holomorphic when read in charts. p. 12

homogenization: The homogeneous polynomial $F(X_0, \ldots, X_n) = X_0^{\deg f} f(X_1/X_0, \ldots, X_n/X_0)$ attached to $f \in k[x_1, \ldots, x_n]$. p. 28

homomorphism of group varieties: A morphism of group varieties respecting their multiplication, inverse and identity. p. 26

inertia group: The subgroup $I_v = \mathrm{Gal}(k_v^{\mathrm{sep}}/k_v^{\mathrm{unr}})$ of Gal_{k_v}, the elements acting trivially on the residue field. p. 95

inflation map: The map $\mathrm{Inf}\colon H^i(G/H, A^H) \to H^i(G, A)$ composing cocycles with $G \to G/H$, for H closed and normal. p. 113

integral model: For a k_v-scheme X, an $\mathcal{O}_v$-scheme $\mathcal{X}$ with an isomorphism $\mathcal{X} \times_{\operatorname{Spec} \mathcal{O}_v} \operatorname{Spec} k_v \to X$. p. 87

integral variety: A k-variety whose rings of sections over non-empty opens are integral domains; equivalently, a reduced and irreducible one. p. 29

intersection number: For plane curves $C = V_+(F)$ and $D = V_+(G)$ without common component, $I_P(C, D) = \dim_k \mathcal{O}_{\mathbb{P}^2_k, P}/(f, g)$. p. 41

invariant differential: The differential $\omega = dx/(2y + a_1x + a_3)$ of a Weierstrass equation. It has no zeros or poles and is invariant under translation. p. 45

invertible module: A finitely generated projective module of constant rank 1 over a ring R; equivalently, an R-module P with $P \otimes_R Q \cong R$ for some Q. p. 28

irreducible variety: A k-variety whose underlying topological space is non-empty and not the union of two proper closed subsets. p. 29

isogeny: A non-zero holomorphic group homomorphism between complex tori. Its algebraic analogue is an isogeny of elliptic curves. p. 15

isogeny of elliptic curves: A morphism $\varphi\colon E_1 \to E_2$ of elliptic curves with $\varphi(O_1) = O_2$. It is automatically a group homomorphism. p. 57

isomorphism of complex tori: An invertible isogeny. p. 15

isomorphism of Riemann surfaces: A bijective holomorphic map whose inverse is holomorphic. p. 12

j-invariant: The quantity $j = c_4^3/\Delta$ of a smooth Weierstrass equation. It classifies elliptic curves up to isomorphism over $\overline{k}$. p. 45

k-variety: A separated scheme of finite type over a field k. p. 25

kernel of reduction: The subgroup $E_1(k_v)$ of k_v-points of E whose reduction is $\bar{O}$. p. 89

Kodaira type: The configuration of the special fiber of the minimal regular proper model over $\bar{\kappa}$, one of I_0, I_n, II, III, IV, I_0^*, I_n^*, IV*, III*, II*. p. 102

Krull topology: The topology on Gal_k in which the subgroups Gal_L, for L finite over k, form a basis of open neighborhoods of 1. p. 109

Kummer map: The connecting homomorphism $\delta\colon E(k)/m \to H^1(k, E[m])$, $P \mapsto (\sigma \mapsto \sigma(Q) - Q)$ with $[m]Q = P$. p. 118

Kummer pairing: For an elliptic curve E over a number field k with $E[m](\bar{k}) \subseteq E(k)$, the pairing $E(k) \times \mathrm{Gal}_k \to E[m](\bar{k})$, $(P, \sigma) \mapsto \sigma(Q) - Q$ with $mQ = P$. p. 128

lattice: A subgroup of $\mathbb{C}$, free of rank 2, that spans $\mathbb{C}$ as an $\mathbb{R}$-vector space. p. 13

lattice in a real vector space: A subgroup generated by a basis of a finite-dimensional real vector space. p. 136

lattice in an algebra: A finitely generated $\mathbb{Z}$-submodule of a finite-dimensional $\mathbb{Q}$-algebra B that spans B over $\mathbb{Q}$. p. 67

line at infinity: The line $Z = 0$ of $\mathbb{P}^2_k$, the complement of the affine patch $Z \neq 0$. p. 45

local degree: For a place v of a number field k, the degree $n_v = [k_v : \mathbb{Q}_v]$ of the completion over the completion of $\mathbb{Q}$. p. 139

local field: The completion of a global field at a place. p. 85

local height: For a divisor $D = \sum n_Q Q$ on a nice curve C over k_v, a continuous function λ on $C(k_v)\backslash \operatorname{supp} D$ such that $\lambda + n_Q \log|u_Q|_v$ is bounded near each k_v-point Q of the support, for a uniformizer u_Q. p. 155

local ramification index: For a finite extension K' of k_v, the integer e with $v' = e\,v$ on k_v, where v' is the normalized valuation of K'. p. 93

logarithmic height: The function $h = \log H$ on $\mathbb{P}^N(\overline{\mathbb{Q}})$. p. 139

m-torsion subgroup: The kernel $(\mathbb{C}/\Lambda)[m]$ of multiplication by m. It is isomorphic to $(\mathbb{Z}/m\mathbb{Z})^2$. p. 16

Mahler measure: For $f = a_0 \prod_j (T - \alpha_j) \in \mathbb{C}[T]$, the number $M(f) = |a_0| \prod_j \max(1, |\alpha_j|)$. p. 142

maximal ideal: The ideal $\mathfrak{m}_v = \{x \in k_v : v(x) > 0\}$ of $\mathcal{O}_v$. p. 85

maximal order: An order not properly contained in another order. p. 67

maximal unramified extension: The union k_v^{unr} of the finite unramified extensions of k_v inside k_v^{sep}. p. 95

meromorphic differential on a Riemann surface: A family of expressions $f_\phi(z)\,dz$, one for each chart ϕ, with f_ϕ meromorphic, related on overlaps by the chain rule. p. 19

meromorphic function: A holomorphic map from a Riemann surface to $\mathbb{P}^1(\mathbb{C})$ that is not identically ∞. p. 18

minimal regular proper model: The regular proper model $\mathcal{C}$ of E to which every other regular proper model maps, extending the identity of E. Its smooth locus is the Néron model. p. 99

minimal Weierstrass equation: A Weierstrass equation for an elliptic curve over k_v with coefficients in $\mathcal{O}_v$ whose discriminant has the least valuation among all such equations; also called v-minimal. p. 86

minimal Weierstrass model: The Weierstrass model of a minimal Weierstrass equation. p. 87

morphism of degree d: A map $\mathbb{P}^N \to \mathbb{P}^M$ given by homogeneous polynomials of degree d with no common zero other than 0. p. 144

multiplication by m map: The isogeny $[m]\colon \mathbb{C}/\Lambda \to \mathbb{C}/\Lambda$, $z + \Lambda \mapsto mz + \Lambda$. p. 15

multiplicative group: The k-variety $\mathbb{G}_{\mathrm{m}} = \operatorname{Spec} k[x, x^{-1}]$, with $\mathbb{G}_{\mathrm{m}}(R) = R^\times$ for every k-algebra R. p. 26

multiplicative reduction: An elliptic curve over k_v has multiplicative reduction if E_π has a node. Then E_π^{ns} is $\mathbb{G}_{\mathrm{m}}$ or a twisted multiplicative group. p. 92

multiplicity of a component: For a component Γ_i of the special fiber of a regular model, the valuation d_i of π_v at the generic point of Γ_i. p. 101

Néron model: For an elliptic curve E over k_v, a smooth $\mathcal{O}_v$-scheme $\mathcal{E}$ with generic fiber E such that every morphism from the generic fiber of a smooth $\mathcal{O}_v$-scheme $\mathcal{T}$ to E extends uniquely to $\mathcal{T} \to \mathcal{E}$. p. 88

Newton polygon: For $\sum_i c_i T^i$ over a p-adic field, the lower boundary of the convex hull of the points $(i, w(c_i))$. Its slopes are minus the valuations of the roots. p. 141

nice variety: A smooth, projective, geometrically integral variety. p. 31

nonsingular polynomial: A homogeneous $F \in \mathbb{C}[X, Y, Z]$ such that F and its three partial derivatives have no common zero in $\mathbb{P}^2(\mathbb{C})$. p. 21

nonsplit multiplicative reduction: Multiplicative reduction in which the slopes of the tangent lines at the node do not lie in $\kappa(v)$. p. 92

Northcott property: A height h on X has it if $\{x \in X : h(x) \leqslant C\}$ is finite for every C. p. 134
number field: A finite extension of $\mathbb{Q}$. p. 85
open unit disk: The Riemann surface $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$. p. 12
order: A lattice in a finite-dimensional $\mathbb{Q}$-algebra that is also a subring. p. 67
ordinary: An elliptic curve in characteristic p that is not supersingular; equivalently, one with $E[p](\overline{k}) \cong \mathbb{Z}/p\mathbb{Z}$. p. 80
Picard group: The group $\operatorname{Pic} C$ of divisors on a curve C modulo principal divisors. p. 36
place: For a global field k, an equivalence class of non-trivial absolute values on k. The set of places is $\mathcal{P}_k$. p. 85
point at infinity: The rational point $O = (0 : 1 : 0)$, the only $\overline{k}$-point of a Weierstrass cubic on the line at infinity, where it is a flex. p. 47
pointed set: A set with a distinguished element, such as the non-abelian $H^1(G, A)$. p. 112
potential good reduction: An elliptic curve over k_v has potential good reduction if it acquires good reduction over a finite extension of k_v; equivalently $j \in \mathcal{O}_v$. p. 93
principal divisor: The divisor $\operatorname{div}(f) = \sum_P \operatorname{ord}_P(f)P$ of a non-zero rational function. p. 36
profinite group: An inverse limit of finite groups, such as Gal_k: a compact, Hausdorff and totally disconnected topological group. p. 109
projective closure: The Zariski closure in $\mathbb{P}^n_k$ of a closed subvariety $X = V(I)$ of $\mathbb{A}^n_k = U_0$. It is cut out by the homogeneous ideal generated by the homogenizations of all elements of I. p. 28
projective plane: The space $\mathbb{P}^2(\mathbb{C}) = \big(\mathbb{C}^3 - \{(0,0,0)\}\big)/\mathbb{C}^\times$, with points written $(X : Y : Z)$. p. 21
projective plane curve: The zero locus $V_+(F) \subseteq \mathbb{P}^2_k$ of a non-constant homogeneous polynomial with no repeated factor over $\overline{k}$. p. 41
quadratic height: A height q on an abelian group satisfying the parallelogram law $q(P+Q)+q(P-Q) = 2q(P)+2q(Q)$. p. 134
quadratic twist: For $d \in k^\times$ and $\operatorname{char} k \neq 2$, the curve $E^d\colon y^2 = x^3 + d^2Ax + d^3B$ attached to $E\colon y^2 = x^3 + Ax + B$. It has the same j-invariant, and its isomorphism class depends only on d in $k^\times/k^{\times 2}$. p. 49
quasi-quadratic: A height h on an abelian group is quasi-quadratic if $h(P+Q)+h(P-Q)-2h(P)-2h(Q)$ is bounded. p. 134
quaternion algebra: An algebra over a field F with basis $1, i, j, ij$ and $i^2 = a$, $j^2 = b$, $ji = -ij$ for some $a, b \in F^\times$; written $(a, b \mid F)$. p. 66
ramification divisor: For a separable morphism φ, the divisor $\operatorname{div}(\varphi^*\omega) - \varphi^* \operatorname{div}(\omega)$. p. 40
ramification index: For a non-constant morphism of curves, $e_\varphi(P) = \operatorname{ord}_P(\varphi^* t)$ with t a uniformizer at $\varphi(P)$. p. 35
ramified Galois set: A set with an action of Gal_{k_v} on which the inertia group I_v acts non-trivially. p. 95
rank: The number of elements of a $\mathbb{Z}$-basis of a free abelian group. p. 13
rational map: For an integral variety X, a morphism to Y from a non-empty open subvariety of X, up to agreement on overlaps. p. 30
rational point: A point of a k-variety X with values in k, i.e. an element of $X(k) = \operatorname{Hom}_k(\operatorname{Spec} k, X)$. p. 25
reduced norm: The element $\operatorname{nrd}(\alpha) = \alpha\bar{\alpha}$ of F, for α in a quaternion algebra over F. It is multiplicative. p. 66
reduced trace: The element $\operatorname{trd}(\alpha) = \alpha + \bar{\alpha}$ of F, for α in a quaternion algebra over F. p. 66
reduced variety: A k-variety whose rings of sections have no non-zero nilpotents. p. 29
reduction map: The map $E(k_v) \to E_\pi(\kappa(v))$, $P \mapsto \bar{P}$, reducing primitive homogeneous coordinates modulo $\mathfrak{m}_v$. p. 89
reduction modulo π: For an elliptic curve E over k_v, the special fiber E_π of a minimal Weierstrass model, a Weierstrass cubic over $\kappa(v)$, well defined up to isomorphism. p. 88
regular differential: A differential ω that is zero or has $\operatorname{div}(\omega) \geqslant 0$. These form a k-vector space. p. 38
regular proper model: An integral model of an elliptic curve over k_v that is a regular integral scheme, flat and proper over $\mathcal{O}_v$. p. 99
regular variety: A variety whose local rings are all regular local rings. p. 31
regulator of a quadratic height: The determinant of the Gram matrix $(\langle P_i, P_j\rangle)$ of a positive definite canonical height, for elements P_i mapping to a basis of A/A_{tors}. p. 137

relative Frobenius morphism: For $C \subseteq \mathbb{P}^n_k$ with q a power of $\operatorname{char} k > 0$, the morphism $C \to C^{(q)}$ raising homogeneous coordinates to the q-th power, where $C^{(q)}$ is defined by the equations of C with coefficients raised to the q-th power. p. 35

residue field: The quotient of a local ring by its maximal ideal: $k(P)$ for a closed point P, and the finite field $\kappa(v) = \mathcal{O}_v/\mathfrak{m}_v$ for a non-archimedean place v. p. 85

restriction map: The map $\operatorname{Res}\colon H^i(G, A) \to H^i(H, A)$ restricting cocycles to a closed subgroup H. p. 113

Riemann sphere: The projective line $\mathbb{P}^1(\mathbb{C})$, a compact Riemann surface covered by two charts. p. 12

Riemann surface: A connected Hausdorff topological space with a complex structure. p. 12

ring of integers: For a non-archimedean place v, the ring $\mathcal{O}_v = \{x \in k_v : v(x) \geqslant 0\}$. p. 85

scheme: A locally ringed space covered by open subsets isomorphic to affine schemes $\operatorname{Spec} A$. p. 25

second countable: A topological space with a countable basis for its topology. p. 12

Segre embedding: The map $\mathbb{P}^N \times \mathbb{P}^M \to \mathbb{P}^{NM+N+M}$ sending $([x_i], [y_j])$ to $[x_i y_j]$. p. 145

Selmer elliptic curve: For $d \in k^\times$ and $\operatorname{char} k \neq 3$, the plane cubic $C_d\colon X^3 + Y^3 = dZ^3$ with base point $O = (1 : -1 : 0)$. p. 69

semistable reduction: Good or multiplicative reduction. p. 92

separated: A scheme X over S whose diagonal $X \to X \times_S X$ is a closed immersion. p. 25

simply connected: A connected topological space whose fundamental group is trivial. p. 13

smooth projective plane curve: The zero locus in $\mathbb{P}^2(\mathbb{C})$ of a nonsingular homogeneous polynomial. p. 21

smooth variety: A variety which, locally cut out of $\mathbb{A}^n_k$ with pure dimension d, has Jacobian matrix of rank $n - d$ at every $\overline{k}$-point; equivalently, $X_{\overline{k}}$ is regular. p. 31

special fiber: For an $\mathcal{O}_v$-scheme $\mathcal{X}$, the $\kappa(v)$-scheme $\mathcal{X}_\pi = \mathcal{X} \times_{\operatorname{Spec} \mathcal{O}_v} \operatorname{Spec} \kappa(v)$. p. 87

special point: The closed point $\mathfrak{m}_v$ of $\operatorname{Spec} \mathcal{O}_v$, its only one, with residue field $\kappa(v)$. p. 87

split multiplicative reduction: Multiplicative reduction in which the slopes of the tangent lines at the node lie in $\kappa(v)$. Then $E^{\mathrm{ns}}_\pi \cong \mathbb{G}_{\mathrm{m}}$. p. 92

split quaternion algebra: A quaternion algebra over F isomorphic to $M_2(F)$; equivalently, one that is not a division ring. p. 66

split torus: A torus isomorphic to $\mathbb{G}^n_{\mathrm{m}}$ over k. p. 27

standard involution: On a quaternion algebra, the conjugation $t + xi + yj + zij \mapsto t - xi - yj - zij$; in general, an F-linear anti-automorphism of order at most 2 with $\alpha\bar{\alpha} \in F$ for all α. p. 66

supersingular: An elliptic curve in characteristic p with $E[p](\overline{k}) = 0$; equivalently, one whose endomorphism ring over $\overline{k}$ is an order in a quaternion algebra. p. 80

Tamagawa number: $c_v = \#\Phi(\kappa(v)) = [E(k_v) : E_0(k_v)]$. p. 103

Tate module: For a prime $\ell \neq \operatorname{char} k$, the $\mathbb{Z}_\ell$-module $T_\ell(E) = \varprojlim_n E[\ell^n](k^{\mathrm{sep}})$, free of rank 2. p. 62

torsor: A k-variety with a right action of a smooth algebraic group G that becomes isomorphic to G with right translation over k^{sep}. p. 124

torus: A group variety T over k with $T_{\overline{k}} \cong (\mathbb{G}^n_{\mathrm{m}})_{\overline{k}}$ for some $n \geqslant 0$. p. 27

trace of Frobenius: For an elliptic curve $E/\mathbb{F}_q$, the integer $a = q + 1 - \#E(\mathbb{F}_q)$, which is the trace of Frobenius on $T_\ell(E)$. p. 76

transition map: For charts ϕ_1 and ϕ_2, the map $\phi_2 \circ \phi_1^{-1}\colon \phi_1(U_1 \cap U_2) \to \phi_2(U_1 \cap U_2)$. p. 11

translation-invariant differential: On $\mathbb{C}^\times$ or a complex torus, a meromorphic differential ω with $\tau_a^*\omega = \omega$ for every translation τ_a. p. 19

twist: A k-variety Y that becomes isomorphic to X over k^{sep}. p. 118

twisted cubic: The affine curve $V(x_2 - x_1^2, x_3 - x_1^3) \subseteq \mathbb{A}^3_k$, the image of $t \mapsto (t, t^2, t^3)$. p. 28

twisted multiplicative group: For the quadratic algebra L_d, the conic $\mathbb{G}^{(d)}_{\mathrm{m}}\colon \operatorname{Nm}(x + yt) = 1$, with $\mathbb{G}^{(d)}_{\mathrm{m}}(R)$ the norm-one units of $L_d \otimes_k R$; isomorphic to $\mathbb{G}_{\mathrm{m}}$ over L_d. p. 26

uniformizer: A function t with $\operatorname{ord}_P(t) = 1$ at a regular closed point P. p. 33

unit group: The group $\mathcal{O}_v^\times = \{x \in k_v : v(x) = 0\}$ of units of $\mathcal{O}_v$. p. 85

unramified at a place: A Galois extension F of a number field k is unramified at a non-archimedean place v if the inertia group I_v fixes F. p. 127

unramified class: A class in $H^1(k_v, A)$ whose restriction to the inertia group I_v is 0. p. 120

unramified extension: A finite extension of k_v with ramification index 1. p. 93

unramified Galois set: A set with an action of Gal_{k_v} on which the inertia group I_v acts trivially. p. 95

unramified outside: A class in $H^1(k, A)$ is unramified outside a finite set T of places if its restriction to k_v is unramified for every non-archimedean $v \notin T$. p. 121

upper half plane: The Riemann surface $\mathbb{H} = \{z \in \mathbb{C} : \mathrm{Im}(z) > 0\}$. p. 12

v-adic distance: For a k_v-point Q of a nice curve C and a function t_Q with a zero of order e at Q and no other zeros, the function $d_v(P, Q) = \min(|t_Q(P)|_v^{1/e}, 1)$ on $C(k_v)$. p. 152

valuation of the minimal discriminant: The value $v(\Delta)$ of a minimal Weierstrass equation for an elliptic curve over k_v. p. 86

Verschiebung: In characteristic p, the dual $\mathsf{V}^r = \widehat{\mathsf{F}^r}$ of the p^r-th power Frobenius isogeny $\mathsf{F}^r \colon E \to E^{(p^r)}$. p. 59

weak Mordell–Weil theorem: For an elliptic curve E over a number field k and $m \geqslant 2$, the group $E(k)/mE(k)$ is finite. p. 130

Weierstrass coordinates: Functions x, y on an elliptic curve with poles of order 2 and 3 at O only, giving an isomorphism onto a Weierstrass cubic. p. 47

Weierstrass equation: A projective plane cubic $Y^2Z + a_1XYZ + a_3YZ^2 = X^3 + a_2X^2Z + a_4XZ^2 + a_6Z^3$. p. 45

Weierstrass model: The closed subscheme of $\mathbb{P}^2_{\mathcal{O}_v}$ cut out by a Weierstrass equation with coefficients in $\mathcal{O}_v$; an integral model of the elliptic curve it defines over k_v. p. 87

Weierstrass $\wp$-function: For a lattice Λ, the meromorphic function $\wp_\Lambda(z) = z^{-2} + \sum_{\omega \in \Lambda - \{0\}} \left[(z-\omega)^{-2} - \omega^{-2}\right]$ on $\mathbb{C}$. p. 22

Weil pairing: For N not divisible by $\mathrm{char}\, k$, the bilinear, alternating, non-degenerate, Galois-equivariant pairing $e_N \colon E[N](k^{\mathrm{sep}}) \times E[N](k^{\mathrm{sep}}) \to \mu_N(k^{\mathrm{sep}})$. p. 64

Weil reciprocity: For non-zero rational functions f, g on a nice curve whose divisors have disjoint support, the identity $f(\mathrm{div}\, g) = g(\mathrm{div}\, f)$, where $f(D) = \prod_P f(P)^{n_P}$ for $D = \sum n_P P$. p. 37

zeta function: For a nice curve C over $\mathbb{F}_q$, the power series $Z(C/\mathbb{F}_q; T) = \exp\left(\sum_{n \geqslant 1} \#C(\mathbb{F}_{q^n}) T^n/n\right)$. p. 74

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